MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: Mansourah Massarah Gold Project

Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION

3210 FOUNDATION DESIGN OF CONVEYOR 3210-CNV-001

DESIGN CALCULATION

AS BUILT

MD-216-3210-EG-CV-CAL-1001

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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2007	Saudi Code for Concrete Structures

TABLE OF CONTENTS

DEFINITIONS AND ABBREVIATIONS 3

1 INTENT 4

2 EXECUTIVE SUMMARY 4

3 SCOPE 4

4 CONCRETE STRUCTURE 4

4.1 Design 4
4.2 Materials 4
4.3 Foundation of structures 4
4.4 Minimum concrete cover 4
4.5 Allowable soil bearing capacity 4
4.6 Reference 4

5 INPUT LOADS FROM SUPERSTRUCTURE 4-7

5.1 Support Reaction from TR-1 4
5.2 Support Reaction from TR-2 5
5.3 Support Reaction from TR-3 5
5.4 Support Reaction from VGTU 6
5.5 Load Combinations 7

6 GEOMETRY 7-8
6.1 Geometry of Ground Portals 7
6.2 Geometry of TR-1 & TR-2 Foundation 8
6.3 Geometry of TR-3 Foundation 8
6.4 Geometry of VGTU Foundation 8

7 DESIGN OF CONCRETE COMPONENTS 9-24

7.1 Design summary of Pedestals 9
7.2 Design summary of Tie Beams 9
7.3 Design summary of Foundations 9

ATTACHMENT 1 MECHANICAL GENERAL ARRANGEMENT DRAWING 25

ATTACHMENT 2 STRUCTURAL GENERAL ARRANGEMENT DRAWING CONTAINING BASE PLATE DETAIL 30

ATTACHMENT 3 STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR GP 32

ATTACHMENT 4 STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR TR-1 70

ATTACHMENT 5 STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR TR-2 100

ATTACHMENT 6 STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR TR-3 130

ATTACHMENT 7 STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR VGTU 160

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
RL	:	Reduced Level
C.L.	:	Centre Line
CJ	:	Construction Joint
FDN	:	Foundation
TOC	:	Top of concrete
FGL	:	Finished Ground Level
FFL	:	Finished Floor Level
BOF	:	Bottom of footing
TYP	:	Typical
TOC	:	Top of concrete
IP	:	Insert Plate
RCC	:	Reinforced Cement Concrete
MKD.	:	Marked
THK.	:	Thick

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1. INTENT

This document is prepared for Concrete design calculation of Conveyor Gallery 3210-CNV-001.
This conveyor is between Crushed Ore Storage Bin and Ball Mill and belt width of the conveyor is 1200 mm.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

This document is prepared for foundation design of Conveyor 3210-CNV-001. All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
The column and beam reinforcement design have been performed in Staad-Pro Software. The footings of Ground Portal GR-1 to GR-4 & VGTU have been performed in MS Excel.
Foundation of TR-1, TR-2 & TR-3 has been designed by FEM analysis in Staad-Pro Software.

4. CONCRETE STRUCTURE

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 and ACI 318M.

4.2 Materials

Concrete Strength of 28 days
· Foundation and Structural Members: 35 MPa
· Grade slab and pavement: 30 MPa
· Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade
· Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
· Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundations for Structures

Foundation shall be Isolated and combined Shallow Concrete foundation. The design calculations are prepared as per recommendation of standard SBC-304 Clause 13.3.

4.4 Minimum Concrete Cover

Items	Minimum Cover(mm)
Foundation (Side & Bottom)	75
Foundation (Top)	50
Column & Pedestal	50
Beam	50
Slab	40

4.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 500 kN/m2 at 1m & 3m depth below FGL (ie. RL 911.500m & RL 909.500m respectively) refer Table 4 - BH-17,18,19,25,29 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.6 Reference

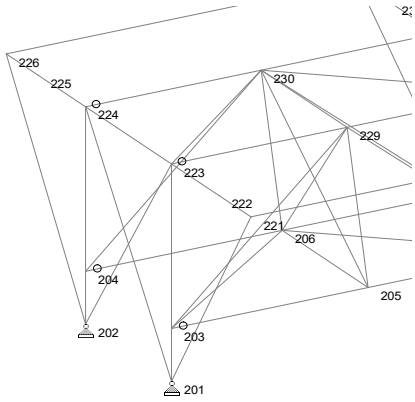
Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

5 INPUT LOADS FROM SUPERSTRUCTURE

For load calculation of superstructure refer Document No: MD-216-3210-EG-ST-CAL-1001

5.1 Support Reaction from TR-1

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
201	1 DL	-0.2	90.7	-12.1
	2 LL	0.1	33.4	-0.8
	3 ML - CONVEYOR MATERI	0.0	25.8	2.3
	4 WL (+Z)	0.5	-32.1	-20.7
	5 WL (-Z)	-0.5	32.1	20.8
	6 TL1	0.0	-6.3	-3.2
	7 TL2	0.0	6.3	3.2
	8 EARTHQUAKE LOAD (+X	-1.1	0.1	0.0
	9 EARTHQUAKE LOAD (+Z	0.0	-0.9	-0.5
202	1 DL	0.2	27.9	8.1
	2 LL	-0.1	40.6	0.8
	3 ML - CONVEYOR MATERI	0.0	25.7	-2.3
	4 WL (+Z)	-0.5	32.1	-17.1
	5 WL (-Z)	0.5	-32.1	17.1
	6 TL1	0.0	6.3	-2.9
	7 TL2	0.0	-6.3	2.9
	8 EARTHQUAKE LOAD (+X	-1.1	0.4	-0.1
	9 EARTHQUAKE LOAD (+Z	0.0	0.9	-0.5



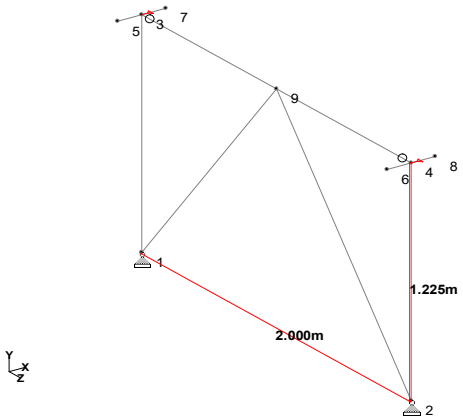
Staad model of 12m Gallery End portal which is acting as TR1

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5.2 Support Reaction from TR-2

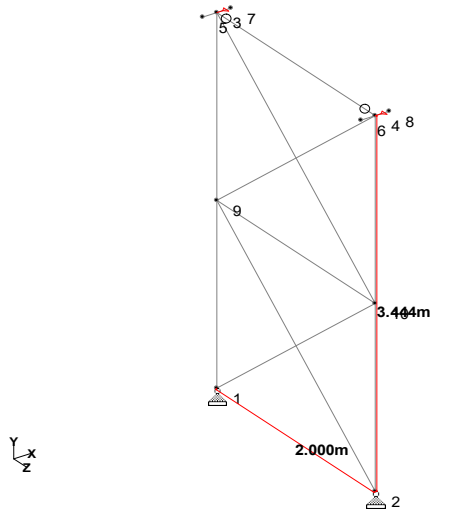
Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
1	1 DEAD LOAD	-3.5	186.8	-4.6
	2 LIVE LOAD	-2.1	60.9	0.1
	3 MATERIAL LOAD	-0.7	41.6	0.0
	4 WL (+Z)	1.6	-84.5	-29.5
	5 WL (-Z)	-1.6	84.5	29.5
	6 WL (+X)	-0.4	0.0	0.0
	7 WL (-X)	0.4	0.0	0.0
	8 TL1	0.0	0.1	0.1
	9 TL2	0.0	-0.1	-0.1
	10 EL (+X)	0.0	0.0	0.0
	11 EL (+Z)	0.0	-6.4	-3.5
2	1 DEAD LOAD	-2.0	54.6	-5.0
	2 LIVE LOAD	-0.4	58.8	0.0
	3 MATERIAL LOAD	-0.8	42.2	0.0
	4 WL (+Z)	-1.6	84.5	-29.5
	5 WL (-Z)	1.6	-84.5	29.5
	6 WL (+X)	-0.4	0.0	0.0
	7 WL (-X)	0.4	0.0	0.0
	8 TL1	0.0	-0.1	-0.1
	9 TL2	0.0	0.1	0.1
	10 EL (+X)	-0.1	0.1	0.0
	11 EL (+Z)	0.0	6.4	-3.5



Staad model of TR2 superstructure

5.3 Support Reaction from TR-3

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
11	1 DEAD LOAD	-0.9	245.6	0.1
	2 LIVE LOAD	-0.2	89.9	3.1
	3 MATERIAL LOAD	-0.3	56.8	2.2
	4 WL (+Z)	0.9	-227.6	-43.0
	5 WL (-Z)	-0.9	227.6	43.0
	6 WL (+X)	-1.0	0.0	0.0
	7 WL (-X)	1.0	0.0	0.0
	8 TL1	-0.3	7.2	16.6
	9 TL2	0.3	-7.2	-16.6
	10 EL (+X)	0.0	-0.1	0.0
	11 EL (+Z)	0.0	-7.1	-1.2
14	1 DEAD LOAD	-0.6	94.5	-13.0
	2 LIVE LOAD	-0.7	78.1	-3.2
	3 MATERIAL LOAD	-0.3	56.4	-2.2
	4 WL (+Z)	-0.9	227.6	-43.5
	5 WL (-Z)	0.9	-227.6	43.6
	6 WL (+X)	-1.0	0.0	0.0
	7 WL (-X)	1.0	0.0	0.0
	8 TL1	0.3	-7.2	-16.6
	9 TL2	-0.3	7.2	16.6
	10 EL (+X)	0.0	0.3	0.0
	11 EL (+Z)	0.0	7.1	-1.2



Staad model of TR3 superstructure

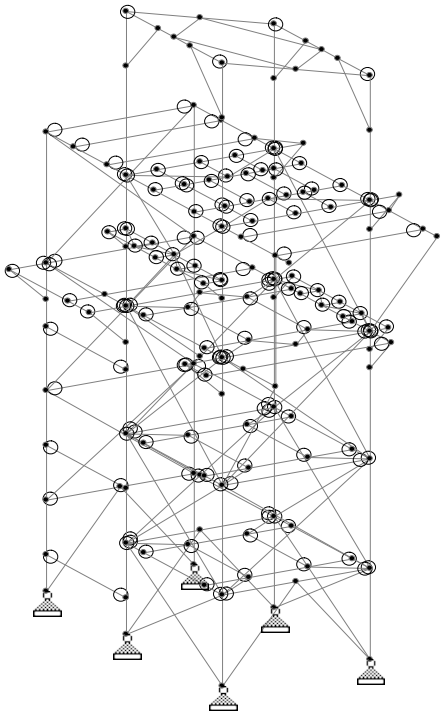
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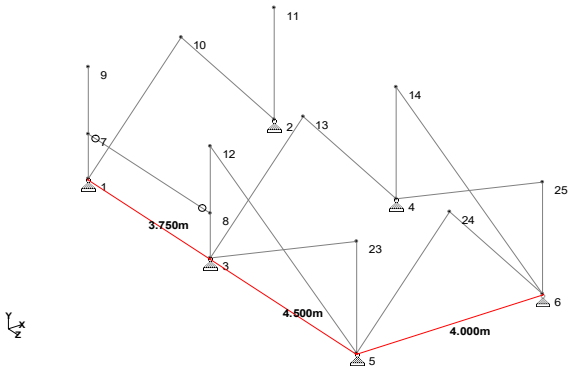
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5.4 Support Reaction from VGTU

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
1	1 DEAD LOAD	0.3	33.1	0.1
	2 LIVE LOAD	0.0	59.2	0.1
	3 MATERIAL LOAD	0.0	0.0	0.0
	4 WL (+Z)	0.0	-0.4	-0.6
	5 WL (-Z)	0.0	0.4	0.6
	6 WL (+X)	-13.9	-43.8	-0.1
	7 WL (-X)	13.9	43.8	0.1
	8 TL1	0.1	0.0	2.7
	9 TL2	-0.1	0.0	-2.7
	10 EL (+X)	-0.1	-0.3	0.0
	11 EL (+Z)	0.0	0.0	0.0
2	1 DEAD LOAD	-0.3	31.5	0.0
	2 LIVE LOAD	0.0	60.9	0.0
	3 MATERIAL LOAD	0.0	0.0	0.0
	4 WL (+Z)	0.0	-0.3	-0.9
	5 WL (-Z)	0.0	0.3	0.9
	6 WL (+X)	-13.9	43.9	0.0
	7 WL (-X)	13.9	-43.9	0.0
	8 TL1	0.0	0.0	1.4
	9 TL2	0.0	0.0	-1.4
	10 EL (+X)	-0.1	0.3	0.0
	11 EL (+Z)	0.0	0.0	0.0
3	1 DEAD LOAD	20.4	251.3	10.6
	2 LIVE LOAD	1.0	182.5	10.3
	3 MATERIAL LOAD	0.1	21.5	1.3
	4 WL (+Z)	-1.5	-191.2	-43.4
	5 WL (-Z)	1.5	191.2	43.4
	6 WL (+X)	-50.6	-232.0	-9.3
	7 WL (-X)	50.6	232.0	9.3
	8 TL1	0.3	-10.1	34.4
	9 TL2	-0.3	10.1	-34.4
	10 EL (+X)	-1.8	-8.7	-0.3
	11 EL (+Z)	-0.1	-9.1	-1.7
4	1 DEAD LOAD	19.0	94.5	2.6
	2 LIVE LOAD	0.0	190.9	10.5
	3 MATERIAL LOAD	0.1	32.3	1.8
	4 WL (+Z)	-0.7	-202.9	-48.6
	5 WL (-Z)	0.7	202.9	48.6
	6 WL (+X)	-51.1	231.7	8.6
	7 WL (-X)	51.1	-231.7	-8.6
	8 TL1	-2.3	7.1	38.0
	9 TL2	2.3	-7.1	-38.0
	10 EL (+X)	-1.8	8.5	0.3
	11 EL (+Z)	0.0	-9.0	-2.0
5	1 DEAD LOAD	21.1	254.9	-15.7
	2 LIVE LOAD	0.0	164.3	-9.3
	3 MATERIAL LOAD	-0.1	19.8	-1.2
	4 WL (+Z)	1.5	191.6	-43.4
	5 WL (-Z)	-1.5	-191.6	43.4
	6 WL (+X)	-53.4	-260.5	12.3
	7 WL (-X)	53.4	260.5	-12.3
	8 TL1	2.2	10.1	-39.1
	9 TL2	-2.2	-10.1	39.1
	10 EL (+X)	-1.7	-8.7	0.4
	11 EL (+Z)	0.1	9.1	-1.7
6	1 DEAD LOAD	19.5	70.5	-9.8
	2 LIVE LOAD	-0.9	191.6	-11.5
	3 MATERIAL LOAD	-0.1	34.1	-2.0
	4 WL (+Z)	0.7	203.2	-49.4
	5 WL (-Z)	-0.7	-203.2	49.4
	6 WL (+X)	-53.6	260.7	-11.5
	7 WL (-X)	53.6	-260.7	11.5
	8 TL1	-0.2	-7.1	-37.4
	9 TL2	0.2	7.1	37.4
	10 EL (+X)	-1.7	8.8	-0.4
	11 EL (+Z)	0.0	9.0	-2.0



Staad model of VGTU superstructure



Node number and orientation of VGTU superstructure

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5.5 LOAD COMBINATIONS

5.5.1 Load Combination for Base pressure calculation

(As per Section 2.4 of SBC 301 -2007)

Load Cases	Reference
LOAD COMB 301 1.0 DL	(Eq. 2.4.1-1 of SBC 301)
LOAD COMB 302 1.0 (DL + TL1) + 1.0 (LL+ML)	(Eq. 2.4.1-2 of SBC 301)
LOAD COMB 303 1.0 (DL + TL1) + 1.0 (LL+ML)	
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75 (LL+ML)	(Eq. 2.4.1-6 of SBC 301)
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75 (LL+ML)	
LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75 (LL+ML)	
LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75 (LL+ML)	
LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75 (LL+ML)	
LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75 (LL+ML)	
LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75 (LL+ML)	
LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75 (LL+ML)	
LOAD COMB 312 0.6 DL + 1.0 WL(+Z)	(Eq. 2.4.1-7 of SBC 301)
LOAD COMB 313 0.6 DL + 1.0 WL(-Z)	
LOAD COMB 314 0.6 DL + 1.0 WL(+X)	
LOAD COMB 315 0.6 DL + 1.0 WL(-X)	
LOAD COMB 316 0.6 DL + 0.7 EL(+X)	(Eq. 2.4.1-8 of SBC 301)
LOAD COMB 317 0.6 DL + 0.7 EL(-X)	
LOAD COMB 318 0.6 DL + 0.7 EL(+Z)	
LOAD COMB 319 0.6 DL + 0.7 EL(-Z)	

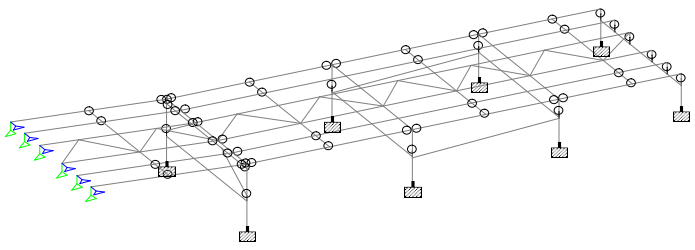
5.5.2 Load Combination for Concrete design

Load Case	Reference
LOAD COMB 401 1.4 DL	(Eq. 2.3.2-1 of SBC 301)
LOAD COMB 402 1.4 (DL + TL1) + 1.7 (LL+ML)	(Eq. 2.3.2-2 of SBC 301)
LOAD COMB 403 1.4 (DL + TL1) + 1.7 (LL+ML)	
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5 (LL+ML)	(Eq. 2.3.2-4 of SBC 301)
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5 (LL+ML)	
LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5 (LL+ML)	
LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5 (LL+ML)	
LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5 (LL+ML)	(Eq. 2.3.2-5 of SBC 301)
LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5 (LL+ML)	
LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5 (LL+ML)	
LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5 (LL+ML)	
LOAD COMB 412 0.9 DL + 1.6 WL(+Z)	(Eq. 2.3.2-6 of SBC 301)
LOAD COMB 413 0.9 DL + 1.6 WL(-Z)	
LOAD COMB 414 0.9 DL + 1.6 WL(+X)	
LOAD COMB 415 0.9 DL + 1.6 WL(-X)	
LOAD COMB 416 0.9 DL + 1.0 EL(+X)	(Eq. 2.3.2-7 of SBC 301)
LOAD COMB 417 0.9 DL + 1.0 EL(-X)	
LOAD COMB 418 0.9 DL + 1.0 EL(+Z)	
LOAD COMB 419 0.9 DL + 1.0 EL(-Z)	

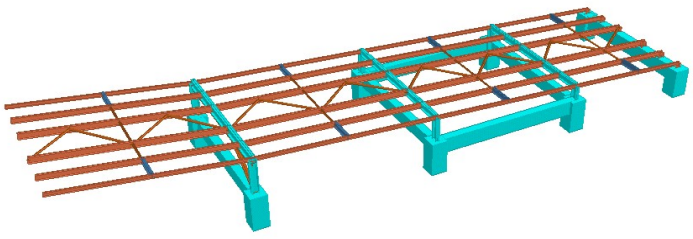
Eq. 2.3.2-3 of SBC 301 - 2007 has not been considered as it would not produce worst load combination in the absence of Roof Live Load & Rain Load

6 GEOMETRY

6.1 Geometry of Ground Portals



Staad model of Ground Portals

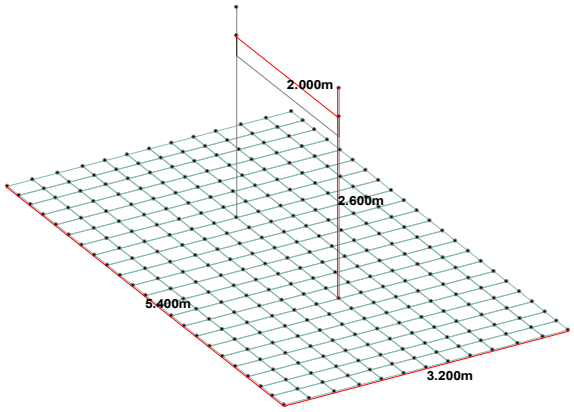


Isometric view of Ground Portals

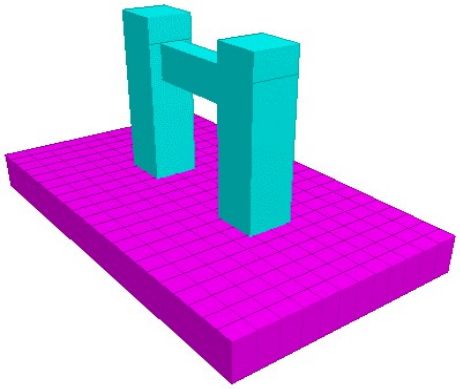
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
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Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
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Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

6.2 Geometry of TR-1 & TR-2 Foundation

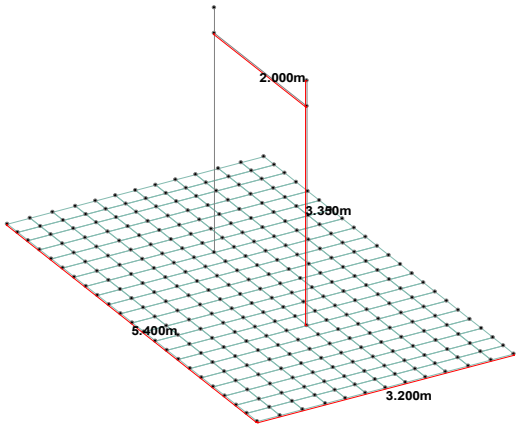


Staad Model of TR1 & TR2 Foundation mkd F2

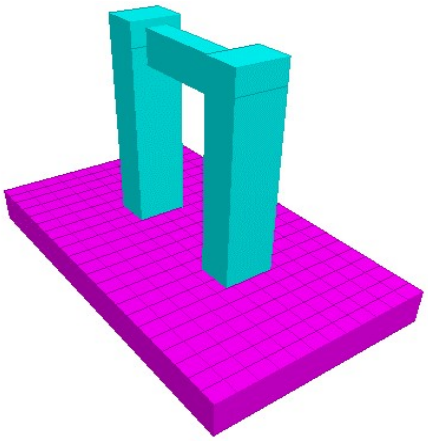


Isometric view of TR1 & TR2 Foundation mkd F2

6.3 Geometry of TR-3 Foundation

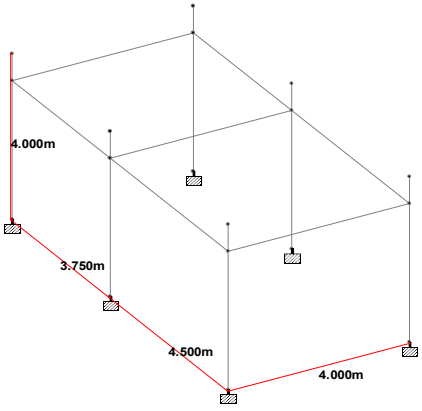


Staad Model of TR3 Foundation mkd F3

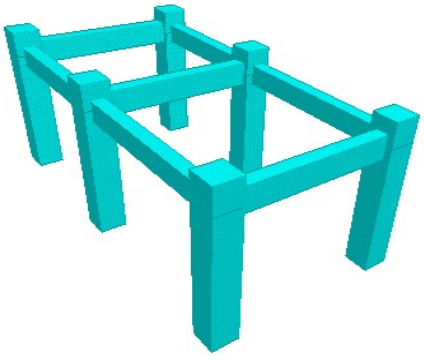


Isometric view of of TR3 Foundation mkd F3

6.4 Geometry of VGTU Foundation



Staad Model of VGTU Foundation



Isometric view of VGTU foundation

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Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

7. DESIGN OF CONCRETE COMPONENTS

All foundations have been designed using worksheet in accordance to SBC-304.

All beams & columns have been designed in Staad-Pro by ACI 318M.

For Design Results, Refer Attachemnt 3 to 8 - Staad Output file.

7.1 Design Summary of Pedestals

7.1.1 Column Reinforcement Summary

Column mark	Width (Along conv)	Depth (Across conv)	Reinforcement	Stirrup	Remarks	Numbers
P1 & P1X	600	600	12 No.- D20	3 Nos D10 @ 200c/c	GP1	2
P2	600	600	12 No.- D20	3 Nos D10 @ 200c/c	GP2	2
P3	600	600	12 No.- D20	3 Nos D10 @ 200c/c	GP3	2
P4	600	600	12 No.- D20	3 Nos D10 @ 200c/c	GP4	2
P5	600	600	12 No.- D20	3 Nos D10 @ 200c/c	TR1	2
P6	700	500	12 No.- D20	3 Nos D10 @ 200c/c	TR2	2
P7	700	500	12 No.- D20	3 Nos D10 @ 200c/c	TR3	2
P8	600	600	12 No.- D20	3 Nos D10 @ 200c/c	VGTU Grid A-1, A-2, B-1, B-2	4
P9	600	600	12 No.- D20	3 Nos D10 @ 200c/c	VGTU A-3, B-3	2
P10	450	450	8 No.- D20	2 Nos D10 @ 200c/c	VGTU Post	2

7.2 Design Summary of Tie Beam

7.2.1 Beam Reinforcement Summary

Beam mark	Width (mm)	Depth (mm)	Top Reinforcement	Bottom Reinforcement	Stirrup	Numbers
TB1	350	600	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	1
TB2	300	500	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	3
TB3	300	600	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	2
TB4	300	500	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	2
TB5	300	500	3 Nos D20	3 Nos D20	2L D10 STRPS @ 200c/c	1
TB6	300	500	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	4
TB7	300	600	3 Nos D20	3 Nos D20	2L D10 STRPS @ 200c/c	3

7.3 Design of Foundation

7.3.1 Foundation marking, Max/min Base pressure and Reinforcement Summary

Structure Name	Foundation mark	Length (Along conv) (mm)	Width (Across conv) (mm)	Thickness (mm)	Max pressure (kN/m^2)	Top reinforcement	Bottom reinforcement	F.O.S.	Numbers
GP1	F1	1500	2000	300	78.3	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.15	2
GP2	**F1**	**1500**	**2000**	**300**	**105**	**D12 @ 200c/c B/W**	**D16 @ 200c/c B/W**	**0.20**	**2**
GP3	F1	1500	2000	300	101	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.19	2
GP4	F1	1500	2000	300	64.5	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.12	2
TR1	F2	3200	5400	550	73.5	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.13	1
TR2	**F2**	**3200**	**5400**	**550**	**90**	**D12 @ 200c/c B/W**	**D16 @ 200c/c B/W**	**0.16**	**1**
TR3	**F3**	**3200**	**5400**	**550**	**125**	**D12 @ 200c/c B/W**	**D16 @ 200c/c B/W**	**0.23**	**1**
VGTU Grid A-1	**F4**	**3000**	**3500**	**450**	**152**	**D12 @ 150c/c B/W**	**D16 @ 150c/c B/W**	**0.27**	**1**
VGTU Grid A-2	F4	3000	3500	450	145	D12 @ 150c/c B/W	D16 @ 150c/c B/W	0.26	1
VGTU Grid A-3	**F5**	**2200**	**2200**	**450**	**123**	**D12 @ 200c/c B/W**	**D16 @ 200c/c B/W**	**0.22**	**1**
VGTU Grid B-1	F4	3000	3500	450	118	D12 @ 150c/c B/W	D16 @ 150c/c B/W	0.21	1
VGTU Grid B-2	F4	3000	3500	450	112	D12 @ 150c/c B/W	D16 @ 150c/c B/W	0.20	1
VGTU Grid B-3	F5	2200	2200	450	120	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.22	1

The design of the most critical foundation of the group is attached in the following pages.

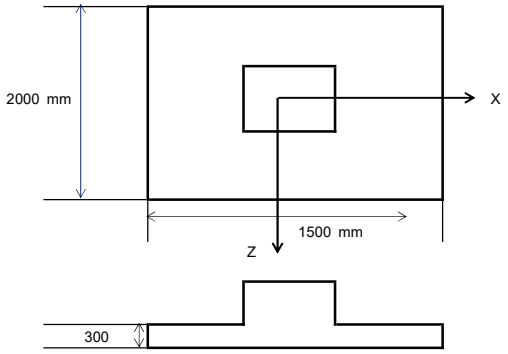
Global Project ID	**900707**
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Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
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Ma'aden Revision No:	A01	Outotec / L&T Revision	0
Date	17-04-2020	Date	17-04-2020
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Checked	SBY	Checked	SBY
Approved	PRI	Approved	PRI

7.3.2 Design of Footing mkd. F1 for supporting ground portal GP2

Finished Ground level	F.G.L	=	-0.500	m	Staad Node no	=	77 & 78
Bottom of foundation	B.O.F	=	-1.500	m	Weight of soil	=	33.26 KN
Staad support reaction level		=	-1.500	m	Weight of foundation	=	21.60 KN
Ground Water Level	G.W.L	=	-10.000	m	Uplift	=	0.00 KN
Surcharge	Ws	=	0.000	KN/m^2	Surcharge	=	0 KN
Length of footing	Lz	=	2.00	m	Pressure on top for tension	=	18.3 KN/m^2
Width of footing	Lx	=	1.50	m			
Thickness of footing	D	=	0.30	m			
Depth of pedestal	Bx	=	0.60	m			
Width of pedestal	Bz	=	0.60	m			
Height of Load point from bottom of footing	H	=	0.000	m			
Section Modulus of Pedestal	Zxx	=	1.00	m^3			
Section Modulus of Pedestal	Zzz	=	0.75	m^3			
Allowable Net Bearing Capacity		=	500	KN/m^2			
Gross Bearing Capacity		=	518	KN/m^2			
Grade of concrete		=	35	N/mm^2			
Grade of steel		=	420	N/mm^2			
Clear cover		=	75	mm			
Top Reinfocement		=	12	mm			
Bottom Reinforcement		=	16	mm			
Co-efficient of friction between soil & concrete		=	0.3				
Unit wt of soil		=	18	KN/m^3			
Unit wt of concrete		=	24	KN/m^3			

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301 1.0 DL	8.7	62.532	-9.2	-8.23	0.0	3.0	117.4	51.43	26.84	1	0.10
302 1.0 (DL + TL1) -	8.1	97.401	-13.9	-30.22	0.005	4.0	152.3	86.31	15.20	1	0.17
303 1.0 (DL + TL2) -	9.1	98.848	-18.4	-28.02	0.09	2.5	153.7	82.53	19.94	1	0.16
304 1.0 DL+ 0.75 W	8.6	85.452	-5.1	-13.06	0.017	3.2	140.3	64.05	29.49	1	0.12
305 1.0 DL+ 0.75 W	8.6	93	-23.7	-34.74	0.064	3.2	147.9	88.29	10.28	1	0.17
306 0.6 DL + 1.0 WL	5.2	32.487	6.9	9.51	0.0	1.8	87.4	41.04	17.19	1	0.08
307 0.6 DL + 1.0 WL	5.2	42.551	-17.9	-19.39	0.043	1.8	97.4	54.33	10.62	1	0.10
301 1.0 DL	8.7	65.82	13.6	13.45	-0.015	3.1	120.7	57.79	22.67	1	0.11
302 1.0 (DL + TL1) -	8.1	102.862	25.4	43.44	-0.143	4.1	157.7	101.42	3.73	1	0.20
303 1.0 (DL + TL2) -	9.1	100.042	16.1	25.56	0.054	2.5	154.9	80.51	22.76	1	0.16
304 1.0 DL+ 0.75 W	8.6	96.324	28.3	40.12	-0.1	3.2	151.2	94.83	5.96	1	0.18
305 1.0 DL+ 0.75 W	8.6	88.763	9.6	18.35	-0.013	3.2	143.6	70.51	25.24	1	0.14
306 0.6 DL + 1.0 WL	5.2	44.533	20.6	22.58	-0.041	1.9	99.4	58.20	8.06	1	0.11
307 0.6 DL + 1.0 WL	5.2	34.451	-4.3	-6.44	0.0	1.8	89.3	38.66	20.88	1	0.07

Max P_{max} = 101.42 KN/m^2
Max P_{min} = 3.73 KN/m^2
Resultant F.O.S = 0.20

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401 1.4 DL	12.2	87.5	-12.8	-11.53	0.0	4.26	46.39	11.97
402 1.4 (DL + TL1) -	11.3	147.0	-21.6	-48.58	0.015	5.66	105.13	-7.11
403 1.4 (DL + TL2) -	12.8	149.1	-27.8	-45.49	0.134	3.50	99.84	-0.46
404 1.2 DL+ 1.6 WL	10.4	84.8	5.4	2.80	-0.013	3.72	36.02	20.50
405 1.2 DL+ 1.6 WL	10.4	100.9	-34.4	-43.45	0.087	3.78	82.12	-14.86
406 0.9 DL + 1.6 WL	7.8	48.2	11.6	15.72	-0.033	2.71	35.41	-3.25
407 0.9 DL + 1.6 WL	7.8	64.3	-28.1	-30.54	0.067	2.77	55.68	-12.79
401 1.4 DL	12.1	92.1	19.1	18.83	-0.021	4.32	55.30	6.13
402 1.4 (DL + TL1) -	11.3	154.7	37.7	67.13	-0.208	5.73	126.34	-23.21
403 1.4 (DL + TL2) -	12.8	150.7	24.7	42.10	0.066	3.54	97.06	3.44
404 1.2 DL+ 1.6 WL	10.4	104.9	39.8	49.88	-0.084	3.82	89.93	-20.02
405 1.2 DL+ 1.6 WL	10.4	88.7	0.0	3.45	0.018	3.77	38.05	21.11
406 0.9 DL + 1.6 WL	7.8	67.3	32.2	35.32	-0.065	2.80	61.49	-16.62
407 0.9 DL + 1.6 WL	7.8	51.2	-7.6	-11.12	0.037	2.75	31.84	2.27

Max Net factored B.P = 126.34 KN/m^2

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DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.45 m

Moment about Z: Muz = 12.79 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uzlim} = 321.51 KNm

As $M_{uzlim} > M_{uz}$ — Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 156.6 mm^2

Length of cantilever along Z = 0.7 m

Moment about X: Mux = 30.95 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uxlim} = 275.84 KNm

As $M_{uxlim} > M_{ux}$ — Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 412.9 mm^2

Provide Bottom Reinforcement = ***16 mm dia @ 200 c/c***

Effective Depth Along X = 217 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or

A_{stmin} = 0.0014bd

A_{stmin} = 390.6 mm^2

A_{req} = 156.6 mm^2

Required Spacing = **514.8** mm

Effective Depth Along X = 201 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or

A_{stmin} = 0.0014bd

A_{stmin} = 361.8 mm^2

A_{req} = 412.9 mm^2

Required Spacing = **487.0** mm

Ast prov = 1005.3 mm^2

Top Reinforcement

Length of cantilever along X = 0.45 m

Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uzlim} = 327.46 KNm

As $M_{uzlim} > M_{uz}$ — Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 0.0 mm^2

Length of cantilever along Z = 0.7 m

Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uxlim} = 292.56 KNm

As $M_{uxlim} > M_{ux}$ — Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 0.0 mm^2

Provide top Reinforcement = ***12 mm dia @ 200 c/c***

Effective Depth Along X = 219 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

or

A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 197.1 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **573.8** mm

Effective Depth Along Z = 207 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

or

A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

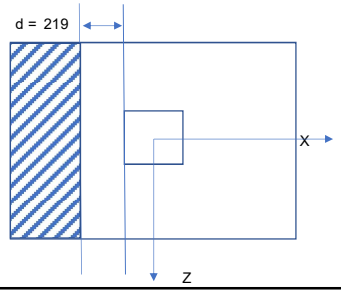
A_{stmin} = 186.3 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **607.1** mm

Ast prov = 565.5 mm^2

ONE WAY SHEAR CKECK



Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01		Outotec / L&T Revision	0
Date	17-04-2020		Date	17-04-2020
Revision Text	FIRST ISSUE		Revision Text	FIRST ISSUE
Prepared By	APL		Prepared By	APL
Checked	SBY		Checked	SBY
Approved	PRI		Approved	PRI

Along X Direction Cantilever

L	=	0.231	m		Least of
Reinforcement ratio	ρ_w	=	0.0046		
Effective Depth Along X	d	=	219	mm	
Shear Force at d distance	V_u	=	29.18	KN	
Allowable shear strength	φVc	=	168.29	KN	O.K
Utility ratio		=	0.17		

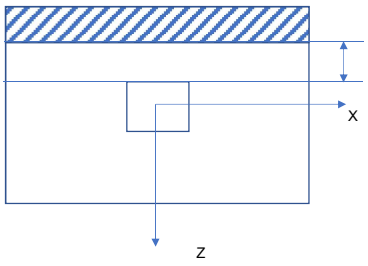
i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete; Mu = 3.37 KNm; φ = 0.75 for shear; φVc = 179.8 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$; φVc = 168.3 KN — *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$ = 281.8 KN

ONE WAY SHEAR CKECK



Along Z Direction Cantilever

L	=	0.493	m		Least of
Reinforcement ratio	ρ_w	=	0.0049		
Effective Depth Along Z	d	=	207	mm	
Shear Force at d distance	V_u	=	62.29	KN	
Allowable shear strength	φVc	=	157.72	KN	O.K
Utility ratio		=	0.39		

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

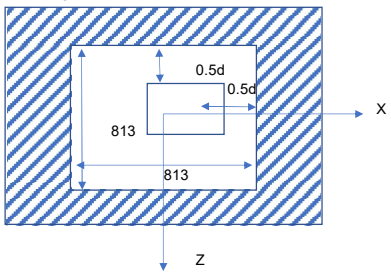
λ = 1 Normal concrete; Mu = 15.35 KNm; φ = 0.75 for shear; φVc = 157.7 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$; φVc = 159.8 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$; φVc = 266.4 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area	=	3	m2
Punching area	=	0.66	m2
λ	=	1	
β	=	1	
αs	=	40	
b0	=	3252.0	mm

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 3.02 N/mm²

$v_c = 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 2.27 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	3.25	m	
Depth at section	d	=	213	mm	
Shear Force at section	Vu	=	295.5	KN	
Allowable shear strength	φVc	=	1014.24	KN	O.K
Utility ratio			0.29		

Vc = $v_c b_0 d$ = 1352.32 KN

φ = 0.75 for shear

φV_n = φVc

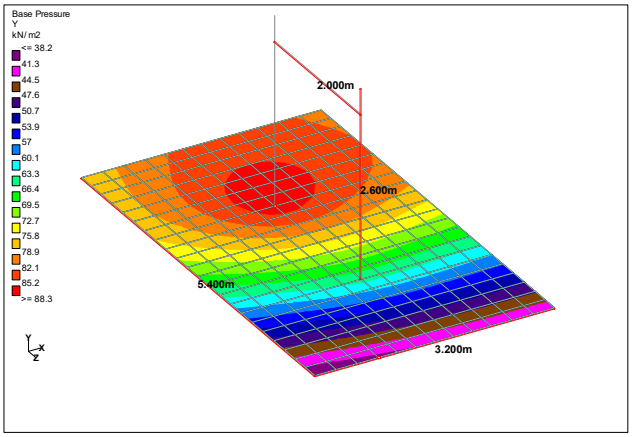
φVc = 1014.2 KN

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	17-04-2020	19-05-2020	02-11-2023	Date	17-04-2020	19-05-2020	02-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK/APL	RPK/APL	Prepared By	RPK	RPK/APL	RPK/APL
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

7.3.3 Design of combined footing mkd. as F2 for Trestle TR2

For Calculation of Base pressure and design forces for column & foundation Raft, Refer STAAD File **3210-CNV-001-TR2-Fdn.STD**



Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +913.500m)
Finished Ground level	F.G.L	=	-1.25	m	(Corresponds to RL +912.250m)
Bottom of foundation	B.O.F	=	-3.50	m	
Top of pedestal (staad support)	T.O.F	=	-0.90	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	3.200	m	
Width of footing	Lx	=	5.400	m	
Thickness of footing	D	=	0.55	m	
Depth of pedestal	Bx	=	0.60	m	
Width of pedestal	Bz	=	0.60	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	540.5	KN/m^2	
Soil load on footing in staad		=	30.6	KN/m^2	staad input

1) Check for Gross Base Pressure :

Maximum Base Pressure		=	88.3	KN/m^2	Load Case 305
Minimum Base Pressure		=	10.3	KN/m^2	Load Case 313

2) Design of foundation Raft :

Slab Thickness =	550 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	75 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75

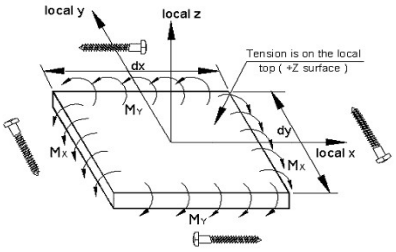
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	10.670	kN-m	Corresponding Mxy	3.330	kN-m	*(Plate No. - 24 , L/C No. - 413)*
Min. Mx	-49.930	kN-m	Corresponding Mxy	-5.660	kN-m	*(Plate No. - 99 , L/C No. - 403)*
Max. My	33.750	kN-m	Corresponding Mxy	2.630	kN-m	*(Plate No. - 25 , L/C No. - 413)*
Min. My	-74.080	kN-m	Corresponding Mxy	2.340	kN-m	*(Plate No. - 114 , L/C No. - 405)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	10.67 + 3.33 =	14.000	kN-m
My1 =	My + abs(Mxy) =	33.75 + 2.63 =	36.380	kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	10.67 + 0.329 =	10.999	kN-m
My2 =	My+abs(Mxy^2/Mx) =	33.75 + 0.478 =	34.228	kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd =	14 kN-m
Myd =	36.38 kN-m

ALONG LOCAL X - DIRECTION :

Mu =	14 kN-m						
Φ*Kn =	Mu/bd^2				deff. =	469	mm
Mu/bd^2 =	14X10^6/(1000X469^2) =	0.064	N/mm^2	<	6.9	N/mm^2	
Pt =	0.0002	Reqd. Ast =	0.0002X1000X469/100 =	0.94	cm^2		
Provided	12 mm @	200 C/C	(Provide Ast =	5.65	cm^2)		
			Total Provide Ast =	5.65	cm^2	O.K.	
			Total Provide Pt =	0.0012			

ALONG LOCAL Y - DIRECTION :

Mu =	36.38 kN-m						
Φ*Kn =	Mu/bd^2				deff. =	457	mm
Mu/bd^2 =	36.38X10^6/(1000X457^2) =	0.165	N/mm^2	<	6.9	N/mm^2	
Pt =	0.0004	Reqd. Ast =	0.0004X1000X469/100 =	1.83	cm^2		
Provided	12 mm @	200 C/C	(Provide Ast =	5.65	cm^2)		
			Total Provide Ast =	5.65	cm^2	O.K.	
			Total Provide Pt =	0.0012			

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	17-04-2020	19-05-2020	02-11-2023	Date	17-04-2020	19-05-2020	02-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK/APL	RPK/APL	Prepared By	RPK	RPK/APL	RPK/APL
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -49.93 - 5.66 = -55.59 kN-m
My1 = My - abs(Mxy) = -74.08 - 2.34 = -76.42 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -49.93 - 0.432 = -50.362 kN-m
My2 = My-abs(Mxy^2/Mx) = -74.08 - 0.11 = -73.97 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -55.59 kN-m
Myd = -76.42 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 55.59 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 55.59X10^6/(1000X467^2) = 0.255 N/mm^2 deff. = 467 mm < 6.9 N/mm^2
Pt = 0.0007 Reqd. Ast = 0.0007X1000X467/100 = 3.27 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0022

ALONG LOCAL Y - DIRECTION :

Mu = 76.42 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 76.42X10^6/(1000X451^2) = 0.376 N/mm^2 deff. = 451 mm < 6.9 N/mm^2
Pt = 0.0010 Reqd. Ast = 0.001X1000X451/100 = 4.51 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0022

SHEAR CHECK

(Staad Group : SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = 0.12 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 247 , L/C No. - 405)*
Vu = 56.04 kN

Provided Pt = 0.0022
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0022X56.04X10^3X467/55.59/10^6)X1000X467/10^3
= 450.272 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0022)X1000X467/10^3
= 459.515 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X467/10^3
= 801.215 kN
HENCE, Vc = 450.272 kN
Φt*Vc = 0.75X450.272 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 337.704 kN > 56.04T , Hence O.K.

ALONG LOCAL Y - DIRECTION :

SQy = 0.15 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 146 , L/C No. - 405)*
Vu = 67.65 kN

Pt provided = 0.0022
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0022X67.65X10^3X451/76.42/10^3)X1000X451/10^3
= 433.639 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0022)X1000X451/10^3
= 443.772 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X451/10^3
= 773.764 kN
HENCE, Vc = 433.639 kN
Φt*Vc = 0.75X433.639 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 325.229 kN > 67.65T , Hence O.K.

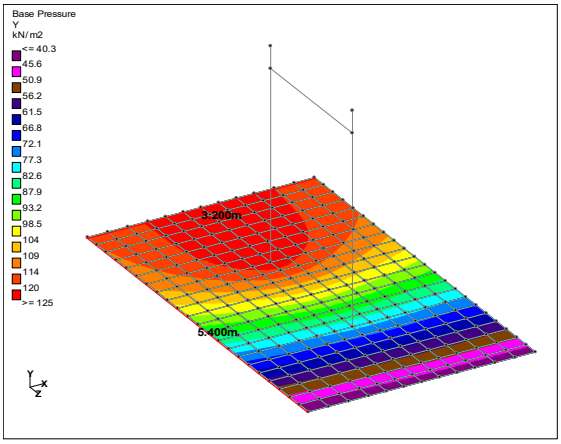
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	17-04-2020	19-05-2020	02-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK/APL	RPK/APL
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	17-04-2020	19-05-2020	02-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK/APL	RPK/APL
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

7.3.4 Design of combined footing mkd. as F3 for Trestle TR3

For Calculation of Base pressure and design forces for column & foundation Raft, Refer STAAD File **3130-CNV-002-TR3_Fdn.STD**



Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +913.500m)
Finished Ground level	F.G.L	=	-0.5	m	(Corresponds to RL +913.000m)
Bottom of foundation	B.O.F	=	-3.5	m	
Top of pedestal (staad support)	T.O.F	=	-0.15	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	3.200	m	
Width of footing	Lx	=	5.400	m	
Thickness of footing	D	=	0.55	m	
Depth of pedestal	Bx	=	0.60	m	
Width of pedestal	Bz	=	0.60	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	554	KN/m^2	
Soil load on footing in staad		=	44.1	KN/m^2	staad input

1) Check for Gross Base Pressure :

Maximum Base Pressure		=	125.0	KN/m^2	Load Case 305
Minimum Base Pressure		=	-4.2	KN/m^2	Load Case 313

2) Design of foundation Raft :

Slab Thickness =	550	mm	Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Width =	1000	mm	Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Clear Cover =	75	mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Compressive Strength of Concrete , fc' =	35	N/mm^2	trength Reduction Factor of Shear Reinforcement , Φt =	0.75	

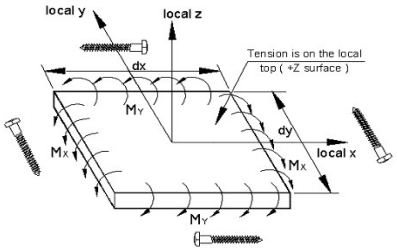
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	30.710	kN-m	Corresponding Mxy	1.890	kN-m	*(Plate No. - 24 , L/C No. - 413)*
Min. Mx	-80.410	kN-m	Corresponding Mxy	13.740	kN-m	*(Plate No. - 115 , L/C No. - 405)*
Max. My	78.680	kN-m	Corresponding Mxy	2.480	kN-m	*(Plate No. - 25 , L/C No. - 413)*
Min. My	-132.120	kN-m	Corresponding Mxy	5.000	kN-m	*(Plate No. - 114 , L/C No. - 405)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	30.71 + 1.89 =	32.600	kN-m
My1 =	My + abs(Mxy) =	78.68 + 2.48 =	81.160	kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	30.71 + 0.045 =	30.755	kN-m
My2 =	My+abs(Mxy^2/Mx) =	78.68 + 0.135 =	78.815	kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd =	32.6	kN-m					
Myd =	81.16	kN-m					

ALONG LOCAL X - DIRECTION :

Mu =	32.6	kN-m						
Φ*Kn =	Mu/bd^2					deff. =	469	mm
Mu/bd^2 =	32.6X10^6/(1000X469^2) =		0.148	N/mm^2	<	6.9	N/mm^2	
Pt =	0.0004		Reqd. Ast =	).0004X1000X469/100 =		1.88	cm^2	
Provided	12	mm @	200	C/C	(Provide Ast =	5.65	$)cm^2$	
					Total Provide Ast =	5.65	cm^2	O.K.
					Total Provide Pt =	0.0012		

ALONG LOCAL Y - DIRECTION :

Mu =	81.16	kN-m						
Φ*Kn =	Mu/bd^2					deff. =	457	mm
Mu/bd^2 =	81.16X10^6/(1000X457^2) =		0.369	N/mm^2	<	6.9	N/mm^2	
Pt =	0.0010		Reqd. Ast =	0.001X1000X469/100 =		4.57	cm^2	
Provided	12	mm @	200	C/C	(Provide Ast =	5.65	$)cm^2$	
					Total Provide Ast =	5.65	cm^2	O.K.
					Total Provide Pt =	0.0012		

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	17-04-2020	19-05-2020	02-11-2023				Date	17-04-2020	19-05-2020	02-11-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	RPK	RPK/APL	RPK/APL				Prepared By	RPK	RPK/APL	RPK/APL			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -80.41 - 13.74 = -94.15 kN-m
My1 = My - abs(Mxy) = -132.12 - 5 = -137.12 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -80.41 - 1.429 = -81.839 kN-m
My2 = My-abs(Mxy^2/Mx) = -132.12 - 0.311 = -131.809 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -94.15 kN-m
Myd = -137.12 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 94.15 kN-m
Φ*Kn = Mu/bd^2 deff. = 467 mm
Mu/bd^2 = 94.15X10^6/(1000X467^2) = 0.432 N/mm^2 < 6.9 N/mm^2
Pt = 0.0012 Reqd. Ast =)0.0012X1000X467/100 = 5.6 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0022

ALONG LOCAL Y - DIRECTION :

Mu = 137.12 kN-m
Φ*Kn = Mu/bd^2 deff. = 451 mm
Mu/bd^2 = 137.12X10^6/(1000X451^2) = 0.674 N/mm^2 < 6.9 N/mm^2
Pt = 0.0018 Reqd. Ast =)0.0018X1000X451/100 = 8.12 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0022

SHEAR CHECK

(Staad Group : SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.19 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 315 , L/C No. - 405)*
Vu = 88.73 kN

Provided Pt = 0.0022
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0022X88.73X10^3X467/94.15/10^6)X1000X467/10^3
= 449.736 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0022)X1000X467/10^3
= 459.515 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X467/10^3
= 801.215 kN
HENCE, Vc = 449.736 kN
Φt*Vc = 0.75X449.736 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 337.302 kN > 88.73T , Hence O.K.

ALONG LOCAL Y - DIRECTION :

SQy = 0.27 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 146 , L/C No. - 405)*
Vu = 121.77 kN

Pt provided = 0.0022
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0022X121.77X10^3X451/137.12/10^3)X1000X451/10^3
= 433.66 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0022)X1000X451/10^3
= 443.772 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X451/10^3
= 773.764 kN
HENCE, Vc = 433.66 kN
Φt*Vc = 0.75X433.66 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 325.245 kN > 121.77T , Hence O.K.

Global Project ID	**900707**
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Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01
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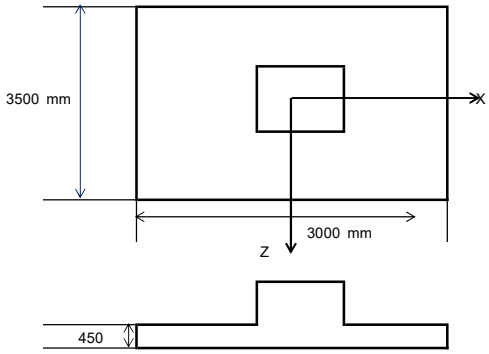
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7.3.5 Design of footing mkd. F4 for VGTU Column(Grid 3210/A - 3210/1)

Finished Ground level	F.G.L	=	-0.500	m
Bottom of foundation	B.O.F	=	-3.500	m
Top of footing (staad support)	T.O.F	=	-3.050	m
Ground Water Level	G.W.L	=	-10.000	m
Surcharge	Ws	=	0.000	KN/m^2
Length of footing	Lz	=	3.50	m
Width of footing	Lx	=	3.00	m
Thickness of footing	D	=	0.45	m
Depth of pedestal	Bx	=	0.60	m
Width of pedestal	Bz	=	0.60	m
Height of Load point from bottom of footing	H	=	0.225	m
Section Modulus of Pedestal	Zxx	=	6.13	m^3
Section Modulus of Pedestal	Zzz	=	5.25	m^3
Allowable Net Bearing Capacity		=	500	KN/m^2
Gross Bearing Capacity		=	554	KN/m^2
Grade of concrete		=	35	N/mm^2
Grade of steel		=	420	N/mm^2
Clear cover		=	50	mm
Top Reinfocement		=	12	mm
Bottom Reinforcement		=	16	mm
Co-efficient of friction between soil & concrete		=	0.3	
Unit wt of soil		=	18	KN/m^3
Unit wt of concrete		=	24	KN/m^3

Staad support node number		=	14	
Weight of soil		=	465.43	KN
Weight of foundation		=	113.40	KN
Uplift		=	0.00	KN
Surcharge		=	0	KN
Pressure on top for tension		=	55.1	KN/m^2

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	-22.2	322.17	0.8	4.39	-0.2	56.1	901.0	97.22	74.40	1	0.18
302	-22.2	517.99	-8.7	-3.57	-0.21	56.6	1096.8	115.82	93.10	1	0.21
303	-21.1	494.61	6.0	8.19	-0.23	52.8	1073.4	113.62	90.85	1	0.21
304	-22.5	614.81	15.1	56.84	-0.71	57.0	1193.6	133.81	93.55	1	0.24
305	-21.0	305.75	-16.8	-51.17	0.29	53.1	884.6	102.71	65.78	1	0.19
306	18.3	233.12	0.7	3.81	-0.22	-51.1	811.9	87.68	66.98	1	0.16
307	-61.8	687.42	-2.3	1.86	-0.21	161.1	1266.2	151.59	89.60	1	0.27
308	-20.9	455.01	-0.8	2.86	-0.22	52.7	1033.8	108.97	87.96	1	0.20
309	-22.7	465.52	-0.9	2.80	-0.2	57.4	1044.3	110.84	88.08	1	0.20
310	-21.8	465.35	-0.4	4.31	-0.23	55.1	1044.2	110.65	88.25	1	0.20
311	-21.7	455.18	-1.3	1.35	-0.2	55.0	1034.0	109.16	87.79	1	0.20
312	-14.3	399.36	21.7	74.63	-0.78	36.3	978.2	112.26	74.06	1	0.20
313	-12.3	-12.71	-20.8	-69.37	0.55	31.1	566.1	71.17	36.66	1	0.13
314	40.1	-109.56	2.5	3.93	-0.12	-107.8	469.3	65.86	23.52	1	0.12
315	-66.7	496.17	-1.5	1.34	-0.12	175.1	1075.0	135.96	68.80	1	0.25
316	-12.1	186.29	0.5	2.68	-0.13	30.6	765.1	79.13	66.61	1	0.14
317	-14.5	200.31	0.4	2.59	-0.1	36.8	779.1	81.64	66.77	1	0.15
318	-13.3	200.09	1.0	4.60	-0.1	33.8	778.9	81.37	67.00	1	0.15
319	-13.3	186.52	-0.1	0.67	-0.09	33.6	765.3	79.40	66.38	1	0.14

Max P_{max} = 151.59 KN/m^2

Max P_{min} = 23.52 KN/m^2

Resultant F.O.S = 0.27

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Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	-31.1	451.0	1.1	6.15	-0.3	78.61	58.93	26.98
402	-30.9	780.4	-12.9	-5.61	-0.3	78.73	90.24	58.41
403	-29.4	747.7	7.7	10.84	-0.33	73.41	86.96	55.45
404	-27.9	808.4	33.8	119.43	-1.31	70.79	109.97	44.00
405	-24.8	149.0	-34.1	-110.99	0.82	62.50	44.22	-15.83
406	59.1	-5.9	3.0	6.30	-0.25	-159.70	30.88	-32.01
407	-111.8	963.3	-3.3	2.15	-0.24	292.97	147.89	35.58
408	-24.7	468.7	-0.1	4.28	-0.27	62.19	57.18	32.09
409	-28.0	488.7	-0.2	4.17	-0.23	71.07	60.76	32.32
410	-26.4	488.4	0.7	7.04	-0.28	66.77	60.38	32.64
411	-26.3	469.0	-1.0	1.41	-0.21	66.49	57.56	31.77
412	-21.5	619.6	34.7	119.16	-1.24	54.69	88.88	29.14
413	-18.4	-39.7	-33.3	-111.27	0.89	46.41	23.23	-30.78
414	65.5	-194.6	3.9	6.02	-0.18	-175.80	15.93	-53.00
415	-105.4	774.5	-2.5	1.87	-0.17	276.86	126.81	20.72
416	-18.3	279.9	0.8	4.00	-0.19	46.08	36.09	17.23
417	-21.6	300.0	0.6	3.89	-0.16	54.97	39.67	17.46
418	-20.0	299.7	1.5	6.76	-0.21	50.67	39.29	17.78
419	-19.9	280.3	-0.1	1.13	-0.14	50.39	36.47	16.91

Max Net factored B.P = 147.89 KN/m^2

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 1.2 m

Moment about Z: Muz = 106.48 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 1049.16 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 727.3 mm^2

Effective Depth Along X = 392 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or $A_{stmin} = 0.0014bd$

A_{stmin} = 705.6 mm^2

A_{req} = 727.3 mm^2

Required Spacing = **276.5** mm

Length of cantilever along Z = 1.45 m

Moment about X: Mux = 155.47 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$ = 965.26 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$ = 1115.9 mm^2

Effective Depth Along X = 376 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or $A_{stmin} = 0.0014bd$

A_{stmin} = 676.8 mm^2

A_{req} = 1115.9 mm^2

Required Spacing = **180.2** mm

Provide Bottom Reinforcement = **16 mm dia @ 150** c/c

Ast prov = 1340.4 mm^2

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Top Reinforcement

Length of cantilever along X = 1.2 m

Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 1059.89 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along X = 394 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

or

A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 354.6 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **318.9** mm

Length of cantilever along Z = 1.45 m

Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 996.31 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along Z = 382 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

or

A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 343.8 mm^2

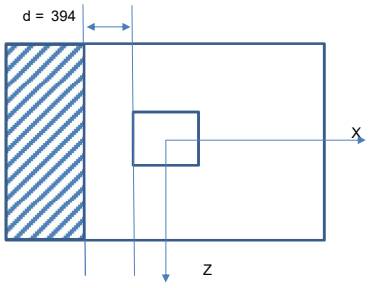
A_{req} = 0.0 mm^2

Required Spacing = **329.0** mm

Provide top Reinforcement = **12 mm dia @ 150 c/c**

Ast prov = 754.0 mm^2

ONE WAY SHEAR CKECK



Along X Direction Cantilever: L = 0.806 m

Reinforcement ratio ρ_w = 0.0034

Effective Depth Along X d = 394 mm

Shear Force at d distance V_u = 119.20 KN

Allowable shear strength φVc = 296.42 KN O.K

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete

Mu = 48.04 KNm

φ = 0.75 for shear

φVc = 296.4 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$

φVc = 296.8 KN *Table 22.5.5.1 SBC 304*

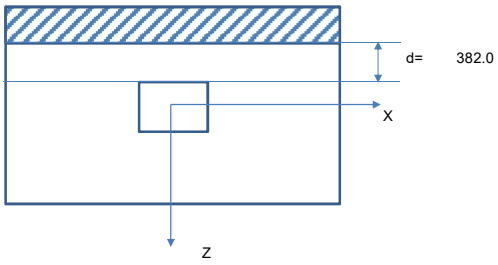
iii) $V_c = 0.29\lambda\sqrt{f_c'} b_w d$

= 507.0 KN

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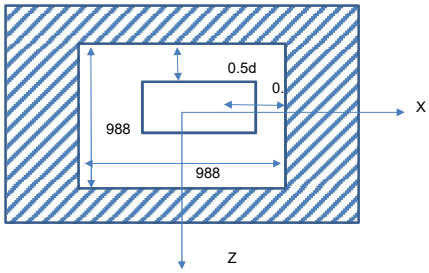
ONE WAY SHEAR CKECK



Along Z Direction Cantilever	L	=	1.068	m		Least of	i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio	ρ_w	=	0.0035							*Table 22.5.5.1 SBC 304*
Effective Depth Along Z	d	=	382	mm				λ	=	1 Normal concrete
Shear Force at d distance	V_u	=	157.95	KN				Mu	=	84.35 KNm
Allowable shear strength	φVc	=	283.42	KN	O.K			φ	=	0.75 for shear
								φVc	=	283.4 KN
							ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
								φVc	=	288.3 KN
							iii)	V_c	=	$0.29\lambda\sqrt{f_c'}b_w d$
								φVc	=	491.5 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area	=	10.5	m2	
Punching area	=	0.98	m2	
λ	=	1		
β	=	1		
αs	=	40		
b0	=	3952.0	mm	

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

v_c	$0.33\lambda\sqrt{f_c'}$	=	1.95 N/mm²
v_c	$0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	3.02 N/mm²
v_c	$0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	2.91 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column fac	u	=	3.95	m		Vc	=	$v_c b_0 d$
Depth at section	d	=	388	mm			=	2993.62 KN
Shear Force at section	Vu	=	1408.5	KN		φ	=	0.75 for shear
Allowable shear strength	φVc	=	2245.21	KN	O.K	$φV_n$	=	φVc
						φVc	=	2245.2 KN

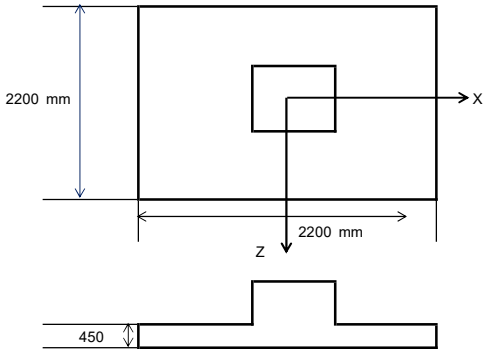
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7.3.6 Design of footing F5 for VGTU Column(Grid 3210/A - 3210/3)

Finished Ground level	F.G.L	=	-0.500	m	Staad support node number	=	11	
Bottom of foundation	B.O.F	=	-3.500	m	Weight of soil	=	205.63 KN	*inputs
Top of footing (staad support)	T.O.F	=	-3.050	m	Weight of foundation	=	52.27 KN	
Ground Water Level	G.W.L	=	-10.000	m	Uplift	=	0.00 KN	
Surcharge	Ws	=	0.000	KN/m^2	Surcharge	=	0 KN	
Length of footing	Lz	=	2.20	m	Pressure on top for tension	=	53.3 KN/m^2	
Width of footing	Lx	=	2.20	m				
Thickness of footing	D	=	0.45	m				
Depth of pedestal	Bx	=	0.60	m				
Width of pedestal	Bz	=	0.60	m				
Height of Load point from bottom of footing	H	=	0.225	m				
Section Modulus of Pedestal	Zxx	=	1.77	m^3				
Section Modulus of Pedestal	Zzz	=	1.77	m^3				
Allowable Net Bearing Capacity		=	500	KN/m^2				
Gross Bearing Capacity		=	554	KN/m^2				
Grade of concrete		=	35	N/mm^2				
Grade of steel		=	420	N/mm^2				
Clear cover		=	50	mm				
Top Reinfocement		=	12	mm				
Bottom Reinforcement		=	16	mm				
Co-efficient of friction between soil & concrete		=	0.3					
Unit wt of soil		=	18	KN/m^3				
Unit wt of concrete		=	24	KN/m^3				



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$ β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	-3.9	83.9	-0.6	1.73	-2.2	7.5	341.8	75.84	65.40	1	0.14
302	-3.8	145.66	-0.9	2.19	-2.08	7.4	403.6	88.80	77.97	1	0.16
303	-4.0	143	-1.9	-0.75	-2.58	7.8	400.9	87.63	78.04	1	0.16
304	-3.7	114.96	23.6	64.29	-2.55	7.1	372.9	117.28	36.80	1	0.21
305	-4.1	143.44	-26.0	-62.36	-2.03	8.0	401.3	122.58	43.26	1	0.22
306	9.4	84.88	-0.8	1.22	0.51	-27.2	342.8	86.86	54.79	1	0.16
307	-17.2	173.56	-1.6	0.71	-5.09	42.4	431.5	113.43	64.86	1	0.20
308	-3.8	128.93	-1.2	0.99	-2.2	7.2	386.8	84.56	75.29	1	0.15
309	-4.0	129.52	-1.2	0.95	-2.38	7.9	387.4	85.04	75.05	1	0.15
310	-3.9	128.82	-0.5	2.72	-2.3	7.6	386.7	85.69	74.11	1	0.15
311	-3.9	129.62	-1.9	-0.78	-2.28	7.6	387.5	84.78	75.35	1	0.15
312	-2.1	31.32	32.7	85.48	-1.65	3.9	289.2	110.13	9.39	1	0.20
313	-2.6	69.3	-33.4	-83.40	-0.96	5.1	327.2	117.48	17.73	1	0.21
314	15.4	-8.78	0.2	1.38	2.42	-41.9	249.1	75.86	27.09	1	0.14
315	-20.0	109.46	-0.9	0.70	-5.04	50.9	367.4	105.00	46.81	1	0.19
316	-2.2	49.94	-0.3	1.07	-1.18	4.1	307.8	66.50	60.71	1	0.12
317	-2.5	50.73	-0.4	1.00	-1.43	5.0	308.6	67.14	60.40	1	0.12
318	-2.3	49.8	0.6	3.37	-1.3	4.5	307.7	68.00	59.15	1	0.12
319	-2.3	50.87	-1.3	-1.30	-1.3	4.5	308.8	67.08	60.51	1	0.12

Max P_{max} = 122.58 KN/m^2

Max P_{min} = 9.39 KN/m^2

Resultant F.O.S = 0.22

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.	0	1	2
Date	17-04-2020	19-05-2020	02-11-2023	Date	17-04-2020	19-05-2020	02-11-2023
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Prepared By	RPK	RPK/APL	RPK/APL	Prepared By	RPK	RPK/APL	RPK/APL
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	-5.4	117.5	-0.8	2.42	-3.1	10.54	31.57	16.96
402	-5.4	222.1	-1.5	2.75	-2.96	10.41	53.30	38.46
403	-5.6	218.3	-2.9	-1.37	-3.65	10.88	52.01	38.21
404	-4.3	100.5	51.8	136.67	-3.24	8.10	102.33	-60.82
405	-5.1	161.2	-54.0	-133.52	-2.13	10.03	114.20	-47.57
406	23.7	36.3	-0.2	2.11	3.28	-65.20	45.43	-30.43
407	-33.0	225.5	-2.0	1.02	-8.66	83.33	94.12	-0.95
408	-4.4	130.3	-1.1	1.61	-2.51	8.41	32.57	21.28
409	-4.9	131.5	-1.2	1.52	-2.87	9.72	33.49	20.83
410	-4.7	130.1	0.2	4.90	-2.71	9.04	34.74	19.04
411	-4.7	131.7	-2.4	-1.76	-2.67	9.10	33.32	21.08
412	-3.1	45.1	52.3	136.66	-2.51	5.82	89.60	-70.97
413	-3.9	105.9	-53.4	-133.53	-1.41	7.75	101.47	-57.74
414	24.8	-19.1	0.4	2.10	4.01	-67.48	35.27	-43.15
415	-31.8	170.1	-1.5	1.01	-7.93	81.05	81.38	-11.10
416	-3.3	74.9	-0.5	1.60	-1.79	6.13	19.84	11.13
417	-3.8	76.1	-0.6	1.51	-2.14	7.42	20.75	10.68
418	-3.5	74.8	0.8	4.88	-1.98	6.74	22.00	8.89
419	-3.5	76.3	-1.9	-1.78	-1.95	6.81	20.60	10.92

Max Net factored B.P = 114.20 KN/m^2

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.8 m

Moment about Z: Muz = 36.54 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 1049.16 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 247.4 mm^2

Effective Depth Along X = 392 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of

or $A_{stmin} = 0.0014bd$

A_{stmin} = 705.6 mm^2

A_{req} = 247.4 mm^2

Required Spacing = **285.0** mm

Length of cantilever along Z = 0.8 m

Moment about X: Mux = 36.54 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 965.26 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 258.1 mm^2

Effective Depth Along X = 376 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of

or $A_{stmin} = 0.0014bd$

A_{stmin} = 676.8 mm^2

A_{req} = 258.1 mm^2

Required Spacing = **297.1** mm

Provide Bottom Reinforcement = **16 mm dia @ 200** c/c

Ast prov = 1005.3 mm^2

Global Project ID	**900707**
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Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01
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Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
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Top Reinforcement

Length of cantilever along X = 0.8 m
Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 1059.89 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along X = 394 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd
Table 24.4.3.2 SBC 304

A_{stmin} = 354.6 mm^2
A_{req} = 0.0 mm^2
Required Spacing = **318.9** mm

Length of cantilever along Z = 0.8 m
Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 996.31 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along Z = 382 mm

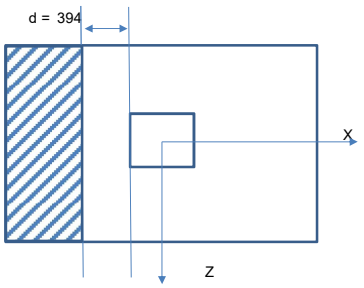
For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd
Table 24.4.3.2 SBC 304

A_{stmin} = 343.8 mm^2
A_{req} = 0.0 mm^2
Required Spacing = **329.0** mm

Provide top Reinforcement = **12 mm dia @ 200 c/c**
Ast prov = 565.5 mm^2

ONE WAY SHEAR CKECK



Along X Direction Cantilever: L = 0.406 m
Reinforcement ratio: ρ_w = 0.0026
Effective Depth Along X: d = 394 mm
Shear Force at d distance: V_u = 46.37 KN
Allowable shear strength: φVc = 292.53 KN O.K

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$
Table 22.5.5.1 SBC 304
λ = 1 Normal concrete
Mu = 9.41 KNm
φ = 0.75 for shear
φVc = 304.6 KN

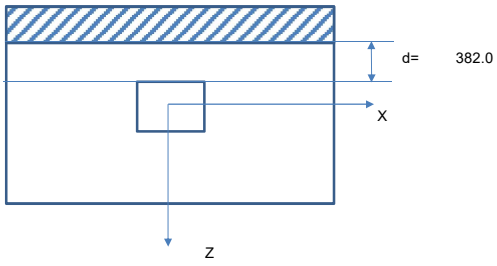
ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$
φVc = 292.5 KN *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'} b_w d$
= 507.0 KN

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Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.	0	1	2
Date	17-04-2020	19-05-2020	02-11-2023	Date	17-04-2020	19-05-2020	02-11-2023
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Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

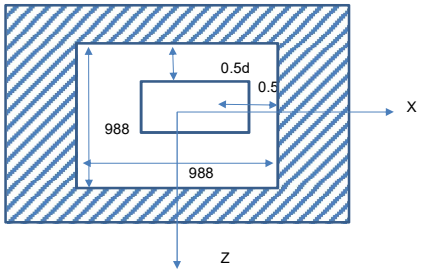
ONE WAY SHEAR CKECK



Along Z Direction	Cantilever	L	=	0.418	m		Least of	i)	V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0026						*Table 22.5.5.1 SBC 304*
Effective Depth Along Z		d	=	382	mm				λ	= 1 Normal concrete
Shear Force at d distance		V_u	=	47.74	KN				Mu	= 9.98 KNm
Allowable shear strength		φVc	=	284.01	KN	O.K			φ	= 0.75 for shear
									φVc	= 294.6 KN
								ii)	V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
									φVc	= 284.0 KN
								iii)	V_c	= $0.29\lambda\sqrt{f_c'}b_w d$
									φVc	= 491.5 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area	=	4.84	m2	
Punching area	=	0.98	m2	
λ	=	1		
β	=	1		
αs	=	40		
b0	=	3952.0	mm	

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

v_c	= $0.33\lambda\sqrt{f_c'}$	=	1.95 N/mm²
v_c	= $0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	3.02 N/mm²
v_c	= $0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	2.91 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

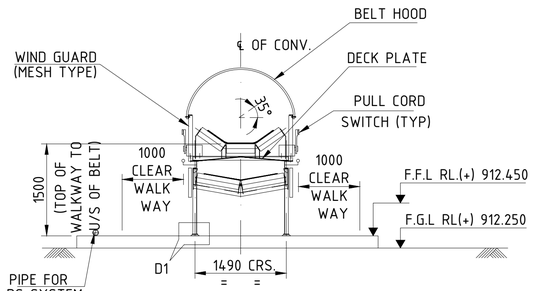
Perimeter at 0.5d from column face	u	=	3.95	m		Vc	= $v_c b_0 d$
Depth at section	d	=	388	mm			= 2993.62 KN
Shear Force at section	Vu	=	441.3	KN		φ	= 0.75 for shear
Allowable shear strength	φVc	=	2245.21	KN	O.K	φV_n	= φVc
						φVc	= 2245.2 KN

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
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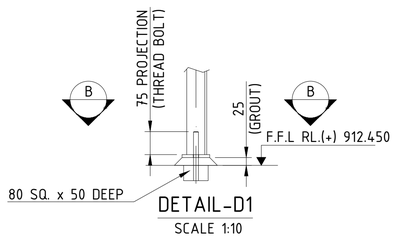
ATTACHMENT 1 **MECHANICAL GENERAL ARRANGEMENT DRAWINGS**

DRAWING NUMBER: MD-216-3210-EG-ME-ELV-1001 | REV: A01



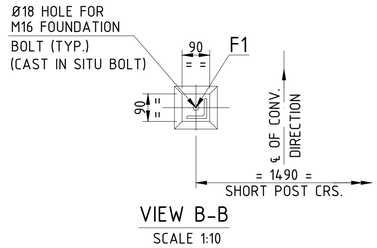
TYPICAL CONVEYOR SECTION ON GROUND (1200 B/W)

SCALE 1:50



DETAIL-D1

SCALE 1:10



VIEW B-B

SCALE 1:10

CONVEYOR CHARACTERISTICS

	PARTICULARS	SPECIFICATION	QUANTITY
MISCELLANEOUS EQUIPMENT	TAKE-UP / TRAVEL	VGTU. / 4000 mm	
	TAKE-UP LOAD	~5760 kg.(APPROXIMATELY) RCC BLOCKS, 40 kg EACH	
	SCRAPER AT DISCHARGE PULLEY	EXTERNAL / MAIN & PRE CLEANER	1 NO. + 1 NO.
	SCRAPER AT TAKE-UP	INTERNAL V-PLOUGH	1 NO.
	SCRAPER AT TAIL PULLEY	INTERNAL V-PLOUGH	1 NO.
	BELT SWAY SWITCH (ZS)	@ 50 m INTERVAL	3 PAIRS
	PULL CORD SWITCH (HS)	@ 30 m INTERVAL	5 PAIRS
	ZERO SPEED SWITCH (SS)	AT TAIL PULLEY	1 NO.
	BELT RIP SWITCH (YS)	AFTER FEEDING ZONE	1 NO.
	DECK PLATE	3.15 mm THK. CONTINUOUS	1 LOT.
	BELT HOOD	GALVANIZED CORRUGATED SHEET	1 LOT.

ABBREVIATION

CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
BRG.:	BEARING	OPNG.:	OPENING

SIGN CONVENTION

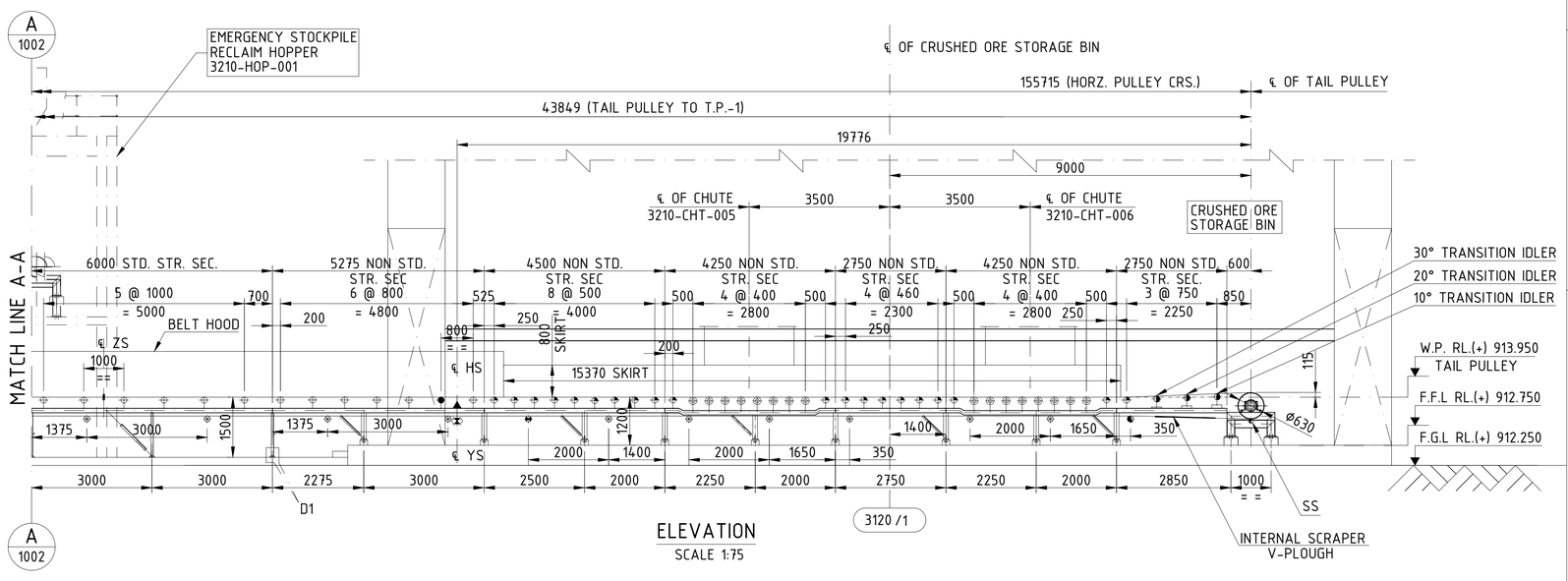
UPWARD (-)
DOWNWARD (+)
TOWARDS DRIVE PULLEY=(+)
TOWARDS TAIL PULLEY=(-)
REVERSIBLE LOAD=(±)

Note: Wind & Seismic load to be considered during civil & structural design

TABLE-1 : LOAD FOR CONVEYOR SHORT POST

FOR CONV. 3210-CNV-001

ZONE	POINT	VERTICAL LOAD IN "t"		HORIZONTAL LOAD IN "t"	
		DEAD LOAD	LIVE LOAD	ALONG	ACROSS
NORMAL ZONE	F1	(+)0.55	(+)0.45	(±)0.11	(±)0.09



ELEVATION

SCALE 1:75

CONVEYOR CHARACTERISTICS

	PARTICULARS		SPECIFICATION	QUANTITY
CONVEYOR	EQUIPMENT TAG NO.		3210-CNV-001	1 NO.
	EQUIPMENT DESCRIPTION		BALL MILL FEED CONVEYOR 1	
	LOCATION		CRUSHED ORE STORAGE BIN TO BALL MILL	
	CAPACITY RATED / DESIGN	(t/h)	500 / 550	
	BELT WIDTH	(mm)	1200	
	BELT SPEED	(m/sec)	1.195	
	TRIPPER FIXED / MOVABLE		N.A.	
	CONVEYOR LENGTH / LIFT	(mm)	155715 / 17130	
MATERIAL	MATERIAL		GOLD ORE : FRESH ORE / OXIDE ORE	
	BULK DENSITY	(t/m³)	1.6	
	MAX. LUMP SIZE	(mm)	(-)10	
	MOISTURE CONTENT	(%)	2	
	SURCHARGE ANGLE	(deg.)	20	
BELTING	MAX. RUNNING TENSION	(T1 kN)	71.648	
	SLACK SIDE TENSION	(T2 kN)	34.007	
	TAIL TENSION	(T0 kN)	36.442	
	BELT DESIGNATION		EP-1000 / 4 (HEAVY DUTY)	
	TOP / BOTTOM COVER THICKNESS	(mm)	6 / 3	
	COVER GRADE / FABRIC		DIN-X / EP	
PULLEYS	DISCHARGE WITH DRIVE	(MKD. x DIA.)	D-5 x 800 mm	1 NO.
	DISCHARGE WITHOUT DRIVE	(MKD. x DIA.)	N.A.	
	SNUB	(MKD. x DIA.)	ND-7 x 500 mm	1 NO.
	TAIL	(MKD. x DIA.)	ND-6 x 630 mm	1 NO.
	TAKE-UP	(MKD. x DIA.)	ND-6 x 630 mm	1 NO.
	DEFLECTOR	(MKD. x DIA.)	DR-3 x 273 mm	1 NO.
	TRIPPER DISCHARGE	(MKD. x DIA.)	N.A.	
	TRIPPER BEND	(MKD. x DIA.)	N.A.	
	BEND	(MKD. x DIA.)	ND-6 x 630 mm	2 NOS.
	LAGGING	(mm)	DRIVE: 16thk CERAMIC	
			NON-DRIVE: 16thk PLAIN RUBBER	
BEARING/SLEEVE PLUMMER BLOCK	DISCHARGE WITH DRIVE		TBA	2 SETS
	DISCHARGE WITHOUT DRIVE		N.A.	
	SNUB		TBA	2 SETS
	TAIL		TBA	2 SETS
	TAKE-UP		TBA	2 SETS
	DEFLECTOR		TBA	2 SETS
	BEND		TBA	4 SETS
	TRIPPER DISCHARGE		N.A.	
	TRIPPER BEND		N.A.	
IDLER	TROUGHING	(DIA. x deg.)	152.4 x 35°	114 NOS.
	TROUGHING RETRACTABLE	(DIA. x deg.)	152.4 x 45°	29 NOS.
	TROUGHING TRAINER	(DIA. x deg.)	152.4 x 35°	12 NOS.
	RETURN	(DIA. x deg.)	152.4 x 10°	48 NOS.
	SELF ALIGNING RETURN TRAINER INVERTED IDLER	(DIA. x deg.)	152.4 x 10°	5 NOS.
	IMPACT RETRACTABLE	(DIA. x deg.)	190 x 45°	22 NOS.
	TRANSITION (ADJUSTABLE)	(DIA. x deg.)	152.4 x 10°/20°/30°	(1+2+2) NOS.
	RETURN SELF CLEANING	(DIA. x deg.)	190 x 0°	2 NOS.
	NON MAGNETIC TROUGHING	(DIA. x deg.)	152.4 x 35°	2 NOS.
	WEIGHING IDLER	(DIA. x deg.)	152.4 x 35°	4 NOS.
	SPECIAL MACHINED IDLER	(DIA. x deg.)	152.4 x 35°	8 NOS.
	RETURN FLAT IDLER	(DIA. x deg.)	152.4 x 0°	4 NOS.
DRIVE	MOTOR: RATING	kW/rpm (SYN)	75 / 1800	1 NO.
	MOTOR: FRAME SIZE		TBA	
	MOTOR: TYPE		SQUIRREL CAGE	
	MOTOR: SUPPLY		4.16kV, 60Hz	
	REDUCER: TYPE		BEVEL HELICAL	1 NO.
	REDUCER: MODEL NO./COOLING/RATIO		TBA	
	REDUCER: RATIO		63	
	REDUCER: HOLD BACK		YES	
	BRAKE: TYPE		ELECTRO HYDRAULIC THRUSTER	1 NO.
	BRAKE: MODEL NO.		TBA	
	BRAKE: DRAM DIAMETER		TBA	
	COUPLING: LOW SPEED HIGH SPEED: TYPE		FLUID COUPLING	1 NO.
	COUPLING: LOW SPEED HIGH SPEED: MODEL NO.		TBA	
	COUPLING: TYPE		GEAR	1 NO.
	COUPLING: MODEL NO.		TBA	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-ELV-1003	CONV.3210-CNV-001-SECTION								
MD-216-3210-EG-ME-ELV-1002	CONV.3210-CNV-001-ELEVATION								
MD-216-3210-EG-ME-LPX-1001	CONV.3210-CNV-001-PLAN VIEW	A01	28/02/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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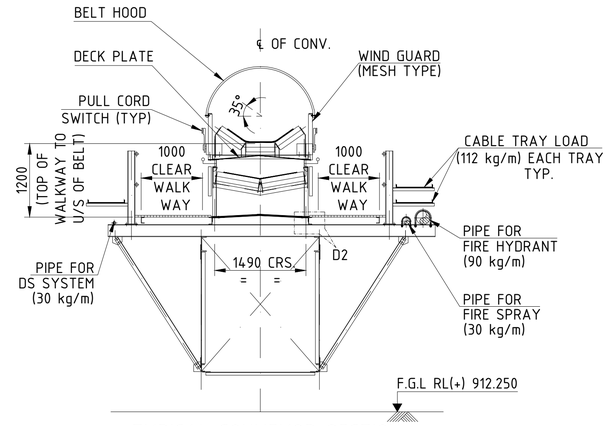
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OUTOTEC DRAWING NUMBER: 900707__GAU01__US01__MTL01__32110 | DOCUMENT ID: OU100669129 | REVISION: 0

DRAWN	HKD	25/02/2020
CHECKED	BRD	26/02/2020
ENGINEER	BD	23/02/2020
APPROVED	BGR	28/02/2020
EMPLOYER		
SCALE	1:10, 1:50, 1:75	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GRINDING & CLASSIFICATION
CONV.3210-CNV-001 - ELEVATION

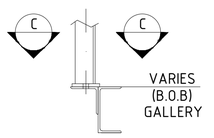
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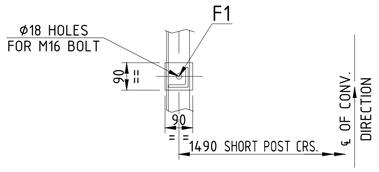
TYPICAL CONVEYOR SECTION ON GALLERY (1200 B/W)

SCALE 1:50



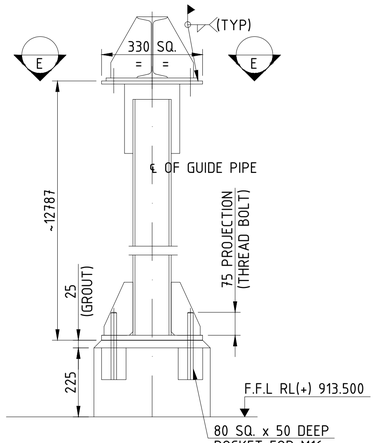
DETAIL-D2

SCALE 1:10



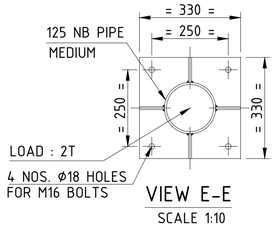
VIEW C-C

SCALE 1:10



VIEW H-H

SCALE 1:10



VIEW E-E

SCALE 1:10

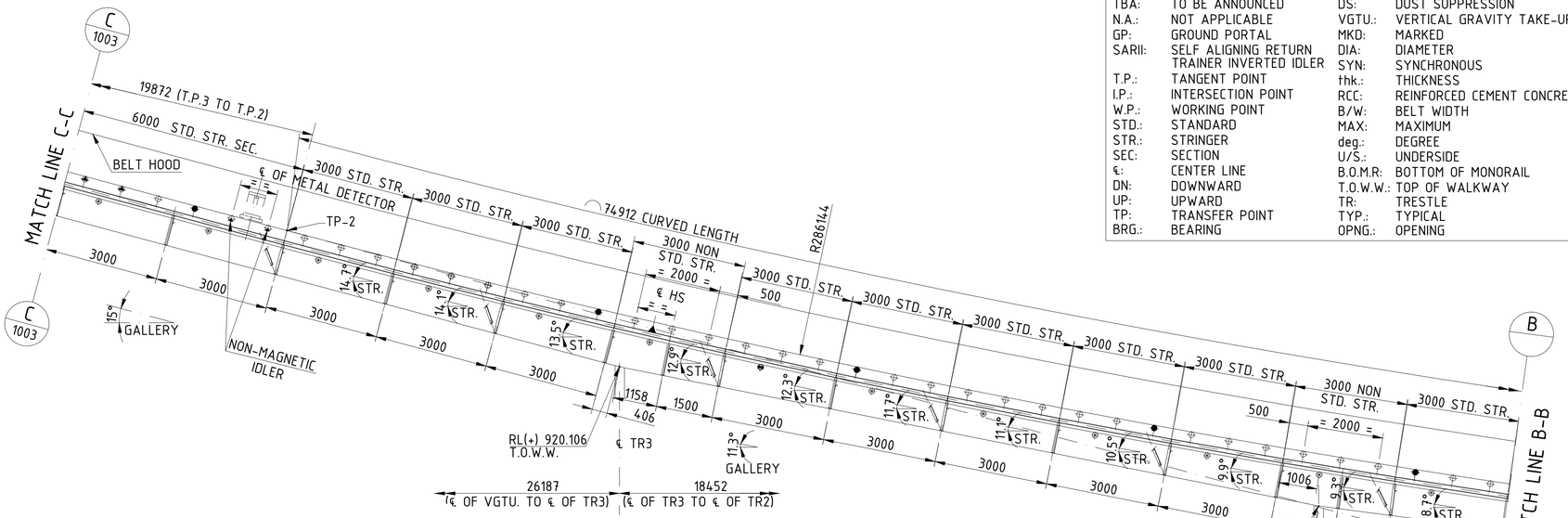
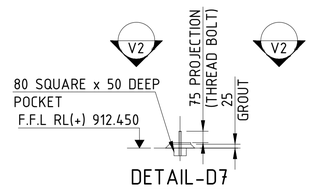


TABLE-1A : LOAD FOR CONVEYOR SHORT POST

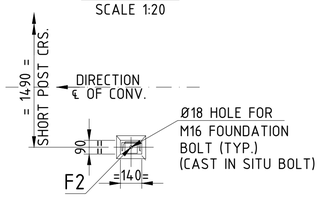
FOR CONV. 3210-CNV-001

ZONE	POINT	VERTICAL LOAD IN "t"		HORIZONTAL LOAD IN "t"	
		DEAD LOAD	LIVE LOAD	ALONG	ACROSS
NORMAL ZONE	F1	(+)0.55	(+)0.45	(±)0.15	(±)0.08
IMPACT ZONE	F2	(+)0.85	(+)0.65	(±)0.17	(±)0.09

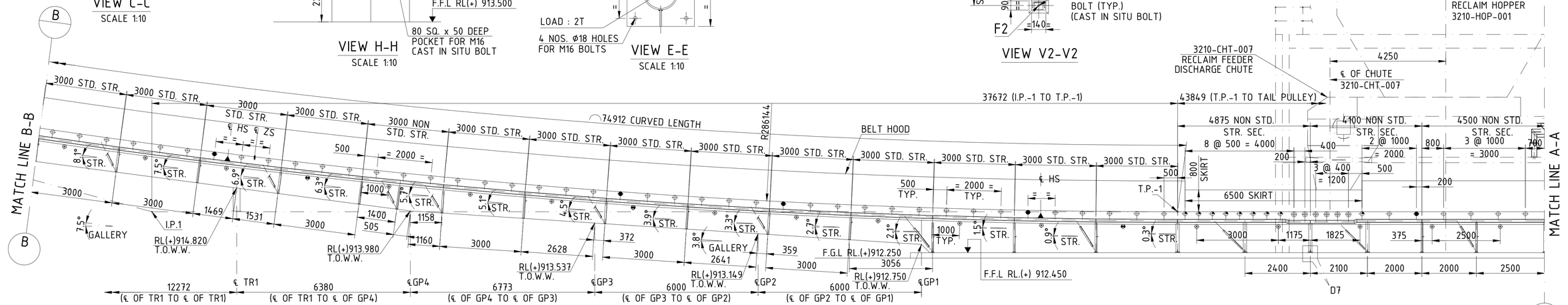


DETAIL-D7

SCALE 1:20



VIEW V2-V2



ABBREVIATION

CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
BRG.:	BEARING	OPNG.:	OPENING

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-ELV-1003	CONV.3210-CNV-001-SECTION								
MD-216-3210-EG-ME-ELV-1001	CONV.3210-CNV-001-ELEVATION								
MD-216-3210-EG-ME-LPX-1001	CONV.3210-CNV-001-PLAN VIEW	A01	28/02/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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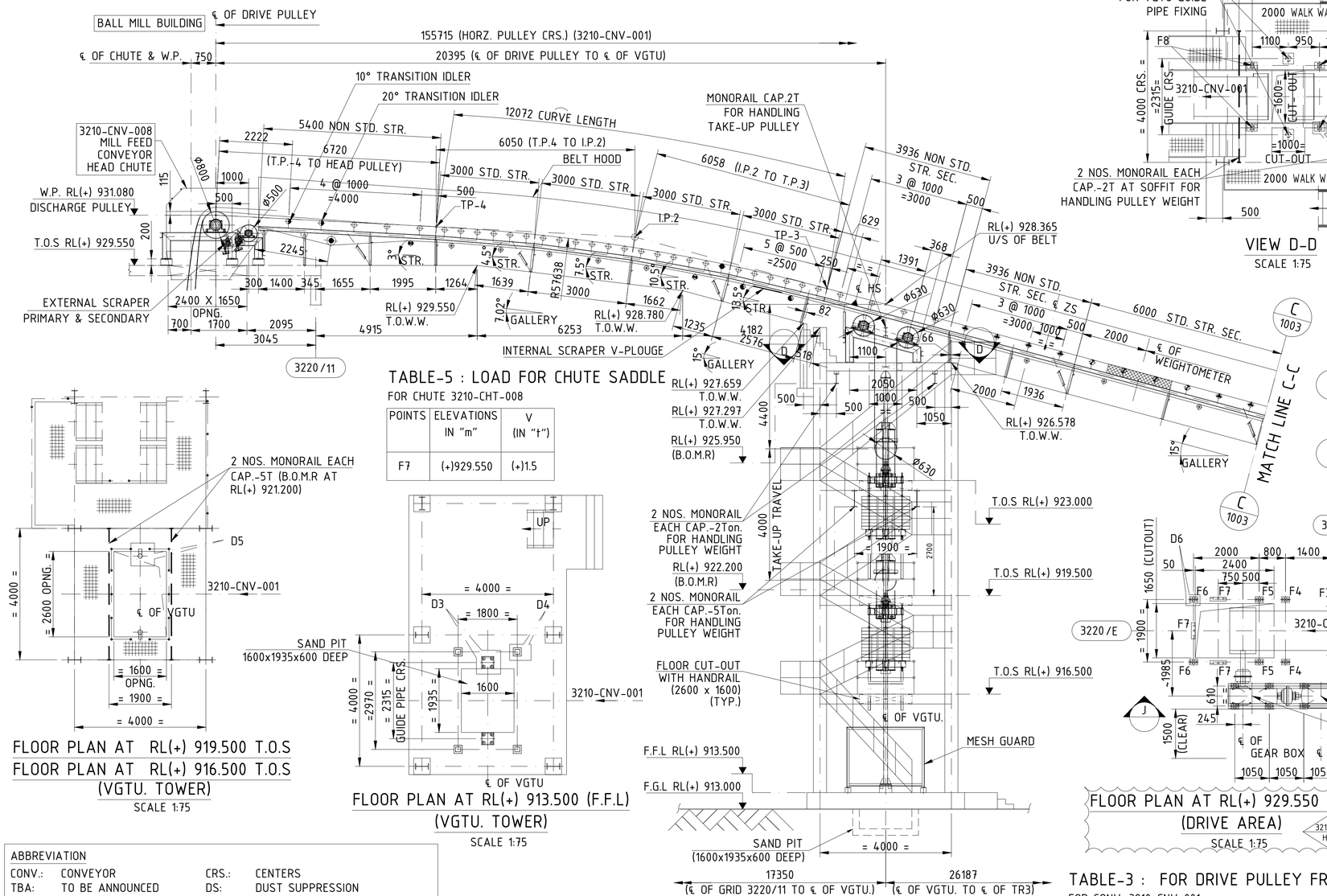
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OUTOTEC DRAWING NUMBER: 900707__GAU01__US01__MTL01__32111 | DOCUMENT ID: OU100669130 | REVISION: 0

DRAWN	HKD	25/02/2020
CHECKED	BRD	26/02/2020
ENGINEER	BD	23/02/2020
APPROVED	BGR	28/02/2020
EMPLOYER		
SCALE	1:10, 1:50, 1:75	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GRINDING & CLASSIFICATION
CONV.3210-CNV-001 - ELEVATION

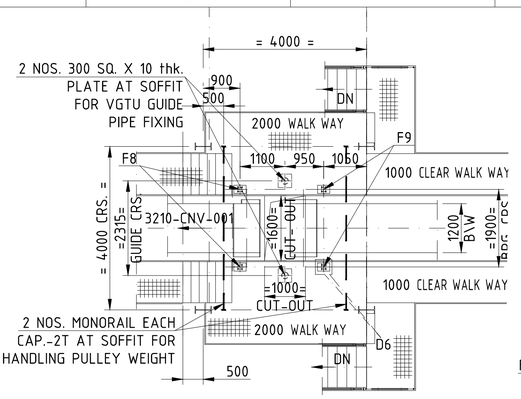
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DRAWING NUMBER MD-216-3210-EG-ME-ELV-1003 REV A01

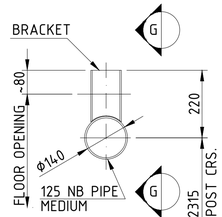


FLOOR PLAN AT RL(+) 919.500 T.O.S
FLOOR PLAN AT RL(+) 916.500 T.O.S
(VGTU. TOWER)
SCALE 1:75

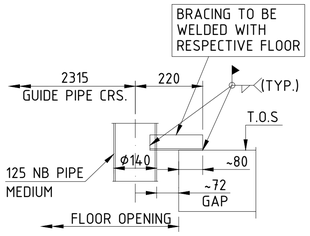
FLOOR PLAN AT RL(+) 913.500 (F.F.L)
(VGTU. TOWER)
SCALE 1:75



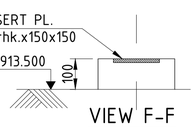
VIEW D-D
SCALE 1:75



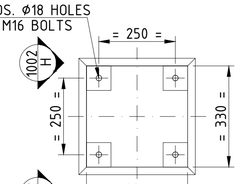
DETAIL-D5
SCALE 1:10



VIEW G-G
SCALE 1:10



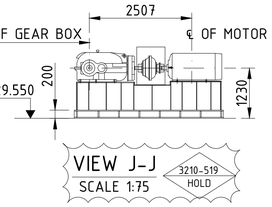
VIEW F-F
SCALE 1:10



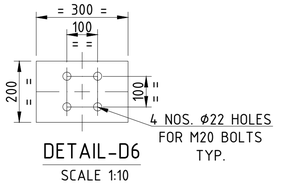
DETAIL-D3
SCALE 1:10



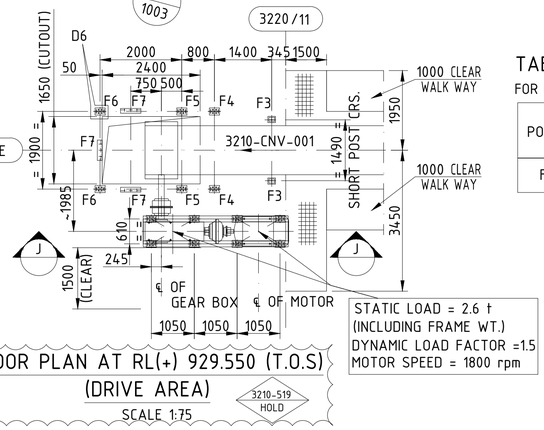
DETAIL-D4
SCALE 1:10



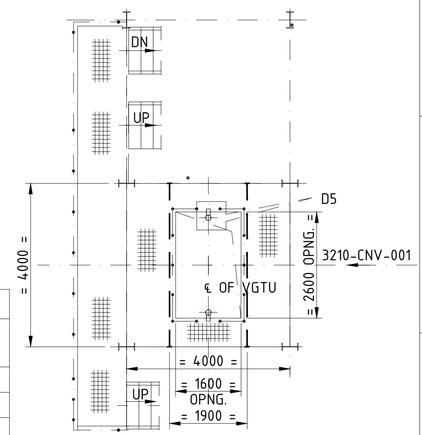
VIEW J-J
SCALE 1:75
3210-519 HOLD



DETAIL-D6
SCALE 1:10



FLOOR PLAN AT RL(+) 929.550 (T.O.S)
(DRIVE AREA)
SCALE 1:75
3210-519 HOLD



FLOOR PLAN AT RL(+) 923.000 T.O.S
(VGTU. TOWER)
SCALE 1:75

TABLE-5 : LOAD FOR CHUTE SADDLE

FOR CHUTE 3210-CHT-008

POINTS	ELEVATIONS IN "m"	V (IN "t")
F7	(+)929.550	(+)1.5

TABLE-4: LOAD FOR CONVEYOR SHORT POST

FOR CONV. 3210-CNV-001

POINT	ELEVATION IN "m"	VERTICAL LOAD (IN "t")		HORIZONTAL LOAD (IN "t")	
		DEAD LOAD	LIVE LOAD	ALONG	ACROSS
F3	(+)929.550	(+) 0.55	(+) 0.45	(±) 0.11	(±) 0.09

TABLE-3 : FOR DRIVE PULLEY FRAME LOAD

FOR CONV. 3210-CNV-001

POINTS	ELEVATIONS IN "m"	LOAD CASE I			LOAD CASE II		
		V (IN "t")	H ALONG (IN "t")	H ACROSS (IN "t")	V (IN "t")	H ALONG (IN "t")	H ACROSS (IN "t")
F4	(+)929.550	(+)5.6	(–)2.9	(±)0.6	(+)7.2	(–)3.7	(±)0.7
F5		(+)1.3	(–)2.4	(±)0.2	(–)0.4	(–)2.5	(±)0.2
F6		(+)2.3	(–)0.7	(±)0.3	(–)1.4	(–)1.8	(±)0.2

NOTE:
CASE I : CONVEYOR NORMAL RUNNING + MATERIAL LOAD + DEAD LOAD
CASE II : CONVEYOR STARTING + CHUTE EMPTY + DEAD LOAD

TABLE-2 : LOAD FOR TAKE UP BEND FRAME

FOR CONV. 3210-CNV-001

POINTS	VERTICAL LOAD (IN "t")	HORIZONTAL LOAD	
		ALONG (IN "t")	ACROSS (IN "t")
F8	(+) 3.50	(+) 1.50	(±) 0.40
F9	(+) 3.50	(–) 1.50	(±) 0.40

ABBREVIATION

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TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
BRG.:	BEARING	OPNG.:	OPENING

SIGN CONVENTION

UPWARD (–)
DOWNWARD (+)
TOWARDS DRIVE PULLEY=(+)
TOWARDS TAIL PULLEY=(–)
REVERSIBLE LOAD=(±)

Note: Wind & Seismic load to be considered during civil & structural design

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-ELV-1002	CONV.3210-CNV-001-ELEVATION								
MD-216-3210-EG-ME-ELV-1001	CONV.3210-CNV-001-ELEVATION								
MD-216-3210-EG-ME-LPX-1001	CONV.3210-CNV-001-PLAN VIEW	A01	28/02/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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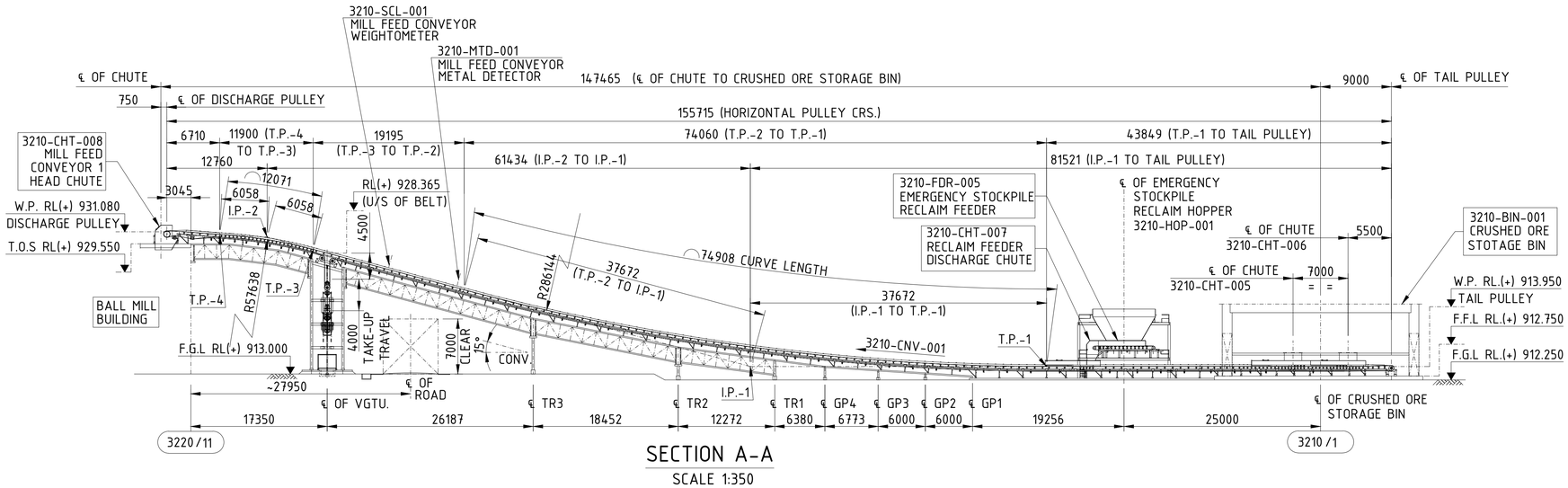
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OUTOTEC DRAWING NUMBER	900707 __GAU01 __US01 __ MTL01__32112	DOCUMENT ID OU100669131 / REVISION 0

DRAWN	HKD	25/02/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	BRD	26/02/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	BD	23/02/2020	GRINDING & CLASSIFICATION
APPROVED	BGR	28/02/2020	CONV.3210-CNV-001 - SECTION
EMPLOYER			
SCALE	1:10, 1:50, 1:75	A1	DRAWING NUMBER MD-216-3210-EG-ME-ELV-1003 / REV A01

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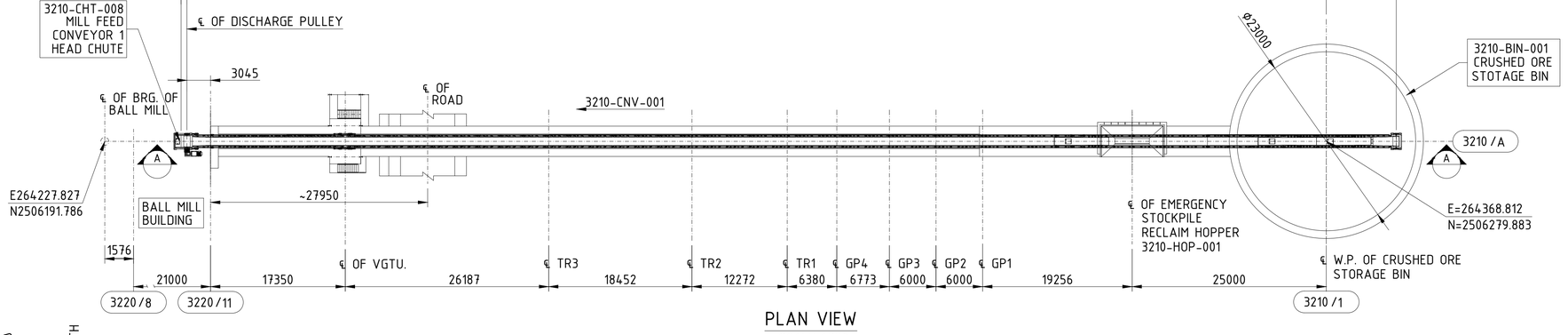


SECTION A-A
SCALE 1:350

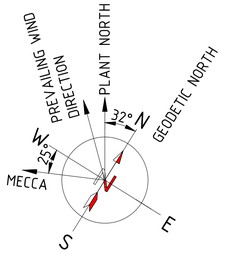
ABBREVIATION			
CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
BRG.:	BEARING	OPNG.:	OPENING

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)912.750 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.250 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN INDICATIVE.

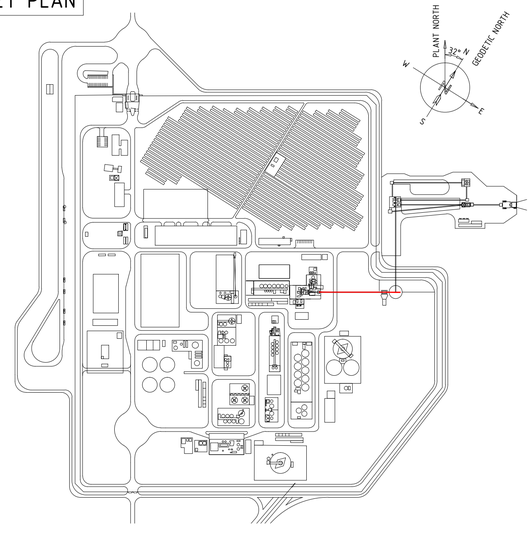
- CONCRETE FLOOR
- CHECKERED PLATE
- GRATING
- BRICK WALL
- CONCRETE WALL



PLAN VIEW
SCALE 1:350



KEY PLAN



REV	DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
	MD-216-3210-EG-ME-ELV-1003	CONV.3210-CNV-001-SECTION								
	MD-216-3210-EG-ME-ELV-1002	CONV.3210-CNV-001-ELEVATION								
	MD-216-3210-EG-ME-ELV-1001	CONV.3210-CNV-001-ELEVATION	A01	28/02/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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Saudi Arabian Mining Company شركة معادن العربية السعودية

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DRAWN	HKD	25/02/2020
CHECKED	BRD	26/02/2020
ENGINEER	BD	23/02/2020
APPROVED	BGR	28/02/2020
EMPLOYER		
SCALE	1:50	A1

LARSEN & TOUBRO DRAWING NUMBER: 019045_100_H_G_16116_001 | REVISION: 0
OUTOTEC DRAWING NUMBER: 900707__GAU01__US01__MTL01__32101 | DOCUMENT ID: OU100669115 | REVISION: 0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GRINDING & CLASSIFICATION
CONV.3210-CNV-001 - PLAN VIEW

DRAWING NUMBER: MD-216-3210-EG-ME-LPX-1001 | REV: A01

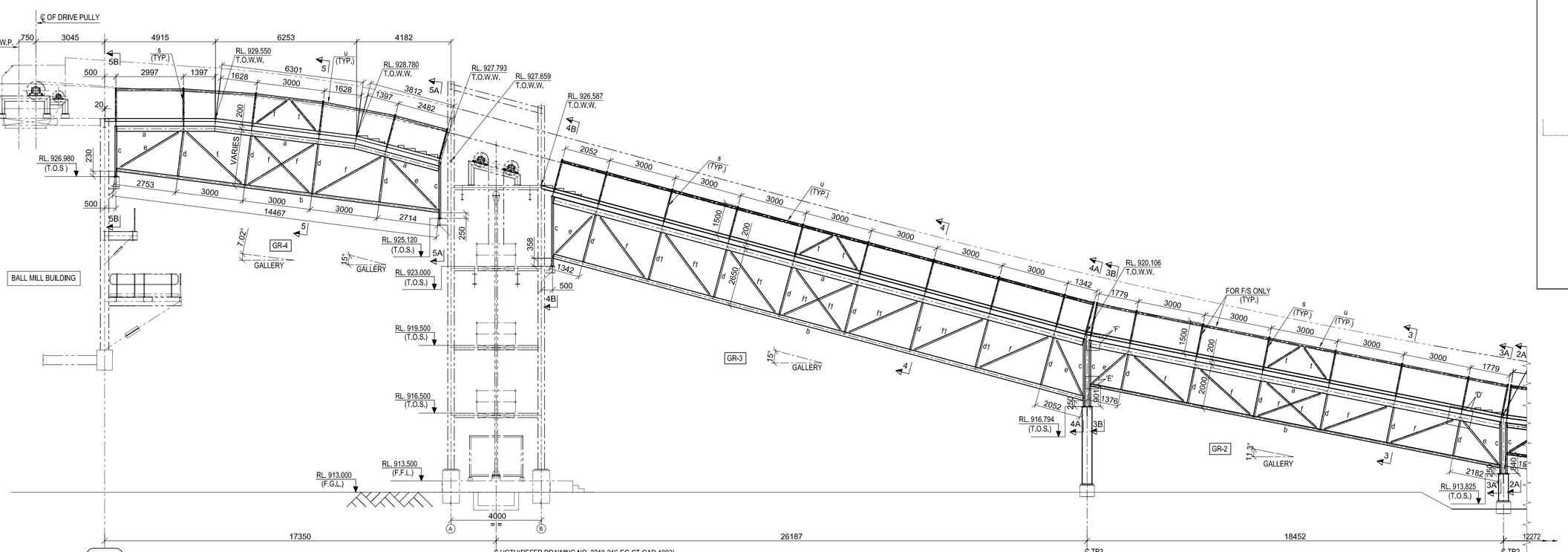
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01			
Date	17-04-2020	19-05-2020	02-11-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	RPK/APL	RPK/APL	RPK/APL			
Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI			

Outotec / L&T Revision No.:	0	1	2		
Date	17-04-2020	19-05-2020	02-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK/APL	RPK/APL	RPK/APL		
Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI		

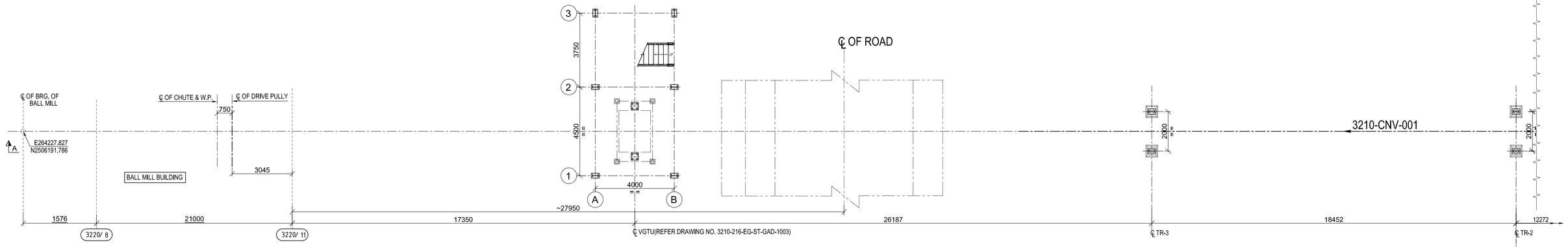
ATTACHMENT 2 **STRUCTURAL GENERAL ARRANGEMENT DRAWING CONTAINING BASE PLATE DETAIL**

DRAWING NUMBER: MD - 216 - 3210 - EG - ST - GAD - 1001
REV: A01



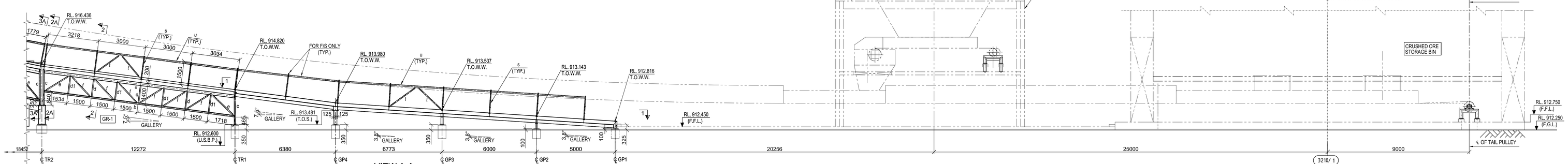
VIEW A-A
ELEVATION OF CONVEYOR MKD. 3210-CNV-001
(SCALE 1:100)

MATCH LINE A



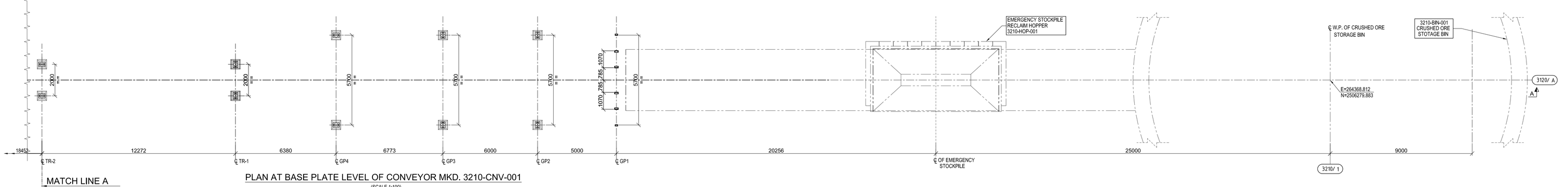
PLAN AT BASE PLATE LEVEL OF CONVEYOR MKD. 3210-CNV-001
(SCALE 1:100)

MATCH LINE A

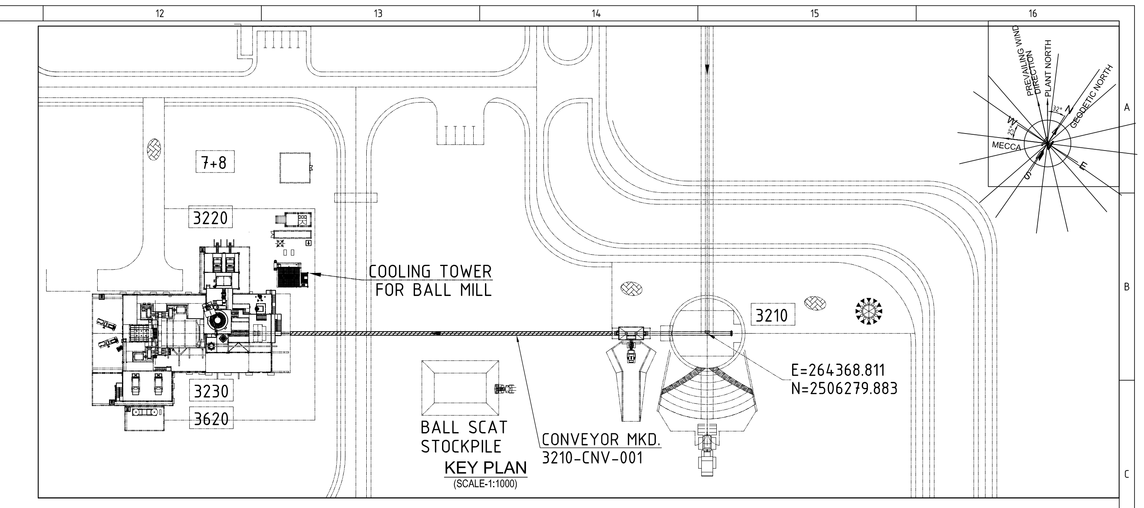


VIEW A-A
ELEVATION OF CONVEYOR MKD. 3210-CNV-001
(SCALE 1:100)

MATCH LINE A



PLAN AT BASE PLATE LEVEL OF CONVEYOR MKD. 3210-CNV-001
(SCALE 1:100)



LEGENDS :-

RL - REDUCED LEVEL
℄ - CENTRE LINE
T.O.S. - TOP OF STEEL.
F.G.L. - FINISHED GROUND LEVEL.
F.F.L. - FINISHED FLOOR LEVEL.
TYP. - TYPICAL.
B.O.M.R. - BOTTOM OF MONORAIL.
MKD. - MARKED.
MC - MOMENT CONNECTION
THK. - THICK
F/S - FAR SIDE
N/S - NEAR SIDE
W.P.-WORKING POINT
U.S.B.P. - UNDER SIDE OF BASE PLATE
U.N.O. - UNLESS NOTED OTHERWISE
T.O.W.W. - TOP OF WALKWAY
REF. - REFER
DRG. NO. - DRAWING NUMBER

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETER AND LEVELS & COORDINATES ARE IN METER UNO.
2. GRADE OF STEEL USED IS S275R AS PER EN10025.
3. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING MD - 216 - 3210 -EG -ST - GAD - 1002 TO MD - 216 - 3210 -EG -ST - GAD - 1005.
4. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO MD - 216 - 1000 -EG -ST - CRT - 1001

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-LPX-1001	CONV.3210-CNV-001 - PLAN VIEW								
MD-216-3210-EG-ME-ELV-1001	CONV.3210-CNV-001 - ELEVATION								
MD-216-3210-EG-ME-ELV-1002	CONV.3210-CNV-001 - ELEVATION								
MD-216-3210-EG-ME-ELV-1003	CONV.3210-CNV-001 - SECTION	A01	15/04/2020	ISSUED FOR REFERENCE	SUS	SBY	APL	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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Saudi Arabian Mining Company شركة معادن العربية السعودية

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LARSEN & TOUBRO DRAWING NUMBER: 01904 5_100_S_G_26025_011 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707 _ GAU01 _ US01 _ MTL01 _ 32131 — DOCUMENT ID: OU100680971 — REVISION: 0

DRAWN	SUS	15/04/2020
CHECKED	SBY	16/04/2020
ENGINEER	APL	13/04/2020
APPROVED	PRI	17/04/2020
EMPLOYER		
SCALE	1:100	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
STRUCTURAL GA OF 3210-CNV-001, SH1
PLAN, ELEVATIONS & DETAILS

DRAWING NUMBER: MD - 216 - 3210 - EG - ST - GAD - 1001 — REV: A01

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01			
Date	17-04-2020	19-05-2020	02-11-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	RPK/APL	RPK/APL	RPK/APL			
Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI			

Outotec / L&T Revision No.:	0	1	2		
Date	17-04-2020	19-05-2020	02-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK/APL	RPK/APL	RPK/APL		
Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI		

ATTACHMENT 3 **STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR GROUND PORTAL**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 19-May-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 1
JOB PART GROUND PORTAL FDN
JOB REF CONVEYOR 3210-CNV-001
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 -18.773 -0.15 -2.165; 2 -18.773 -0.15 3.735; 3 -12 -0.15 -2.165;
4 -12 -0.15 3.735; 5 -6 -0.15 -2.165; 6 -6 -0.15 3.735; 7 -1 -0.15 -2.165;
8 -1 -0.15 -1.17; 9 -1 -0.15 0; 10 -1 -0.15 1.57; 11 -1 -0.15 2.74;
12 -1 -0.15 3.735; 14 -3 0.196559 -2.165; 15 -3 0.196559 -1.17;
16 -3 0.196559 0; 17 -3 0.196559 1.57; 18 -3 0.196559 2.74;
19 -3 0.196559 3.735; 20 -4.5 0.294838 0; 21 -6 0.393118 -2.165;
22 -6 0.393118 -1.17; 23 -6 0.393118 0; 24 -6 0.393118 1.57;
25 -6 0.393118 2.74; 26 -6 0.393118 3.735; 27 -7.5 0.491397 0;
28 -9 0.589677 -2.165; 29 -9 0.589677 -1.17; 30 -9 0.589677 0;
31 -9 0.589677 1.57; 32 -9 0.589677 2.74; 33 -9 0.589677 3.735;
34 -10.5 0.687956 0; 35 -12 0.786236 -2.165; 36 -12 0.786236 -1.17;
37 -12 0.786236 0; 38 -12 0.786236 1.57; 39 -12 0.786236 2.74;
40 -12 0.786236 3.735; 41 -13.6933 0.897177 0; 42 -15.3865 1.00812 -2.165;
43 -15.3865 1.00812 -1.17; 44 -15.3865 1.00812 0; 45 -15.3865 1.00812 1.57;
46 -15.3865 1.00812 2.74; 47 -15.3865 1.00812 3.735; 48 -17.0798 1.11906 0;
49 -18.773 1.23 -2.165; 50 -18.773 1.23 -1.17; 51 -18.773 1.23 0;
52 -18.773 1.23 1.57; 53 -18.773 1.23 2.74; 54 -18.773 1.23 3.735;
55 -20.368 1.44 0; 56 -21.963 1.65 -2.165; 57 -21.963 1.65 -1.17;
58 -21.963 1.65 0; 59 -21.963 1.65 1.57; 60 -21.963 1.65 2.74;
61 -21.963 1.65 3.735; 62 -23.558 1.86 0; 63 -25.153 2.07 -2.165;
64 -25.153 2.07 -1.17; 65 -25.153 2.07 0; 66 -25.153 2.07 1.57;
67 -25.153 2.07 2.74; 68 -25.153 2.07 3.735; 69 -18.773 1.03 -2.165;
70 -18.773 1.03 3.735; 71 -18.773 1.03 -0.265; 72 -18.773 1.03 1.835;
73 -18.773 -1.5 -2.165; 74 -18.773 -1.5 3.735; 75 -12 -1.5 -2.165;
76 -12 -1.5 3.735; 77 -6 -1.5 -2.165; 78 -6 -1.5 3.735; 79 -1 -1.35 -2.165;
80 -1 -1.35 3.735; 81 -1 0.06 -2.165; 82 -1 0.06 -1.17; 83 -1 0.06 0;
84 -1 0.06 1.57; 85 -1 0.06 2.74; 86 -1 0.06 3.735;
MEMBER INCIDENCES
1 49 50; 2 50 51; 3 51 52; 4 52 53; 5 53 54; 6 3 35; 7 4 40; 8 35 36; 9 36 37;
10 37 38; 11 38 39; 12 39 40; 13 5 21; 14 6 26; 15 21 22; 16 22 23; 17 23 24;
18 24 25; 19 25 26; 20 1 69; 21 69 49; 22 2 70; 23 70 54; 31 65 62; 32 62 58;
33 58 55; 34 55 51; 35 51 48; 36 48 44; 37 44 41; 38 41 37; 39 37 34; 40 34 30;
41 30 27; 42 27 23; 43 23 20; 44 20 16; 45 83 9; 47 66 59; 49 52 45; 51 38 31;
52 31 24; 53 24 17; 54 84 10; 71 64 57; 73 50 43; 75 36 29; 76 29 22; 77 22 15;
78 82 8; 79 67 60; 81 53 46; 83 39 32; 84 32 25; 85 25 18; 86 85 11; 101 63 56;
103 49 42; 105 35 28; 106 28 21; 107 21 14; 108 81 7; 109 68 61; 111 54 47;
113 40 33; 114 33 26; 115 26 19; 116 86 12; 133 56 57; 134 42 43; 135 28 29;
136 14 15; 137 60 61; 138 46 47; 139 32 33; 140 18 19; 151 69 71; 152 71 72;
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153 72 70; 154 1 71; 155 2 72; 161 66 62; 162 59 62; 163 59 55; 164 52 55;
165 52 48; 166 45 48; 167 45 41; 168 38 41; 169 38 34; 170 31 34; 171 31 27;
172 24 27; 173 24 20; 174 17 20; 175 17 83; 201 57 58; 202 58 59; 203 59 60;
204 43 44; 205 44 45; 206 45 46; 207 29 30; 208 30 31; 209 31 32; 210 15 16;
211 16 17; 212 17 18; 214 59 52; 216 45 38; 218 57 50; 220 43 36; 222 60 53;
224 46 39; 226 56 49; 228 42 35; 230 61 54; 232 47 40; 301 73 1; 302 74 2;
303 75 3; 304 76 4; 305 77 5; 306 78 6; 307 79 7; 308 80 12; 309 3 5; 310 4 6;
311 1 2; 312 3 4; 313 5 6; 314 7 8; 315 8 9; 316 9 10; 317 10 11; 318 11 12;
319 14 81; 320 15 82; 321 16 83; 322 17 84; 323 18 85; 324 19 86;
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
1 TO 5 TAPERED 0.3 0.008 0.3 0.15 0.016 0.15 0.016
6 TO 19 TAPERED 0.25 0.01 0.25 0.15 0.016 0.15 0.016
20 TO 23 TAPERED 0.25 0.01 0.25 0.15 0.016 0.15 0.016
31 TO 45 47 49 51 TO 54 214 216 321 322 TABLE ST UPN240
73 75 TO 78 81 83 TO 86 220 224 320 323 TABLE ST UPN200
71 79 218 222 TABLE ST UPN200
101 103 105 TO 109 111 113 TO 116 226 228 230 232 319 324 TABLE ST UPN140
133 TO 140 TABLE ST IPE200
151 TO 153 TABLE LD L90X90X8 SP 0.01
154 155 TABLE LD L65X65X6 SP 0.01
161 TO 175 TABLE ST L65X65X6
201 TO 212 TABLE ST L50X50X6
MEMBER PROPERTY EUROPEAN
301 TO 308 PRIS YD 0.6 ZD 0.6
309 310 PRIS YD 0.6 ZD 0.3
311 TO 313 PRIS YD 0.5 ZD 0.3
314 TO 318 PRIS YD 0.6 ZD 0.35
CONSTANTS
BETA 90 MEMB 6 7 13 14
BETA 356.2 MEMB 8 TO 12 15 TO 19
BETA 352.5 MEMB 1 TO 5
MATERIAL STEEL MEMB 1 TO 23 31 TO 45 47 49 51 TO 54 71 73 75 TO 79 81 -
83 TO 86 101 103 105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 175 -
201 TO 212 214 216 218 220 222 224 226 228 230 232 319 TO 324
MATERIAL CONCRETE MEMB 301 TO 318
SUPPORTS
```

```
63 TO 68 FIXED BUT FX MX MY MZ
73 TO 80 FIXED
MEMBER TRUSS
154 155 161 TO 175 201 TO 212
MEMBER RELEASE
35 49 73 81 START MY MZ
1 103 105 107 111 113 115 133 TO 140 151 START MY MZ
5 34 106 114 133 TO 140 153 214 218 222 226 228 230 232 END MY MZ
6 7 13 14 20 22 START MP 0.99
45 54 78 86 108 116 END MP 0.99
*1 103 105 107 111 113 115 133 TO 140 151 START MP 0.99
*5 34 106 114 133 TO 140 153 214 218 222 226 228 230 232 END MP 0.99
MEMBER OFFSET
311 TO 313 START 0 -0.25 0
311 TO 313 END 0 -0.25 0
309 310 START 0 -0.3 0
309 310 END 0 -0.3 0
314 TO 318 START 0 -0.3 0
314 TO 318 END 0 -0.3 0
LOAD 1 DL
SELFWEIGHT Y -1.1
*WALKWAY LOAD @30KG/SQM INCLUDING SEAL PLATE LOAD @ 30KG/SQM & HANDRAIL LOAD @15KG/M
*WALKWAY WIDTH 1000MM & BELT WIDTH 1200MM
MEMBER LOAD
71 73 75 TO 77 79 81 83 TO 85 218 220 222 224 320 323 UNI GY -0.33
31 TO 44 47 49 51 TO 53 214 216 321 322 UNI GY -0.48
*CABLE TRAY LOAD @112KG/TRAY/M RUN * 3 NOS OF TRAY = 336KG/M RUN
*CONCENTRATED FORCE = 336*3=10kN & MOMENT= 10*0.4 = 4kNM
1 8 15 133 TO 136 CON GY -10 0.42
1 8 15 133 TO 136 CMOM GX -4
*LIGHTING POLE LOAD = 100KG/SUPPORT POINT(ASSUMED)
1 8 15 133 TO 136 CON GY -1 0.595
5 12 19 137 TO 140 CON GY -1 0.2
*PIPE DEAD LOAD
1 8 15 133 TO 136 CON GY -2.03 0.2
5 12 19 137 TO 140 CON GY -3.57 0.595
*CONVEYOR DEAD LOAD VERTICAL
321 322 CON GY -5 0.351
43 53 CON GY -5 0.357
41 52 CON GY -5 0.363
39 51 CON GY -5 0.369
37 216 CON GY -5 0.763
35 49 CON GY -5 1.156
34 CON GY -5 1.103
214 CON GY -5 2.71
33 214 CON GY -5 1.373
31 47 CON GY -5 1.59
*CONVEYOR DEAD LOAD HORIZONTAL ALONG LENGTH
321 322 CON GX 1.5 0.351
43 53 CON GX 1.5 0.357
41 52 CON GX 1.5 0.363
39 51 CON GX 1.5 0.369
37 216 CON GX 1.5 0.763
35 49 CON GX 1.5 1.156
```

```
34 CON GX 1.5 1.103
214 CON GX 1.5 2.71
33 214 CON GX 1.5 1.373
31 47 CON GX 1.5 1.59
**CONVEYOR DEAD LOAD HORIZONTAL ACROSS LENGTH
321 322 CON GZ 0.8 0.351
43 53 CON GZ 0.8 0.357
41 52 CON GZ 0.8 0.363
39 51 CON GZ 0.8 0.369
37 216 CON GZ 0.8 0.763
35 49 CON GZ 0.8 1.156
34 CON GZ 0.8 1.103
214 CON GZ 0.8 2.71
33 214 CON GZ 0.8 1.373
31 47 CON GZ 0.8 1.59
************************************************************
LOAD 2 LL
*WALKWAY LOAD @400KG/SQ. MTER(FOR GALLERY)
MEMBER LOAD
71 73 75 TO 77 79 81 83 TO 85 218 220 222 224 320 323 UNI GY -2
31 TO 44 47 49 51 TO 53 214 216 321 322 UNI GY -2
************************************************************
LOAD 3 ML
*CONVEYOR MATERIAL LOAD
MEMBER LOAD
321 322 CON GY -5.5 0.351
43 53 CON GY -5.5 0.357
41 52 CON GY -5.5 0.363
39 51 CON GY -5.5 0.369
37 216 CON GY -5.5 0.763
35 49 CON GY -5.5 1.156
34 CON GY -5.5 1.103
214 CON GY -5.5 2.71
33 214 CON GY -5.5 1.373
31 47 CON GY -5.5 1.59
*Spillage Load @75kg/ Sq. m on seal plate
31 TO 44 47 49 51 TO 53 214 216 321 322 UNI GY -0.64
************************************************************
LOAD 4 WIND LOAD IN +Z DIRN
MEMBER LOAD
321 322 CON GZ 2.58 0.351
43 53 CON GZ 2.58 0.357
41 52 CON GZ 2.58 0.363
39 51 CON GZ 2.58 0.369
37 216 CON GZ 2.58 0.763
35 49 CON GZ 2.58 1.156
34 CON GZ 2.58 1.103
214 CON GZ 2.58 2.71
33 214 CON GZ 2.58 1.373
31 47 CON GZ 2.58 1.59
*****
321 322 CMOM GX 2.66 0.351
43 53 CMOM GX 2.66 0.357
41 52 CMOM GX 2.66 0.363
```

```
39 51 CMOM GX 2.66 0.369
37 216 CMOM GX 2.66 0.763
35 49 CMOM GX 2.66 1.156
34 CMOM GX 2.66 1.103
214 CMOM GX 2.66 2.71
33 214 CMOM GX 2.66 1.373
31 47 CMOM GX 2.66 1.59
*TRESTLE BODY WIND
6 7 13 14 20 TO 23 UNI GZ 1.2
**WIND ON CABLE TRAY SUPPORT
*178 179 181 TO 184 433 443 453 CON GZ 1.11 0.395
*178 179 181 TO 184 433 443 453 CMOM GX 0.81
************************************************************
LOAD 5 WIND LOAD IN -Z DIRN
MEMBER LOAD
321 322 CON GZ -2.58 0.351
43 53 CON GZ -2.58 0.357
41 52 CON GZ -2.58 0.363
39 51 CON GZ -2.58 0.369
37 216 CON GZ -2.58 0.763
35 49 CON GZ -2.58 1.156
34 CON GZ -2.58 1.103
214 CON GZ -2.58 2.71
33 214 CON GZ -2.58 1.373
31 47 CON GZ -2.58 1.59
*****
321 322 CMOM GX -2.66 0.351
43 53 CMOM GX -2.66 0.357
41 52 CMOM GX -2.66 0.363
39 51 CMOM GX -2.66 0.369
37 216 CMOM GX -2.66 0.763
35 49 CMOM GX -2.66 1.156
34 CMOM GX -2.66 1.103
214 CMOM GX -2.66 2.71
33 214 CMOM GX -2.66 1.373
31 47 CMOM GX -2.66 1.59
*WIND ON TRESTLE BODY
6 7 13 14 20 TO 23 UNI GZ -1.2
**WIND ON CABLE TRAY SUPPORT
*178 179 181 TO 184 433 443 453 CON GZ -1.11 0.395
*178 179 181 TO 184 433 443 453 CMOM GX -0.81
************************************************************
LOAD 6 TL1
TEMPERATURE LOAD
1 TO 23 31 TO 45 47 49 51 TO 54 71 73 75 TO 79 81 83 TO 86 101 103 -
105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 175 201 TO 212 319 -
320 TO 324 TEMP 25
************************************************************
LOAD 7 TL2
TEMPERATURE LOAD
1 TO 23 31 TO 45 47 49 51 TO 54 71 73 75 TO 79 81 83 TO 86 101 103 -
105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 175 201 TO 212 319 -
320 TO 324 TEMP -25
************************************************************
```

```
LOAD 11 G
REPEAT LOAD
1 1.0
LOAD 12 G + QN + MN
REPEAT LOAD
1 1.0 2 1.0 3 1.0
************************************************************
****LOAD COMBINATION for structural design
*LOAD COMB 201 1.4 DL
*1 1.4
*LOAD COMB 202 1.2 (DL + TL1) + 1.6  (LL+ML)
*1 1.2 6 1.2 2 1.6 3 1.6
*LOAD COMB 203 1.2 (DL + TL2) + 1.6  (LL+ML)
*1 1.2 7 1.2 2 1.6 3 1.6
*LOAD COMB 204 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
*1 1.2 4 1.6 2 0.5 3 0.5
*LOAD COMB 205 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
*1 1.2 5 1.6 2 0.5 3 0.5
*LOAD COMB 206 0.9 DL + 1.6 WL(+Z)
*1 0.9 4 1.6
*LOAD COMB 207 0.9 DL + 1.6 WL(-Z)
*1 0.9 5 1.6
************************************************************
****LOAD COMBINATION FOR BEARING PRESSURE CHECK
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 6 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL2) + 1.0  (LL+ML)
1 1.0 7 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 4 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 5 0.75 2 0.75 3 0.75
LOAD COMB 306 0.6 DL + 1.0 WL(+Z)
1 0.6 4 1.0
LOAD COMB 307 0.6 DL + 1.0 WL(-Z)
1 0.6 5 1.0
************************************************************
****LOAD COMBINATION FOR FOUNDATION DESIGN
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 6 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 (DL + TL2) + 1.7  (LL+ML)
1 1.4 7 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
1 1.2 4 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
1 1.2 5 1.6 2 0.5 3 0.5
LOAD COMB 406 0.9 DL + 1.6 WL(+Z)
1 0.9 4 1.6
LOAD COMB 407 0.9 DL + 1.6 WL(-Z)
1 0.9 5 1.6
```

```
**************************************************************
PERFORM ANALYSIS
*STEEL TAKE OFF LIST 1 TO 23 31 TO 47 49 51 TO 54 71 73 75 TO 79 81 83 TO 86 -
*101 103 105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 176 -
*201 TO 212 214 216 218 220 222 224 226 228 230 232
*LOAD LIST 201 TO 207
*PARAMETER 1
*CODE LRFD
*FYLD 275000 MEMB 1 TO 23 31 TO 47 49 51 TO 54 71 73 75 TO 79 81 83 TO 86 101 -
*103 105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 176 201 TO 212 -
*214 216 218 220 222 224 226 228 230 232
*FU 410000 MEMB 1 TO 23 31 TO 47 49 51 TO 54 71 73 75 TO 79 81 83 TO 86 101 -
*103 105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 176 201 TO 212 -
*214 216 218 220 222 224 226 228 230 232
*TRACK 0 MEMB 1 TO 23 31 TO 47 49 51 TO 54 71 73 75 TO 79 81 83 TO 86 101 103 -
*105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 176 201 TO 212 214 -
*216 218 220 222 224 226 228 230 232
*BEAM 1 MEMB 1 TO 23 31 TO 47 49 51 TO 54 71 73 75 TO 79 81 83 TO 86 101 103 -
*105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 176 201 TO 212 214 -
*216 218 220 222 224 226 228 230 232
*TMAIN 300 MEMB 1 TO 23 31 TO 47 49 51 TO 54 71 73 75 TO 79 81 83 TO 86 101 -
*103 105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 176 201 TO 212 -
*214 216 218 220 222 224 226 228 230 232
*MAIN 200 MEMB 1 TO 23 31 TO 47 49 51 TO 54 71 73 75 TO 79 81 83 TO 86 101 -
*103 105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 176 201 TO 212 -
*214 216 218 220 222 224 226 228 230 232
*NSF 0.8 MEMB 1 TO 23 31 TO 47 49 51 TO 54 71 73 75 TO 79 81 83 TO 86 101 103 -
*105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 176 201 TO 212 214 -
*216 218 220 222 224 226 228 230 232
*KZ 0.85 MEMB 161 TO 176
*KY 0.85 MEMB 161 TO 176
*UNT 1.57 MEMB 8 TO 12 15 TO 19
*LY 1.57 MEMB 1 TO 5 8 TO 12 15 TO 19
*LX 1.57 MEMB 1 TO 5 8 TO 12 15 TO 19
*LZ 5.7 MEMB 1 TO 5 8 TO 12 15 TO 19
**LX 2.1 MEMB 151 TO 153
**LY 2.1 MEMB 151 TO 153
**LZ 2.1 MEMB 151 TO 153
*LZ 6 MEMB 39 TO 46 51 TO 54 75 TO 78 83 TO 86 105 TO 108 113 TO 116
*LZ 6.79 MEMB 35 TO 38 49 73 81 103 111 216 220 224 228 232
*LZ 6.44 MEMB 31 TO 34 47 71 79 101 109 214 218 222 226 230
*LY 3 MEMB 39 TO 46 51 TO 54 75 TO 78 83 TO 86 105 TO 108 113 TO 116
*LY 3.4 MEMB 31 TO 38 47 49 71 73 79 81 101 103 109 111 214 216 218 220 222 -
*224 226 228 230 232
*UNT 3 MEMB 39 TO 46 51 TO 54 75 TO 78 83 TO 86 105 TO 108 113 TO 116
*UNT 3.4 MEMB 31 TO 38 47 49 71 73 79 81 101 103 109 111 214 216 218 220 222 -
*224 226 228 230 232
*CHECK CODE MEMB 1 TO 23 31 TO 47 49 51 TO 54 71 73 75 TO 79 81 83 TO 86 101 -
*103 105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 176 201 TO 212 -
*214 216 218 220 222 224 226 228 230 232
LOAD LIST 401 TO 407
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 301 TO 318
```

```
FYMAIN 420000 MEMB 301 TO 318
TRACK 1 MEMB 301 TO 318
CLB 0.05 MEMB 301 TO 318
CLT 0.05 MEMB 301 TO 318
CLS 0.05 MEMB 301 TO 318
MAXMAIN 20 MEMB 301 TO 308
MINMAIN 20 MEMB 301 TO 308
MAXMAIN 16 MEMB 309 TO 318
MINMAIN 16 MEMB 309 TO 318
MINSEC 8 MEMB 301 TO 318
REINF 0 MEMB 301 TO 308
DESIGN COLUMN 301 TO 308
DESIGN BEAM 309 TO 318
END CONCRETE DESIGN
FINISH
```

```
*****************************************************
*                                                   *
*           STAAD.Pro CONNECT Edition               *
*           Version  21.00.01.12                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    MAY 19, 2020                   *
*           Time=    15: 3:56                       *
*                                                   *
*  Licensed to: MMH KOLKATTA                        *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE:  D:\F Drive\Maaden Gold\Conveyor CNV-3210-001\Civil\Documents\Staad\R1\3210-CNV
     2. START JOB INFORMATION
     3. ENGINEER DATE 19-MAY-20
     4. JOB NAME MMGP-SAUDI ARABIA
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045/900707
     7. JOB REV 1
     8. JOB PART GROUND PORTAL FDN
     9. JOB REF CONVEYOR 3210-CNV-001
    10. ENGINEER NAME APL
    11. CHECKER NAME SBY
    12. APPROVED NAME PRI
    13. END JOB INFORMATION
    14. INPUT WIDTH 79
    15. UNIT METER KN
    16. JOINT COORDINATES
    17. 1 -18.773 -0.15 -2.165; 2 -18.773 -0.15 3.735; 3 -12 -0.15 -2.165
    18. 4 -12 -0.15 3.735; 5 -6 -0.15 -2.165; 6 -6 -0.15 3.735; 7 -1 -0.15 -2.165
    19. 8 -1 -0.15 -1.17; 9 -1 -0.15 0; 10 -1 -0.15 1.57; 11 -1 -0.15 2.74
    20. 12 -1 -0.15 3.735; 14 -3 0.196559 -2.165; 15 -3 0.196559 -1.17
    21. 16 -3 0.196559 0; 17 -3 0.196559 1.57; 18 -3 0.196559 2.74
    22. 19 -3 0.196559 3.735; 20 -4.5 0.294838 0; 21 -6 0.393118 -2.165
    23. 22 -6 0.393118 -1.17; 23 -6 0.393118 0; 24 -6 0.393118 1.57
    24. 25 -6 0.393118 2.74; 26 -6 0.393118 3.735; 27 -7.5 0.491397 0
    25. 28 -9 0.589677 -2.165; 29 -9 0.589677 -1.17; 30 -9 0.589677 0
    26. 31 -9 0.589677 1.57; 32 -9 0.589677 2.74; 33 -9 0.589677 3.735
    27. 34 -10.5 0.687956 0; 35 -12 0.786236 -2.165; 36 -12 0.786236 -1.17
    28. 37 -12 0.786236 0; 38 -12 0.786236 1.57; 39 -12 0.786236 2.74
    29. 40 -12 0.786236 3.735; 41 -13.6933 0.897177 0; 42 -15.3865 1.00812 -2.165
    30. 43 -15.3865 1.00812 -1.17; 44 -15.3865 1.00812 0; 45 -15.3865 1.00812 1.57
    31. 46 -15.3865 1.00812 2.74; 47 -15.3865 1.00812 3.735; 48 -17.0798 1.11906 0
    32. 49 -18.773 1.23 -2.165; 50 -18.773 1.23 -1.17; 51 -18.773 1.23 0
    33. 52 -18.773 1.23 1.57; 53 -18.773 1.23 2.74; 54 -18.773 1.23 3.735
    34. 55 -20.368 1.44 0; 56 -21.963 1.65 -2.165; 57 -21.963 1.65 -1.17
    35. 58 -21.963 1.65 0; 59 -21.963 1.65 1.57; 60 -21.963 1.65 2.74
    36. 61 -21.963 1.65 3.735; 62 -23.558 1.86 0; 63 -25.153 2.07 -2.165
    37. 64 -25.153 2.07 -1.17; 65 -25.153 2.07 0; 66 -25.153 2.07 1.57
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39. 70 -18.773 1.03 3.735; 71 -18.773 1.03 -0.265; 72 -18.773 1.03 1.835
40. 73 -18.773 -1.5 -2.165; 74 -18.773 -1.5 3.735; 75 -12 -1.5 -2.165
41. 76 -12 -1.5 3.735; 77 -6 -1.5 -2.165; 78 -6 -1.5 3.735; 79 -1 -1.35 -2.165
42. 80 -1 -1.35 3.735; 81 -1 0.06 -2.165; 82 -1 0.06 -1.17; 83 -1 0.06 0
43. 84 -1 0.06 1.57; 85 -1 0.06 2.74; 86 -1 0.06 3.735
44. MEMBER INCIDENCES
45. 1 49 50; 2 50 51; 3 51 52; 4 52 53; 5 53 54; 6 3 35; 7 4 40; 8 35 36; 9 36 37
46. 10 37 38; 11 38 39; 12 39 40; 13 5 21; 14 6 26; 15 21 22; 16 22 23; 17 23 24
47. 18 24 25; 19 25 26; 20 1 69; 21 69 49; 22 2 70; 23 70 54; 31 65 62; 32 62 58
48. 33 58 55; 34 55 51; 35 51 48; 36 48 44; 37 44 41; 38 41 37; 39 37 34; 40 34 30
49. 41 30 27; 42 27 23; 43 23 20; 44 20 16; 45 83 9; 47 66 59; 49 52 45; 51 38 31
50. 52 31 24; 53 24 17; 54 84 10; 71 64 57; 73 50 43; 75 36 29; 76 29 22; 77 22 15
51. 78 82 8; 79 67 60; 81 53 46; 83 39 32; 84 32 25; 85 25 18; 86 85 11; 101 63 56
52. 103 49 42; 105 35 28; 106 28 21; 107 21 14; 108 81 7; 109 68 61; 111 54 47
53. 113 40 33; 114 33 26; 115 26 19; 116 86 12; 133 56 57; 134 42 43; 135 28 29
54. 136 14 15; 137 60 61; 138 46 47; 139 32 33; 140 18 19; 151 69 71; 152 71 72
55. 153 72 70; 154 1 71; 155 2 72; 161 66 62; 162 59 62; 163 59 55; 164 52 55
56. 165 52 48; 166 45 48; 167 45 41; 168 38 41; 169 38 34; 170 31 34; 171 31 27
57. 172 24 27; 173 24 20; 174 17 20; 175 17 83; 201 57 58; 202 58 59; 203 59 60
58. 204 43 44; 205 44 45; 206 45 46; 207 29 30; 208 30 31; 209 31 32; 210 15 16
59. 211 16 17; 212 17 18; 214 59 52; 216 45 38; 218 57 50; 220 43 36; 222 60 53
60. 224 46 39; 226 56 49; 228 42 35; 230 61 54; 232 47 40; 301 73 1; 302 74 2
61. 303 75 3; 304 76 4; 305 77 5; 306 78 6; 307 79 7; 308 80 12; 309 3 5; 310 4 6
62. 311 1 2; 312 3 4; 313 5 6; 314 7 8; 315 8 9; 316 9 10; 317 10 11; 318 11 12
63. 319 14 81; 320 15 82; 321 16 83; 322 17 84; 323 18 85; 324 19 86
64. DEFINE MATERIAL START
65. ISOTROPIC STEEL
66. E 2.05E+008
67. POISSON 0.3
68. DENSITY 76.8195
69. ALPHA 1.2E-005
70. DAMP 0.03
71. TYPE STEEL
72. STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
73. ISOTROPIC CONCRETE
74. E 2.17185E+007
75. POISSON 0.17
76. DENSITY 23.5616
77. ALPHA 1E-005
78. DAMP 0.05
79. TYPE CONCRETE
80. STRENGTH FCU 27579
81. END DEFINE MATERIAL
82. MEMBER PROPERTY EUROPEAN
83. 1 TO 5 TAPERED 0.3 0.008 0.3 0.15 0.016 0.15 0.016
84. 6 TO 19 TAPERED 0.25 0.01 0.25 0.15 0.016 0.15 0.016
85. 20 TO 23 TAPERED 0.25 0.01 0.25 0.15 0.016 0.15 0.016
86. 31 TO 45 47 49 51 TO 54 214 216 321 322 TABLE ST UPN240
87. 73 75 TO 78 81 83 TO 86 220 224 320 323 TABLE ST UPN200
88. 71 79 218 222 TABLE ST UPN200
89. 101 103 105 TO 109 111 113 TO 116 226 228 230 232 319 324 TABLE ST UPN140
90. 133 TO 140 TABLE ST IPE200
91. 151 TO 153 TABLE LD L90X90X8 SP 0.01
92. 154 155 TABLE LD L65X65X6 SP 0.01
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 95. MEMBER PROPERTY EUROPEAN
 96. 301 TO 308 PRIS YD 0.6 ZD 0.6
 97. 309 310 PRIS YD 0.6 ZD 0.3
 98. 311 TO 313 PRIS YD 0.5 ZD 0.3
 99. 314 TO 318 PRIS YD 0.6 ZD 0.35
100. CONSTANTS
101. BETA 90 MEMB 6 7 13 14
102. BETA 356.2 MEMB 8 TO 12 15 TO 19
103. BETA 352.5 MEMB 1 TO 5
104. MATERIAL STEEL MEMB 1 TO 23 31 TO 45 47 49 51 TO 54 71 73 75 TO 79 81 -
105. 83 TO 86 101 103 105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 175 -
106. 201 TO 212 214 216 218 220 222 224 226 228 230 232 319 TO 324
107. MATERIAL CONCRETE MEMB 301 TO 318
108. SUPPORTS
109. 63 TO 68 FIXED BUT FX MX MY MZ
110. 73 TO 80 FIXED
111. MEMBER TRUSS
112. 154 155 161 TO 175 201 TO 212
113. MEMBER RELEASE
114. 35 49 73 81 START MY MZ
115. 1 103 105 107 111 113 115 133 TO 140 151 START MY MZ
116. 5 34 106 114 133 TO 140 153 214 218 222 226 228 230 232 END MY MZ
117. 6 7 13 14 20 22 START MP 0.99
118. 45 54 78 86 108 116 END MP 0.99
119. *1 103 105 107 111 113 115 133 TO 140 151 START MP 0.99
120. *5 34 106 114 133 TO 140 153 214 218 222 226 228 230 232 END MP 0.99
121. MEMBER OFFSET
122. 311 TO 313 START 0 -0.25 0
123. 311 TO 313 END 0 -0.25 0
124. 309 310 START 0 -0.3 0
125. 309 310 END 0 -0.3 0
126. 314 TO 318 START 0 -0.3 0
127. 314 TO 318 END 0 -0.3 0
128. LOAD 1 DL
129. SELFWEIGHT Y -1.1
130. *WALKWAY LOAD @30KG/SQM INCLUDING SEAL PLATE LOAD @ 30KG/SQM _HANDRAIL LOAD @1
131. *WALKWAY WIDTH 1000MM _BELT WIDTH 1200MM
132. MEMBER LOAD
133. 71 73 75 TO 77 79 81 83 TO 85 218 220 222 224 320 323 UNI GY -0.33
134. 31 TO 44 47 49 51 TO 53 214 216 321 322 UNI GY -0.48
135. *CABLE TRAY LOAD @112KG/TRAY/M RUN * 3 NOS OF TRAY = 336KG/M RUN
136. *CONCENTRATED FORCE = 336*3=10KN _MOMENT= 10*0.4 = 4KNM
137. 1 8 15 133 TO 136 CON GY -10 0.42
138. 1 8 15 133 TO 136 CMOM GX -4
139. *LIGHTING POLE LOAD = 100KG/SUPPORT POINT(ASSUMED)
140. 1 8 15 133 TO 136 CON GY -1 0.595
141. 5 12 19 137 TO 140 CON GY -1 0.2
142. *PIPE DEAD LOAD
143. 1 8 15 133 TO 136 CON GY -2.03 0.2
144. 5 12 19 137 TO 140 CON GY -3.57 0.595
145. *CONVEYOR DEAD LOAD VERTICAL
146. 321 322 CON GY -5 0.351
147. 43 53 CON GY -5 0.357
148. 41 52 CON GY -5 0.363
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151. 35 49 CON GY -5 1.156
152. 34 CON GY -5 1.103
153. 214 CON GY -5 2.71
154. 33 214 CON GY -5 1.373
155. 31 47 CON GY -5 1.59
156. *CONVEYOR DEAD LOAD HORIZONTAL ALONG LENGTH
157. 321 322 CON GX 1.5 0.351
158. 43 53 CON GX 1.5 0.357
159. 41 52 CON GX 1.5 0.363
160. 39 51 CON GX 1.5 0.369
161. 37 216 CON GX 1.5 0.763
162. 35 49 CON GX 1.5 1.156
163. 34 CON GX 1.5 1.103
164. 214 CON GX 1.5 2.71
165. 33 214 CON GX 1.5 1.373
166. 31 47 CON GX 1.5 1.59
167. **CONVEYOR DEAD LOAD HORIZONTAL ACROSS LENGTH
168. 321 322 CON GZ 0.8 0.351
169. 43 53 CON GZ 0.8 0.357
170. 41 52 CON GZ 0.8 0.363
171. 39 51 CON GZ 0.8 0.369
172. 37 216 CON GZ 0.8 0.763
173. 35 49 CON GZ 0.8 1.156
174. 34 CON GZ 0.8 1.103
175. 214 CON GZ 0.8 2.71
176. 33 214 CON GZ 0.8 1.373
177. 31 47 CON GZ 0.8 1.59
178. ************************************************************
179. LOAD 2 LL
180. *WALKWAY LOAD @400KG/SQ. MTER(FOR GALLERY)
181. MEMBER LOAD
182. 71 73 75 TO 77 79 81 83 TO 85 218 220 222 224 320 323 UNI GY -2
183. 31 TO 44 47 49 51 TO 53 214 216 321 322 UNI GY -2
184. ************************************************************
185. LOAD 3 ML
186. *CONVEYOR MATERIAL LOAD
187. MEMBER LOAD
188. 321 322 CON GY -5.5 0.351
189. 43 53 CON GY -5.5 0.357
190. 41 52 CON GY -5.5 0.363
191. 39 51 CON GY -5.5 0.369
192. 37 216 CON GY -5.5 0.763
193. 35 49 CON GY -5.5 1.156
194. 34 CON GY -5.5 1.103
195. 214 CON GY -5.5 2.71
196. 33 214 CON GY -5.5 1.373
197. 31 47 CON GY -5.5 1.59
198. *SPILLAGE LOAD @75KG/ SQ. M ON SEAL PLATE
199. 31 TO 44 47 49 51 TO 53 214 216 321 322 UNI GY -0.64
200. ************************************************************
201. LOAD 4 WIND LOAD IN +Z DIRN
202. MEMBER LOAD
203. 321 322 CON GZ 2.58 0.351
204. 43 53 CON GZ 2.58 0.357
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207. 37 216 CON GZ 2.58 0.763
208. 35 49 CON GZ 2.58 1.156
209. 34 CON GZ 2.58 1.103
210. 214 CON GZ 2.58 2.71
211. 33 214 CON GZ 2.58 1.373
212. 31 47 CON GZ 2.58 1.59
213. *****
214. 321 322 CMOM GX 2.66 0.351
215. 43 53 CMOM GX 2.66 0.357
216. 41 52 CMOM GX 2.66 0.363
217. 39 51 CMOM GX 2.66 0.369
218. 37 216 CMOM GX 2.66 0.763
219. 35 49 CMOM GX 2.66 1.156
220. 34 CMOM GX 2.66 1.103
221. 214 CMOM GX 2.66 2.71
222. 33 214 CMOM GX 2.66 1.373
223. 31 47 CMOM GX 2.66 1.59
224. *TRESTLE BODY WIND
225. 6 7 13 14 20 TO 23 UNI GZ 1.2
226. **WIND ON CABLE TRAY SUPPORT
227. *178 179 181 TO 184 433 443 453 CON GZ 1.11 0.395
228. *178 179 181 TO 184 433 443 453 CMOM GX 0.81
229. ************************************************************
230. LOAD 5 WIND LOAD IN -Z DIRN
231. MEMBER LOAD
232. 321 322 CON GZ -2.58 0.351
233. 43 53 CON GZ -2.58 0.357
234. 41 52 CON GZ -2.58 0.363
235. 39 51 CON GZ -2.58 0.369
236. 37 216 CON GZ -2.58 0.763
237. 35 49 CON GZ -2.58 1.156
238. 34 CON GZ -2.58 1.103
239. 214 CON GZ -2.58 2.71
240. 33 214 CON GZ -2.58 1.373
241. 31 47 CON GZ -2.58 1.59
242. *****
243. 321 322 CMOM GX -2.66 0.351
244. 43 53 CMOM GX -2.66 0.357
245. 41 52 CMOM GX -2.66 0.363
246. 39 51 CMOM GX -2.66 0.369
247. 37 216 CMOM GX -2.66 0.763
248. 35 49 CMOM GX -2.66 1.156
249. 34 CMOM GX -2.66 1.103
250. 214 CMOM GX -2.66 2.71
251. 33 214 CMOM GX -2.66 1.373
252. 31 47 CMOM GX -2.66 1.59
253. *WIND ON TRESTLE BODY
254. 6 7 13 14 20 TO 23 UNI GZ -1.2
255. **WIND ON CABLE TRAY SUPPORT
256. *178 179 181 TO 184 433 443 453 CON GZ -1.11 0.395
257. *178 179 181 TO 184 433 443 453 CMOM GX -0.81
258. ************************************************************
259. LOAD 6 TL1
260. TEMPERATURE LOAD
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263. 320 TO 324 TEMP 25
264. ********************************************************************
265. LOAD 7 TL2
266. TEMPERATURE LOAD
267. 1 TO 23 31 TO 45 47 49 51 TO 54 71 73 75 TO 79 81 83 TO 86 101 103 -
268. 105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 175 201 TO 212 319 -
269. 320 TO 324 TEMP -25
270. ********************************************************************
271. LOAD 11 G
272. REPEAT LOAD
273. 1 1.0
274. LOAD 12 G + QN + MN
275. REPEAT LOAD
276. 1 1.0 2 1.0 3 1.0
277. ********************************************************************
278. ****LOAD COMBINATION FOR STRUCTURAL DESIGN
279. *LOAD COMB 201 1.4 DL
280. *1 1.4
281. *LOAD COMB 202 1.2 (DL + TL1) + 1.6  (LL+ML)
282. *1 1.2 6 1.2 2 1.6 3 1.6
283. *LOAD COMB 203 1.2 (DL + TL2) + 1.6  (LL+ML)
284. *1 1.2 7 1.2 2 1.6 3 1.6
285. *LOAD COMB 204 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
286. *1 1.2 4 1.6 2 0.5 3 0.5
287. *LOAD COMB 205 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
288. *1 1.2 5 1.6 2 0.5 3 0.5
289. *LOAD COMB 206 0.9 DL + 1.6 WL(+Z)
290. *1 0.9 4 1.6
291. *LOAD COMB 207 0.9 DL + 1.6 WL(-Z)
292. *1 0.9 5 1.6
293. ********************************************************************
294. ****LOAD COMBINATION FOR BEARING PRESSURE CHECK
295. LOAD COMB 301 1.0 DL
296. 1 1.0
297. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
298. 1 1.0 6 1.0 2 1.0 3 1.0
299. LOAD COMB 303 1.0 (DL + TL2) + 1.0  (LL+ML)
300. 1 1.0 7 1.0 2 1.0 3 1.0
301. LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
302. 1 1.0 4 0.75 2 0.75 3 0.75
303. LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
304. 1 1.0 5 0.75 2 0.75 3 0.75
305. LOAD COMB 306 0.6 DL + 1.0 WL(+Z)
306. 1 0.6 4 1.0
307. LOAD COMB 307 0.6 DL + 1.0 WL(-Z)
308. 1 0.6 5 1.0
309. ********************************************************************
310. ****LOAD COMBINATION FOR FOUNDATION DESIGN
311. LOAD COMB 401 1.4 DL
312. 1 1.4
313. LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
314. 1 1.4 6 1.4 2 1.7 3 1.7
315. LOAD COMB 403 1.4 (DL + TL2) + 1.7  (LL+ML)
316. 1 1.4 7 1.4 2 1.7 3 1.7
```

```
319. LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
320. 1 1.2 5 1.6 2 0.5 3 0.5
321. LOAD COMB 406 0.9 DL + 1.6 WL(+Z)
322. 1 0.9 4 1.6
323. LOAD COMB 407 0.9 DL + 1.6 WL(-Z)
324. 1 0.9 5 1.6
325. ************************************************************
326. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**85**	**NUMBER OF MEMBERS**	**142**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**14**

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 9, TOTAL DEGREES OF FREEDOM = 450
TOTAL LOAD COMBINATION CASES = 14 SO FAR.

```
327. *STEEL TAKE OFF LIST 1 TO 23 31 TO 47 49 51 TO 54 71 73 75 TO 79 81 83 TO 86 -
328. *101 103 105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 176 -
329. *201 TO 212 214 216 218 220 222 224 226 228 230 232
330. *LOAD LIST 201 TO 207
331. *PARAMETER 1
332. *CODE LRFD
333. *FYLD 275000 MEMB 1 TO 23 31 TO 47 49 51 TO 54 71 73 75 TO 79 81 83 TO 86 101 -
334. *103 105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 176 201 TO 212 -
335. *214 216 218 220 222 224 226 228 230 232
336. *FU 410000 MEMB 1 TO 23 31 TO 47 49 51 TO 54 71 73 75 TO 79 81 83 TO 86 101 -
337. *103 105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 176 201 TO 212 -
338. *214 216 218 220 222 224 226 228 230 232
339. *TRACK 0 MEMB 1 TO 23 31 TO 47 49 51 TO 54 71 73 75 TO 79 81 83 TO 86 101 103 -
340. *105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 176 201 TO 212 214 -
341. *216 218 220 222 224 226 228 230 232
342. *BEAM 1 MEMB 1 TO 23 31 TO 47 49 51 TO 54 71 73 75 TO 79 81 83 TO 86 101 103 -
343. *105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 176 201 TO 212 214 -
344. *216 218 220 222 224 226 228 230 232
345. *TMAIN 300 MEMB 1 TO 23 31 TO 47 49 51 TO 54 71 73 75 TO 79 81 83 TO 86 101 -
346. *103 105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 176 201 TO 212 -
347. *214 216 218 220 222 224 226 228 230 232
348. *MAIN 200 MEMB 1 TO 23 31 TO 47 49 51 TO 54 71 73 75 TO 79 81 83 TO 86 101 -
349. *103 105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 176 201 TO 212 -
350. *214 216 218 220 222 224 226 228 230 232
351. *NSF 0.8 MEMB 1 TO 23 31 TO 47 49 51 TO 54 71 73 75 TO 79 81 83 TO 86 101 103 -
352. *105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 176 201 TO 212 214 -
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353. *216 218 220 222 224 226 228 230 232
354. *KZ 0.85 MEMB 161 TO 176
355. *KY 0.85 MEMB 161 TO 176
356. *UNT 1.57 MEMB 8 TO 12 15 TO 19
357. *LY 1.57 MEMB 1 TO 5 8 TO 12 15 TO 19
358. *LX 1.57 MEMB 1 TO 5 8 TO 12 15 TO 19
359. *LZ 5.7 MEMB 1 TO 5 8 TO 12 15 TO 19
360. **LX 2.1 MEMB 151 TO 153
361. **LY 2.1 MEMB 151 TO 153
362. **LZ 2.1 MEMB 151 TO 153
363. *LZ 6 MEMB 39 TO 46 51 TO 54 75 TO 78 83 TO 86 105 TO 108 113 TO 116
364. *LZ 6.79 MEMB 35 TO 38 49 73 81 103 111 216 220 224 228 232
365. *LZ 6.44 MEMB 31 TO 34 47 71 79 101 109 214 218 222 226 230
366. *LY 3 MEMB 39 TO 46 51 TO 54 75 TO 78 83 TO 86 105 TO 108 113 TO 116
367. *LY 3.4 MEMB 31 TO 38 47 49 71 73 79 81 101 103 109 111 214 216 218 220 222 -
368. *224 226 228 230 232
369. *UNT 3 MEMB 39 TO 46 51 TO 54 75 TO 78 83 TO 86 105 TO 108 113 TO 116
370. *UNT 3.4 MEMB 31 TO 38 47 49 71 73 79 81 101 103 109 111 214 216 218 220 222 -
371. *224 226 228 230 232
372. *CHECK CODE MEMB 1 TO 23 31 TO 47 49 51 TO 54 71 73 75 TO 79 81 83 TO 86 101 -
373. *103 105 TO 109 111 113 TO 116 133 TO 140 151 TO 155 161 TO 176 201 TO 212 -
374. *214 216 218 220 222 224 226 228 230 232
375. LOAD LIST 401 TO 407
376. START CONCRETE DESIGN
377. CODE ACI 2008
378. FC 35000 MEMB 301 TO 318
379. FYMAIN 420000 MEMB 301 TO 318
380. TRACK 1 MEMB 301 TO 318
381. TRACK 2 MEMB 310 312
382. CLB 0.05 MEMB 301 TO 318
383. CLT 0.05 MEMB 301 TO 318
384. CLS 0.05 MEMB 301 TO 318
385. MAXMAIN 20 MEMB 301 TO 308
386. MINMAIN 20 MEMB 301 TO 308
387. MAXMAIN 16 MEMB 309 TO 318
388. MINMAIN 16 MEMB 309 TO 318
389. MINSEC 8 MEMB 301 TO 318
390. REINF 0 MEMB 301 TO 308
391. DESIGN COLUMN 301 TO 308
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COLUMN NO. 301 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	140.21	1.33	0.00703

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	140.21	25.60	0.13525

==========

COLUMN NO. 302 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    103.56      0.50    0.00356
 ---------------------------------------------------------

    COLUMN INTERACTION:  MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    103.56     26.33    0.18835
 ---------------------------------------------------------


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     COLUMN  NO.   303  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-----------------------------------------------------------

 12 - 20 MM               1.047      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION:  MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    170.13     32.18    0.14010
 ---------------------------------------------------------

    COLUMN INTERACTION:  MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    170.13     11.82    0.05145
 ---------------------------------------------------------


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FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	136.48	31.77	0.17241

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	136.48	13.65	0.07407

==========

COLUMN NO. 305 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	149.28	32.26	0.16007

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    149.28      9.53    0.04730
 ---------------------------------------------------------
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COLUMN NO. 306 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
------------------------------------------------------------

 12 - 20 MM               1.047      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    116.24     31.89    0.20321
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    116.24     11.21    0.07144
 ---------------------------------------------------------
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COLUMN NO. 307 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM
```

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	70.30	3.89	0.04616

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	70.30	58.30	0.69118

==========

COLUMN NO. 308 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	58.57	2.99	0.04254

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

**********END OF COLUMN DESIGN RESULTS**********

392. DESIGN BEAM 309 TO 318

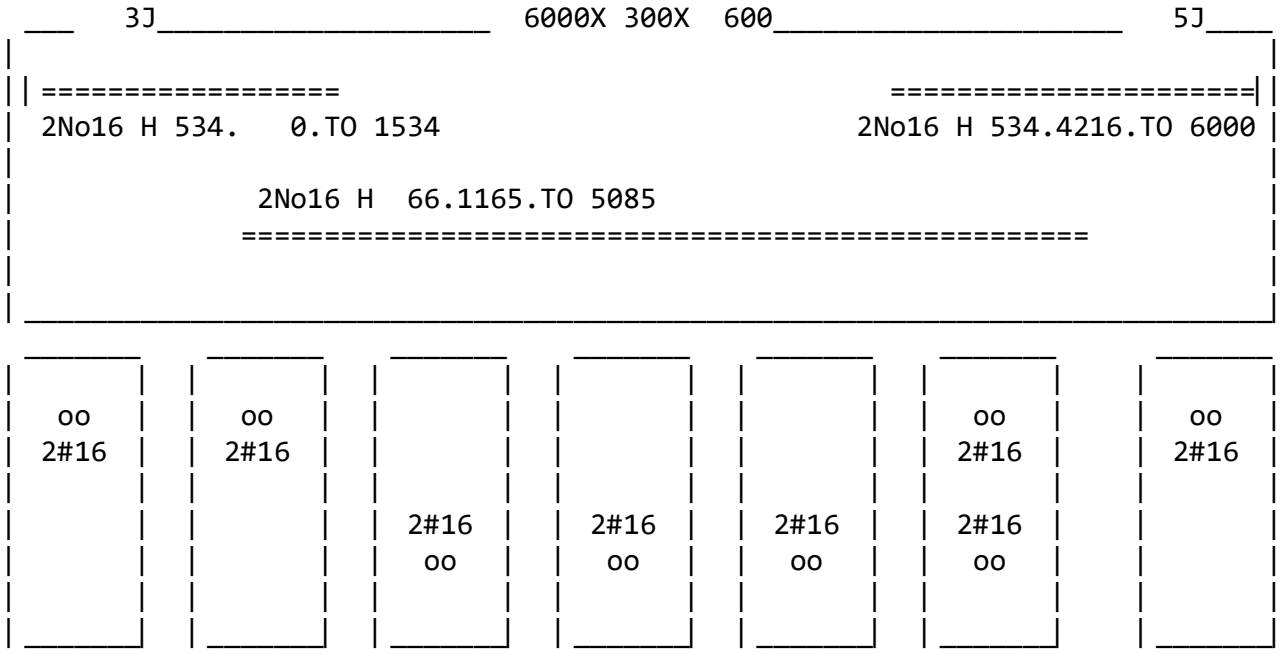
BEAM NO. 309 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	1165.	5085.	NO	NO
	CRITICAL POS MOMENT= 12.64 KN-MET AT 3000.MM, LOAD 402					
	REQD STEEL= 84.MM2, RHO=0.0005, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 63898.6 cm^4					
2	534.	2 - 16MM	0.	1534.	YES	NO
	CRITICAL NEG MOMENT= 17.56 KN-MET AT 0.MM, LOAD 402					
	REQD STEEL= 116.MM2, RHO=0.0007, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 63898.6 cm^4					
3	534.	2 - 16MM	4216.	6000.	NO	YES
	CRITICAL NEG MOMENT= 16.65 KN-MET AT 6000.MM, LOAD 403					
	REQD STEEL= 110.MM2, RHO=0.0007, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 63898.6 cm^4					

B E A M N O. 309 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 16.25 KNS Vc= 152.24 KNS Vs= 0.00 KNS
Tu= 0.09 KN-MET Tc= 6.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 15.96 KNS Vc= 152.70 KNS Vs= 0.00 KNS



BEAM NO. 310 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	1165.	4585.	NO	NO
	CRITICAL POS MOMENT= 12.59 KN-MET AT 3000.MM, LOAD 402					
	REQD STEEL= 83.MM2, RHO=0.0005, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 63898.6 cm^4					
2	534.	2 - 16MM	0.	1534.	YES	NO
	CRITICAL NEG MOMENT= 17.30 KN-MET AT 0.MM, LOAD 402					
	REQD STEEL= 114.MM2, RHO=0.0007, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 63898.6 cm^4					
3	534.	2 - 16MM	4216.	6000.	NO	YES

CRITICAL NEG MOMENT= 17.04 KN-MET AT 6000.MM, LOAD 403
REQD STEEL= 113.MM2, RHO=0.0007, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

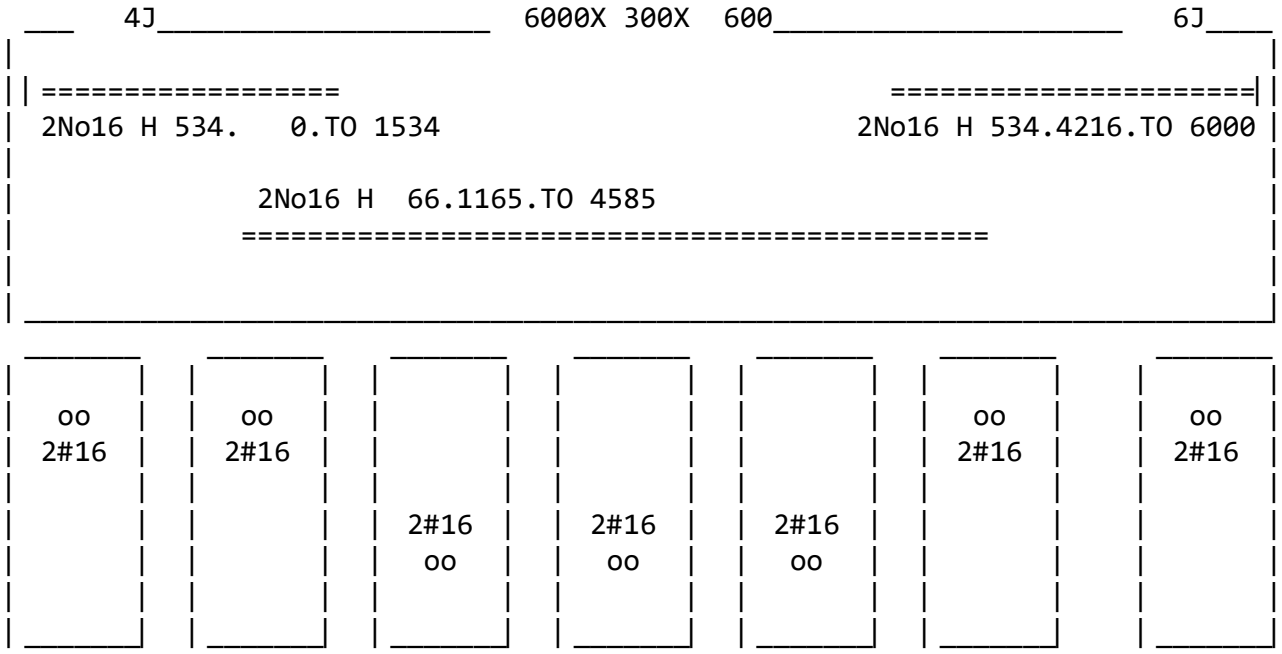
Cracked Moment of Inertia Iz at above location = 63898.6 cm^4

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)		MOMENTS(+VE/-VE) (KNS-MET)		LOAD(+VE/-VE)	
0.	0./	87.	0./	17.	0/	402
500.	0./	41.	0./	8.	0/	402
1000.	0./	4.	0./	1.	0/	402
1500.	27./	0.	5./	0.	403/	0
2000.	47./	0.	9./	0.	403/	0
2500.	59./	0.	12./	0.	403/	0
3000.	63./	0.	13./	0.	402/	0
3500.	60./	0.	12./	0.	402/	0
4000.	48./	0.	9./	0.	402/	0
4500.	28./	0.	5./	0.	402/	0
5000.	0./	3.	0./	1.	0/	403
5500.	0./	40.	0./	8.	0/	403
6000.	0./	86.	0./	17.	0/	403

B E A M N O. 310 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 16.15 KNS Vc= 152.27 KNS Vs= 0.00 KNS
Tu= 0.08 KN-MET Tc= 6.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 16.07 KNS Vc= 152.39 KNS Vs= 0.00 KNS
Tu= 0.08 KN-MET Tc= 6.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 311 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5900. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

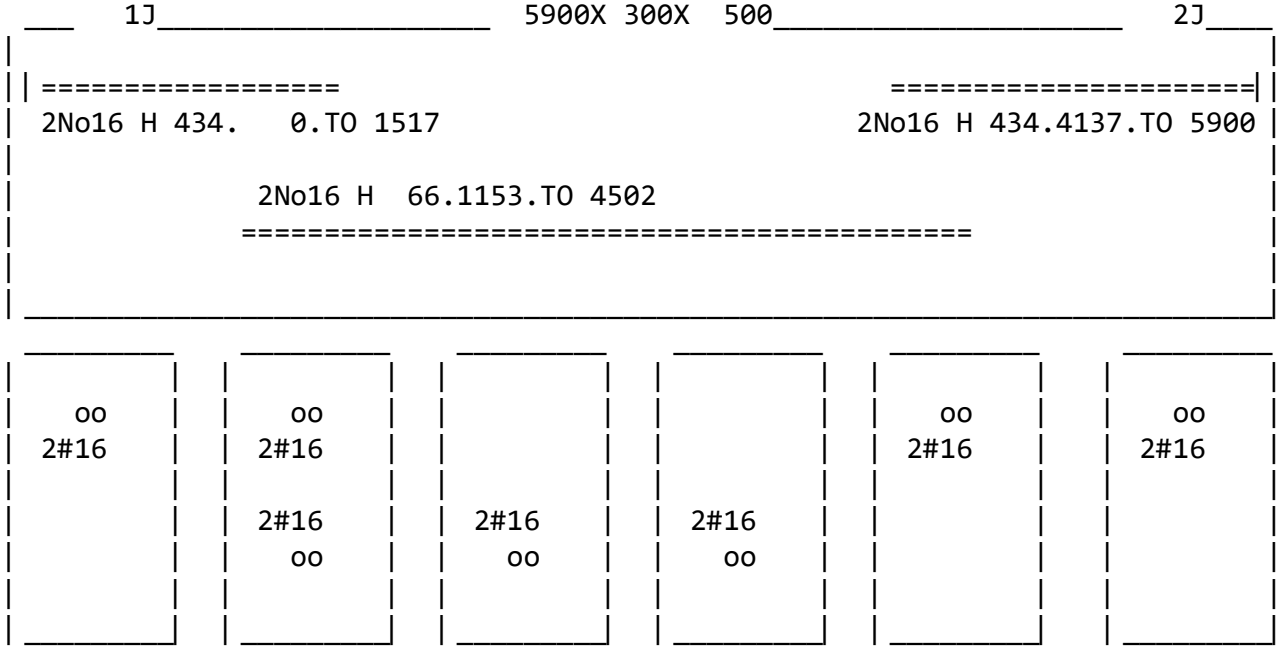
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	1153.	4502.	NO	NO
	CRITICAL POS MOMENT= 9.23 KN-MET AT 2950.MM, LOAD 403					
	REQD STEEL= 75.MM2, RHO=0.0006, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					
2	434.	2 - 16MM	0.	1517.	YES	NO
	CRITICAL NEG MOMENT= 14.94 KN-MET AT 0.MM, LOAD 402					
	REQD STEEL= 122.MM2, RHO=0.0009, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					
3	434.	2 - 16MM	4137.	5900.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     15.30 KN-MET  AT  5900.MM, LOAD  403|
|   REQD STEEL=   125.MM2, RHO=0.0010, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS        |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    41100.8 cm^4
```

B E A M N O. 311 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   13.60 KNS  Vc=  122.78 KNS  Vs=    0.00 KNS
 Tu=    0.02 KN-MET  Tc=    5.3 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   13.78 KNS  Vc=  122.71 KNS  Vs=    0.00 KNS
 Tu=    0.02 KN-MET  Tc=    5.3 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 312 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5900. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END

```
|-------------------------------------------------------------|
|   CRITICAL POS MOMENT=      7.55 KN-MET  AT  2950.MM, LOAD  401|
|   REQD STEEL=    61.MM2, RHO=0.0005, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS        |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                      |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    41100.8 cm^4

  2        434.     2 - 16MM         0.         2009.        YES  NO
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     20.36 KN-MET  AT     0.MM, LOAD  402|
|   REQD STEEL=   166.MM2, RHO=0.0013, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS        |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                      |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    41100.8 cm^4

  3        434.     2 - 16MM      3646.         5900.        NO   YES
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     21.24 KN-MET  AT  5900.MM, LOAD  403|
|   REQD STEEL=   173.MM2, RHO=0.0013, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS        |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                      |
|-------------------------------------------------------------|

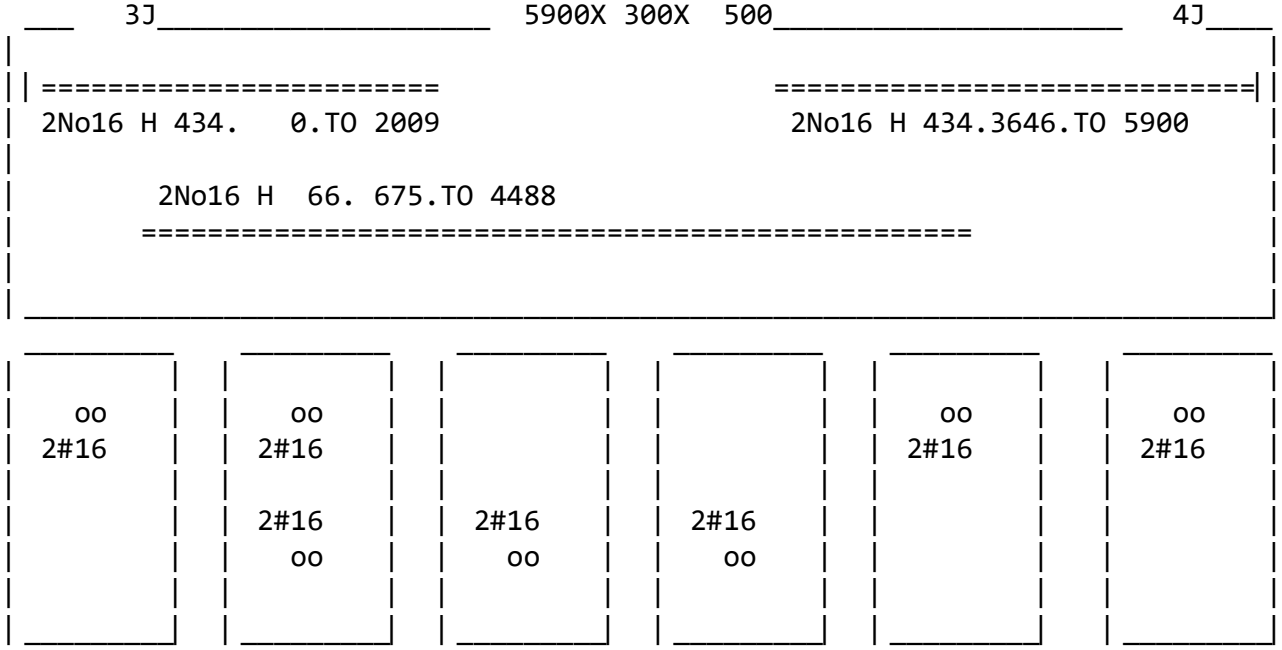
  Cracked Moment of Inertia Iz at above location =    41100.8 cm^4


REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )         (SQ.  MM )               (KNS-MET )

    0.          0./     127.          0./       20.        0/  402
  492.          0./      81.          0./       13.        0/  402
  983.          1./      43.          0./        7.      406/  402
 1475.         16./      14.          3./        2.      406/  402
 1967.         32./       0.          5./        0.      401/    0
 2458.         43./       0.          7./        0.      401/    0
 2950.         47./       0.          8./        0.      401/    0
 3442.         42./       0.          7./        0.      401/    0
 3933.         29./       0.          5./        0.      401/    0
 4425.         13./      15.          2./        2.      407/  403
 4917.          0./      46.          0./        7.        0/  403
 5408.          0./      85.          0./       14.        0/  403
 5900.          0./     132.          0./       21.        0/  403
```

B E A M N O. 312 D E S I G N R E S U L T S - SHEAR

AT END SUPPORT - Vu= 13.94 KNS Vc= 120.25 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 4.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 313 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5900. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	678.	4976.	NO	NO

CRITICAL POS MOMENT= 7.21 KN-MET AT 2950.MM, LOAD 401
REQD STEEL= 59.MM2, RHO=0.0005, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 41100.8 cm^4

2	434.	2 - 16MM	0.	2009.	YES	NO

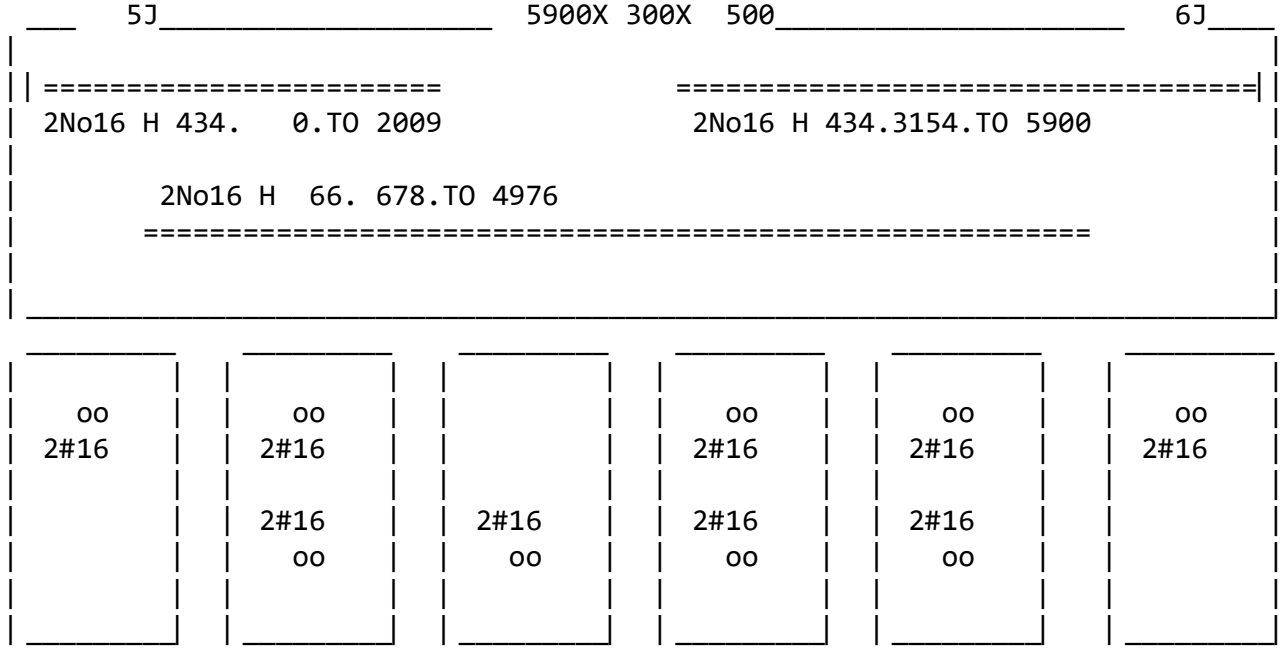
CRITICAL NEG MOMENT= 19.73 KN-MET AT 0.MM, LOAD 403
REQD STEEL= 161.MM2, RHO=0.0012, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     22.73 KN-MET  AT  5900.MM, LOAD  402|
|   REQD STEEL=   186.MM2, RHO=0.0014, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS        |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    41100.8 cm^4
```

B E A M N O. 313 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   13.48 KNS  Vc=  118.01 KNS  Vs=    0.00 KNS
 Tu=    0.02 KN-MET  Tc=    4.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   13.91 KNS  Vc=  118.01 KNS  Vs=    0.00 KNS
 Tu=    0.02 KN-MET  Tc=    4.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 314 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 995. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END

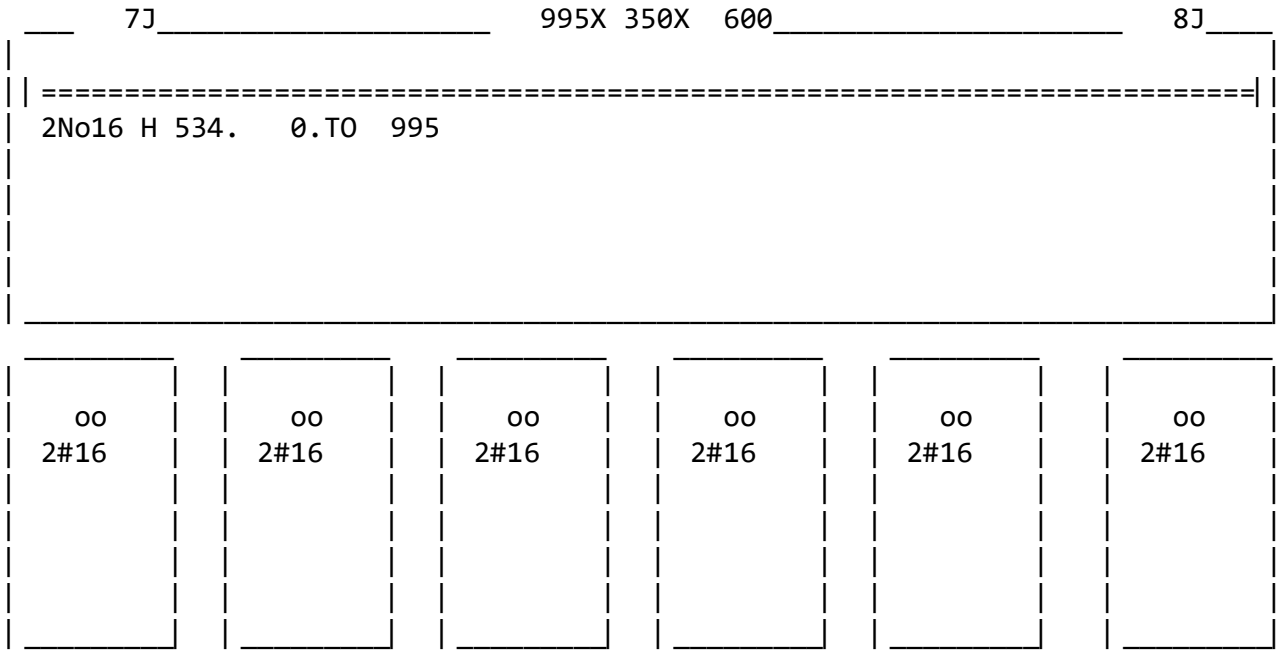
```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     55.45 KN-MET  AT     0.MM, LOAD  403|
|   REQD STEEL=   369.MM2, RHO=0.0020, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  218. MMS        |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 65065.9 cm^4

B E A M N O. 314 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 314 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 314 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



==========

BEAM NO. 315 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1170. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END

```
|-------------------------------------------------------------|
|   CRITICAL POS MOMENT=     41.75 KN-MET  AT  1170.MM, LOAD  403|
|   REQD STEEL=   277.MM2, RHO=0.0015, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  218. MMS        |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 65065.9 cm^4

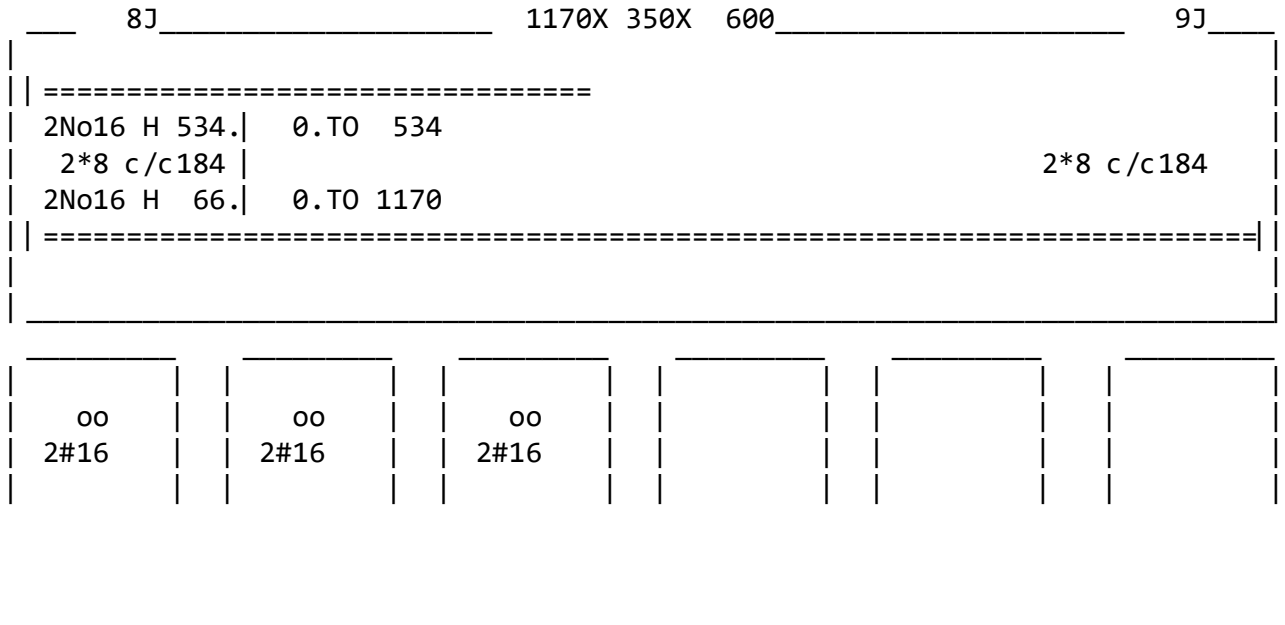
```
  2        534.     2 - 16MM         0.          534.        YES  NO
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=      2.50 KN-MET  AT     0.MM, LOAD  403|
|   REQD STEEL=    16.MM2, RHO=0.0001, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  218. MMS        |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 65065.9 cm^4

B E A M N O. 315 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 37.64 KNS Vc= 184.00 KNS Vs= 0.00 KNS
Tu= 10.18 KN-MET Tc= 8.9 KN-MET Ts= 13.6 KN-MET LOAD 402
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 184. MM C/C FOR 51. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.34 SQ.CM.

AT END SUPPORT - Vu= 36.86 KNS Vc= 181.24 KNS Vs= 0.00 KNS
Tu= 10.18 KN-MET Tc= 8.9 KN-MET Ts= 13.6 KN-MET LOAD 402
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 184. MM C/C FOR 51. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.34 SQ.CM.



BEAM NO. 316 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1570. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

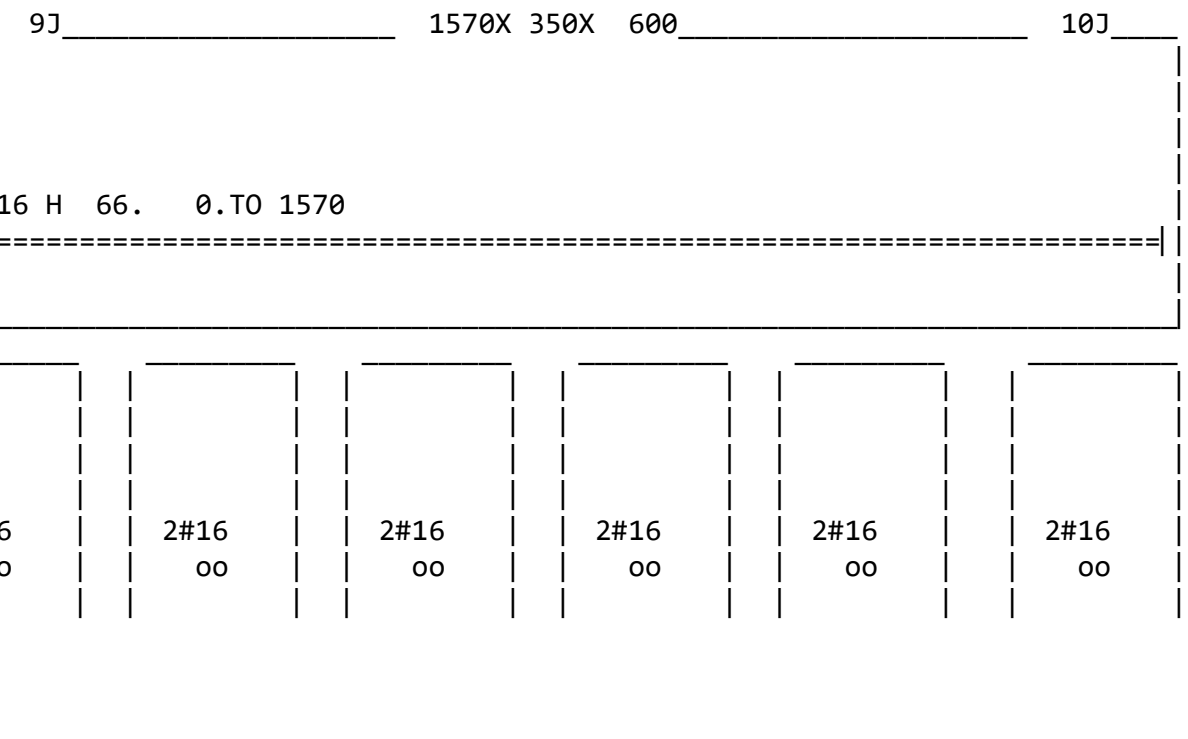
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	1570.	YES	YES

CRITICAL POS MOMENT= 44.35 KN-MET AT 916.MM, LOAD 403
REQD STEEL= 295.MM2, RHO=0.0016, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 218. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 65065.9 cm^4

B E A M N O. 316 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 2.05 KNS Vc= 174.73 KNS Vs= 0.00 KNS
Tu= 0.52 KN-MET Tc= 8.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 1.77 KNS Vc= 174.69 KNS Vs= 0.00 KNS
Tu= 0.52 KN-MET Tc= 8.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



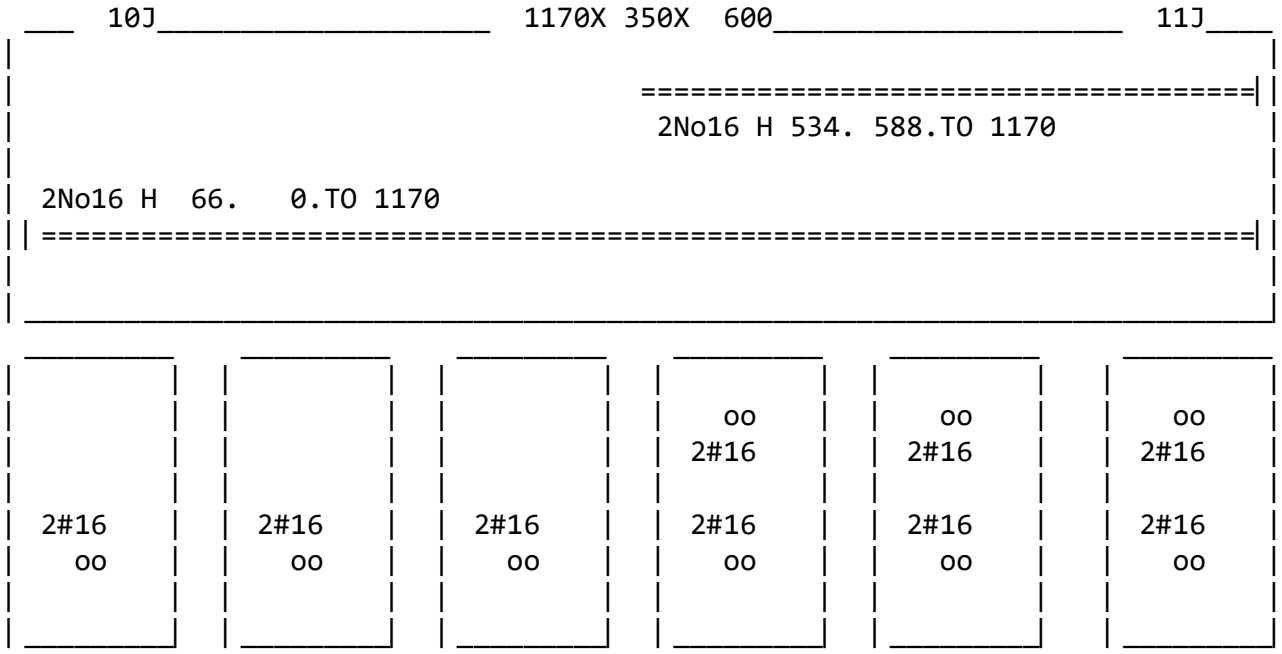
BEAM NO. 317 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1170. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	1170.	YES	YES
	CRITICAL POS MOMENT= 42.62 KN-MET AT 0.MM, LOAD 403					
	REQD STEEL= 283.MM2, RHO=0.0015, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 218. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 65065.9 cm^4					
2	534.	2 - 16MM	588.	1170.	NO	YES
	CRITICAL NEG MOMENT= 1.55 KN-MET AT 1170.MM, LOAD 402					
	REQD STEEL= 10.MM2, RHO=0.0001, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 218. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 65065.9 cm^4					

B E A M N O. 317 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 19.71 KNS Vc= 180.66 KNS Vs= 0.00 KNS
Tu= 6.08 KN-MET Tc= 8.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 20.49 KNS Vc= 183.04 KNS Vs= 0.00 KNS
Tu= 6.08 KN-MET Tc= 8.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



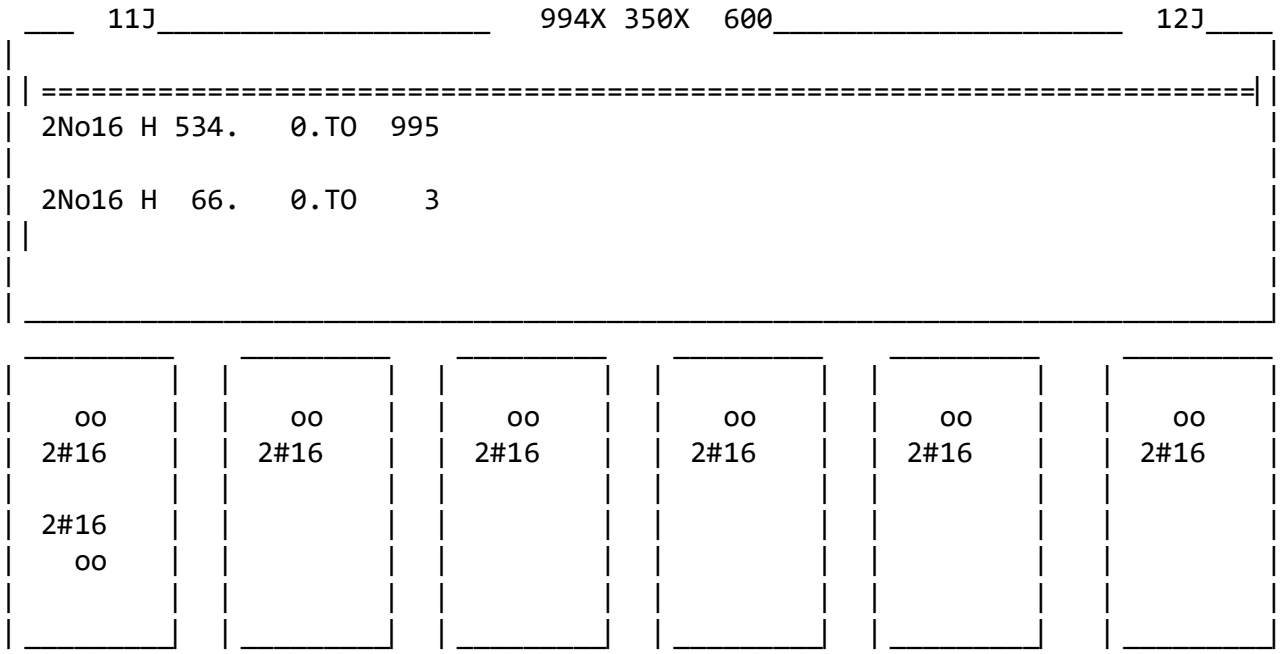
BEAM NO. 318 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 995. MM FY - 420. FC - 35. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	3.	YES	NO
	CRITICAL POS MOMENT= 0.48 KN-MET AT 0.MM, LOAD 406					
	REQD STEEL= 3.MM2, RHO=0.0000, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 218. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 65065.9 cm^4					
2	534.	2 - 16MM	0.	995.	YES	YES
	CRITICAL NEG MOMENT= 56.51 KN-MET AT 995.MM, LOAD 403					
	REQD STEEL= 376.MM2, RHO=0.0020, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 218. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 65065.9 cm^4					

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 318 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 318 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



**********END OF BEAM DESIGN**********

393. END CONCRETE DESIGN
394. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 19,2020 TIME= 15: 3:59 ****

```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	17-04-2020	19-05-2020	02-11-2023				Date	17-04-2020	19-05-2020	02-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK/APL	RPK/APL	RPK/APL				Prepared By	RPK/APL	RPK/APL	RPK/APL		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 4 **STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR TR-1**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 19-May-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 1
JOB PART TRESTLE TR1 FOUNDATION MKD. F2
JOB REF FOUNDATIONS FOR CONVEYOR 3210-CNV-001
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -2.25 0; 2 0 -2.25 2; 3 0 1.49012e-008 0; 4 0 1.49012e-008 2; 5 0 0.35 0;
6 0 0.35 2; 7 0 -2.25 -1.7; 8 0 -2.25 3.7; 9 1.6 -2.25 -1.7; 10 1.6 -2.25 3.7;
11 -1.6 -2.25 -1.7; 12 -1.6 -2.25 3.7; 13 -0.75 -2.25 3.7; 14 -0.75 -2.25 2.75;
15 -1.6 -2.25 2.75; 16 -0.5 -2.25 3.7; 17 -0.5 -2.25 2.75; 18 -0.25 -2.25 3.7;
19 -0.25 -2.25 2.75; 20 0 -2.25 2.75; 21 0.25 -2.25 3.7; 22 0.25 -2.25 2.75;
23 0.5 -2.25 3.7; 24 0.5 -2.25 2.75; 25 0.75 -2.25 3.7; 26 0.75 -2.25 2.75;
27 1.6 -2.25 2.75; 28 -0.75 -2.25 2.5; 29 -1.6 -2.25 2.5; 30 -0.5 -2.25 2.5;
31 -0.25 -2.25 2.5; 32 0 -2.25 2.5; 33 0.25 -2.25 2.5; 34 0.5 -2.25 2.5;
35 0.75 -2.25 2.5; 36 1.6 -2.25 2.5; 37 -0.75 -2.25 2.25; 38 -1.6 -2.25 2.25;
39 -0.5 -2.25 2.25; 40 -0.25 -2.25 2.25; 41 0 -2.25 2.25; 42 0.25 -2.25 2.25;
43 0.5 -2.25 2.25; 44 0.75 -2.25 2.25; 45 1.6 -2.25 2.25; 46 -0.75 -2.25 2;
47 -1.6 -2.25 2; 48 -0.5 -2.25 2; 49 -0.25 -2.25 2; 50 0.25 -2.25 2;
51 0.5 -2.25 2; 52 0.75 -2.25 2; 53 1.6 -2.25 2; 54 -0.75 -2.25 1.75;
55 -1.6 -2.25 1.75; 56 -0.5 -2.25 1.75; 57 -0.25 -2.25 1.75; 58 0 -2.25 1.75;
59 0.25 -2.25 1.75; 60 0.5 -2.25 1.75; 61 0.75 -2.25 1.75; 62 1.6 -2.25 1.75;
63 -0.75 -2.25 1.5; 64 -1.6 -2.25 1.5; 65 -0.5 -2.25 1.5; 66 -0.25 -2.25 1.5;
67 0 -2.25 1.5; 68 0.25 -2.25 1.5; 69 0.5 -2.25 1.5; 70 0.75 -2.25 1.5;
71 1.6 -2.25 1.5; 72 -0.75 -2.25 1.25; 73 -1.6 -2.25 1.25; 74 -0.5 -2.25 1.25;
75 -0.25 -2.25 1.25; 76 0 -2.25 1.25; 77 0.25 -2.25 1.25; 78 0.5 -2.25 1.25;
79 0.75 -2.25 1.25; 80 1.6 -2.25 1.25; 81 -0.75 -2.25 1; 82 -1.6 -2.25 1;
83 -0.5 -2.25 1; 84 -0.25 -2.25 1; 85 0 -2.25 1; 86 0.25 -2.25 1;
87 0.5 -2.25 1; 88 0.75 -2.25 1; 89 1.6 -2.25 1; 90 -0.75 -2.25 0.75;
91 -1.6 -2.25 0.75; 92 -0.5 -2.25 0.75; 93 -0.25 -2.25 0.75; 94 0 -2.25 0.75;
95 0.25 -2.25 0.75; 96 0.5 -2.25 0.75; 97 0.75 -2.25 0.75; 98 1.6 -2.25 0.75;
99 -0.75 -2.25 0.5; 100 -1.6 -2.25 0.5; 101 -0.5 -2.25 0.5;
102 -0.25 -2.25 0.5; 103 0 -2.25 0.5; 104 0.25 -2.25 0.5; 105 0.5 -2.25 0.5;
106 0.75 -2.25 0.5; 107 1.6 -2.25 0.5; 108 -0.75 -2.25 0.25;
109 -1.6 -2.25 0.25; 110 -0.5 -2.25 0.25; 111 -0.25 -2.25 0.25;
112 0 -2.25 0.25; 113 0.25 -2.25 0.25; 114 0.5 -2.25 0.25; 115 0.75 -2.25 0.25;
116 1.6 -2.25 0.25; 117 -0.75 -2.25 0; 118 -1.6 -2.25 0; 119 -0.5 -2.25 0;
120 -0.25 -2.25 0; 121 0.25 -2.25 0; 122 0.5 -2.25 0; 123 0.75 -2.25 0;
124 1.6 -2.25 0; 125 -0.75 -2.25 -0.25; 126 -1.6 -2.25 -0.25;
127 -0.5 -2.25 -0.25; 128 -0.25 -2.25 -0.25; 129 0 -2.25 -0.25;
130 0.25 -2.25 -0.25; 131 0.5 -2.25 -0.25; 132 0.75 -2.25 -0.25;
133 1.6 -2.25 -0.25; 134 -0.75 -2.25 -0.5; 135 -1.6 -2.25 -0.5;
136 -0.5 -2.25 -0.5; 137 -0.25 -2.25 -0.5; 138 0 -2.25 -0.5;
139 0.25 -2.25 -0.5; 140 0.5 -2.25 -0.5; 141 0.75 -2.25 -0.5;
142 1.6 -2.25 -0.5; 143 -0.75 -2.25 -0.75; 144 -1.6 -2.25 -0.75;
145 -0.5 -2.25 -0.75; 146 -0.25 -2.25 -0.75; 147 0 -2.25 -0.75;
148 0.25 -2.25 -0.75; 149 0.5 -2.25 -0.75; 150 0.75 -2.25 -0.75;
151 1.6 -2.25 -0.75; 152 -0.75 -2.25 -1.7; 153 -0.5 -2.25 -1.7;
```

```
154 -0.25 -2.25 -1.7; 155 0.25 -2.25 -1.7; 156 0.5 -2.25 -1.7;
157 0.75 -2.25 -1.7; 158 -0.75 -2.25 -1; 159 -1.6 -2.25 -1;
160 -0.75 -2.25 -1.25; 161 -1.6 -2.25 -1.25; 162 -0.5 -2.25 -1;
163 -0.5 -2.25 -1.25; 164 -0.25 -2.25 -1; 165 -0.25 -2.25 -1.25;
166 0 -2.25 -1; 167 0 -2.25 -1.25; 168 0.25 -2.25 -1; 169 0.25 -2.25 -1.25;
170 0.5 -2.25 -1; 171 0.5 -2.25 -1.25; 172 0.75 -2.25 -1; 173 0.75 -2.25 -1.25;
174 1.6 -2.25 -1; 175 1.6 -2.25 -1.25; 176 -0.75 -2.25 3.25;
177 -1.6 -2.25 3.25; 178 -0.75 -2.25 3; 179 -1.6 -2.25 3; 180 -0.5 -2.25 3.25;
181 -0.5 -2.25 3; 182 -0.25 -2.25 3.25; 183 -0.25 -2.25 3; 184 0 -2.25 3.25;
185 0 -2.25 3; 186 0.25 -2.25 3.25; 187 0.25 -2.25 3; 188 0.5 -2.25 3.25;
189 0.5 -2.25 3; 190 0.75 -2.25 3.25; 191 0.75 -2.25 3; 192 1.6 -2.25 3.25;
193 1.6 -2.25 3; 194 -0.75 -2.25 -1.475; 195 -1.6 -2.25 -1.475;
196 -0.5 -2.25 -1.475; 197 -0.25 -2.25 -1.475; 198 0 -2.25 -1.475;
199 0.25 -2.25 -1.475; 200 0.5 -2.25 -1.475; 201 0.75 -2.25 -1.475;
202 1.6 -2.25 -1.475; 203 -0.75 -2.25 3.475; 204 -1.6 -2.25 3.475;
205 -0.5 -2.25 3.475; 206 -0.25 -2.25 3.475; 207 0 -2.25 3.475;
208 0.25 -2.25 3.475; 209 0.5 -2.25 3.475; 210 0.75 -2.25 3.475;
211 1.6 -2.25 3.475; 212 1.03333 -2.25 2.75; 213 1.03333 -2.25 2.5;
214 1.31667 -2.25 2.75; 215 1.31667 -2.25 2.5; 216 1.03333 -2.25 2.25;
217 1.31667 -2.25 2.25; 218 1.03333 -2.25 2; 219 1.31667 -2.25 2;
220 1.03333 -2.25 1.75; 221 1.31667 -2.25 1.75; 222 1.03333 -2.25 1.5;
223 1.31667 -2.25 1.5; 224 1.03333 -2.25 1.25; 225 1.31667 -2.25 1.25;
226 1.03333 -2.25 1; 227 1.31667 -2.25 1; 228 1.03333 -2.25 0.75;
229 1.31667 -2.25 0.75; 230 1.03333 -2.25 0.5; 231 1.31667 -2.25 0.5;
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238 1.03333 -2.25 -0.5; 239 1.31667 -2.25 -0.5; 240 1.03333 -2.25 -0.75;
241 1.31667 -2.25 -0.75; 242 1.03333 -2.25 -1; 243 1.31667 -2.25 -1;
244 1.03333 -2.25 -1.25; 245 1.31667 -2.25 -1.25; 246 1.03333 -2.25 3.25;
247 1.03333 -2.25 3; 248 1.31667 -2.25 3.25; 249 1.31667 -2.25 3;
250 1.03333 -2.25 -1.475; 251 1.31667 -2.25 -1.475; 252 1.03333 -2.25 -1.7;
253 1.31667 -2.25 -1.7; 254 1.03333 -2.25 3.7; 255 1.03333 -2.25 3.475;
256 1.31667 -2.25 3.7; 257 1.31667 -2.25 3.475; 258 -1.31667 -2.25 2.75;
259 -1.31667 -2.25 2.5; 260 -1.03333 -2.25 2.75; 261 -1.03333 -2.25 2.5;
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265 -1.03333 -2.25 2; 266 -1.31667 -2.25 1.75; 267 -1.03333 -2.25 1.75;
268 -1.31667 -2.25 1.5; 269 -1.03333 -2.25 1.5; 270 -1.31667 -2.25 1.25;
271 -1.03333 -2.25 1.25; 272 -1.31667 -2.25 1; 273 -1.03333 -2.25 1;
274 -1.31667 -2.25 0.75; 275 -1.03333 -2.25 0.75; 276 -1.31667 -2.25 0.5;
277 -1.03333 -2.25 0.5; 278 -1.31667 -2.25 0.25; 279 -1.03333 -2.25 0.25;
280 -1.31667 -2.25 0; 281 -1.03333 -2.25 0; 282 -1.31667 -2.25 -0.25;
283 -1.03333 -2.25 -0.25; 284 -1.31667 -2.25 -0.5; 285 -1.03333 -2.25 -0.5;
286 -1.31667 -2.25 -0.75; 287 -1.03333 -2.25 -0.75; 288 -1.31667 -2.25 -1;
289 -1.03333 -2.25 -1; 290 -1.31667 -2.25 -1.25; 291 -1.03333 -2.25 -1.25;
292 -1.31667 -2.25 3.25; 293 -1.31667 -2.25 3; 294 -1.03333 -2.25 3.25;
295 -1.03333 -2.25 3; 296 -1.31667 -2.25 -1.475; 297 -1.03333 -2.25 -1.475;
298 -1.31667 -2.25 -1.7; 299 -1.03333 -2.25 -1.7; 300 -1.31667 -2.25 3.7;
301 -1.31667 -2.25 3.475; 302 -1.03333 -2.25 3.7; 303 -1.03333 -2.25 3.475;
MEMBER INCIDENCES
1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4;
ELEMENT INCIDENCES SHELL
15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33;
20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41;
27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49;
```

34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56;
41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63;
48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69;
55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77;
60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85;
67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93;
74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99;
80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104;
84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111;
90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117;
96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121;
100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127;
105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131;
111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137;
114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140;
119 134 136 145 143; 120 136 137 146 145; 121 137 138 147 146;
122 138 139 148 147; 123 139 140 149 148; 124 140 141 150 149;
137 143 145 162 158; 138 158 162 163 160; 140 145 146 164 162;
141 162 164 165 163; 143 146 147 166 164; 144 164 166 167 165;
146 147 148 168 166; 147 166 168 169 167; 149 148 149 170 168;
150 168 170 171 169; 152 149 150 172 170; 153 170 172 173 171;
162 176 180 181 178; 163 178 181 17 14; 165 180 182 183 181; 166 181 183 19 17;
168 182 184 185 183; 169 183 185 20 19; 171 184 186 187 185; 172 185 187 22 20;
174 186 188 189 187; 175 187 189 24 22; 177 188 190 191 189; 178 189 191 26 24;
184 160 163 196 194; 185 194 196 153 152; 186 163 165 197 196;
187 196 197 154 153; 188 165 167 198 197; 189 197 198 7 154;
190 167 169 199 198; 191 198 199 155 7; 192 169 171 200 199;
193 199 200 156 155; 194 171 173 201 200; 195 200 201 157 156;
200 13 16 205 203; 201 203 205 180 176; 202 16 18 206 205; 203 205 206 182 180;
204 18 8 207 206; 205 206 207 184 182; 206 8 21 208 207; 207 207 208 186 184;
208 21 23 209 208; 209 208 209 188 186; 210 23 25 210 209; 211 209 210 190 188;
214 26 212 213 35; 215 212 214 215 213; 216 214 27 36 215; 217 35 213 216 44;
218 213 215 217 216; 219 215 36 45 217; 220 44 216 218 52; 221 216 217 219 218;
222 217 45 53 219; 223 52 218 220 61; 224 218 219 221 220; 225 219 53 62 221;
226 61 220 222 70; 227 220 221 223 222; 228 221 62 71 223; 229 70 222 224 79;
230 222 223 225 224; 231 223 71 80 225; 232 79 224 226 88; 233 224 225 227 226;
234 225 80 89 227; 235 88 226 228 97; 236 226 227 229 228; 237 227 89 98 229;
238 97 228 230 106; 239 228 229 231 230; 240 229 98 107 231;
241 106 230 232 115; 242 230 231 233 232; 243 231 107 116 233;
244 115 232 234 123; 245 232 233 235 234; 246 233 116 124 235;
247 123 234 236 132; 248 234 235 237 236; 249 235 124 133 237;
250 132 236 238 141; 251 236 237 239 238; 252 237 133 142 239;
253 141 238 240 150; 254 238 239 241 240; 255 239 142 151 241;
256 150 240 242 172; 257 240 241 243 242; 258 241 151 174 243;
259 172 242 244 173; 260 242 243 245 244; 261 243 174 175 245;
262 190 246 247 191; 263 246 248 249 247; 264 248 192 193 249;
265 191 247 212 26; 266 247 249 214 212; 267 249 193 27 214;
268 173 244 250 201; 269 244 245 251 250; 270 245 175 202 251;
271 201 250 252 157; 272 250 251 253 252; 273 251 202 9 253;
274 25 254 255 210; 275 254 256 257 255; 276 256 10 211 257;
277 210 255 246 190; 278 255 257 248 246; 279 257 211 192 248;
280 15 258 259 29; 281 258 260 261 259; 282 260 14 28 261; 283 29 259 262 38;
284 259 261 263 262; 285 261 28 37 263; 286 38 262 264 47; 287 262 263 265 264;
288 263 37 46 265; 289 47 264 266 55; 290 264 265 267 266; 291 265 46 54 267;

```
292 55 266 268 64; 293 266 267 269 268; 294 267 54 63 269; 295 64 268 270 73;
296 268 269 271 270; 297 269 63 72 271; 298 73 270 272 82; 299 270 271 273 272;
300 271 72 81 273; 301 82 272 274 91; 302 272 273 275 274; 303 273 81 90 275;
304 91 274 276 100; 305 274 275 277 276; 306 275 90 99 277;
307 100 276 278 109; 308 276 277 279 278; 309 277 99 108 279;
310 109 278 280 118; 311 278 279 281 280; 312 279 108 117 281;
313 118 280 282 126; 314 280 281 283 282; 315 281 117 125 283;
316 126 282 284 135; 317 282 283 285 284; 318 283 125 134 285;
319 135 284 286 144; 320 284 285 287 286; 321 285 134 143 287;
322 144 286 288 159; 323 286 287 289 288; 324 287 143 158 289;
325 159 288 290 161; 326 288 289 291 290; 327 289 158 160 291;
328 177 292 293 179; 329 292 294 295 293; 330 294 176 178 295;
331 179 293 258 15; 332 293 295 260 258; 333 295 178 14 260;
334 161 290 296 195; 335 290 291 297 296; 336 291 160 194 297;
337 195 296 298 11; 338 296 297 299 298; 339 297 194 152 299;
340 12 300 301 204; 341 300 302 303 301; 342 302 13 203 303;
343 204 301 292 177; 344 301 303 294 292; 345 303 203 176 294;
START GROUP DEFINITION
ELEMENT
_MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
108 111 112 115 116 119 120 123 124 137 138 140 141 149 150 152 153 162 163 -
165 166 174 175 177 178 184 TO 187 192 TO 195 200 TO 203 208 TO 211 -
214 TO 345
_MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
111 TO 116 119 TO 124 137 138 140 141 143 144 146 147 149 150 152 153 162 -
163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
214 TO 219 226 TO 243 250 TO 285 292 TO 309 316 TO 345
_SHEAR 63 TO 68 71 TO 76 137 138 140 141 143 144 146 147 149 150 152 153 162 -
163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
214 TO 345
END GROUP DEFINITION
ELEMENT PROPERTY
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
178 184 TO 195 200 TO 211 214 TO 345 THICKNESS 0.55
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.6 ZD 0.6
5 PRIS YD 0.5 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
*****************************************
SUPPORTS
```

```
1 2 7 TO 303 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
*****************************************
MEMBER OFFSET
5 START 0 -0.25 0
5 END 0 -0.25 0
LOAD 1 DEAD LOAD    DL
SELFWEIGHT Y -1
JOINT LOAD
5   FX  0.21    FY  -91.307 FZ  12.186
6   FX  -0.21   FY  -28.04  FZ  -8.203
5   FX  0   FY  -17 FZ  0
6   FX  0   FY  -13 FZ  3.3
ELEMENT LOAD
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
178 184 TO 195 200 TO 211 214 TO 345 PR GY -30.6
*****************************************
LOAD 2 LIVE LOAD    LL
JOINT LOAD
5   FX  -0.051  FY  -33.409 FZ  0.724
6   FX  0.051   FY  -40.629 FZ  -0.716
5   FX  0   FY  -13 FZ  0
6   FX  0   FY  -13 FZ  0
*****************************************
LOAD 3 MATERIAL LOAD    LL
JOINT LOAD
5   FX  -0.025  FY  -25.768 FZ  -2.326
6   FX  0.025   FY  -25.701 FZ  2.33
5   FX  0   FY  -8  FZ  0
6   FX  0   FY  -8  FZ  0
*****************************************
LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
JOINT LOAD
5   FX  -0.458  FY  32.112  FZ  20.724
6   FX  0.458   FY  -32.112 FZ  17.111
5   FX  0   FY  0   FZ  0.2
6   FX  0   FY  0   FZ  11
*****************************************
LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
JOINT LOAD
5   FX  0.458   FY  -32.112 FZ  -20.767
6   FX  -0.458  FY  32.112  FZ  -17.069
5   FX  0   FY  0   FZ  -0.2
6   FX  0   FY  0   FZ  -11
*****************************************
LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
JOINT LOAD
5 FX 1.515 FY 0 FZ 0
6 FX 1.515 FY 0 FZ 0
*****************************************
LOAD 7 WIND LOAD ACROSS THE CONV.   WL (-X)
JOINT LOAD
5 FX -1.515 FY 0 FZ 0
```

```
6 FX -1.515 FY 0 FZ 0
*****************************************
LOAD 8 TEMP LOAD    TL1
JOINT LOAD
5   FX  -0.034  FY  6.315   FZ  3.192
6   FX  0.034   FY  -6.315  FZ  2.913
5   FX  0   FY  0   FZ  -0.2
6   FX  0   FY  0   FZ  7.4
*****************************************
LOAD 9 TEMP LOAD    TL2
JOINT LOAD
5   FX  0.034   FY  -6.315  FZ  -3.192
6   FX  -0.034  FY  6.315   FZ  -2.913
5   FX  0   FY  0   FZ  0.2
6   FX  0   FY  0   FZ  -7.4
*****************************************
LOAD 10 SEISMIC LOAD ALONG THE CONV.    SL (+X)
JOINT LOAD
5   FX  1.138   FY  -0.144  FZ  0.032
6   FX  1.136   FY  -0.374  FZ  0.069
*****************************************
LOAD 11 SEISMIC LOAD ACROSS THE CONV.   SL (+Z)
JOINT LOAD
5   FX  -0.02   FY  0.947   FZ  0.539
6   FX  0.02    FY  -0.947  FZ  0.453
*****************************************
*****Load combination for bearing capacity check
LOAD 301 1.0 DL
REPEAT LOAD
1 1.0
LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 8 1.0 2 1.0 3 1.0
LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 9 1.0 2 1.0 3 1.0
LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 4 0.75 2 0.75 3 0.75
LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 5 0.75 2 0.75 3 0.75
LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 6 0.75 2 0.75 3 0.75
LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 7 0.75 2 0.75 3 0.75
LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 0.525 2 0.75 3 0.75
LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 -0.525 2 0.75 3 0.75
```

```
LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 0.525 2 0.75 3 0.75
LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD 312 0.6 DL + 1.0 WL(+Z)
REPEAT LOAD
1 0.6 4 1.0
LOAD 313 0.6 DL + 1.0 WL(-Z)
REPEAT LOAD
1 0.6 5 1.0
LOAD 314 0.6 DL + 1.0 WL(+X)
REPEAT LOAD
1 0.6 6 1.0
LOAD 315 0.6 DL + 1.0 WL(-X)
REPEAT LOAD
1 0.6 7 1.0
LOAD 316 0.6 DL + 0.7 EL(+X)
REPEAT LOAD
1 0.6 10 0.7
LOAD 317 0.6 DL + 0.7 EL(-X)
REPEAT LOAD
1 0.6 10 -0.7
LOAD 318 0.6 DL + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 11 0.7
LOAD 319 0.6 DL + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 11 -0.7
*****Load combination for foundation design
LOAD 401 1.4 DL
REPEAT LOAD
1 1.4
LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 8 1.4 2 1.7 3 1.7
LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 9 1.4 2 1.7 3 1.7
LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 4 1.6 2 0.5 3 0.5
LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 5 1.6 2 0.5 3 0.5
LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 6 1.6 2 0.5 3 0.5
LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 7 1.6 2 0.5 3 0.5
LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
REPEAT LOAD
```

```
1 1.2 10 1.0 2 0.5 3 0.5
LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 1.0 2 0.5 3 0.5
LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD 412 0.9 DL + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 1.6
LOAD 413 0.9 DL + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 5 1.6
LOAD 414 0.9 DL + 1.6 WL(+X)
REPEAT LOAD
1 0.9 6 1.6
LOAD 415 0.9 DL + 1.6 WL(-X)
REPEAT LOAD
1 0.9 7 1.6
LOAD 416 0.9 DL + 1.0 EL(+X)
REPEAT LOAD
1 0.9 10 1.0
LOAD 417 0.9 DL + 1.0 EL(-X)
REPEAT LOAD
1 0.9 10 -1.0
LOAD 418 0.9 DL + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 11 1.0
LOAD 419 0.9 DL + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 11 -1.0
*****************************************
PERFORM ANALYSIS
*****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 1 TO 5
FYMAIN 420000 MEMB 1 TO 5
TRACK 1 ALL
CLB 0.05 MEMB 1 TO 5
CLT 0.05 MEMB 1 TO 5
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro CONNECT Edition            *
*           Version  21.00.01.12                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAY 19, 2020                *
*           Time=    15: 7:51                    *
*                                                *
*  Licensed to: MMH KOLKATTA                     *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE:  D:\F Drive\Maaden Gold\Conveyor CNV-3210-001\Civil\Documents\Staad\R1\3210-CNV
     2. START JOB INFORMATION
     3. ENGINEER DATE 19-MAY-20
     4. JOB NAME MMGP-SAUDI ARABIA
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045/900707
     7. JOB REV 1
     8. JOB PART TRESTLE TR1 FOUNDATION MKD. F2
     9. JOB REF FOUNDATIONS FOR CONVEYOR 3210-CNV-001
    10. END JOB INFORMATION
    11. INPUT WIDTH 79
    12. UNIT METER KN
    13. JOINT COORDINATES
    14. 1 0 -2.25 0; 2 0 -2.25 2; 3 0 1.49012E-008 0; 4 0 1.49012E-008 2; 5 0 0.35 0
    15. 6 0 0.35 2; 7 0 -2.25 -1.7; 8 0 -2.25 3.7; 9 1.6 -2.25 -1.7; 10 1.6 -2.25 3.7
    16. 11 -1.6 -2.25 -1.7; 12 -1.6 -2.25 3.7; 13 -0.75 -2.25 3.7; 14 -0.75 -2.25 2.75
    17. 15 -1.6 -2.25 2.75; 16 -0.5 -2.25 3.7; 17 -0.5 -2.25 2.75; 18 -0.25 -2.25 3.7
    18. 19 -0.25 -2.25 2.75; 20 0 -2.25 2.75; 21 0.25 -2.25 3.7; 22 0.25 -2.25 2.75
    19. 23 0.5 -2.25 3.7; 24 0.5 -2.25 2.75; 25 0.75 -2.25 3.7; 26 0.75 -2.25 2.75
    20. 27 1.6 -2.25 2.75; 28 -0.75 -2.25 2.5; 29 -1.6 -2.25 2.5; 30 -0.5 -2.25 2.5
    21. 31 -0.25 -2.25 2.5; 32 0 -2.25 2.5; 33 0.25 -2.25 2.5; 34 0.5 -2.25 2.5
    22. 35 0.75 -2.25 2.5; 36 1.6 -2.25 2.5; 37 -0.75 -2.25 2.25; 38 -1.6 -2.25 2.25
    23. 39 -0.5 -2.25 2.25; 40 -0.25 -2.25 2.25; 41 0 -2.25 2.25; 42 0.25 -2.25 2.25
    24. 43 0.5 -2.25 2.25; 44 0.75 -2.25 2.25; 45 1.6 -2.25 2.25; 46 -0.75 -2.25 2
    25. 47 -1.6 -2.25 2; 48 -0.5 -2.25 2; 49 -0.25 -2.25 2; 50 0.25 -2.25 2
    26. 51 0.5 -2.25 2; 52 0.75 -2.25 2; 53 1.6 -2.25 2; 54 -0.75 -2.25 1.75
    27. 55 -1.6 -2.25 1.75; 56 -0.5 -2.25 1.75; 57 -0.25 -2.25 1.75; 58 0 -2.25 1.75
    28. 59 0.25 -2.25 1.75; 60 0.5 -2.25 1.75; 61 0.75 -2.25 1.75; 62 1.6 -2.25 1.75
    29. 63 -0.75 -2.25 1.5; 64 -1.6 -2.25 1.5; 65 -0.5 -2.25 1.5; 66 -0.25 -2.25 1.5
    30. 67 0 -2.25 1.5; 68 0.25 -2.25 1.5; 69 0.5 -2.25 1.5; 70 0.75 -2.25 1.5
    31. 71 1.6 -2.25 1.5; 72 -0.75 -2.25 1.25; 73 -1.6 -2.25 1.25; 74 -0.5 -2.25 1.25
    32. 75 -0.25 -2.25 1.25; 76 0 -2.25 1.25; 77 0.25 -2.25 1.25; 78 0.5 -2.25 1.25
    33. 79 0.75 -2.25 1.25; 80 1.6 -2.25 1.25; 81 -0.75 -2.25 1; 82 -1.6 -2.25 1
    34. 83 -0.5 -2.25 1; 84 -0.25 -2.25 1; 85 0 -2.25 1; 86 0.25 -2.25 1
    35. 87 0.5 -2.25 1; 88 0.75 -2.25 1; 89 1.6 -2.25 1; 90 -0.75 -2.25 0.75
    36. 91 -1.6 -2.25 0.75; 92 -0.5 -2.25 0.75; 93 -0.25 -2.25 0.75; 94 0 -2.25 0.75
    37. 95 0.25 -2.25 0.75; 96 0.5 -2.25 0.75; 97 0.75 -2.25 0.75; 98 1.6 -2.25 0.75
```

39. 102 -0.25 -2.25 0.5; 103 0 -2.25 0.5; 104 0.25 -2.25 0.5; 105 0.5 -2.25 0.5
40. 106 0.75 -2.25 0.5; 107 1.6 -2.25 0.5; 108 -0.75 -2.25 0.25
41. 109 -1.6 -2.25 0.25; 110 -0.5 -2.25 0.25; 111 -0.25 -2.25 0.25
42. 112 0 -2.25 0.25; 113 0.25 -2.25 0.25; 114 0.5 -2.25 0.25; 115 0.75 -2.25 0.25
43. 116 1.6 -2.25 0.25; 117 -0.75 -2.25 0; 118 -1.6 -2.25 0; 119 -0.5 -2.25 0
44. 120 -0.25 -2.25 0; 121 0.25 -2.25 0; 122 0.5 -2.25 0; 123 0.75 -2.25 0
45. 124 1.6 -2.25 0; 125 -0.75 -2.25 -0.25; 126 -1.6 -2.25 -0.25
46. 127 -0.5 -2.25 -0.25; 128 -0.25 -2.25 -0.25; 129 0 -2.25 -0.25
47. 130 0.25 -2.25 -0.25; 131 0.5 -2.25 -0.25; 132 0.75 -2.25 -0.25
48. 133 1.6 -2.25 -0.25; 134 -0.75 -2.25 -0.5; 135 -1.6 -2.25 -0.5
49. 136 -0.5 -2.25 -0.5; 137 -0.25 -2.25 -0.5; 138 0 -2.25 -0.5
50. 139 0.25 -2.25 -0.5; 140 0.5 -2.25 -0.5; 141 0.75 -2.25 -0.5
51. 142 1.6 -2.25 -0.5; 143 -0.75 -2.25 -0.75; 144 -1.6 -2.25 -0.75
52. 145 -0.5 -2.25 -0.75; 146 -0.25 -2.25 -0.75; 147 0 -2.25 -0.75
53. 148 0.25 -2.25 -0.75; 149 0.5 -2.25 -0.75; 150 0.75 -2.25 -0.75
54. 151 1.6 -2.25 -0.75; 152 -0.75 -2.25 -1.7; 153 -0.5 -2.25 -1.7
55. 154 -0.25 -2.25 -1.7; 155 0.25 -2.25 -1.7; 156 0.5 -2.25 -1.7
56. 157 0.75 -2.25 -1.7; 158 -0.75 -2.25 -1; 159 -1.6 -2.25 -1.
57. 160 -0.75 -2.25 -1.25; 161 -1.6 -2.25 -1.25; 162 -0.5 -2.25 -1.
58. 163 -0.5 -2.25 -1.25; 164 -0.25 -2.25 -1; 165 -0.25 -2.25 -1.25
59. 166 0 -2.25 -1; 167 0 -2.25 -1.25; 168 0.25 -2.25 -1; 169 0.25 -2.25 -1.25
60. 170 0.5 -2.25 -1; 171 0.5 -2.25 -1.25; 172 0.75 -2.25 -1; 173 0.75 -2.25 -1.25
61. 174 1.6 -2.25 -1; 175 1.6 -2.25 -1.25; 176 -0.75 -2.25 3.25
62. 177 -1.6 -2.25 3.25; 178 -0.75 -2.25 3; 179 -1.6 -2.25 3; 180 -0.5 -2.25 3.25
63. 181 -0.5 -2.25 3; 182 -0.25 -2.25 3.25; 183 -0.25 -2.25 3; 184 0 -2.25 3.25
64. 185 0 -2.25 3; 186 0.25 -2.25 3.25; 187 0.25 -2.25 3; 188 0.5 -2.25 3.25
65. 189 0.5 -2.25 3; 190 0.75 -2.25 3.25; 191 0.75 -2.25 3; 192 1.6 -2.25 3.25
66. 193 1.6 -2.25 3; 194 -0.75 -2.25 -1.475; 195 -1.6 -2.25 -1.475
67. 196 -0.5 -2.25 -1.475; 197 -0.25 -2.25 -1.475; 198 0 -2.25 -1.475
68. 199 0.25 -2.25 -1.475; 200 0.5 -2.25 -1.475; 201 0.75 -2.25 -1.475
69. 202 1.6 -2.25 -1.475; 203 -0.75 -2.25 3.475; 204 -1.6 -2.25 3.475
70. 205 -0.5 -2.25 3.475; 206 -0.25 -2.25 3.475; 207 0 -2.25 3.475
71. 208 0.25 -2.25 3.475; 209 0.5 -2.25 3.475; 210 0.75 -2.25 3.475
72. 211 1.6 -2.25 3.475; 212 1.03333 -2.25 2.75; 213 1.03333 -2.25 2.5
73. 214 1.31667 -2.25 2.75; 215 1.31667 -2.25 2.5; 216 1.03333 -2.25 2.25
74. 217 1.31667 -2.25 2.25; 218 1.03333 -2.25 2; 219 1.31667 -2.25 2
75. 220 1.03333 -2.25 1.75; 221 1.31667 -2.25 1.75; 222 1.03333 -2.25 1.5
76. 223 1.31667 -2.25 1.5; 224 1.03333 -2.25 1.25; 225 1.31667 -2.25 1.25
77. 226 1.03333 -2.25 1; 227 1.31667 -2.25 1; 228 1.03333 -2.25 0.75
78. 229 1.31667 -2.25 0.75; 230 1.03333 -2.25 0.5; 231 1.31667 -2.25 0.5
79. 232 1.03333 -2.25 0.25; 233 1.31667 -2.25 0.25; 234 1.03333 -2.25 0
80. 235 1.31667 -2.25 0; 236 1.03333 -2.25 -0.25; 237 1.31667 -2.25 -0.25
81. 238 1.03333 -2.25 -0.5; 239 1.31667 -2.25 -0.5; 240 1.03333 -2.25 -0.75
82. 241 1.31667 -2.25 -0.75; 242 1.03333 -2.25 -1; 243 1.31667 -2.25 -1.
83. 244 1.03333 -2.25 -1.25; 245 1.31667 -2.25 -1.25; 246 1.03333 -2.25 3.25
84. 247 1.03333 -2.25 3; 248 1.31667 -2.25 3.25; 249 1.31667 -2.25 3
85. 250 1.03333 -2.25 -1.475; 251 1.31667 -2.25 -1.475; 252 1.03333 -2.25 -1.7
86. 253 1.31667 -2.25 -1.7; 254 1.03333 -2.25 3.7; 255 1.03333 -2.25 3.475
87. 256 1.31667 -2.25 3.7; 257 1.31667 -2.25 3.475; 258 -1.31667 -2.25 2.75
88. 259 -1.31667 -2.25 2.5; 260 -1.03333 -2.25 2.75; 261 -1.03333 -2.25 2.5
89. 262 -1.31667 -2.25 2.25; 263 -1.03333 -2.25 2.25; 264 -1.31667 -2.25 2
90. 265 -1.03333 -2.25 2; 266 -1.31667 -2.25 1.75; 267 -1.03333 -2.25 1.75
91. 268 -1.31667 -2.25 1.5; 269 -1.03333 -2.25 1.5; 270 -1.31667 -2.25 1.25
92. 271 -1.03333 -2.25 1.25; 272 -1.31667 -2.25 1; 273 -1.03333 -2.25 1

```
 95. 280 -1.31667 -2.25 0; 281 -1.03333 -2.25 0; 282 -1.31667 -2.25 -0.25
 96. 283 -1.03333 -2.25 -0.25; 284 -1.31667 -2.25 -0.5; 285 -1.03333 -2.25 -0.5
 97. 286 -1.31667 -2.25 -0.75; 287 -1.03333 -2.25 -0.75; 288 -1.31667 -2.25 -1.
 98. 289 -1.03333 -2.25 -1; 290 -1.31667 -2.25 -1.25; 291 -1.03333 -2.25 -1.25
 99. 292 -1.31667 -2.25 3.25; 293 -1.31667 -2.25 3; 294 -1.03333 -2.25 3.25
100. 295 -1.03333 -2.25 3; 296 -1.31667 -2.25 -1.475; 297 -1.03333 -2.25 -1.475
101. 298 -1.31667 -2.25 -1.7; 299 -1.03333 -2.25 -1.7; 300 -1.31667 -2.25 3.7
102. 301 -1.31667 -2.25 3.475; 302 -1.03333 -2.25 3.7; 303 -1.03333 -2.25 3.475
103. MEMBER INCIDENCES
104. 1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4
105. ELEMENT INCIDENCES SHELL
106. 15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33
107. 20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41
108. 27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49
109. 34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56
110. 41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63
111. 48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69
112. 55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77
113. 60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85
114. 67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93
115. 74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99
116. 80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104
117. 84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111
118. 90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117
119. 96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121
120. 100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127
121. 105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131
122. 111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137
123. 114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140
124. 119 134 136 145 143; 120 136 137 146 145; 121 137 138 147 146
125. 122 138 139 148 147; 123 139 140 149 148; 124 140 141 150 149
126. 137 143 145 162 158; 138 158 162 163 160; 140 145 146 164 162
127. 141 162 164 165 163; 143 146 147 166 164; 144 164 166 167 165
128. 146 147 148 168 166; 147 166 168 169 167; 149 148 149 170 168
129. 150 168 170 171 169; 152 149 150 172 170; 153 170 172 173 171
130. 162 176 180 181 178; 163 178 181 17 14; 165 180 182 183 181; 166 181 183 19 17
131. 168 182 184 185 183; 169 183 185 20 19; 171 184 186 187 185; 172 185 187 22 20
132. 174 186 188 189 187; 175 187 189 24 22; 177 188 190 191 189; 178 189 191 26 24
133. 184 160 163 196 194; 185 194 196 153 152; 186 163 165 197 196
134. 187 196 197 154 153; 188 165 167 198 197; 189 197 198 7 154
135. 190 167 169 199 198; 191 198 199 155 7; 192 169 171 200 199
136. 193 199 200 156 155; 194 171 173 201 200; 195 200 201 157 156
137. 200 13 16 205 203; 201 203 205 180 176; 202 16 18 206 205; 203 205 206 182 180
138. 204 18 8 207 206; 205 206 207 184 182; 206 8 21 208 207; 207 207 208 186 184
139. 208 21 23 209 208; 209 208 209 188 186; 210 23 25 210 209; 211 209 210 190 188
140. 214 26 212 213 35; 215 212 214 215 213; 216 214 27 36 215; 217 35 213 216 44
141. 218 213 215 217 216; 219 215 36 45 217; 220 44 216 218 52; 221 216 217 219 218
142. 222 217 45 53 219; 223 52 218 220 61; 224 218 219 221 220; 225 219 53 62 221
143. 226 61 220 222 70; 227 220 221 223 222; 228 221 62 71 223; 229 70 222 224 79
144. 230 222 223 225 224; 231 223 71 80 225; 232 79 224 226 88; 233 224 225 227 226
145. 234 225 80 89 227; 235 88 226 228 97; 236 226 227 229 228; 237 227 89 98 229
146. 238 97 228 230 106; 239 228 229 231 230; 240 229 98 107 231
147. 241 106 230 232 115; 242 230 231 233 232; 243 231 107 116 233
148. 244 115 232 234 123; 245 232 233 235 234; 246 233 116 124 235
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151. 253 141 238 240 150; 254 238 239 241 240; 255 239 142 151 241
152. 256 150 240 242 172; 257 240 241 243 242; 258 241 151 174 243
153. 259 172 242 244 173; 260 242 243 245 244; 261 243 174 175 245
154. 262 190 246 247 191; 263 246 248 249 247; 264 248 192 193 249
155. 265 191 247 212 26; 266 247 249 214 212; 267 249 193 27 214
156. 268 173 244 250 201; 269 244 245 251 250; 270 245 175 202 251
157. 271 201 250 252 157; 272 250 251 253 252; 273 251 202 9 253
158. 274 25 254 255 210; 275 254 256 257 255; 276 256 10 211 257
159. 277 210 255 246 190; 278 255 257 248 246; 279 257 211 192 248
160. 280 15 258 259 29; 281 258 260 261 259; 282 260 14 28 261; 283 29 259 262 38
161. 284 259 261 263 262; 285 261 28 37 263; 286 38 262 264 47; 287 262 263 265 264
162. 288 263 37 46 265; 289 47 264 266 55; 290 264 265 267 266; 291 265 46 54 267
163. 292 55 266 268 64; 293 266 267 269 268; 294 267 54 63 269; 295 64 268 270 73
164. 296 268 269 271 270; 297 269 63 72 271; 298 73 270 272 82; 299 270 271 273 272
165. 300 271 72 81 273; 301 82 272 274 91; 302 272 273 275 274; 303 273 81 90 275
166. 304 91 274 276 100; 305 274 275 277 276; 306 275 90 99 277
167. 307 100 276 278 109; 308 276 277 279 278; 309 277 99 108 279
168. 310 109 278 280 118; 311 278 279 281 280; 312 279 108 117 281
169. 313 118 280 282 126; 314 280 281 283 282; 315 281 117 125 283
170. 316 126 282 284 135; 317 282 283 285 284; 318 283 125 134 285
171. 319 135 284 286 144; 320 284 285 287 286; 321 285 134 143 287
172. 322 144 286 288 159; 323 286 287 289 288; 324 287 143 158 289
173. 325 159 288 290 161; 326 288 289 291 290; 327 289 158 160 291
174. 328 177 292 293 179; 329 292 294 295 293; 330 294 176 178 295
175. 331 179 293 258 15; 332 293 295 260 258; 333 295 178 14 260
176. 334 161 290 296 195; 335 290 291 297 296; 336 291 160 194 297
177. 337 195 296 298 11; 338 296 297 299 298; 339 297 194 152 299
178. 340 12 300 301 204; 341 300 302 303 301; 342 302 13 203 303
179. 343 204 301 292 177; 344 301 303 294 292; 345 303 203 176 294
180. START GROUP DEFINITION
181. ELEMENT
182. _MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
183. 63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
184. 108 111 112 115 116 119 120 123 124 137 138 140 141 149 150 152 153 162 163 -
185. 165 166 174 175 177 178 184 TO 187 192 TO 195 200 TO 203 208 TO 211 -
186. 214 TO 345
187. _MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
188. 111 TO 116 119 TO 124 137 138 140 141 143 144 146 147 149 150 152 153 162 -
189. 163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
190. 214 TO 219 226 TO 243 250 TO 285 292 TO 309 316 TO 345
191. _SHEAR 63 TO 68 71 TO 76 137 138 140 141 143 144 146 147 149 150 152 153 162 -
192. 163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
193. 214 TO 345
194. END GROUP DEFINITION
195. ELEMENT PROPERTY
196. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
197. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
198. 143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
199. 178 184 TO 195 200 TO 211 214 TO 345 THICKNESS 0.55
200. DEFINE MATERIAL START
201. ISOTROPIC CONCRETE
202. E 2.17185E+007
203. POISSON 0.17
204. DENSITY 23.5616
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207. TYPE CONCRETE
208. STRENGTH FCU 27579
209. END DEFINE MATERIAL
210. MEMBER PROPERTY AMERICAN
211. 1 TO 4 PRIS YD 0.6 ZD 0.6
212. 5 PRIS YD 0.5 ZD 0.3
213. CONSTANTS
214. MATERIAL CONCRETE ALL
215. ****************************************
216. SUPPORTS
217. 1 2 7 TO 303 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
```

JOINT	INFLUENCE AREA
1	0.062500
2	0.062500
7	0.028125
8	0.028125
9	0.015937
10	0.015937
11	0.015937
12	0.015937
13	0.030000
14	0.066666
15	0.035416
16	0.028125
17	0.062500
18	0.028125
19	0.062500
20	0.062500
21	0.028125
22	0.062500
23	0.028125
24	0.062500
25	0.030000
26	0.066666
27	0.035416
28	0.066666
29	0.035416
30	0.062500
31	0.062500
32	0.062500
33	0.062500
34	0.062500
35	0.066666
36	0.035416
37	0.066666
38	0.035416
39	0.062500
40	0.062500
41	0.062500
42	0.062500
43	0.062500

46	0.066666
47	0.035416
48	0.062500
49	0.062500
50	0.062500
51	0.062500
52	0.066666
53	0.035416
54	0.066666
55	0.035416
56	0.062500
57	0.062500
58	0.062500
59	0.062500
60	0.062500
61	0.066666
62	0.035416
63	0.066666
64	0.035416
65	0.062500
66	0.062500
67	0.062500
68	0.062500
69	0.062500
70	0.066666
71	0.035416
72	0.066666
73	0.035416
74	0.062500
75	0.062500
76	0.062500
77	0.062500
78	0.062500
79	0.066666
80	0.035416
81	0.066666
82	0.035416
83	0.062500
84	0.062500
85	0.062500
86	0.062500
87	0.062500
88	0.066666
89	0.035416
90	0.066666
91	0.035416
92	0.062500
93	0.062500
94	0.062500
95	0.062500
96	0.062500
97	0.066666
98	0.035416
99	0.066666

102	0.062500
103	0.062500
104	0.062500
105	0.062500
106	0.066666
107	0.035416
108	0.066666
109	0.035416
110	0.062500
111	0.062500
112	0.062500
113	0.062500
114	0.062500
115	0.066666
116	0.035416
117	0.066666
118	0.035416
119	0.062500
120	0.062500
121	0.062500
122	0.062500
123	0.066666
124	0.035416
125	0.066666
126	0.035416
127	0.062500
128	0.062500
129	0.062500
130	0.062500
131	0.062500
132	0.066666
133	0.035416
134	0.066666
135	0.035416
136	0.062500
137	0.062500
138	0.062500
139	0.062500
140	0.062500
141	0.066666
142	0.035416
143	0.066666
144	0.035416
145	0.062500
146	0.062500
147	0.062500
148	0.062500
149	0.062500
150	0.066666
151	0.035416
152	0.030000
153	0.028125
154	0.028125
155	0.028125

158	0.066666
159	0.035416
160	0.063333
161	0.033645
162	0.062500
163	0.059375
164	0.062500
165	0.059375
166	0.062500
167	0.059375
168	0.062500
169	0.059375
170	0.062500
171	0.059375
172	0.066666
173	0.063333
174	0.035416
175	0.033645
176	0.063333
177	0.033645
178	0.066666
179	0.035416
180	0.059375
181	0.062500
182	0.059375
183	0.062500
184	0.059375
185	0.062500
186	0.059375
187	0.062500
188	0.059375
189	0.062500
190	0.063333
191	0.066666
192	0.033645
193	0.035416
194	0.060000
195	0.031875
196	0.056250
197	0.056250
198	0.056250
199	0.056250
200	0.056250
201	0.060000
202	0.031875
203	0.060000
204	0.031875
205	0.056250
206	0.056250
207	0.056250
208	0.056250
209	0.056250
210	0.060000
211	0.031875

214	0.070834
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216	0.070834
217	0.070834
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219	0.070834
220	0.070834
221	0.070834
222	0.070834
223	0.070834
224	0.070834
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237	0.070834
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239	0.070834
240	0.070834
241	0.070834
242	0.070834
243	0.070834
244	0.067292
245	0.067292
246	0.067292
247	0.070834
248	0.067292
249	0.070834
250	0.063750
251	0.063750
252	0.031875
253	0.031875
254	0.031875
255	0.063750
256	0.031875
257	0.063750
258	0.070834
259	0.070834
260	0.070834
261	0.070834
262	0.070834
263	0.070834
264	0.070834
265	0.070834
266	0.070834
267	0.070834

270	0.070834
271	0.070834
272	0.070834
273	0.070834
274	0.070834
275	0.070834
276	0.070834
277	0.070834
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285	0.070834
286	0.070834
287	0.070834
288	0.070834
289	0.070834
290	0.067292
291	0.067292
292	0.067292
293	0.070834
294	0.067292
295	0.070834
296	0.063750
297	0.063750
298	0.031875
299	0.031875
300	0.031875
301	0.063750
302	0.031875
303	0.063750
TOTAL	17.280014

```
218. ****************************************
219. MEMBER OFFSET
220. 5 START 0 -0.25 0
221. 5 END 0 -0.25 0
222. LOAD 1 DEAD LOAD DL
223. SELFWEIGHT Y -1
224. JOINT LOAD
225. 5 FX 0.21 FY -91.307 FZ 12.186
226. 6 FX -0.21 FY -28.04 FZ -8.203
227. 5 FX 0 FY -17 FZ 0
228. 6 FX 0 FY -13 FZ 3.3
229. ELEMENT LOAD
230. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
231. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
232. 143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
233. 178 184 TO 195 200 TO 211 214 TO 345 PR GY -30.6
234. ****************************************
235. LOAD 2 LIVE LOAD LL
236. JOINT LOAD
```

```
239. 5 FX 0 FY -13 FZ 0
240. 6 FX 0 FY -13 FZ 0
241. ****************************************
242. LOAD 3 MATERIAL LOAD LL
243. JOINT LOAD
244. 5 FX -0.025 FY -25.768 FZ -2.326
245. 6 FX 0.025 FY -25.701 FZ 2.33
246. 5 FX 0 FY -8 FZ 0
247. 6 FX 0 FY -8 FZ 0
248. ****************************************
249. LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
250. JOINT LOAD
251. 5 FX -0.458 FY 32.112 FZ 20.724
252. 6 FX 0.458 FY -32.112 FZ 17.111
253. 5 FX 0 FY 0 FZ 0.2
254. 6 FX 0 FY 0 FZ 11
255. ****************************************
256. LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
257. JOINT LOAD
258. 5 FX 0.458 FY -32.112 FZ -20.767
259. 6 FX -0.458 FY 32.112 FZ -17.069
260. 5 FX 0 FY 0 FZ -0.2
261. 6 FX 0 FY 0 FZ -11
262. ****************************************
263. LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
264. JOINT LOAD
265. 5 FX 1.515 FY 0 FZ 0
266. 6 FX 1.515 FY 0 FZ 0
267. ****************************************
268. LOAD 7 WIND LOAD ACROSS THE CONV.  WL (-X)
269. JOINT LOAD
270. 5 FX -1.515 FY 0 FZ 0
271. 6 FX -1.515 FY 0 FZ 0
272. ****************************************
273. LOAD 8 TEMP LOAD   TL1
274. JOINT LOAD
275. 5 FX -0.034 FY 6.315 FZ 3.192
276. 6 FX 0.034 FY -6.315 FZ 2.913
277. 5 FX 0 FY 0 FZ -0.2
278. 6 FX 0 FY 0 FZ 7.4
279. ****************************************
280. LOAD 9 TEMP LOAD   TL2
281. JOINT LOAD
282. 5 FX 0.034 FY -6.315 FZ -3.192
283. 6 FX -0.034 FY 6.315 FZ -2.913
284. 5 FX 0 FY 0 FZ 0.2
285. 6 FX 0 FY 0 FZ -7.4
286. ****************************************
287. LOAD 10 SEISMIC LOAD ALONG THE CONV. SL (+X)
288. JOINT LOAD
289. 5 FX 1.138 FY -0.144 FZ 0.032
290. 6 FX 1.136 FY -0.374 FZ 0.069
291. ****************************************
292. LOAD 11 SEISMIC LOAD ACROSS THE CONV. SL (+Z)
```

```
295. 6 FX 0.02 FY -0.947 FZ 0.453
296. ****************************************
297. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
298. LOAD 301 1.0 DL
299. REPEAT LOAD
300. 1 1.0
301. LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
302. REPEAT LOAD
303. 1 1.0 8 1.0 2 1.0 3 1.0
304. LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
305. REPEAT LOAD
306. 1 1.0 9 1.0 2 1.0 3 1.0
307. LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
308. REPEAT LOAD
309. 1 1.0 4 0.75 2 0.75 3 0.75
310. LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
311. REPEAT LOAD
312. 1 1.0 5 0.75 2 0.75 3 0.75
313. LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
314. REPEAT LOAD
315. 1 1.0 6 0.75 2 0.75 3 0.75
316. LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
317. REPEAT LOAD
318. 1 1.0 7 0.75 2 0.75 3 0.75
319. LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
320. REPEAT LOAD
321. 1 1.0 10 0.525 2 0.75 3 0.75
322. LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
323. REPEAT LOAD
324. 1 1.0 10 -0.525 2 0.75 3 0.75
325. LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
326. REPEAT LOAD
327. 1 1.0 11 0.525 2 0.75 3 0.75
328. LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
329. REPEAT LOAD
330. 1 1.0 11 -0.525 2 0.75 3 0.75
331. LOAD 312 0.6 DL + 1.0 WL(+Z)
332. REPEAT LOAD
333. 1 0.6 4 1.0
334. LOAD 313 0.6 DL + 1.0 WL(-Z)
335. REPEAT LOAD
336. 1 0.6 5 1.0
337. LOAD 314 0.6 DL + 1.0 WL(+X)
338. REPEAT LOAD
339. 1 0.6 6 1.0
340. LOAD 315 0.6 DL + 1.0 WL(-X)
341. REPEAT LOAD
342. 1 0.6 7 1.0
343. LOAD 316 0.6 DL + 0.7 EL(+X)
344. REPEAT LOAD
345. 1 0.6 10 0.7
346. LOAD 317 0.6 DL + 0.7 EL(-X)
347. REPEAT LOAD
348. 1 0.6 10 -0.7
```

```
351. 1 0.6 11 0.7
352. LOAD 319 0.6 DL + 0.7 EL(-Z)
353. REPEAT LOAD
354. 1 0.6 11 -0.7
355. *****LOAD COMBINATION FOR FOUNDATION DESIGN
356. LOAD 401 1.4 DL
357. REPEAT LOAD
358. 1 1.4
359. LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
360. REPEAT LOAD
361. 1 1.4 8 1.4 2 1.7 3 1.7
362. LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
363. REPEAT LOAD
364. 1 1.4 9 1.4 2 1.7 3 1.7
365. LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
366. REPEAT LOAD
367. 1 1.2 4 1.6 2 0.5 3 0.5
368. LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
369. REPEAT LOAD
370. 1 1.2 5 1.6 2 0.5 3 0.5
371. LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
372. REPEAT LOAD
373. 1 1.2 6 1.6 2 0.5 3 0.5
374. LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
375. REPEAT LOAD
376. 1 1.2 7 1.6 2 0.5 3 0.5
377. LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
378. REPEAT LOAD
379. 1 1.2 10 1.0 2 0.5 3 0.5
380. LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
381. REPEAT LOAD
382. 1 1.2 10 -1.0 2 0.5 3 0.5
383. LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
384. REPEAT LOAD
385. 1 1.2 11 1.0 2 0.5 3 0.5
386. LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
387. REPEAT LOAD
388. 1 1.2 11 -1.0 2 0.5 3 0.5
389. LOAD 412 0.9 DL + 1.6 WL(+Z)
390. REPEAT LOAD
391. 1 0.9 4 1.6
392. LOAD 413 0.9 DL + 1.6 WL(-Z)
393. REPEAT LOAD
394. 1 0.9 5 1.6
395. LOAD 414 0.9 DL + 1.6 WL(+X)
396. REPEAT LOAD
397. 1 0.9 6 1.6
398. LOAD 415 0.9 DL + 1.6 WL(-X)
399. REPEAT LOAD
400. 1 0.9 7 1.6
401. LOAD 416 0.9 DL + 1.0 EL(+X)
402. REPEAT LOAD
403. 1 0.9 10 1.0
404. LOAD 417 0.9 DL + 1.0 EL(-X)
```

```
407. LOAD 418 0.9 DL + 1.0 EL(+Z)
408. REPEAT LOAD
409. 1 0.9 11 1.0
410. LOAD 419 0.9 DL + 1.0 EL(-Z)
411. REPEAT LOAD
412. 1 0.9 11 -1.0
413. ****************************************
414. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**303**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**264**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**299**

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 49, TOTAL DEGREES OF FREEDOM = 921
TOTAL LOAD COMBINATION CASES = 0 SO FAR.

```
415. ****************************************
416. LOAD LIST 401 TO 419
417. START CONCRETE DESIGN
```

```
418. CODE ACI 2008
419. FC 35000 MEMB 1 TO 5
420. FYMAIN 420000 MEMB 1 TO 5
421. TRACK 1 ALL
422. CLB 0.05 MEMB 1 TO 5
423. CLT 0.05 MEMB 1 TO 5
424. CLS 0.05 ALL
425. MAXMAIN 25 ALL
426. MINMAIN 16 ALL
427. MINSEC 8 ALL
428. REINF 0 MEMB 1 TO 4
429. DESIGN COLUMN 1 TO 4
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	236.45	0.10	0.00018

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	236.45	5.95	0.01119

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    233.28      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION:  MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    233.28      0.00    0.00000
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     3  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-----------------------------------------------------------

 12 - 20 MM               1.047      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION:  MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    113.23      0.10    0.00038
 ---------------------------------------------------------

    COLUMN INTERACTION:  MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    113.23     22.12    0.08683
 ---------------------------------------------------------


=====================================================================
```

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	88.39	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	88.39	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

430. DESIGN BEAM 5

BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

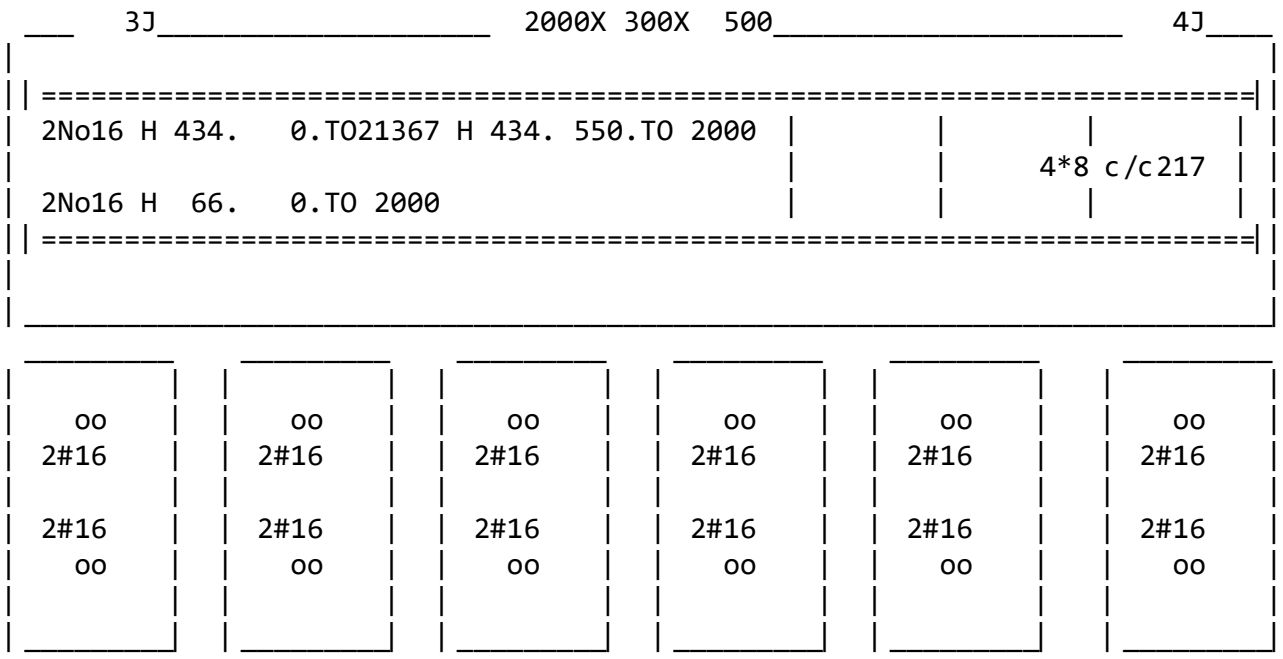
LEN - 2000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	2000.	YES	YES
	CRITICAL POS MOMENT= 44.76 KN-MET AT 0.MM, LOAD 404					
	REQD STEEL= 368.MM2, RHO=0.0028, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					
2	434.	2 - 16MM	0.	1367.	YES	NO
	CRITICAL NEG MOMENT= 34.53 KN-MET AT 0.MM, LOAD 413					
	REQD STEEL= 283.MM2, RHO=0.0022, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					
3	434.	2 - 16MM	550.	2000.	NO	YES
	CRITICAL NEG MOMENT= 46.88 KN-MET AT 2000.MM, LOAD 404					
	REQD STEEL= 386.MM2, RHO=0.0030, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 4.24 KNS Vc= 128.09 KNS Vs= 0.00 KNS
Tu= 0.08 KN-MET Tc= 5.4 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.

AT END SUPPORT - Vu= 48.22 KNS Vc= 127.14 KNS Vs= 0.00 KNS
Tu= 0.14 KN-MET Tc= 5.3 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 217. MM C/C FOR 566. MM



**********END OF BEAM DESIGN**********

431. END CONCRETE DESIGN
432. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 19,2020 TIME= 15: 7:59 ****

```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
```

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	17-04-2020	19-05-2020	02-11-2023				Date	17-04-2020	19-05-2020	02-11-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	RPK/APL	RPK/APL	RPK/APL				Prepared By	RPK/APL	RPK/APL	RPK/APL			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 5 **STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR TR-2**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 16-Apr-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0.35 0; 2 0 0.35 2; 3 0 1.49012e-008 0; 4 0 1.49012e-008 2; 101 0 -2.25 0;
102 0 -2.25 2; 103 0 -2.25 -1.7; 104 0 -2.25 3.7; 105 1.6 -2.25 -1.7;
106 1.6 -2.25 3.7; 107 -1.6 -2.25 -1.7; 108 -1.6 -2.25 3.7;
109 -0.75 -2.25 3.7; 110 -0.75 -2.25 2.75; 111 -1.6 -2.25 2.75;
112 -0.5 -2.25 3.7; 113 -0.5 -2.25 2.75; 114 -0.25 -2.25 3.7;
115 -0.25 -2.25 2.75; 116 0 -2.25 2.75; 117 0.25 -2.25 3.7;
118 0.25 -2.25 2.75; 119 0.5 -2.25 3.7; 120 0.5 -2.25 2.75; 121 0.75 -2.25 3.7;
122 0.75 -2.25 2.75; 123 1.6 -2.25 2.75; 124 -0.75 -2.25 2.5;
125 -1.6 -2.25 2.5; 126 -0.5 -2.25 2.5; 127 -0.25 -2.25 2.5; 128 0 -2.25 2.5;
129 0.25 -2.25 2.5; 130 0.5 -2.25 2.5; 131 0.75 -2.25 2.5; 132 1.6 -2.25 2.5;
133 -0.75 -2.25 2.25; 134 -1.6 -2.25 2.25; 135 -0.5 -2.25 2.25;
136 -0.25 -2.25 2.25; 137 0 -2.25 2.25; 138 0.25 -2.25 2.25;
139 0.5 -2.25 2.25; 140 0.75 -2.25 2.25; 141 1.6 -2.25 2.25; 142 -0.75 -2.25 2;
143 -1.6 -2.25 2; 144 -0.5 -2.25 2; 145 -0.25 -2.25 2; 146 0.25 -2.25 2;
147 0.5 -2.25 2; 148 0.75 -2.25 2; 149 1.6 -2.25 2; 150 -0.75 -2.25 1.75;
151 -1.6 -2.25 1.75; 152 -0.5 -2.25 1.75; 153 -0.25 -2.25 1.75;
154 0 -2.25 1.75; 155 0.25 -2.25 1.75; 156 0.5 -2.25 1.75; 157 0.75 -2.25 1.75;
158 1.6 -2.25 1.75; 159 -0.75 -2.25 1.5; 160 -1.6 -2.25 1.5;
161 -0.5 -2.25 1.5; 162 -0.25 -2.25 1.5; 163 0 -2.25 1.5; 164 0.25 -2.25 1.5;
165 0.5 -2.25 1.5; 166 0.75 -2.25 1.5; 167 1.6 -2.25 1.5; 168 -0.75 -2.25 1.25;
169 -1.6 -2.25 1.25; 170 -0.5 -2.25 1.25; 171 -0.25 -2.25 1.25;
172 0 -2.25 1.25; 173 0.25 -2.25 1.25; 174 0.5 -2.25 1.25; 175 0.75 -2.25 1.25;
176 1.6 -2.25 1.25; 177 -0.75 -2.25 1; 178 -1.6 -2.25 1; 179 -0.5 -2.25 1;
180 -0.25 -2.25 1; 181 0 -2.25 1; 182 0.25 -2.25 1; 183 0.5 -2.25 1;
184 0.75 -2.25 1; 185 1.6 -2.25 1; 186 -0.75 -2.25 0.75; 187 -1.6 -2.25 0.75;
188 -0.5 -2.25 0.75; 189 -0.25 -2.25 0.75; 190 0 -2.25 0.75;
191 0.25 -2.25 0.75; 192 0.5 -2.25 0.75; 193 0.75 -2.25 0.75;
194 1.6 -2.25 0.75; 195 -0.75 -2.25 0.5; 196 -1.6 -2.25 0.5;
197 -0.5 -2.25 0.5; 198 -0.25 -2.25 0.5; 199 0 -2.25 0.5; 200 0.25 -2.25 0.5;
201 0.5 -2.25 0.5; 202 0.75 -2.25 0.5; 203 1.6 -2.25 0.5; 204 -0.75 -2.25 0.25;
205 -1.6 -2.25 0.25; 206 -0.5 -2.25 0.25; 207 -0.25 -2.25 0.25;
208 0 -2.25 0.25; 209 0.25 -2.25 0.25; 210 0.5 -2.25 0.25; 211 0.75 -2.25 0.25;
212 1.6 -2.25 0.25; 213 -0.75 -2.25 0; 214 -1.6 -2.25 0; 215 -0.5 -2.25 0;
216 -0.25 -2.25 0; 217 0.25 -2.25 0; 218 0.5 -2.25 0; 219 0.75 -2.25 0;
220 1.6 -2.25 0; 221 -0.75 -2.25 -0.25; 222 -1.6 -2.25 -0.25;
223 -0.5 -2.25 -0.25; 224 -0.25 -2.25 -0.25; 225 0 -2.25 -0.25;
226 0.25 -2.25 -0.25; 227 0.5 -2.25 -0.25; 228 0.75 -2.25 -0.25;
229 1.6 -2.25 -0.25; 230 -0.75 -2.25 -0.5; 231 -1.6 -2.25 -0.5;
232 -0.5 -2.25 -0.5; 233 -0.25 -2.25 -0.5; 234 0 -2.25 -0.5;
235 0.25 -2.25 -0.5; 236 0.5 -2.25 -0.5; 237 0.75 -2.25 -0.5;
238 1.6 -2.25 -0.5; 239 -0.75 -2.25 -0.75; 240 -1.6 -2.25 -0.75;
241 -0.5 -2.25 -0.75; 242 -0.25 -2.25 -0.75; 243 0 -2.25 -0.75;
244 0.25 -2.25 -0.75; 245 0.5 -2.25 -0.75; 246 0.75 -2.25 -0.75;
247 1.6 -2.25 -0.75; 248 -0.75 -2.25 -1.7; 249 -0.5 -2.25 -1.7;
250 -0.25 -2.25 -1.7; 251 0.25 -2.25 -1.7; 252 0.5 -2.25 -1.7;
253 0.75 -2.25 -1.7; 254 -0.75 -2.25 -1; 255 -1.6 -2.25 -1;
256 -0.75 -2.25 -1.25; 257 -1.6 -2.25 -1.25; 258 -0.5 -2.25 -1;
```

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262 0 -2.25 -1; 263 0 -2.25 -1.25; 264 0.25 -2.25 -1; 265 0.25 -2.25 -1.25;
266 0.5 -2.25 -1; 267 0.5 -2.25 -1.25; 268 0.75 -2.25 -1; 269 0.75 -2.25 -1.25;
270 1.6 -2.25 -1; 271 1.6 -2.25 -1.25; 272 -0.75 -2.25 3.25;
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277 -0.5 -2.25 3; 278 -0.25 -2.25 3.25; 279 -0.25 -2.25 3; 280 0 -2.25 3.25;
281 0 -2.25 3; 282 0.25 -2.25 3.25; 283 0.25 -2.25 3; 284 0.5 -2.25 3.25;
285 0.5 -2.25 3; 286 0.75 -2.25 3.25; 287 0.75 -2.25 3; 288 1.6 -2.25 3.25;
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292 -0.5 -2.25 -1.475; 293 -0.25 -2.25 -1.475; 294 0 -2.25 -1.475;
295 0.25 -2.25 -1.475; 296 0.5 -2.25 -1.475; 297 0.75 -2.25 -1.475;
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301 -0.5 -2.25 3.475; 302 -0.25 -2.25 3.475; 303 0 -2.25 3.475;
304 0.25 -2.25 3.475; 305 0.5 -2.25 3.475; 306 0.75 -2.25 3.475;
307 1.6 -2.25 3.475; 308 1.03333 -2.25 2.75; 309 1.03333 -2.25 2.5;
310 1.31667 -2.25 2.75; 311 1.31667 -2.25 2.5; 312 1.03333 -2.25 2.25;
313 1.31667 -2.25 2.25; 314 1.03333 -2.25 2; 315 1.31667 -2.25 2;
316 1.03333 -2.25 1.75; 317 1.31667 -2.25 1.75; 318 1.03333 -2.25 1.5;
319 1.31667 -2.25 1.5; 320 1.03333 -2.25 1.25; 321 1.31667 -2.25 1.25;
322 1.03333 -2.25 1; 323 1.31667 -2.25 1; 324 1.03333 -2.25 0.75;
325 1.31667 -2.25 0.75; 326 1.03333 -2.25 0.5; 327 1.31667 -2.25 0.5;
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331 1.31667 -2.25 0; 332 1.03333 -2.25 -0.25; 333 1.31667 -2.25 -0.25;
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343 1.03333 -2.25 3; 344 1.31667 -2.25 3.25; 345 1.31667 -2.25 3;
346 1.03333 -2.25 -1.475; 347 1.31667 -2.25 -1.475; 348 1.03333 -2.25 -1.7;
349 1.31667 -2.25 -1.7; 350 1.03333 -2.25 3.7; 351 1.03333 -2.25 3.475;
352 1.31667 -2.25 3.7; 353 1.31667 -2.25 3.475; 354 -1.31667 -2.25 2.75;
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367 -1.03333 -2.25 1.25; 368 -1.31667 -2.25 1; 369 -1.03333 -2.25 1;
370 -1.31667 -2.25 0.75; 371 -1.03333 -2.25 0.75; 372 -1.31667 -2.25 0.5;
373 -1.03333 -2.25 0.5; 374 -1.31667 -2.25 0.25; 375 -1.03333 -2.25 0.25;
376 -1.31667 -2.25 0; 377 -1.03333 -2.25 0; 378 -1.31667 -2.25 -0.25;
379 -1.03333 -2.25 -0.25; 380 -1.31667 -2.25 -0.5; 381 -1.03333 -2.25 -0.5;
382 -1.31667 -2.25 -0.75; 383 -1.03333 -2.25 -0.75; 384 -1.31667 -2.25 -1;
385 -1.03333 -2.25 -1; 386 -1.31667 -2.25 -1.25; 387 -1.03333 -2.25 -1.25;
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394 -1.31667 -2.25 -1.7; 395 -1.03333 -2.25 -1.7; 396 -1.31667 -2.25 3.7;
397 -1.31667 -2.25 3.475; 398 -1.03333 -2.25 3.7; 399 -1.03333 -2.25 3.475;
MEMBER INCIDENCES
1 101 3; 2 3 1; 3 102 4; 4 4 2; 5 3 4;
ELEMENT INCIDENCES SHELL
15 110 113 126 124; 16 113 115 127 126; 17 115 116 128 127; 18 116 118 129 128;
19 118 120 130 129; 20 120 122 131 130; 23 124 126 135 133; 24 126 127 136 135;
25 127 128 137 136; 26 128 129 138 137; 27 129 130 139 138; 28 130 131 140 139;
31 133 135 144 142; 32 135 136 145 144; 33 136 137 102 145; 34 137 138 146 102;
35 138 139 147 146; 36 139 140 148 147; 39 142 144 152 150; 40 144 145 153 152;
41 145 102 154 153; 42 102 146 155 154; 43 146 147 156 155; 44 147 148 157 156;

47 150 152 161 159; 48 152 153 162 161; 49 153 154 163 162; 50 154 155 164 163;
51 155 156 165 164; 52 156 157 166 165; 55 159 161 170 168; 56 161 162 171 170;
57 162 163 172 171; 58 163 164 173 172; 59 164 165 174 173; 60 165 166 175 174;
63 168 170 179 177; 64 170 171 180 179; 65 171 172 181 180; 66 172 173 182 181;
67 173 174 183 182; 68 174 175 184 183; 71 177 179 188 186; 72 179 180 189 188;
73 180 181 190 189; 74 181 182 191 190; 75 182 183 192 191; 76 183 184 193 192;
79 186 188 197 195; 80 188 189 198 197; 81 189 190 199 198; 82 190 191 200 199;
83 191 192 201 200; 84 192 193 202 201; 87 195 197 206 204; 88 197 198 207 206;
89 198 199 208 207; 90 199 200 209 208; 91 200 201 210 209; 92 201 202 211 210;
95 204 206 215 213; 96 206 207 216 215; 97 207 208 101 216; 98 208 209 217 101;
99 209 210 218 217; 100 210 211 219 218; 103 213 215 223 221;
104 215 216 224 223; 105 216 101 225 224; 106 101 217 226 225;
107 217 218 227 226; 108 218 219 228 227; 111 221 223 232 230;
112 223 224 233 232; 113 224 225 234 233; 114 225 226 235 234;
115 226 227 236 235; 116 227 228 237 236; 119 230 232 241 239;
120 232 233 242 241; 121 233 234 243 242; 122 234 235 244 243;
123 235 236 245 244; 124 236 237 246 245; 137 239 241 258 254;
138 254 258 259 256; 140 241 242 260 258; 141 258 260 261 259;
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147 262 264 265 263; 149 244 245 266 264; 150 264 266 267 265;
152 245 246 268 266; 153 266 268 269 267; 162 272 276 277 274;
163 274 277 113 110; 165 276 278 279 277; 166 277 279 115 113;
168 278 280 281 279; 169 279 281 116 115; 171 280 282 283 281;
172 281 283 118 116; 174 282 284 285 283; 175 283 285 120 118;
177 284 286 287 285; 178 285 287 122 120; 184 256 259 292 290;
185 290 292 249 248; 186 259 261 293 292; 187 292 293 250 249;
188 261 263 294 293; 189 293 294 103 250; 190 263 265 295 294;
191 294 295 251 103; 192 265 267 296 295; 193 295 296 252 251;
194 267 269 297 296; 195 296 297 253 252; 200 109 112 301 299;
201 299 301 276 272; 202 112 114 302 301; 203 301 302 278 276;
204 114 104 303 302; 205 302 303 280 278; 206 104 117 304 303;
207 303 304 282 280; 208 117 119 305 304; 209 304 305 284 282;
210 119 121 306 305; 211 305 306 286 284; 214 122 308 309 131;
215 308 310 311 309; 216 310 123 132 311; 217 131 309 312 140;
218 309 311 313 312; 219 311 132 141 313; 220 140 312 314 148;
221 312 313 315 314; 222 313 141 149 315; 223 148 314 316 157;
224 314 315 317 316; 225 315 149 158 317; 226 157 316 318 166;
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230 318 319 321 320; 231 319 167 176 321; 232 175 320 322 184;
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236 322 323 325 324; 237 323 185 194 325; 238 193 324 326 202;
239 324 325 327 326; 240 325 194 203 327; 241 202 326 328 211;
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248 330 331 333 332; 249 331 220 229 333; 250 228 332 334 237;
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254 334 335 337 336; 255 335 238 247 337; 256 246 336 338 268;
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266 343 345 310 308; 267 345 289 123 310; 268 269 340 346 297;
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272 346 347 349 348; 273 347 298 105 349; 274 121 350 351 306;
275 350 352 353 351; 276 352 106 307 353; 277 306 351 342 286;

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278 351 353 344 342; 279 353 307 288 344; 280 111 354 355 125;
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284 355 357 359 358; 285 357 124 133 359; 286 134 358 360 143;
287 358 359 361 360; 288 359 133 142 361; 289 143 360 362 151;
290 360 361 363 362; 291 361 142 150 363; 292 151 362 364 160;
293 362 363 365 364; 294 363 150 159 365; 295 160 364 366 169;
296 364 365 367 366; 297 365 159 168 367; 298 169 366 368 178;
299 366 367 369 368; 300 367 168 177 369; 301 178 368 370 187;
302 368 369 371 370; 303 369 177 186 371; 304 187 370 372 196;
305 370 371 373 372; 306 371 186 195 373; 307 196 372 374 205;
308 372 373 375 374; 309 373 195 204 375; 310 205 374 376 214;
311 374 375 377 376; 312 375 204 213 377; 313 214 376 378 222;
314 376 377 379 378; 315 377 213 221 379; 316 222 378 380 231;
317 378 379 381 380; 318 379 221 230 381; 319 231 380 382 240;
320 380 381 383 382; 321 381 230 239 383; 322 240 382 384 255;
323 382 383 385 384; 324 383 239 254 385; 325 255 384 386 257;
326 384 385 387 386; 327 385 254 256 387; 328 273 388 389 275;
329 388 390 391 389; 330 390 272 274 391; 331 275 389 354 111;
332 389 391 356 354; 333 391 274 110 356; 334 257 386 392 291;
335 386 387 393 392; 336 387 256 290 393; 337 291 392 394 107;
338 392 393 395 394; 339 393 290 248 395; 340 108 396 397 300;
341 396 398 399 397; 342 398 109 299 399; 343 300 397 388 273;
344 397 399 390 388; 345 399 299 272 390;
START GROUP DEFINITION
ELEMENT
_MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
108 111 112 115 116 119 120 123 124 137 138 140 141 149 150 152 153 162 163 -
165 166 174 175 177 178 184 TO 187 192 TO 195 200 TO 203 208 TO 211 -
214 TO 345
_MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
111 TO 116 119 TO 124 137 138 140 141 143 144 146 147 149 150 152 153 162 -
163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
214 TO 219 226 TO 243 250 TO 285 292 TO 309 316 TO 345
_SHEAR 63 TO 68 71 TO 76 137 138 140 141 143 144 146 147 149 150 152 153 162 -
163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
214 TO 345
END GROUP DEFINITION
ELEMENT PROPERTY
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
178 184 TO 195 200 TO 211 214 TO 345 THICKNESS 0.55
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
```

```
1 TO 4 PRIS YD 0.7 ZD 0.5
5 PRIS YD 0.5 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
*****************************************
SUPPORTS
101 TO 399 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
*****************************************
LOAD 1 DEAD LOAD
SELFWEIGHT Y -1
ELEMENT LOAD
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
178 184 TO 195 200 TO 211 214 TO 345 PR GY -31
JOINT LOAD
1 FX 3.5 FY -186.8 FZ 4.6
2 FX 2 FY -54.6 FZ 5
LOAD 2 LIVE LOAD
JOINT LOAD
1 FX 2.1 FY -60.9 FZ -0.1
2 FX 0.4 FY -58.8 FZ 0
LOAD 3 MATERIAL LOAD
JOINT LOAD
1 FX 0.7 FY -41.6 FZ 0
2 FX 0.8 FY -42.2 FZ 0
LOAD 4 WIND LOAD +Z
JOINT LOAD
1 FX -1.6 FY 84.5 FZ 29.5
2 FX 1.6 FY -84.5 FZ 29.5
LOAD 5 WIND LOAD -Z
JOINT LOAD
1 FX 1.59 FY -84.48 FZ -29.45
2 FX -1.59 FY 84.48 FZ -29.45
LOAD 6 WIND LOAD (+X)
JOINT LOAD
1 FX 0.37 FY 0 FZ 0
2 FX 0.37 FY 0 FZ 0
LOAD 7 WIND LOAD (-X)
JOINT LOAD
1 FX -0.37 FY 0 FZ 0
2 FX -0.37 FY 0 FZ 0
LOAD 8 TEMP LOAD 1
JOINT LOAD
1 FX -0.02 FY -0.11 FZ -0.07
2 FX 0.02 FY 0.11 FZ 0.08
LOAD 9 TEMP LOAD 2
JOINT LOAD
1 FX 0.02 FY 0.11 FZ 0.07
2 FX -0.02 FY -0.11 FZ -0.08
LOAD 10 EARTHQUAKE LOAD +X
JOINT LOAD
1 FX 0.04 FY -0.01 FZ -0.02
2 FX 0.07 FY -0.11 FZ -0.02
```

```
LOAD 11 EARTHQUAKE LOAD +Z
JOINT LOAD
1 FX 0.01 FY 6.4 FZ 3.5
2 FX -0.01 FY -6.4 FZ 3.5
*****************************************
*****Load combination for bearing capacity check
LOAD 301 1.0 DL
REPEAT LOAD
1 1.0
LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 8 1.0 2 1.0 3 1.0
LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 9 1.0 2 1.0 3 1.0
LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 4 0.75 2 0.75 3 0.75
LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 5 0.75 2 0.75 3 0.75
LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 6 0.75 2 0.75 3 0.75
LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 7 0.75 2 0.75 3 0.75
LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 0.525 2 0.75 3 0.75
LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 0.525 2 0.75 3 0.75
LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD 312 0.6 DL + 1.0 WL(+Z)
REPEAT LOAD
1 0.6 4 1.0
LOAD 313 0.6 DL + 1.0 WL(-Z)
REPEAT LOAD
1 0.6 5 1.0
LOAD 314 0.6 DL + 1.0 WL(+X)
REPEAT LOAD
1 0.6 6 1.0
LOAD 315 0.6 DL + 1.0 WL(-X)
REPEAT LOAD
1 0.6 7 1.0
LOAD 316 0.6 DL + 0.7 EL(+X)
REPEAT LOAD
1 0.6 10 0.7
```

```
LOAD 317 0.6 DL + 0.7 EL(-X)
REPEAT LOAD
1 0.6 10 -0.7
LOAD 318 0.6 DL + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 11 0.7
LOAD 319 0.6 DL + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 11 -0.7
*****************************************
*****Load combination for foundation design
LOAD 401 1.4 DL
REPEAT LOAD
1 1.4
LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 8 1.4 2 1.7 3 1.7
LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 9 1.4 2 1.7 3 1.7
LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 4 1.6 2 0.5 3 0.5
LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 5 1.6 2 0.5 3 0.5
LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 6 1.6 2 0.5 3 0.5
LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 7 1.6 2 0.5 3 0.5
LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 1.0 2 0.5 3 0.5
LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 1.0 2 0.5 3 0.5
LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD 412 0.9 DL + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 1.6
LOAD 413 0.9 DL + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 5 1.6
LOAD 414 0.9 DL + 1.6 WL(+X)
REPEAT LOAD
1 0.9 6 1.6
LOAD 415 0.9 DL + 1.6 WL(-X)
```

```
REPEAT LOAD
1 0.9 7 1.6
LOAD 416 0.9 DL + 1.0 EL(+X)
REPEAT LOAD
1 0.9 10 1.0
LOAD 417 0.9 DL + 1.0 EL(-X)
REPEAT LOAD
1 0.9 10 -1.0
LOAD 418 0.9 DL + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 11 1.0
LOAD 419 0.9 DL + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 11 -1.0
*****************************************
PERFORM ANALYSIS
*****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 1 TO 5
FYMAIN 420000 MEMB 1 TO 5
TRACK 1 ALL
CLB 0.05 MEMB 1 TO 5
CLT 0.05 MEMB 1 TO 5
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
*****************************************************
*                                                   *
*           STAAD.Pro CONNECT Edition               *
*           Version  21.00.01.12                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    APR 17, 2020                   *
*           Time=    17: 3:49                       *
*                                                   *
*  Licensed to: MMH KOLKATTA                        *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\F Drive\Maaden Gold\Conveyor CNV-3210-001\Civil\Documents\Staad\3210-CNV-001-TR2-Fdn.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 16-APR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 0.35 0; 2 0 0.35 2; 3 0 1.49012E-008 0; 4 0 1.49012E-008 2; 101 0 -2.25 0
     9. 102 0 -2.25 2; 103 0 -2.25 -1.7; 104 0 -2.25 3.7; 105 1.6 -2.25 -1.7
    10. 106 1.6 -2.25 3.7; 107 -1.6 -2.25 -1.7; 108 -1.6 -2.25 3.7
    11. 109 -0.75 -2.25 3.7; 110 -0.75 -2.25 2.75; 111 -1.6 -2.25 2.75
    12. 112 -0.5 -2.25 3.7; 113 -0.5 -2.25 2.75; 114 -0.25 -2.25 3.7
    13. 115 -0.25 -2.25 2.75; 116 0 -2.25 2.75; 117 0.25 -2.25 3.7
    14. 118 0.25 -2.25 2.75; 119 0.5 -2.25 3.7; 120 0.5 -2.25 2.75; 121 0.75 -2.25 3.7
    15. 122 0.75 -2.25 2.75; 123 1.6 -2.25 2.75; 124 -0.75 -2.25 2.5
    16. 125 -1.6 -2.25 2.5; 126 -0.5 -2.25 2.5; 127 -0.25 -2.25 2.5; 128 0 -2.25 2.5
    17. 129 0.25 -2.25 2.5; 130 0.5 -2.25 2.5; 131 0.75 -2.25 2.5; 132 1.6 -2.25 2.5
    18. 133 -0.75 -2.25 2.25; 134 -1.6 -2.25 2.25; 135 -0.5 -2.25 2.25
    19. 136 -0.25 -2.25 2.25; 137 0 -2.25 2.25; 138 0.25 -2.25 2.25
    20. 139 0.5 -2.25 2.25; 140 0.75 -2.25 2.25; 141 1.6 -2.25 2.25; 142 -0.75 -2.25 2
    21. 143 -1.6 -2.25 2; 144 -0.5 -2.25 2; 145 -0.25 -2.25 2; 146 0.25 -2.25 2
    22. 147 0.5 -2.25 2; 148 0.75 -2.25 2; 149 1.6 -2.25 2; 150 -0.75 -2.25 1.75
    23. 151 -1.6 -2.25 1.75; 152 -0.5 -2.25 1.75; 153 -0.25 -2.25 1.75
    24. 154 0 -2.25 1.75; 155 0.25 -2.25 1.75; 156 0.5 -2.25 1.75; 157 0.75 -2.25 1.75
    25. 158 1.6 -2.25 1.75; 159 -0.75 -2.25 1.5; 160 -1.6 -2.25 1.5
    26. 161 -0.5 -2.25 1.5; 162 -0.25 -2.25 1.5; 163 0 -2.25 1.5; 164 0.25 -2.25 1.5
    27. 165 0.5 -2.25 1.5; 166 0.75 -2.25 1.5; 167 1.6 -2.25 1.5; 168 -0.75 -2.25 1.25
    28. 169 -1.6 -2.25 1.25; 170 -0.5 -2.25 1.25; 171 -0.25 -2.25 1.25
    29. 172 0 -2.25 1.25; 173 0.25 -2.25 1.25; 174 0.5 -2.25 1.25; 175 0.75 -2.25 1.25
    30. 176 1.6 -2.25 1.25; 177 -0.75 -2.25 1; 178 -1.6 -2.25 1; 179 -0.5 -2.25 1
    31. 180 -0.25 -2.25 1; 181 0 -2.25 1; 182 0.25 -2.25 1; 183 0.5 -2.25 1
    32. 184 0.75 -2.25 1; 185 1.6 -2.25 1; 186 -0.75 -2.25 0.75; 187 -1.6 -2.25 0.75
    33. 188 -0.5 -2.25 0.75; 189 -0.25 -2.25 0.75; 190 0 -2.25 0.75
    34. 191 0.25 -2.25 0.75; 192 0.5 -2.25 0.75; 193 0.75 -2.25 0.75
    35. 194 1.6 -2.25 0.75; 195 -0.75 -2.25 0.5; 196 -1.6 -2.25 0.5
    36. 197 -0.5 -2.25 0.5; 198 -0.25 -2.25 0.5; 199 0 -2.25 0.5; 200 0.25 -2.25 0.5
    37. 201 0.5 -2.25 0.5; 202 0.75 -2.25 0.5; 203 1.6 -2.25 0.5; 204 -0.75 -2.25 0.25
    38. 205 -1.6 -2.25 0.25; 206 -0.5 -2.25 0.25; 207 -0.25 -2.25 0.25
```

39. 208 0 -2.25 0.25; 209 0.25 -2.25 0.25; 210 0.5 -2.25 0.25; 211 0.75 -2.25 0.25
40. 212 1.6 -2.25 0.25; 213 -0.75 -2.25 0; 214 -1.6 -2.25 0; 215 -0.5 -2.25 0
41. 216 -0.25 -2.25 0; 217 0.25 -2.25 0; 218 0.5 -2.25 0; 219 0.75 -2.25 0
42. 220 1.6 -2.25 0; 221 -0.75 -2.25 -0.25; 222 -1.6 -2.25 -0.25
43. 223 -0.5 -2.25 -0.25; 224 -0.25 -2.25 -0.25; 225 0 -2.25 -0.25
44. 226 0.25 -2.25 -0.25; 227 0.5 -2.25 -0.25; 228 0.75 -2.25 -0.25
45. 229 1.6 -2.25 -0.25; 230 -0.75 -2.25 -0.5; 231 -1.6 -2.25 -0.5
46. 232 -0.5 -2.25 -0.5; 233 -0.25 -2.25 -0.5; 234 0 -2.25 -0.5
47. 235 0.25 -2.25 -0.5; 236 0.5 -2.25 -0.5; 237 0.75 -2.25 -0.5
48. 238 1.6 -2.25 -0.5; 239 -0.75 -2.25 -0.75; 240 -1.6 -2.25 -0.75
49. 241 -0.5 -2.25 -0.75; 242 -0.25 -2.25 -0.75; 243 0 -2.25 -0.75
50. 244 0.25 -2.25 -0.75; 245 0.5 -2.25 -0.75; 246 0.75 -2.25 -0.75
51. 247 1.6 -2.25 -0.75; 248 -0.75 -2.25 -1.7; 249 -0.5 -2.25 -1.7
52. 250 -0.25 -2.25 -1.7; 251 0.25 -2.25 -1.7; 252 0.5 -2.25 -1.7
53. 253 0.75 -2.25 -1.7; 254 -0.75 -2.25 -1; 255 -1.6 -2.25 -1.
54. 256 -0.75 -2.25 -1.25; 257 -1.6 -2.25 -1.25; 258 -0.5 -2.25 -1.
55. 259 -0.5 -2.25 -1.25; 260 -0.25 -2.25 -1; 261 -0.25 -2.25 -1.25
56. 262 0 -2.25 -1; 263 0 -2.25 -1.25; 264 0.25 -2.25 -1; 265 0.25 -2.25 -1.25
57. 266 0.5 -2.25 -1; 267 0.5 -2.25 -1.25; 268 0.75 -2.25 -1; 269 0.75 -2.25 -1.25
58. 270 1.6 -2.25 -1; 271 1.6 -2.25 -1.25; 272 -0.75 -2.25 3.25
59. 273 -1.6 -2.25 3.25; 274 -0.75 -2.25 3; 275 -1.6 -2.25 3; 276 -0.5 -2.25 3.25
60. 277 -0.5 -2.25 3; 278 -0.25 -2.25 3.25; 279 -0.25 -2.25 3; 280 0 -2.25 3.25
61. 281 0 -2.25 3; 282 0.25 -2.25 3.25; 283 0.25 -2.25 3; 284 0.5 -2.25 3.25
62. 285 0.5 -2.25 3; 286 0.75 -2.25 3.25; 287 0.75 -2.25 3; 288 1.6 -2.25 3.25
63. 289 1.6 -2.25 3; 290 -0.75 -2.25 -1.475; 291 -1.6 -2.25 -1.475
64. 292 -0.5 -2.25 -1.475; 293 -0.25 -2.25 -1.475; 294 0 -2.25 -1.475
65. 295 0.25 -2.25 -1.475; 296 0.5 -2.25 -1.475; 297 0.75 -2.25 -1.475
66. 298 1.6 -2.25 -1.475; 299 -0.75 -2.25 3.475; 300 -1.6 -2.25 3.475
67. 301 -0.5 -2.25 3.475; 302 -0.25 -2.25 3.475; 303 0 -2.25 3.475
68. 304 0.25 -2.25 3.475; 305 0.5 -2.25 3.475; 306 0.75 -2.25 3.475
69. 307 1.6 -2.25 3.475; 308 1.03333 -2.25 2.75; 309 1.03333 -2.25 2.5
70. 310 1.31667 -2.25 2.75; 311 1.31667 -2.25 2.5; 312 1.03333 -2.25 2.25
71. 313 1.31667 -2.25 2.25; 314 1.03333 -2.25 2; 315 1.31667 -2.25 2
72. 316 1.03333 -2.25 1.75; 317 1.31667 -2.25 1.75; 318 1.03333 -2.25 1.5
73. 319 1.31667 -2.25 1.5; 320 1.03333 -2.25 1.25; 321 1.31667 -2.25 1.25
74. 322 1.03333 -2.25 1; 323 1.31667 -2.25 1; 324 1.03333 -2.25 0.75
75. 325 1.31667 -2.25 0.75; 326 1.03333 -2.25 0.5; 327 1.31667 -2.25 0.5
76. 328 1.03333 -2.25 0.25; 329 1.31667 -2.25 0.25; 330 1.03333 -2.25 0
77. 331 1.31667 -2.25 0; 332 1.03333 -2.25 -0.25; 333 1.31667 -2.25 -0.25
78. 334 1.03333 -2.25 -0.5; 335 1.31667 -2.25 -0.5; 336 1.03333 -2.25 -0.75
79. 337 1.31667 -2.25 -0.75; 338 1.03333 -2.25 -1; 339 1.31667 -2.25 -1.
80. 340 1.03333 -2.25 -1.25; 341 1.31667 -2.25 -1.25; 342 1.03333 -2.25 3.25
81. 343 1.03333 -2.25 3; 344 1.31667 -2.25 3.25; 345 1.31667 -2.25 3
82. 346 1.03333 -2.25 -1.475; 347 1.31667 -2.25 -1.475; 348 1.03333 -2.25 -1.7
83. 349 1.31667 -2.25 -1.7; 350 1.03333 -2.25 3.7; 351 1.03333 -2.25 3.475
84. 352 1.31667 -2.25 3.7; 353 1.31667 -2.25 3.475; 354 -1.31667 -2.25 2.75
85. 355 -1.31667 -2.25 2.5; 356 -1.03333 -2.25 2.75; 357 -1.03333 -2.25 2.5
86. 358 -1.31667 -2.25 2.25; 359 -1.03333 -2.25 2.25; 360 -1.31667 -2.25 2
87. 361 -1.03333 -2.25 2; 362 -1.31667 -2.25 1.75; 363 -1.03333 -2.25 1.75
88. 364 -1.31667 -2.25 1.5; 365 -1.03333 -2.25 1.5; 366 -1.31667 -2.25 1.25
89. 367 -1.03333 -2.25 1.25; 368 -1.31667 -2.25 1; 369 -1.03333 -2.25 1
90. 370 -1.31667 -2.25 0.75; 371 -1.03333 -2.25 0.75; 372 -1.31667 -2.25 0.5
91. 373 -1.03333 -2.25 0.5; 374 -1.31667 -2.25 0.25; 375 -1.03333 -2.25 0.25
92. 376 -1.31667 -2.25 0; 377 -1.03333 -2.25 0; 378 -1.31667 -2.25 -0.25
93. 379 -1.03333 -2.25 -0.25; 380 -1.31667 -2.25 -0.5; 381 -1.03333 -2.25 -0.5
94. 382 -1.31667 -2.25 -0.75; 383 -1.03333 -2.25 -0.75; 384 -1.31667 -2.25 -1.

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 95. 385 -1.03333 -2.25 -1; 386 -1.31667 -2.25 -1.25; 387 -1.03333 -2.25 -1.25
 96. 388 -1.31667 -2.25 3.25; 389 -1.31667 -2.25 3; 390 -1.03333 -2.25 3.25
 97. 391 -1.03333 -2.25 3; 392 -1.31667 -2.25 -1.475; 393 -1.03333 -2.25 -1.475
 98. 394 -1.31667 -2.25 -1.7; 395 -1.03333 -2.25 -1.7; 396 -1.31667 -2.25 3.7
 99. 397 -1.31667 -2.25 3.475; 398 -1.03333 -2.25 3.7; 399 -1.03333 -2.25 3.475
100. MEMBER INCIDENCES
101. 1 101 3; 2 3 1; 3 102 4; 4 4 2; 5 3 4
102. ELEMENT INCIDENCES SHELL
103. 15 110 113 126 124; 16 113 115 127 126; 17 115 116 128 127; 18 116 118 129 128
104. 19 118 120 130 129; 20 120 122 131 130; 23 124 126 135 133; 24 126 127 136 135
105. 25 127 128 137 136; 26 128 129 138 137; 27 129 130 139 138; 28 130 131 140 139
106. 31 133 135 144 142; 32 135 136 145 144; 33 136 137 102 145; 34 137 138 146 102
107. 35 138 139 147 146; 36 139 140 148 147; 39 142 144 152 150; 40 144 145 153 152
108. 41 145 102 154 153; 42 102 146 155 154; 43 146 147 156 155; 44 147 148 157 156
109. 47 150 152 161 159; 48 152 153 162 161; 49 153 154 163 162; 50 154 155 164 163
110. 51 155 156 165 164; 52 156 157 166 165; 55 159 161 170 168; 56 161 162 171 170
111. 57 162 163 172 171; 58 163 164 173 172; 59 164 165 174 173; 60 165 166 175 174
112. 63 168 170 179 177; 64 170 171 180 179; 65 171 172 181 180; 66 172 173 182 181
113. 67 173 174 183 182; 68 174 175 184 183; 71 177 179 188 186; 72 179 180 189 188
114. 73 180 181 190 189; 74 181 182 191 190; 75 182 183 192 191; 76 183 184 193 192
115. 79 186 188 197 195; 80 188 189 198 197; 81 189 190 199 198; 82 190 191 200 199
116. 83 191 192 201 200; 84 192 193 202 201; 87 195 197 206 204; 88 197 198 207 206
117. 89 198 199 208 207; 90 199 200 209 208; 91 200 201 210 209; 92 201 202 211 210
118. 95 204 206 215 213; 96 206 207 216 215; 97 207 208 101 216; 98 208 209 217 101
119. 99 209 210 218 217; 100 210 211 219 218; 103 213 215 223 221
120. 104 215 216 224 223; 105 216 101 225 224; 106 101 217 226 225
121. 107 217 218 227 226; 108 218 219 228 227; 111 221 223 232 230
122. 112 223 224 233 232; 113 224 225 234 233; 114 225 226 235 234
123. 115 226 227 236 235; 116 227 228 237 236; 119 230 232 241 239
124. 120 232 233 242 241; 121 233 234 243 242; 122 234 235 244 243
125. 123 235 236 245 244; 124 236 237 246 245; 137 239 241 258 254
126. 138 254 258 259 256; 140 241 242 260 258; 141 258 260 261 259
127. 143 242 243 262 260; 144 260 262 263 261; 146 243 244 264 262
128. 147 262 264 265 263; 149 244 245 266 264; 150 264 266 267 265
129. 152 245 246 268 266; 153 266 268 269 267; 162 272 276 277 274
130. 163 274 277 113 110; 165 276 278 279 277; 166 277 279 115 113
131. 168 278 280 281 279; 169 279 281 116 115; 171 280 282 283 281
132. 172 281 283 118 116; 174 282 284 285 283; 175 283 285 120 118
133. 177 284 286 287 285; 178 285 287 122 120; 184 256 259 292 290
134. 185 290 292 249 248; 186 259 261 293 292; 187 292 293 250 249
135. 188 261 263 294 293; 189 293 294 103 250; 190 263 265 295 294
136. 191 294 295 251 103; 192 265 267 296 295; 193 295 296 252 251
137. 194 267 269 297 296; 195 296 297 253 252; 200 109 112 301 299
138. 201 299 301 276 272; 202 112 114 302 301; 203 301 302 278 276
139. 204 114 104 303 302; 205 302 303 280 278; 206 104 117 304 303
140. 207 303 304 282 280; 208 117 119 305 304; 209 304 305 284 282
141. 210 119 121 306 305; 211 305 306 286 284; 214 122 308 309 131
142. 215 308 310 311 309; 216 310 123 132 311; 217 131 309 312 140
143. 218 309 311 313 312; 219 311 132 141 313; 220 140 312 314 148
144. 221 312 313 315 314; 222 313 141 149 315; 223 148 314 316 157
145. 224 314 315 317 316; 225 315 149 158 317; 226 157 316 318 166
146. 227 316 317 319 318; 228 317 158 167 319; 229 166 318 320 175
147. 230 318 319 321 320; 231 319 167 176 321; 232 175 320 322 184
148. 233 320 321 323 322; 234 321 176 185 323; 235 184 322 324 193
149. 236 322 323 325 324; 237 323 185 194 325; 238 193 324 326 202
150. 239 324 325 327 326; 240 325 194 203 327; 241 202 326 328 211
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151. 242 326 327 329 328; 243 327 203 212 329; 244 211 328 330 219
152. 245 328 329 331 330; 246 329 212 220 331; 247 219 330 332 228
153. 248 330 331 333 332; 249 331 220 229 333; 250 228 332 334 237
154. 251 332 333 335 334; 252 333 229 238 335; 253 237 334 336 246
155. 254 334 335 337 336; 255 335 238 247 337; 256 246 336 338 268
156. 257 336 337 339 338; 258 337 247 270 339; 259 268 338 340 269
157. 260 338 339 341 340; 261 339 270 271 341; 262 286 342 343 287
158. 263 342 344 345 343; 264 344 288 289 345; 265 287 343 308 122
159. 266 343 345 310 308; 267 345 289 123 310; 268 269 340 346 297
160. 269 340 341 347 346; 270 341 271 298 347; 271 297 346 348 253
161. 272 346 347 349 348; 273 347 298 105 349; 274 121 350 351 306
162. 275 350 352 353 351; 276 352 106 307 353; 277 306 351 342 286
163. 278 351 353 344 342; 279 353 307 288 344; 280 111 354 355 125
164. 281 354 356 357 355; 282 356 110 124 357; 283 125 355 358 134
165. 284 355 357 359 358; 285 357 124 133 359; 286 134 358 360 143
166. 287 358 359 361 360; 288 359 133 142 361; 289 143 360 362 151
167. 290 360 361 363 362; 291 361 142 150 363; 292 151 362 364 160
168. 293 362 363 365 364; 294 363 150 159 365; 295 160 364 366 169
169. 296 364 365 367 366; 297 365 159 168 367; 298 169 366 368 178
170. 299 366 367 369 368; 300 367 168 177 369; 301 178 368 370 187
171. 302 368 369 371 370; 303 369 177 186 371; 304 187 370 372 196
172. 305 370 371 373 372; 306 371 186 195 373; 307 196 372 374 205
173. 308 372 373 375 374; 309 373 195 204 375; 310 205 374 376 214
174. 311 374 375 377 376; 312 375 204 213 377; 313 214 376 378 222
175. 314 376 377 379 378; 315 377 213 221 379; 316 222 378 380 231
176. 317 378 379 381 380; 318 379 221 230 381; 319 231 380 382 240
177. 320 380 381 383 382; 321 381 230 239 383; 322 240 382 384 255
178. 323 382 383 385 384; 324 383 239 254 385; 325 255 384 386 257
179. 326 384 385 387 386; 327 385 254 256 387; 328 273 388 389 275
180. 329 388 390 391 389; 330 390 272 274 391; 331 275 389 354 111
181. 332 389 391 356 354; 333 391 274 110 356; 334 257 386 392 291
182. 335 386 387 393 392; 336 387 256 290 393; 337 291 392 394 107
183. 338 392 393 395 394; 339 393 290 248 395; 340 108 396 397 300
184. 341 396 398 399 397; 342 398 109 299 399; 343 300 397 388 273
185. 344 397 399 390 388; 345 399 299 272 390
186. START GROUP DEFINITION
187. ELEMENT
188. _MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
189. 63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
190. 108 111 112 115 116 119 120 123 124 137 138 140 141 149 150 152 153 162 163 -
191. 165 166 174 175 177 178 184 TO 187 192 TO 195 200 TO 203 208 TO 211 -
192. 214 TO 345
193. _MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
194. 111 TO 116 119 TO 124 137 138 140 141 143 144 146 147 149 150 152 153 162 -
195. 163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
196. 214 TO 219 226 TO 243 250 TO 285 292 TO 309 316 TO 345
197. _SHEAR 63 TO 68 71 TO 76 137 138 140 141 143 144 146 147 149 150 152 153 162 -
198. 163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
199. 214 TO 345
200. END GROUP DEFINITION
201. ELEMENT PROPERTY
202. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
203. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
204. 143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
205. 178 184 TO 195 200 TO 211 214 TO 345 THICKNESS 0.55
206. DEFINE MATERIAL START
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207. ISOTROPIC CONCRETE
208. E 2.17185E+007
209. POISSON 0.17
210. DENSITY 23.5616
211. ALPHA 1E-005
212. DAMP 0.05
213. TYPE CONCRETE
214. STRENGTH FCU 27579
215. END DEFINE MATERIAL
216. MEMBER PROPERTY AMERICAN
217. 1 TO 4 PRIS YD 0.7 ZD 0.5
218. 5 PRIS YD 0.5 ZD 0.3
219. CONSTANTS
220. MATERIAL CONCRETE ALL
221. ****************************************
222. SUPPORTS
223. 101 TO 399 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
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JOINT	INFLUENCE AREA
101	0.062500
102	0.062500
103	0.028125
104	0.028125
105	0.015937
106	0.015937
107	0.015937
108	0.015937
109	0.030000
110	0.066666
111	0.035416
112	0.028125
113	0.062500
114	0.028125
115	0.062500
116	0.062500
117	0.028125
118	0.062500
119	0.028125
120	0.062500
121	0.030000
122	0.066666
123	0.035416
124	0.066666
125	0.035416
126	0.062500
127	0.062500
128	0.062500
129	0.062500
130	0.062500
131	0.066666
132	0.035416
133	0.066666
134	0.035416
135	0.062500

136	0.062500
137	0.062500
138	0.062500
139	0.062500
140	0.066666
141	0.035416
142	0.066666
143	0.035416
144	0.062500
145	0.062500
146	0.062500
147	0.062500
148	0.066666
149	0.035416
150	0.066666
151	0.035416
152	0.062500
153	0.062500
154	0.062500
155	0.062500
156	0.062500
157	0.066666
158	0.035416
159	0.066666
160	0.035416
161	0.062500
162	0.062500
163	0.062500
164	0.062500
165	0.062500
166	0.066666
167	0.035416
168	0.066666
169	0.035416
170	0.062500
171	0.062500
172	0.062500
173	0.062500
174	0.062500
175	0.066666
176	0.035416
177	0.066666
178	0.035416
179	0.062500
180	0.062500
181	0.062500
182	0.062500
183	0.062500
184	0.066666
185	0.035416
186	0.066666
187	0.035416
188	0.062500
189	0.062500
190	0.062500
191	0.062500

192	0.062500
193	0.066666
194	0.035416
195	0.066666
196	0.035416
197	0.062500
198	0.062500
199	0.062500
200	0.062500
201	0.062500
202	0.066666
203	0.035416
204	0.066666
205	0.035416
206	0.062500
207	0.062500
208	0.062500
209	0.062500
210	0.062500
211	0.066666
212	0.035416
213	0.066666
214	0.035416
215	0.062500
216	0.062500
217	0.062500
218	0.062500
219	0.066666
220	0.035416
221	0.066666
222	0.035416
223	0.062500
224	0.062500
225	0.062500
226	0.062500
227	0.062500
228	0.066666
229	0.035416
230	0.066666
231	0.035416
232	0.062500
233	0.062500
234	0.062500
235	0.062500
236	0.062500
237	0.066666
238	0.035416
239	0.066666
240	0.035416
241	0.062500
242	0.062500
243	0.062500
244	0.062500
245	0.062500
246	0.066666
247	0.035416

248	0.030000
249	0.028125
250	0.028125
251	0.028125
252	0.028125
253	0.030000
254	0.066666
255	0.035416
256	0.063333
257	0.033645
258	0.062500
259	0.059375
260	0.062500
261	0.059375
262	0.062500
263	0.059375
264	0.062500
265	0.059375
266	0.062500
267	0.059375
268	0.066666
269	0.063333
270	0.035416
271	0.033645
272	0.063333
273	0.033645
274	0.066666
275	0.035416
276	0.059375
277	0.062500
278	0.059375
279	0.062500
280	0.059375
281	0.062500
282	0.059375
283	0.062500
284	0.059375
285	0.062500
286	0.063333
287	0.066666
288	0.033645
289	0.035416
290	0.060000
291	0.031875
292	0.056250
293	0.056250
294	0.056250
295	0.056250
296	0.056250
297	0.060000
298	0.031875
299	0.060000
300	0.031875
301	0.056250
302	0.056250
303	0.056250

304	0.056250
305	0.056250
306	0.060000
307	0.031875
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335	0.070834
336	0.070834
337	0.070834
338	0.070834
339	0.070834
340	0.067292
341	0.067292
342	0.067292
343	0.070834
344	0.067292
345	0.070834
346	0.063750
347	0.063750
348	0.031875
349	0.031875
350	0.031875
351	0.063750
352	0.031875
353	0.063750
354	0.070834
355	0.070834
356	0.070834
357	0.070834
358	0.070834
359	0.070834

360	0.070834
361	0.070834
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364	0.070834
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367	0.070834
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369	0.070834
370	0.070834
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373	0.070834
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377	0.070834
378	0.070834
379	0.070834
380	0.070834
381	0.070834
382	0.070834
383	0.070834
384	0.070834
385	0.070834
386	0.067292
387	0.067292
388	0.067292
389	0.070834
390	0.067292
391	0.070834
392	0.063750
393	0.063750
394	0.031875
395	0.031875
396	0.031875
397	0.063750
398	0.031875
399	0.063750
TOTAL	17.280014

```
224. ****************************************
225. LOAD 1 DEAD LOAD
226. SELFWEIGHT Y -1
227. ELEMENT LOAD
228. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
229. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
230. 143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
231. 178 184 TO 195 200 TO 211 214 TO 345 PR GY -31
232. JOINT LOAD
233. 1 FX 3.5 FY -186.8 FZ 4.6
234. 2 FX 2 FY -54.6 FZ 5
235. LOAD 2 LIVE LOAD
236. JOINT LOAD
237. 1 FX 2.1 FY -60.9 FZ -0.1
238. 2 FX 0.4 FY -58.8 FZ 0
```

```
239. LOAD 3 MATERIAL LOAD
240. JOINT LOAD
241. 1 FX 0.7 FY -41.6 FZ 0
242. 2 FX 0.8 FY -42.2 FZ 0
243. LOAD 4 WIND LOAD +Z
244. JOINT LOAD
245. 1 FX -1.6 FY 84.5 FZ 29.5
246. 2 FX 1.6 FY -84.5 FZ 29.5
247. LOAD 5 WIND LOAD -Z
248. JOINT LOAD
249. 1 FX 1.59 FY -84.48 FZ -29.45
250. 2 FX -1.59 FY 84.48 FZ -29.45
251. LOAD 6 WIND LOAD (+X)
252. JOINT LOAD
253. 1 FX 0.37 FY 0 FZ 0
254. 2 FX 0.37 FY 0 FZ 0
255. LOAD 7 WIND LOAD (-X)
256. JOINT LOAD
257. 1 FX -0.37 FY 0 FZ 0
258. 2 FX -0.37 FY 0 FZ 0
259. LOAD 8 TEMP LOAD 1
260. JOINT LOAD
261. 1 FX -0.02 FY -0.11 FZ -0.07
262. 2 FX 0.02 FY 0.11 FZ 0.08
263. LOAD 9 TEMP LOAD 2
264. JOINT LOAD
265. 1 FX 0.02 FY 0.11 FZ 0.07
266. 2 FX -0.02 FY -0.11 FZ -0.08
267. LOAD 10 EARTHQUAKE LOAD +X
268. JOINT LOAD
269. 1 FX 0.04 FY -0.01 FZ -0.02
270. 2 FX 0.07 FY -0.11 FZ -0.02
271. LOAD 11 EARTHQUAKE LOAD +Z
272. JOINT LOAD
273. 1 FX 0.01 FY 6.4 FZ 3.5
274. 2 FX -0.01 FY -6.4 FZ 3.5
275. ****************************************
276. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
277. LOAD 301 1.0 DL
278. REPEAT LOAD
279. 1 1.0
280. LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
281. REPEAT LOAD
282. 1 1.0 8 1.0 2 1.0 3 1.0
283. LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
284. REPEAT LOAD
285. 1 1.0 9 1.0 2 1.0 3 1.0
286. LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
287. REPEAT LOAD
288. 1 1.0 4 0.75 2 0.75 3 0.75
289. LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
290. REPEAT LOAD
291. 1 1.0 5 0.75 2 0.75 3 0.75
292. LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
293. REPEAT LOAD
294. 1 1.0 6 0.75 2 0.75 3 0.75
```

```
295. LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
296. REPEAT LOAD
297. 1 1.0 7 0.75 2 0.75 3 0.75
298. LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
299. REPEAT LOAD
300. 1 1.0 10 0.525 2 0.75 3 0.75
301. LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
302. REPEAT LOAD
303. 1 1.0 10 -0.525 2 0.75 3 0.75
304. LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
305. REPEAT LOAD
306. 1 1.0 11 0.525 2 0.75 3 0.75
307. LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
308. REPEAT LOAD
309. 1 1.0 11 -0.525 2 0.75 3 0.75
310. LOAD 312 0.6 DL + 1.0 WL(+Z)
311. REPEAT LOAD
312. 1 0.6 4 1.0
313. LOAD 313 0.6 DL + 1.0 WL(-Z)
314. REPEAT LOAD
315. 1 0.6 5 1.0
316. LOAD 314 0.6 DL + 1.0 WL(+X)
317. REPEAT LOAD
318. 1 0.6 6 1.0
319. LOAD 315 0.6 DL + 1.0 WL(-X)
320. REPEAT LOAD
321. 1 0.6 7 1.0
322. LOAD 316 0.6 DL + 0.7 EL(+X)
323. REPEAT LOAD
324. 1 0.6 10 0.7
325. LOAD 317 0.6 DL + 0.7 EL(-X)
326. REPEAT LOAD
327. 1 0.6 10 -0.7
328. LOAD 318 0.6 DL + 0.7 EL(+Z)
329. REPEAT LOAD
330. 1 0.6 11 0.7
331. LOAD 319 0.6 DL + 0.7 EL(-Z)
332. REPEAT LOAD
333. 1 0.6 11 -0.7
334. ****************************************
335. *****LOAD COMBINATION FOR FOUNDATION DESIGN
336. LOAD 401 1.4 DL
337. REPEAT LOAD
338. 1 1.4
339. LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
340. REPEAT LOAD
341. 1 1.4 8 1.4 2 1.7 3 1.7
342. LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
343. REPEAT LOAD
344. 1 1.4 9 1.4 2 1.7 3 1.7
345. LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
346. REPEAT LOAD
347. 1 1.2 4 1.6 2 0.5 3 0.5
348. LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
349. REPEAT LOAD
350. 1 1.2 5 1.6 2 0.5 3 0.5
```

```
351. LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
352. REPEAT LOAD
353. 1 1.2 6 1.6 2 0.5 3 0.5
354. LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
355. REPEAT LOAD
356. 1 1.2 7 1.6 2 0.5 3 0.5
357. LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
358. REPEAT LOAD
359. 1 1.2 10 1.0 2 0.5 3 0.5
360. LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
361. REPEAT LOAD
362. 1 1.2 10 -1.0 2 0.5 3 0.5
363. LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
364. REPEAT LOAD
365. 1 1.2 11 1.0 2 0.5 3 0.5
366. LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
367. REPEAT LOAD
368. 1 1.2 11 -1.0 2 0.5 3 0.5
369. LOAD 412 0.9 DL + 1.6 WL(+Z)
370. REPEAT LOAD
371. 1 0.9 4 1.6
372. LOAD 413 0.9 DL + 1.6 WL(-Z)
373. REPEAT LOAD
374. 1 0.9 5 1.6
375. LOAD 414 0.9 DL + 1.6 WL(+X)
376. REPEAT LOAD
377. 1 0.9 6 1.6
378. LOAD 415 0.9 DL + 1.6 WL(-X)
379. REPEAT LOAD
380. 1 0.9 7 1.6
381. LOAD 416 0.9 DL + 1.0 EL(+X)
382. REPEAT LOAD
383. 1 0.9 10 1.0
384. LOAD 417 0.9 DL + 1.0 EL(-X)
385. REPEAT LOAD
386. 1 0.9 10 -1.0
387. LOAD 418 0.9 DL + 1.0 EL(+Z)
388. REPEAT LOAD
389. 1 0.9 11 1.0
390. LOAD 419 0.9 DL + 1.0 EL(-Z)
391. REPEAT LOAD
392. 1 0.9 11 -1.0
393. ****************************************
394. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**303**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**264**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**299**

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 49, TOTAL DEGREES OF FREEDOM = 921
TOTAL LOAD COMBINATION CASES = 0 SO FAR.

```
395. ****************************************
396. LOAD LIST 401 TO 419
397. START CONCRETE DESIGN
```

```
398. CODE ACI 2008
399. FC 35000 MEMB 1 TO 5
400. FYMAIN 420000 MEMB 1 TO 5
401. TRACK 1 ALL
402. CLB 0.05 MEMB 1 TO 5
403. CLT 0.05 MEMB 1 TO 5
404. CLS 0.05 ALL
405. MAXMAIN 25 ALL
406. MINMAIN 16 ALL
407. MINSEC 8 ALL
408. REINF 0 MEMB 1 TO 4
409. DESIGN COLUMN 1 TO 4
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	400.36	2.30	0.00255

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	400.36	8.98	0.00997

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4455.00   1204.46     270.4
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  485.93  -1583.37    402.34      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4260.51    834.62     195.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  337.39  -1583.37    402.34      0.00    0.00000
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     3  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 420.0  FC -  35.0 MPA,  RECT SIZE - 500.0 X 700.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3500.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-----------------------------------------------------------

 12 - 20 MM               1.077      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4455.00   1204.46     270.4
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  485.93  -1583.37    147.24      1.85    0.00557
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4260.51    834.62     195.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  337.39  -1583.37    147.24     15.40    0.04648
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     4  DESIGN PER ACI 318-08 - AXIAL + BENDING
```

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	117.60	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	117.60	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

410. DESIGN BEAM 5

BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

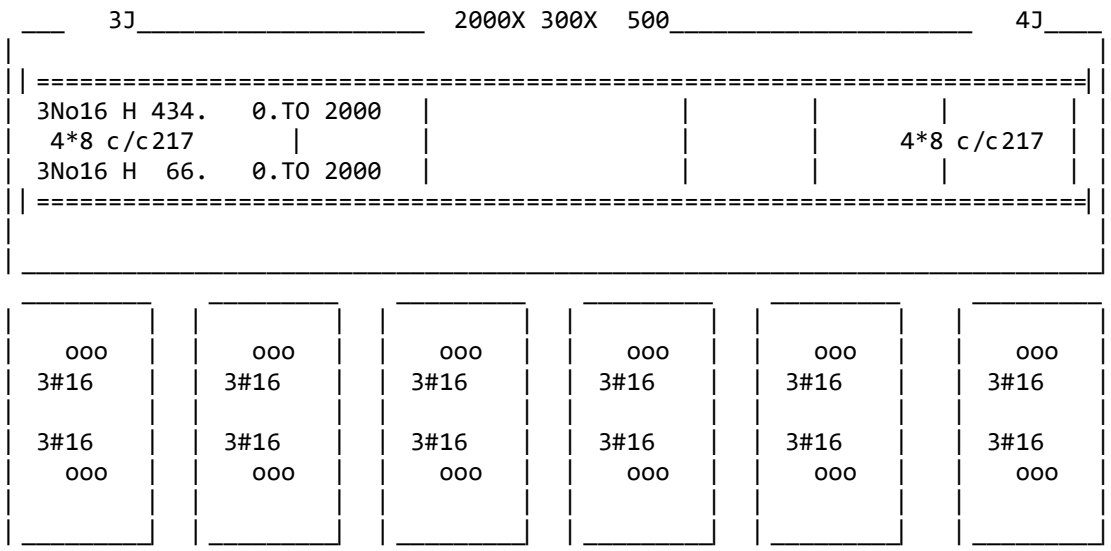
LEN - 2000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	2000.	YES	YES
	CRITICAL POS MOMENT= 57.02 KN-MET AT 0.MM, LOAD 404					
	REQD STEEL= 457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					
2	434.	3 - 16MM	0.	2000.	YES	YES
	CRITICAL NEG MOMENT= 61.21 KN-MET AT 2000.MM, LOAD 404					
	REQD STEEL= 457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 56.71 KNS Vc= 127.61 KNS Vs= 0.00 KNS
Tu= 0.26 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 217. MM C/C FOR 566. MM

AT END SUPPORT - Vu= 61.52 KNS Vc= 127.86 KNS Vs= 0.00 KNS
Tu= 0.26 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 217. MM C/C FOR 566. MM



**********END OF BEAM DESIGN**********

411. END CONCRETE DESIGN
412. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 17,2020 TIME= 17: 3:53 ****

```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
```

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01			
Date	17-04-2020	19-05-2020	02-11-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	RPK/APL	RPK/APL	RPK/APL			
Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI			

Outotec / L&T Revision No.:	0	1	2		
Date	17-04-2020	19-05-2020	02-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK/APL	RPK/APL	RPK/APL		
Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI		

ATTACHMENT 6 **STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR TR-3**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 16-Apr-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0.35 0; 2 0 0.35 2; 3 0 1.49012e-008 0; 4 0 1.49012e-008 2; 101 0 -3 0;
102 0 -3 2; 103 0 -3 -1.7; 104 0 -3 3.7; 105 1.6 -3 -1.7; 106 1.6 -3 3.7;
107 -1.6 -3 -1.7; 108 -1.6 -3 3.7; 109 -0.75 -3 3.7; 110 -0.75 -3 2.75;
111 -1.6 -3 2.75; 112 -0.5 -3 3.7; 113 -0.5 -3 2.75; 114 -0.25 -3 3.7;
115 -0.25 -3 2.75; 116 0 -3 2.75; 117 0.25 -3 3.7; 118 0.25 -3 2.75;
119 0.5 -3 3.7; 120 0.5 -3 2.75; 121 0.75 -3 3.7; 122 0.75 -3 2.75;
123 1.6 -3 2.75; 124 -0.75 -3 2.5; 125 -1.6 -3 2.5; 126 -0.5 -3 2.5;
127 -0.25 -3 2.5; 128 0 -3 2.5; 129 0.25 -3 2.5; 130 0.5 -3 2.5;
131 0.75 -3 2.5; 132 1.6 -3 2.5; 133 -0.75 -3 2.25; 134 -1.6 -3 2.25;
135 -0.5 -3 2.25; 136 -0.25 -3 2.25; 137 0 -3 2.25; 138 0.25 -3 2.25;
139 0.5 -3 2.25; 140 0.75 -3 2.25; 141 1.6 -3 2.25; 142 -0.75 -3 2;
143 -1.6 -3 2; 144 -0.5 -3 2; 145 -0.25 -3 2; 146 0.25 -3 2; 147 0.5 -3 2;
148 0.75 -3 2; 149 1.6 -3 2; 150 -0.75 -3 1.75; 151 -1.6 -3 1.75;
152 -0.5 -3 1.75; 153 -0.25 -3 1.75; 154 0 -3 1.75; 155 0.25 -3 1.75;
156 0.5 -3 1.75; 157 0.75 -3 1.75; 158 1.6 -3 1.75; 159 -0.75 -3 1.5;
160 -1.6 -3 1.5; 161 -0.5 -3 1.5; 162 -0.25 -3 1.5; 163 0 -3 1.5;
164 0.25 -3 1.5; 165 0.5 -3 1.5; 166 0.75 -3 1.5; 167 1.6 -3 1.5;
168 -0.75 -3 1.25; 169 -1.6 -3 1.25; 170 -0.5 -3 1.25; 171 -0.25 -3 1.25;
172 0 -3 1.25; 173 0.25 -3 1.25; 174 0.5 -3 1.25; 175 0.75 -3 1.25;
176 1.6 -3 1.25; 177 -0.75 -3 1; 178 -1.6 -3 1; 179 -0.5 -3 1; 180 -0.25 -3 1;
181 0 -3 1; 182 0.25 -3 1; 183 0.5 -3 1; 184 0.75 -3 1; 185 1.6 -3 1;
186 -0.75 -3 0.75; 187 -1.6 -3 0.75; 188 -0.5 -3 0.75; 189 -0.25 -3 0.75;
190 0 -3 0.75; 191 0.25 -3 0.75; 192 0.5 -3 0.75; 193 0.75 -3 0.75;
194 1.6 -3 0.75; 195 -0.75 -3 0.5; 196 -1.6 -3 0.5; 197 -0.5 -3 0.5;
198 -0.25 -3 0.5; 199 0 -3 0.5; 200 0.25 -3 0.5; 201 0.5 -3 0.5;
202 0.75 -3 0.5; 203 1.6 -3 0.5; 204 -0.75 -3 0.25; 205 -1.6 -3 0.25;
206 -0.5 -3 0.25; 207 -0.25 -3 0.25; 208 0 -3 0.25; 209 0.25 -3 0.25;
210 0.5 -3 0.25; 211 0.75 -3 0.25; 212 1.6 -3 0.25; 213 -0.75 -3 0;
214 -1.6 -3 0; 215 -0.5 -3 0; 216 -0.25 -3 0; 217 0.25 -3 0; 218 0.5 -3 0;
219 0.75 -3 0; 220 1.6 -3 0; 221 -0.75 -3 -0.25; 222 -1.6 -3 -0.25;
223 -0.5 -3 -0.25; 224 -0.25 -3 -0.25; 225 0 -3 -0.25; 226 0.25 -3 -0.25;
227 0.5 -3 -0.25; 228 0.75 -3 -0.25; 229 1.6 -3 -0.25; 230 -0.75 -3 -0.5;
231 -1.6 -3 -0.5; 232 -0.5 -3 -0.5; 233 -0.25 -3 -0.5; 234 0 -3 -0.5;
235 0.25 -3 -0.5; 236 0.5 -3 -0.5; 237 0.75 -3 -0.5; 238 1.6 -3 -0.5;
239 -0.75 -3 -0.75; 240 -1.6 -3 -0.75; 241 -0.5 -3 -0.75; 242 -0.25 -3 -0.75;
243 0 -3 -0.75; 244 0.25 -3 -0.75; 245 0.5 -3 -0.75; 246 0.75 -3 -0.75;
247 1.6 -3 -0.75; 248 -0.75 -3 -1.7; 249 -0.5 -3 -1.7; 250 -0.25 -3 -1.7;
251 0.25 -3 -1.7; 252 0.5 -3 -1.7; 253 0.75 -3 -1.7; 254 -0.75 -3 -1;
255 -1.6 -3 -1; 256 -0.75 -3 -1.25; 257 -1.6 -3 -1.25; 258 -0.5 -3 -1;
259 -0.5 -3 -1.25; 260 -0.25 -3 -1; 261 -0.25 -3 -1.25; 262 0 -3 -1;
263 0 -3 -1.25; 264 0.25 -3 -1; 265 0.25 -3 -1.25; 266 0.5 -3 -1;
267 0.5 -3 -1.25; 268 0.75 -3 -1; 269 0.75 -3 -1.25; 270 1.6 -3 -1;
271 1.6 -3 -1.25; 272 -0.75 -3 3.25; 273 -1.6 -3 3.25; 274 -0.75 -3 3;
275 -1.6 -3 3; 276 -0.5 -3 3.25; 277 -0.5 -3 3; 278 -0.25 -3 3.25;
279 -0.25 -3 3; 280 0 -3 3.25; 281 0 -3 3; 282 0.25 -3 3.25; 283 0.25 -3 3;
284 0.5 -3 3.25; 285 0.5 -3 3; 286 0.75 -3 3.25; 287 0.75 -3 3;
288 1.6 -3 3.25; 289 1.6 -3 3; 290 -0.75 -3 -1.475; 291 -1.6 -3 -1.475;
```

292 -0.5 -3 -1.475; 293 -0.25 -3 -1.475; 294 0 -3 -1.475; 295 0.25 -3 -1.475;
296 0.5 -3 -1.475; 297 0.75 -3 -1.475; 298 1.6 -3 -1.475; 299 -0.75 -3 3.475;
300 -1.6 -3 3.475; 301 -0.5 -3 3.475; 302 -0.25 -3 3.475; 303 0 -3 3.475;
304 0.25 -3 3.475; 305 0.5 -3 3.475; 306 0.75 -3 3.475; 307 1.6 -3 3.475;
308 1.03333 -3 2.75; 309 1.03333 -3 2.5; 310 1.31667 -3 2.75;
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392 -1.31667 -3 -1.475; 393 -1.03333 -3 -1.475; 394 -1.31667 -3 -1.7;
395 -1.03333 -3 -1.7; 396 -1.31667 -3 3.7; 397 -1.31667 -3 3.475;
398 -1.03333 -3 3.7; 399 -1.03333 -3 3.475;
MEMBER INCIDENCES
1 101 3; 2 3 1; 3 102 4; 4 4 2; 5 3 4;
ELEMENT INCIDENCES SHELL
15 110 113 126 124; 16 113 115 127 126; 17 115 116 128 127; 18 116 118 129 128;
19 118 120 130 129; 20 120 122 131 130; 23 124 126 135 133; 24 126 127 136 135;
25 127 128 137 136; 26 128 129 138 137; 27 129 130 139 138; 28 130 131 140 139;
31 133 135 144 142; 32 135 136 145 144; 33 136 137 102 145; 34 137 138 146 102;
35 138 139 147 146; 36 139 140 148 147; 39 142 144 152 150; 40 144 145 153 152;
41 145 102 154 153; 42 102 146 155 154; 43 146 147 156 155; 44 147 148 157 156;
47 150 152 161 159; 48 152 153 162 161; 49 153 154 163 162; 50 154 155 164 163;
51 155 156 165 164; 52 156 157 166 165; 55 159 161 170 168; 56 161 162 171 170;
57 162 163 172 171; 58 163 164 173 172; 59 164 165 174 173; 60 165 166 175 174;
63 168 170 179 177; 64 170 171 180 179; 65 171 172 181 180; 66 172 173 182 181;
67 173 174 183 182; 68 174 175 184 183; 71 177 179 188 186; 72 179 180 189 188;
73 180 181 190 189; 74 181 182 191 190; 75 182 183 192 191; 76 183 184 193 192;
79 186 188 197 195; 80 188 189 198 197; 81 189 190 199 198; 82 190 191 200 199;
83 191 192 201 200; 84 192 193 202 201; 87 195 197 206 204; 88 197 198 207 206;
89 198 199 208 207; 90 199 200 209 208; 91 200 201 210 209; 92 201 202 211 210;
95 204 206 215 213; 96 206 207 216 215; 97 207 208 101 216; 98 208 209 217 101;
99 209 210 218 217; 100 210 211 219 218; 103 213 215 223 221;
104 215 216 224 223; 105 216 101 225 224; 106 101 217 226 225;

107 217 218 227 226; 108 218 219 228 227; 111 221 223 232 230;
112 223 224 233 232; 113 224 225 234 233; 114 225 226 235 234;
115 226 227 236 235; 116 227 228 237 236; 119 230 232 241 239;
120 232 233 242 241; 121 233 234 243 242; 122 234 235 244 243;
123 235 236 245 244; 124 236 237 246 245; 137 239 241 258 254;
138 254 258 259 256; 140 241 242 260 258; 141 258 260 261 259;
143 242 243 262 260; 144 260 262 263 261; 146 243 244 264 262;
147 262 264 265 263; 149 244 245 266 264; 150 264 266 267 265;
152 245 246 268 266; 153 266 268 269 267; 162 272 276 277 274;
163 274 277 113 110; 165 276 278 279 277; 166 277 279 115 113;
168 278 280 281 279; 169 279 281 116 115; 171 280 282 283 281;
172 281 283 118 116; 174 282 284 285 283; 175 283 285 120 118;
177 284 286 287 285; 178 285 287 122 120; 184 256 259 292 290;
185 290 292 249 248; 186 259 261 293 292; 187 292 293 250 249;
188 261 263 294 293; 189 293 294 103 250; 190 263 265 295 294;
191 294 295 251 103; 192 265 267 296 295; 193 295 296 252 251;
194 267 269 297 296; 195 296 297 253 252; 200 109 112 301 299;
201 299 301 276 272; 202 112 114 302 301; 203 301 302 278 276;
204 114 104 303 302; 205 302 303 280 278; 206 104 117 304 303;
207 303 304 282 280; 208 117 119 305 304; 209 304 305 284 282;
210 119 121 306 305; 211 305 306 286 284; 214 122 308 309 131;
215 308 310 311 309; 216 310 123 132 311; 217 131 309 312 140;
218 309 311 313 312; 219 311 132 141 313; 220 140 312 314 148;
221 312 313 315 314; 222 313 141 149 315; 223 148 314 316 157;
224 314 315 317 316; 225 315 149 158 317; 226 157 316 318 166;
227 316 317 319 318; 228 317 158 167 319; 229 166 318 320 175;
230 318 319 321 320; 231 319 167 176 321; 232 175 320 322 184;
233 320 321 323 322; 234 321 176 185 323; 235 184 322 324 193;
236 322 323 325 324; 237 323 185 194 325; 238 193 324 326 202;
239 324 325 327 326; 240 325 194 203 327; 241 202 326 328 211;
242 326 327 329 328; 243 327 203 212 329; 244 211 328 330 219;
245 328 329 331 330; 246 329 212 220 331; 247 219 330 332 228;
248 330 331 333 332; 249 331 220 229 333; 250 228 332 334 237;
251 332 333 335 334; 252 333 229 238 335; 253 237 334 336 246;
254 334 335 337 336; 255 335 238 247 337; 256 246 336 338 268;
257 336 337 339 338; 258 337 247 270 339; 259 268 338 340 269;
260 338 339 341 340; 261 339 270 271 341; 262 286 342 343 287;
263 342 344 345 343; 264 344 288 289 345; 265 287 343 308 122;
266 343 345 310 308; 267 345 289 123 310; 268 269 340 346 297;
269 340 341 347 346; 270 341 271 298 347; 271 297 346 348 253;
272 346 347 349 348; 273 347 298 105 349; 274 121 350 351 306;
275 350 352 353 351; 276 352 106 307 353; 277 306 351 342 286;
278 351 353 344 342; 279 353 307 288 344; 280 111 354 355 125;
281 354 356 357 355; 282 356 110 124 357; 283 125 355 358 134;
284 355 357 359 358; 285 357 124 133 359; 286 134 358 360 143;
287 358 359 361 360; 288 359 133 142 361; 289 143 360 362 151;
290 360 361 363 362; 291 361 142 150 363; 292 151 362 364 160;
293 362 363 365 364; 294 363 150 159 365; 295 160 364 366 169;
296 364 365 367 366; 297 365 159 168 367; 298 169 366 368 178;
299 366 367 369 368; 300 367 168 177 369; 301 178 368 370 187;
302 368 369 371 370; 303 369 177 186 371; 304 187 370 372 196;
305 370 371 373 372; 306 371 186 195 373; 307 196 372 374 205;
308 372 373 375 374; 309 373 195 204 375; 310 205 374 376 214;
311 374 375 377 376; 312 375 204 213 377; 313 214 376 378 222;

```
314 376 377 379 378; 315 377 213 221 379; 316 222 378 380 231;
317 378 379 381 380; 318 379 221 230 381; 319 231 380 382 240;
320 380 381 383 382; 321 381 230 239 383; 322 240 382 384 255;
323 382 383 385 384; 324 383 239 254 385; 325 255 384 386 257;
326 384 385 387 386; 327 385 254 256 387; 328 273 388 389 275;
329 388 390 391 389; 330 390 272 274 391; 331 275 389 354 111;
332 389 391 356 354; 333 391 274 110 356; 334 257 386 392 291;
335 386 387 393 392; 336 387 256 290 393; 337 291 392 394 107;
338 392 393 395 394; 339 393 290 248 395; 340 108 396 397 300;
341 396 398 399 397; 342 398 109 299 399; 343 300 397 388 273;
344 397 399 390 388; 345 399 299 272 390;
START GROUP DEFINITION
ELEMENT
_MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
108 111 112 115 116 119 120 123 124 137 138 140 141 149 150 152 153 162 163 -
165 166 174 175 177 178 184 TO 187 192 TO 195 200 TO 203 208 TO 211 -
214 TO 345
_MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
111 TO 116 119 TO 124 137 138 140 141 143 144 146 147 149 150 152 153 162 -
163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
214 TO 219 226 TO 243 250 TO 285 292 TO 309 316 TO 345
_SHEAR 63 TO 68 71 TO 76 137 138 140 141 143 144 146 147 149 150 152 153 162 -
163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
214 TO 345
END GROUP DEFINITION
ELEMENT PROPERTY
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
178 184 TO 195 200 TO 211 214 TO 345 THICKNESS 0.55
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.7 ZD 0.5
5 PRIS YD 0.5 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
*****************************************
SUPPORTS
101 TO 399 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
*****************************************
LOAD 1 DEAD LOAD
SELFWEIGHT Y -1
ELEMENT LOAD
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
```

```
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
178 184 TO 195 200 TO 211 214 TO 345 PR GY -44.1
JOINT LOAD
1 FX 0.9 FY -245.6 FZ -0.1
2 FX 0.6 FY -94.5 FZ 13
LOAD 2 LIVE LOAD
JOINT LOAD
1 FX 0.2 FY -89.9 FZ -3.1
2 FX 0.7 FY -78.1 FZ 3.2
LOAD 3 MATERIAL LOAD
JOINT LOAD
1 FX 0.3 FY -56.8 FZ -2.2
2 FX 0.3 FY -56.4 FZ 2.2
LOAD 4 WIND LOAD +Z
JOINT LOAD
1 FX -0.9 FY 227.6 FZ 43
2 FX 0.9 FY -227.6 FZ 43.5
LOAD 5 WIND LOAD -Z
JOINT LOAD
1 FX 0.86 FY -227.62 FZ -42.96
2 FX -0.86 FY 227.62 FZ -43.55
LOAD 6 WIND LOAD (+X)
JOINT LOAD
1 FX 1.03 FY 0 FZ 0
2 FX 1.03 FY 0 FZ 0
LOAD 7 WIND LOAD (-X)
JOINT LOAD
1 FX -1.03 FY 0 FZ 0
2 FX -1.03 FY 0 FZ 0
LOAD 8 TEMP LOAD 1
JOINT LOAD
1 FX 0.26 FY -7.15 FZ -16.64
2 FX -0.26 FY 7.15 FZ 16.63
LOAD 9 TEMP LOAD 2
JOINT LOAD
1 FX -0.26 FY 7.15 FZ 16.64
2 FX 0.26 FY -7.15 FZ -16.63
LOAD 10 EARTHQUAKE LOAD +X
JOINT LOAD
1 FX 0.03 FY 0.11 FZ 0
2 FX 0.03 FY -0.28 FZ 0
LOAD 11 EARTHQUAKE LOAD +Z
JOINT LOAD
1 FX -0.01 FY 7.13 FZ 1.19
2 FX 0.01 FY -7.13 FZ 1.2
*****************************************
*****Load combination for bearing capacity check
LOAD 301 1.0 DL
REPEAT LOAD
1 1.0
LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 8 1.0 2 1.0 3 1.0
```

```
LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 9 1.0 2 1.0 3 1.0
LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 4 0.75 2 0.75 3 0.75
LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 5 0.75 2 0.75 3 0.75
LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 6 0.75 2 0.75 3 0.75
LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 7 0.75 2 0.75 3 0.75
LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 0.525 2 0.75 3 0.75
LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 0.525 2 0.75 3 0.75
LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD 312 0.6 DL + 1.0 WL(+Z)
REPEAT LOAD
1 0.6 4 1.0
LOAD 313 0.6 DL + 1.0 WL(-Z)
REPEAT LOAD
1 0.6 5 1.0
LOAD 314 0.6 DL + 1.0 WL(+X)
REPEAT LOAD
1 0.6 6 1.0
LOAD 315 0.6 DL + 1.0 WL(-X)
REPEAT LOAD
1 0.6 7 1.0
LOAD 316 0.6 DL + 0.7 EL(+X)
REPEAT LOAD
1 0.6 10 0.7
LOAD 317 0.6 DL + 0.7 EL(-X)
REPEAT LOAD
1 0.6 10 -0.7
LOAD 318 0.6 DL + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 11 0.7
LOAD 319 0.6 DL + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 11 -0.7
*****************************************
*****Load combination for foundation design
LOAD 401 1.4 DL
```

```
REPEAT LOAD
1 1.4
LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 8 1.4 2 1.7 3 1.7
LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 9 1.4 2 1.7 3 1.7
LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 4 1.6 2 0.5 3 0.5
LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 5 1.6 2 0.5 3 0.5
LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 6 1.6 2 0.5 3 0.5
LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 7 1.6 2 0.5 3 0.5
LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 1.0 2 0.5 3 0.5
LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 1.0 2 0.5 3 0.5
LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD 412 0.9 DL + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 1.6
LOAD 413 0.9 DL + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 5 1.6
LOAD 414 0.9 DL + 1.6 WL(+X)
REPEAT LOAD
1 0.9 6 1.6
LOAD 415 0.9 DL + 1.6 WL(-X)
REPEAT LOAD
1 0.9 7 1.6
LOAD 416 0.9 DL + 1.0 EL(+X)
REPEAT LOAD
1 0.9 10 1.0
LOAD 417 0.9 DL + 1.0 EL(-X)
REPEAT LOAD
1 0.9 10 -1.0
LOAD 418 0.9 DL + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 11 1.0
LOAD 419 0.9 DL + 1.0 EL(-Z)
```

```
REPEAT LOAD
1 0.9 11 -1.0
****************************************
PERFORM ANALYSIS
****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 1 TO 5
FYMAIN 420000 MEMB 1 TO 5
TRACK 1 ALL
CLB 0.05 MEMB 1 TO 5
CLT 0.05 MEMB 1 TO 5
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro CONNECT Edition            *
*           Version  21.00.01.12                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR 17, 2020                *
*           Time=    17: 7:11                    *
*                                                *
*  Licensed to: MMH KOLKATTA                     *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\F Drive\Maaden Gold\Conveyor CNV-3210-001\Civil\Documents\Staad\3210-CNV-001-TR3-Fdn.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 16-APR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 0.35 0; 2 0 0.35 2; 3 0 1.49012E-008 0; 4 0 1.49012E-008 2; 101 0 -3 0
     9. 102 0 -3 2; 103 0 -3 -1.7; 104 0 -3 3.7; 105 1.6 -3 -1.7; 106 1.6 -3 3.7
    10. 107 -1.6 -3 -1.7; 108 -1.6 -3 3.7; 109 -0.75 -3 3.7; 110 -0.75 -3 2.75
    11. 111 -1.6 -3 2.75; 112 -0.5 -3 3.7; 113 -0.5 -3 2.75; 114 -0.25 -3 3.7
    12. 115 -0.25 -3 2.75; 116 0 -3 2.75; 117 0.25 -3 3.7; 118 0.25 -3 2.75
    13. 119 0.5 -3 3.7; 120 0.5 -3 2.75; 121 0.75 -3 3.7; 122 0.75 -3 2.75
    14. 123 1.6 -3 2.75; 124 -0.75 -3 2.5; 125 -1.6 -3 2.5; 126 -0.5 -3 2.5
    15. 127 -0.25 -3 2.5; 128 0 -3 2.5; 129 0.25 -3 2.5; 130 0.5 -3 2.5
    16. 131 0.75 -3 2.5; 132 1.6 -3 2.5; 133 -0.75 -3 2.25; 134 -1.6 -3 2.25
    17. 135 -0.5 -3 2.25; 136 -0.25 -3 2.25; 137 0 -3 2.25; 138 0.25 -3 2.25
    18. 139 0.5 -3 2.25; 140 0.75 -3 2.25; 141 1.6 -3 2.25; 142 -0.75 -3 2
    19. 143 -1.6 -3 2; 144 -0.5 -3 2; 145 -0.25 -3 2; 146 0.25 -3 2; 147 0.5 -3 2
    20. 148 0.75 -3 2; 149 1.6 -3 2; 150 -0.75 -3 1.75; 151 -1.6 -3 1.75
    21. 152 -0.5 -3 1.75; 153 -0.25 -3 1.75; 154 0 -3 1.75; 155 0.25 -3 1.75
    22. 156 0.5 -3 1.75; 157 0.75 -3 1.75; 158 1.6 -3 1.75; 159 -0.75 -3 1.5
    23. 160 -1.6 -3 1.5; 161 -0.5 -3 1.5; 162 -0.25 -3 1.5; 163 0 -3 1.5
    24. 164 0.25 -3 1.5; 165 0.5 -3 1.5; 166 0.75 -3 1.5; 167 1.6 -3 1.5
    25. 168 -0.75 -3 1.25; 169 -1.6 -3 1.25; 170 -0.5 -3 1.25; 171 -0.25 -3 1.25
    26. 172 0 -3 1.25; 173 0.25 -3 1.25; 174 0.5 -3 1.25; 175 0.75 -3 1.25
    27. 176 1.6 -3 1.25; 177 -0.75 -3 1; 178 -1.6 -3 1; 179 -0.5 -3 1; 180 -0.25 -3 1
    28. 181 0 -3 1; 182 0.25 -3 1; 183 0.5 -3 1; 184 0.75 -3 1; 185 1.6 -3 1
    29. 186 -0.75 -3 0.75; 187 -1.6 -3 0.75; 188 -0.5 -3 0.75; 189 -0.25 -3 0.75
    30. 190 0 -3 0.75; 191 0.25 -3 0.75; 192 0.5 -3 0.75; 193 0.75 -3 0.75
    31. 194 1.6 -3 0.75; 195 -0.75 -3 0.5; 196 -1.6 -3 0.5; 197 -0.5 -3 0.5
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    33. 202 0.75 -3 0.5; 203 1.6 -3 0.5; 204 -0.75 -3 0.25; 205 -1.6 -3 0.25
    34. 206 -0.5 -3 0.25; 207 -0.25 -3 0.25; 208 0 -3 0.25; 209 0.25 -3 0.25
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    37. 219 0.75 -3 0; 220 1.6 -3 0; 221 -0.75 -3 -0.25; 222 -1.6 -3 -0.25
    38. 223 -0.5 -3 -0.25; 224 -0.25 -3 -0.25; 225 0 -3 -0.25; 226 0.25 -3 -0.25
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39. 227 0.5 -3 -0.25; 228 0.75 -3 -0.25; 229 1.6 -3 -0.25; 230 -0.75 -3 -0.5
40. 231 -1.6 -3 -0.5; 232 -0.5 -3 -0.5; 233 -0.25 -3 -0.5; 234 0 -3 -0.5
41. 235 0.25 -3 -0.5; 236 0.5 -3 -0.5; 237 0.75 -3 -0.5; 238 1.6 -3 -0.5
42. 239 -0.75 -3 -0.75; 240 -1.6 -3 -0.75; 241 -0.5 -3 -0.75; 242 -0.25 -3 -0.75
43. 243 0 -3 -0.75; 244 0.25 -3 -0.75; 245 0.5 -3 -0.75; 246 0.75 -3 -0.75
44. 247 1.6 -3 -0.75; 248 -0.75 -3 -1.7; 249 -0.5 -3 -1.7; 250 -0.25 -3 -1.7
45. 251 0.25 -3 -1.7; 252 0.5 -3 -1.7; 253 0.75 -3 -1.7; 254 -0.75 -3 -1.
46. 255 -1.6 -3 -1; 256 -0.75 -3 -1.25; 257 -1.6 -3 -1.25; 258 -0.5 -3 -1.
47. 259 -0.5 -3 -1.25; 260 -0.25 -3 -1; 261 -0.25 -3 -1.25; 262 0 -3 -1.
48. 263 0 -3 -1.25; 264 0.25 -3 -1; 265 0.25 -3 -1.25; 266 0.5 -3 -1.
49. 267 0.5 -3 -1.25; 268 0.75 -3 -1; 269 0.75 -3 -1.25; 270 1.6 -3 -1.
50. 271 1.6 -3 -1.25; 272 -0.75 -3 3.25; 273 -1.6 -3 3.25; 274 -0.75 -3 3
51. 275 -1.6 -3 3; 276 -0.5 -3 3.25; 277 -0.5 -3 3; 278 -0.25 -3 3.25
52. 279 -0.25 -3 3; 280 0 -3 3.25; 281 0 -3 3; 282 0.25 -3 3.25; 283 0.25 -3 3
53. 284 0.5 -3 3.25; 285 0.5 -3 3; 286 0.75 -3 3.25; 287 0.75 -3 3
54. 288 1.6 -3 3.25; 289 1.6 -3 3; 290 -0.75 -3 -1.475; 291 -1.6 -3 -1.475
55. 292 -0.5 -3 -1.475; 293 -0.25 -3 -1.475; 294 0 -3 -1.475; 295 0.25 -3 -1.475
56. 296 0.5 -3 -1.475; 297 0.75 -3 -1.475; 298 1.6 -3 -1.475; 299 -0.75 -3 3.475
57. 300 -1.6 -3 3.475; 301 -0.5 -3 3.475; 302 -0.25 -3 3.475; 303 0 -3 3.475
58. 304 0.25 -3 3.475; 305 0.5 -3 3.475; 306 0.75 -3 3.475; 307 1.6 -3 3.475
59. 308 1.03333 -3 2.75; 309 1.03333 -3 2.5; 310 1.31667 -3 2.75
60. 311 1.31667 -3 2.5; 312 1.03333 -3 2.25; 313 1.31667 -3 2.25; 314 1.03333 -3 2
61. 315 1.31667 -3 2; 316 1.03333 -3 1.75; 317 1.31667 -3 1.75; 318 1.03333 -3 1.5
62. 319 1.31667 -3 1.5; 320 1.03333 -3 1.25; 321 1.31667 -3 1.25; 322 1.03333 -3 1
63. 323 1.31667 -3 1; 324 1.03333 -3 0.75; 325 1.31667 -3 0.75; 326 1.03333 -3 0.5
64. 327 1.31667 -3 0.5; 328 1.03333 -3 0.25; 329 1.31667 -3 0.25; 330 1.03333 -3 0
65. 331 1.31667 -3 0; 332 1.03333 -3 -0.25; 333 1.31667 -3 -0.25
66. 334 1.03333 -3 -0.5; 335 1.31667 -3 -0.5; 336 1.03333 -3 -0.75
67. 337 1.31667 -3 -0.75; 338 1.03333 -3 -1; 339 1.31667 -3 -1.
68. 340 1.03333 -3 -1.25; 341 1.31667 -3 -1.25; 342 1.03333 -3 3.25
69. 343 1.03333 -3 3; 344 1.31667 -3 3.25; 345 1.31667 -3 3; 346 1.03333 -3 -1.475
70. 347 1.31667 -3 -1.475; 348 1.03333 -3 -1.7; 349 1.31667 -3 -1.7
71. 350 1.03333 -3 3.7; 351 1.03333 -3 3.475; 352 1.31667 -3 3.7
72. 353 1.31667 -3 3.475; 354 -1.31667 -3 2.75; 355 -1.31667 -3 2.5
73. 356 -1.03333 -3 2.75; 357 -1.03333 -3 2.5; 358 -1.31667 -3 2.25
74. 359 -1.03333 -3 2.25; 360 -1.31667 -3 2; 361 -1.03333 -3 2
75. 362 -1.31667 -3 1.75; 363 -1.03333 -3 1.75; 364 -1.31667 -3 1.5
76. 365 -1.03333 -3 1.5; 366 -1.31667 -3 1.25; 367 -1.03333 -3 1.25
77. 368 -1.31667 -3 1; 369 -1.03333 -3 1; 370 -1.31667 -3 0.75
78. 371 -1.03333 -3 0.75; 372 -1.31667 -3 0.5; 373 -1.03333 -3 0.5
79. 374 -1.31667 -3 0.25; 375 -1.03333 -3 0.25; 376 -1.31667 -3 0
80. 377 -1.03333 -3 0; 378 -1.31667 -3 -0.25; 379 -1.03333 -3 -0.25
81. 380 -1.31667 -3 -0.5; 381 -1.03333 -3 -0.5; 382 -1.31667 -3 -0.75
82. 383 -1.03333 -3 -0.75; 384 -1.31667 -3 -1; 385 -1.03333 -3 -1.
83. 386 -1.31667 -3 -1.25; 387 -1.03333 -3 -1.25; 388 -1.31667 -3 3.25
84. 389 -1.31667 -3 3; 390 -1.03333 -3 3.25; 391 -1.03333 -3 3
85. 392 -1.31667 -3 -1.475; 393 -1.03333 -3 -1.475; 394 -1.31667 -3 -1.7
86. 395 -1.03333 -3 -1.7; 396 -1.31667 -3 3.7; 397 -1.31667 -3 3.475
87. 398 -1.03333 -3 3.7; 399 -1.03333 -3 3.475
88. MEMBER INCIDENCES
89. 1 101 3; 2 3 1; 3 102 4; 4 4 2; 5 3 4
90. ELEMENT INCIDENCES SHELL
91. 15 110 113 126 124; 16 113 115 127 126; 17 115 116 128 127; 18 116 118 129 128
92. 19 118 120 130 129; 20 120 122 131 130; 23 124 126 135 133; 24 126 127 136 135
93. 25 127 128 137 136; 26 128 129 138 137; 27 129 130 139 138; 28 130 131 140 139
94. 31 133 135 144 142; 32 135 136 145 144; 33 136 137 102 145; 34 137 138 146 102
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95. 35 138 139 147 146; 36 139 140 148 147; 39 142 144 152 150; 40 144 145 153 152
96. 41 145 102 154 153; 42 102 146 155 154; 43 146 147 156 155; 44 147 148 157 156
97. 47 150 152 161 159; 48 152 153 162 161; 49 153 154 163 162; 50 154 155 164 163
98. 51 155 156 165 164; 52 156 157 166 165; 55 159 161 170 168; 56 161 162 171 170
99. 57 162 163 172 171; 58 163 164 173 172; 59 164 165 174 173; 60 165 166 175 174
100. 63 168 170 179 177; 64 170 171 180 179; 65 171 172 181 180; 66 172 173 182 181
101. 67 173 174 183 182; 68 174 175 184 183; 71 177 179 188 186; 72 179 180 189 188
102. 73 180 181 190 189; 74 181 182 191 190; 75 182 183 192 191; 76 183 184 193 192
103. 79 186 188 197 195; 80 188 189 198 197; 81 189 190 199 198; 82 190 191 200 199
104. 83 191 192 201 200; 84 192 193 202 201; 87 195 197 206 204; 88 197 198 207 206
105. 89 198 199 208 207; 90 199 200 209 208; 91 200 201 210 209; 92 201 202 211 210
106. 95 204 206 215 213; 96 206 207 216 215; 97 207 208 101 216; 98 208 209 217 101
107. 99 209 210 218 217; 100 210 211 219 218; 103 213 215 223 221
108. 104 215 216 224 223; 105 216 101 225 224; 106 101 217 226 225
109. 107 217 218 227 226; 108 218 219 228 227; 111 221 223 232 230
110. 112 223 224 233 232; 113 224 225 234 233; 114 225 226 235 234
111. 115 226 227 236 235; 116 227 228 237 236; 119 230 232 241 239
112. 120 232 233 242 241; 121 233 234 243 242; 122 234 235 244 243
113. 123 235 236 245 244; 124 236 237 246 245; 137 239 241 258 254
114. 138 254 258 259 256; 140 241 242 260 258; 141 258 260 261 259
115. 143 242 243 262 260; 144 260 262 263 261; 146 243 244 264 262
116. 147 262 264 265 263; 149 244 245 266 264; 150 264 266 267 265
117. 152 245 246 268 266; 153 266 268 269 267; 162 272 276 277 274
118. 163 274 277 113 110; 165 276 278 279 277; 166 277 279 115 113
119. 168 278 280 281 279; 169 279 281 116 115; 171 280 282 283 281
120. 172 281 283 118 116; 174 282 284 285 283; 175 283 285 120 118
121. 177 284 286 287 285; 178 285 287 122 120; 184 256 259 292 290
122. 185 290 292 249 248; 186 259 261 293 292; 187 292 293 250 249
123. 188 261 263 294 293; 189 293 294 103 250; 190 263 265 295 294
124. 191 294 295 251 103; 192 265 267 296 295; 193 295 296 252 251
125. 194 267 269 297 296; 195 296 297 253 252; 200 109 112 301 299
126. 201 299 301 276 272; 202 112 114 302 301; 203 301 302 278 276
127. 204 114 104 303 302; 205 302 303 280 278; 206 104 117 304 303
128. 207 303 304 282 280; 208 117 119 305 304; 209 304 305 284 282
129. 210 119 121 306 305; 211 305 306 286 284; 214 122 308 309 131
130. 215 308 310 311 309; 216 310 123 132 311; 217 131 309 312 140
131. 218 309 311 313 312; 219 311 132 141 313; 220 140 312 314 148
132. 221 312 313 315 314; 222 313 141 149 315; 223 148 314 316 157
133. 224 314 315 317 316; 225 315 149 158 317; 226 157 316 318 166
134. 227 316 317 319 318; 228 317 158 167 319; 229 166 318 320 175
135. 230 318 319 321 320; 231 319 167 176 321; 232 175 320 322 184
136. 233 320 321 323 322; 234 321 176 185 323; 235 184 322 324 193
137. 236 322 323 325 324; 237 323 185 194 325; 238 193 324 326 202
138. 239 324 325 327 326; 240 325 194 203 327; 241 202 326 328 211
139. 242 326 327 329 328; 243 327 203 212 329; 244 211 328 330 219
140. 245 328 329 331 330; 246 329 212 220 331; 247 219 330 332 228
141. 248 330 331 333 332; 249 331 220 229 333; 250 228 332 334 237
142. 251 332 333 335 334; 252 333 229 238 335; 253 237 334 336 246
143. 254 334 335 337 336; 255 335 238 247 337; 256 246 336 338 268
144. 257 336 337 339 338; 258 337 247 270 339; 259 268 338 340 269
145. 260 338 339 341 340; 261 339 270 271 341; 262 286 342 343 287
146. 263 342 344 345 343; 264 344 288 289 345; 265 287 343 308 122
147. 266 343 345 310 308; 267 345 289 123 310; 268 269 340 346 297
148. 269 340 341 347 346; 270 341 271 298 347; 271 297 346 348 253
149. 272 346 347 349 348; 273 347 298 105 349; 274 121 350 351 306
150. 275 350 352 353 351; 276 352 106 307 353; 277 306 351 342 286

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151. 278 351 353 344 342; 279 353 307 288 344; 280 111 354 355 125
152. 281 354 356 357 355; 282 356 110 124 357; 283 125 355 358 134
153. 284 355 357 359 358; 285 357 124 133 359; 286 134 358 360 143
154. 287 358 359 361 360; 288 359 133 142 361; 289 143 360 362 151
155. 290 360 361 363 362; 291 361 142 150 363; 292 151 362 364 160
156. 293 362 363 365 364; 294 363 150 159 365; 295 160 364 366 169
157. 296 364 365 367 366; 297 365 159 168 367; 298 169 366 368 178
158. 299 366 367 369 368; 300 367 168 177 369; 301 178 368 370 187
159. 302 368 369 371 370; 303 369 177 186 371; 304 187 370 372 196
160. 305 370 371 373 372; 306 371 186 195 373; 307 196 372 374 205
161. 308 372 373 375 374; 309 373 195 204 375; 310 205 374 376 214
162. 311 374 375 377 376; 312 375 204 213 377; 313 214 376 378 222
163. 314 376 377 379 378; 315 377 213 221 379; 316 222 378 380 231
164. 317 378 379 381 380; 318 379 221 230 381; 319 231 380 382 240
165. 320 380 381 383 382; 321 381 230 239 383; 322 240 382 384 255
166. 323 382 383 385 384; 324 383 239 254 385; 325 255 384 386 257
167. 326 384 385 387 386; 327 385 254 256 387; 328 273 388 389 275
168. 329 388 390 391 389; 330 390 272 274 391; 331 275 389 354 111
169. 332 389 391 356 354; 333 391 274 110 356; 334 257 386 392 291
170. 335 386 387 393 392; 336 387 256 290 393; 337 291 392 394 107
171. 338 392 393 395 394; 339 393 290 248 395; 340 108 396 397 300
172. 341 396 398 399 397; 342 398 109 299 399; 343 300 397 388 273
173. 344 397 399 390 388; 345 399 299 272 390
174. START GROUP DEFINITION
175. ELEMENT
176. _MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
177. 63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
178. 108 111 112 115 116 119 120 123 124 137 138 140 141 149 150 152 153 162 163 -
179. 165 166 174 175 177 178 184 TO 187 192 TO 195 200 TO 203 208 TO 211 -
180. 214 TO 345
181. _MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
182. 111 TO 116 119 TO 124 137 138 140 141 143 144 146 147 149 150 152 153 162 -
183. 163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
184. 214 TO 219 226 TO 243 250 TO 285 292 TO 309 316 TO 345
185. _SHEAR 63 TO 68 71 TO 76 137 138 140 141 143 144 146 147 149 150 152 153 162 -
186. 163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
187. 214 TO 345
188. END GROUP DEFINITION
189. ELEMENT PROPERTY
190. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
191. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
192. 143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
193. 178 184 TO 195 200 TO 211 214 TO 345 THICKNESS 0.55
194. DEFINE MATERIAL START
195. ISOTROPIC CONCRETE
196. E 2.17185E+007
197. POISSON 0.17
198. DENSITY 23.5616
199. ALPHA 1E-005
200. DAMP 0.05
201. TYPE CONCRETE
202. STRENGTH FCU 27579
203. END DEFINE MATERIAL
204. MEMBER PROPERTY AMERICAN
205. 1 TO 4 PRIS YD 0.7 ZD 0.5
206. 5 PRIS YD 0.5 ZD 0.3
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207. CONSTANTS
208. MATERIAL CONCRETE ALL
209. ****************************************
210. SUPPORTS
211. 101 TO 399 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
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JOINT	INFLUENCE AREA
101	0.062500
102	0.062500
103	0.028125
104	0.028125
105	0.015937
106	0.015937
107	0.015937
108	0.015937
109	0.030000
110	0.066666
111	0.035416
112	0.028125
113	0.062500
114	0.028125
115	0.062500
116	0.062500
117	0.028125
118	0.062500
119	0.028125
120	0.062500
121	0.030000
122	0.066666
123	0.035416
124	0.066666
125	0.035416
126	0.062500
127	0.062500
128	0.062500
129	0.062500
130	0.062500
131	0.066666
132	0.035416
133	0.066666
134	0.035416
135	0.062500
136	0.062500
137	0.062500
138	0.062500
139	0.062500
140	0.066666
141	0.035416
142	0.066666
143	0.035416
144	0.062500
145	0.062500
146	0.062500
147	0.062500

148	0.066666
149	0.035416
150	0.066666
151	0.035416
152	0.062500
153	0.062500
154	0.062500
155	0.062500
156	0.062500
157	0.066666
158	0.035416
159	0.066666
160	0.035416
161	0.062500
162	0.062500
163	0.062500
164	0.062500
165	0.062500
166	0.066666
167	0.035416
168	0.066666
169	0.035416
170	0.062500
171	0.062500
172	0.062500
173	0.062500
174	0.062500
175	0.066666
176	0.035416
177	0.066666
178	0.035416
179	0.062500
180	0.062500
181	0.062500
182	0.062500
183	0.062500
184	0.066666
185	0.035416
186	0.066666
187	0.035416
188	0.062500
189	0.062500
190	0.062500
191	0.062500
192	0.062500
193	0.066666
194	0.035416
195	0.066666
196	0.035416
197	0.062500
198	0.062500
199	0.062500
200	0.062500
201	0.062500
202	0.066666
203	0.035416

204	0.066666
205	0.035416
206	0.062500
207	0.062500
208	0.062500
209	0.062500
210	0.062500
211	0.066666
212	0.035416
213	0.066666
214	0.035416
215	0.062500
216	0.062500
217	0.062500
218	0.062500
219	0.066666
220	0.035416
221	0.066666
222	0.035416
223	0.062500
224	0.062500
225	0.062500
226	0.062500
227	0.062500
228	0.066666
229	0.035416
230	0.066666
231	0.035416
232	0.062500
233	0.062500
234	0.062500
235	0.062500
236	0.062500
237	0.066666
238	0.035416
239	0.066666
240	0.035416
241	0.062500
242	0.062500
243	0.062500
244	0.062500
245	0.062500
246	0.066666
247	0.035416
248	0.030000
249	0.028125
250	0.028125
251	0.028125
252	0.028125
253	0.030000
254	0.066666
255	0.035416
256	0.063333
257	0.033645
258	0.062500
259	0.059375

260	0.062500
261	0.059375
262	0.062500
263	0.059375
264	0.062500
265	0.059375
266	0.062500
267	0.059375
268	0.066666
269	0.063333
270	0.035416
271	0.033645
272	0.063333
273	0.033645
274	0.066666
275	0.035416
276	0.059375
277	0.062500
278	0.059375
279	0.062500
280	0.059375
281	0.062500
282	0.059375
283	0.062500
284	0.059375
285	0.062500
286	0.063333
287	0.066666
288	0.033645
289	0.035416
290	0.060000
291	0.031875
292	0.056250
293	0.056250
294	0.056250
295	0.056250
296	0.056250
297	0.060000
298	0.031875
299	0.060000
300	0.031875
301	0.056250
302	0.056250
303	0.056250
304	0.056250
305	0.056250
306	0.060000
307	0.031875
308	0.070834
309	0.070834
310	0.070834
311	0.070834
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314	0.070834
315	0.070834

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317	0.070834
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334	0.070834
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336	0.070834
337	0.070834
338	0.070834
339	0.070834
340	0.067292
341	0.067292
342	0.067292
343	0.070834
344	0.067292
345	0.070834
346	0.063750
347	0.063750
348	0.031875
349	0.031875
350	0.031875
351	0.063750
352	0.031875
353	0.063750
354	0.070834
355	0.070834
356	0.070834
357	0.070834
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360	0.070834
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367	0.070834
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369	0.070834
370	0.070834
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372	0.070834
373	0.070834
374	0.070834
375	0.070834
376	0.070834
377	0.070834
378	0.070834
379	0.070834
380	0.070834
381	0.070834
382	0.070834
383	0.070834
384	0.070834
385	0.070834
386	0.067292
387	0.067292
388	0.067292
389	0.070834
390	0.067292
391	0.070834
392	0.063750
393	0.063750
394	0.031875
395	0.031875
396	0.031875
397	0.063750
398	0.031875
399	0.063750
TOTAL	17.280014

```
212. ******************************************
213. LOAD 1 DEAD LOAD
214. SELFWEIGHT Y -1
215. ELEMENT LOAD
216. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
217. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
218. 143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
219. 178 184 TO 195 200 TO 211 214 TO 345 PR GY -44.1
220. JOINT LOAD
221. 1 FX 0.9 FY -245.6 FZ -0.1
222. 2 FX 0.6 FY -94.5 FZ 13
223. LOAD 2 LIVE LOAD
224. JOINT LOAD
225. 1 FX 0.2 FY -89.9 FZ -3.1
226. 2 FX 0.7 FY -78.1 FZ 3.2
227. LOAD 3 MATERIAL LOAD
228. JOINT LOAD
229. 1 FX 0.3 FY -56.8 FZ -2.2
230. 2 FX 0.3 FY -56.4 FZ 2.2
231. LOAD 4 WIND LOAD +Z
232. JOINT LOAD
233. 1 FX -0.9 FY 227.6 FZ 43
234. 2 FX 0.9 FY -227.6 FZ 43.5
235. LOAD 5 WIND LOAD -Z
236. JOINT LOAD
237. 1 FX 0.86 FY -227.62 FZ -42.96
238. 2 FX -0.86 FY 227.62 FZ -43.55
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239. LOAD 6 WIND LOAD (+X)
240. JOINT LOAD
241. 1 FX 1.03 FY 0 FZ 0
242. 2 FX 1.03 FY 0 FZ 0
243. LOAD 7 WIND LOAD (-X)
244. JOINT LOAD
245. 1 FX -1.03 FY 0 FZ 0
246. 2 FX -1.03 FY 0 FZ 0
247. LOAD 8 TEMP LOAD 1
248. JOINT LOAD
249. 1 FX 0.26 FY -7.15 FZ -16.64
250. 2 FX -0.26 FY 7.15 FZ 16.63
251. LOAD 9 TEMP LOAD 2
252. JOINT LOAD
253. 1 FX -0.26 FY 7.15 FZ 16.64
254. 2 FX 0.26 FY -7.15 FZ -16.63
255. LOAD 10 EARTHQUAKE LOAD +X
256. JOINT LOAD
257. 1 FX 0.03 FY 0.11 FZ 0
258. 2 FX 0.03 FY -0.28 FZ 0
259. LOAD 11 EARTHQUAKE LOAD +Z
260. JOINT LOAD
261. 1 FX -0.01 FY 7.13 FZ 1.19
262. 2 FX 0.01 FY -7.13 FZ 1.2
263. ****************************************
264. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
265. LOAD 301 1.0 DL
266. REPEAT LOAD
267. 1 1.0
268. LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
269. REPEAT LOAD
270. 1 1.0 8 1.0 2 1.0 3 1.0
271. LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
272. REPEAT LOAD
273. 1 1.0 9 1.0 2 1.0 3 1.0
274. LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
275. REPEAT LOAD
276. 1 1.0 4 0.75 2 0.75 3 0.75
277. LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
278. REPEAT LOAD
279. 1 1.0 5 0.75 2 0.75 3 0.75
280. LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
281. REPEAT LOAD
282. 1 1.0 6 0.75 2 0.75 3 0.75
283. LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
284. REPEAT LOAD
285. 1 1.0 7 0.75 2 0.75 3 0.75
286. LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
287. REPEAT LOAD
288. 1 1.0 10 0.525 2 0.75 3 0.75
289. LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
290. REPEAT LOAD
291. 1 1.0 10 -0.525 2 0.75 3 0.75
292. LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
293. REPEAT LOAD
294. 1 1.0 11 0.525 2 0.75 3 0.75
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295. LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
296. REPEAT LOAD
297. 1 1.0 11 -0.525 2 0.75 3 0.75
298. LOAD 312 0.6 DL + 1.0 WL(+Z)
299. REPEAT LOAD
300. 1 0.6 4 1.0
301. LOAD 313 0.6 DL + 1.0 WL(-Z)
302. REPEAT LOAD
303. 1 0.6 5 1.0
304. LOAD 314 0.6 DL + 1.0 WL(+X)
305. REPEAT LOAD
306. 1 0.6 6 1.0
307. LOAD 315 0.6 DL + 1.0 WL(-X)
308. REPEAT LOAD
309. 1 0.6 7 1.0
310. LOAD 316 0.6 DL + 0.7 EL(+X)
311. REPEAT LOAD
312. 1 0.6 10 0.7
313. LOAD 317 0.6 DL + 0.7 EL(-X)
314. REPEAT LOAD
315. 1 0.6 10 -0.7
316. LOAD 318 0.6 DL + 0.7 EL(+Z)
317. REPEAT LOAD
318. 1 0.6 11 0.7
319. LOAD 319 0.6 DL + 0.7 EL(-Z)
320. REPEAT LOAD
321. 1 0.6 11 -0.7
322. ****************************************
323. *****LOAD COMBINATION FOR FOUNDATION DESIGN
324. LOAD 401 1.4 DL
325. REPEAT LOAD
326. 1 1.4
327. LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
328. REPEAT LOAD
329. 1 1.4 8 1.4 2 1.7 3 1.7
330. LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
331. REPEAT LOAD
332. 1 1.4 9 1.4 2 1.7 3 1.7
333. LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
334. REPEAT LOAD
335. 1 1.2 4 1.6 2 0.5 3 0.5
336. LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
337. REPEAT LOAD
338. 1 1.2 5 1.6 2 0.5 3 0.5
339. LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
340. REPEAT LOAD
341. 1 1.2 6 1.6 2 0.5 3 0.5
342. LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
343. REPEAT LOAD
344. 1 1.2 7 1.6 2 0.5 3 0.5
345. LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
346. REPEAT LOAD
347. 1 1.2 10 1.0 2 0.5 3 0.5
348. LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
349. REPEAT LOAD
350. 1 1.2 10 -1.0 2 0.5 3 0.5
```

```
351. LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
352. REPEAT LOAD
353. 1 1.2 11 1.0 2 0.5 3 0.5
354. LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
355. REPEAT LOAD
356. 1 1.2 11 -1.0 2 0.5 3 0.5
357. LOAD 412 0.9 DL + 1.6 WL(+Z)
358. REPEAT LOAD
359. 1 0.9 4 1.6
360. LOAD 413 0.9 DL + 1.6 WL(-Z)
361. REPEAT LOAD
362. 1 0.9 5 1.6
363. LOAD 414 0.9 DL + 1.6 WL(+X)
364. REPEAT LOAD
365. 1 0.9 6 1.6
366. LOAD 415 0.9 DL + 1.6 WL(-X)
367. REPEAT LOAD
368. 1 0.9 7 1.6
369. LOAD 416 0.9 DL + 1.0 EL(+X)
370. REPEAT LOAD
371. 1 0.9 10 1.0
372. LOAD 417 0.9 DL + 1.0 EL(-X)
373. REPEAT LOAD
374. 1 0.9 10 -1.0
375. LOAD 418 0.9 DL + 1.0 EL(+Z)
376. REPEAT LOAD
377. 1 0.9 11 1.0
378. LOAD 419 0.9 DL + 1.0 EL(-Z)
379. REPEAT LOAD
380. 1 0.9 11 -1.0
381. ****************************************
382. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**303**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**264**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**299**

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 49, TOTAL DEGREES OF FREEDOM = 921
TOTAL LOAD COMBINATION CASES = 0 SO FAR.

```
383. ****************************************
384. LOAD LIST 401 TO 419
```

385. START CONCRETE DESIGN

```
386. CODE ACI 2008
387. FC 35000 MEMB 1 TO 5
388. FYMAIN 420000 MEMB 1 TO 5
389. TRACK 1 ALL
390. CLB 0.05 MEMB 1 TO 5
391. CLT 0.05 MEMB 1 TO 5
392. CLS 0.05 ALL
393. MAXMAIN 25 ALL
394. MINMAIN 16 ALL
395. MINSEC 8 ALL
396. REINF 0 MEMB 1 TO 4
397. DESIGN COLUMN 1 TO 4
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	517.52	0.60	0.00039

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	517.52	20.02	0.01290

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4455.00   1204.46     270.4
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  485.93  -1583.37    528.98      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4260.51    834.62     195.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  337.39  -1583.37    528.98      0.00    0.00000
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     3  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 420.0  FC -  35.0 MPA,  RECT SIZE - 500.0 X 700.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3500.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-----------------------------------------------------------

 12 - 20 MM               1.077      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4455.00   1204.46     270.4
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  485.93  -1583.37    242.66      0.53    0.00073
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4260.51    834.62     195.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  337.39  -1583.37    242.66     20.83    0.02861
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     4  DESIGN PER ACI 318-08 - AXIAL + BENDING
```

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	203.54	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	203.54	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

398. DESIGN BEAM 5

BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

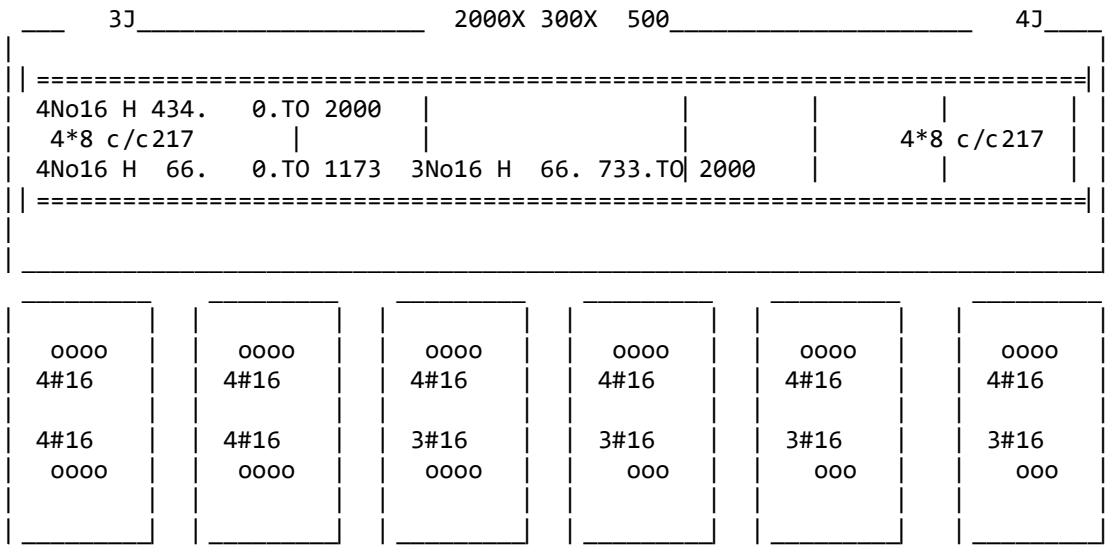
LEN - 2000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	4 - 16MM	0.	1173.	YES	NO
	CRITICAL POS MOMENT= 105.40 KN-MET AT 0.MM, LOAD 404					
	REQD STEEL= 667.MM2, RHO=0.0051, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 56. MMS					
	REQD. DEVELOPMENT LENGTH = 586. MMS					
	Cracked Moment of Inertia Iz at above location = 73606.0 cm^4					
2	66.	3 - 16MM	733.	2000.	NO	YES
	CRITICAL POS MOMENT= 82.08 KN-MET AT 2000.MM, LOAD 413					
	REQD STEEL= 515.MM2, RHO=0.0040, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					
3	434.	4 - 16MM	0.	2000.	YES	YES
	CRITICAL NEG MOMENT= 112.65 KN-MET AT 2000.MM, LOAD 404					
	REQD STEEL= 714.MM2, RHO=0.0055, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 56. MMS					
	REQD. DEVELOPMENT LENGTH = 586. MMS					
	Cracked Moment of Inertia Iz at above location = 73606.0 cm^4					

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 106.63 KNS Vc= 126.02 KNS Vs= 16.14 KNS
Tu= 0.34 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 217. MM C/C FOR 566. MM

AT END SUPPORT - Vu= 111.43 KNS Vc= 126.02 KNS Vs= 22.55 KNS
Tu= 0.34 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 217. MM C/C FOR 566. MM



**********END OF BEAM DESIGN**********

399. END CONCRETE DESIGN
400. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 17,2020 TIME= 17: 7:15 ****

```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
```

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01			
Date	17-04-2020	19-05-2020	02-11-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	RPK/APL	RPK/APL	RPK/APL			
Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI			

Outotec / L&T Revision No.:	0	1	2		
Date	17-04-2020	19-05-2020	02-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK/APL	RPK/APL	RPK/APL		
Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI		

ATTACHMENT 7 **STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR VGTU**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 16-Apr-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART GROUND PORTAL FOUNDATIONS
JOB REF FOUNDATIONS FOR CONVEYOR 3130-CNV-003
ENGINEER NAME RPK
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 1 -3.75; 2 4 1 -3.75; 3 0 1 0; 4 4 1 0; 5 0 1 4.5; 6 4 1 4.5;
11 0 -3 -3.75; 12 4 -3 -3.75; 13 0 -3 0; 14 0 -3 4.5; 15 4 -3 0; 16 4 -3 4.5;
105 0 0.35 0; 106 0 0.35 4.5; 107 4 0.35 0; 108 4 0.35 4.5; 114 0 0.35 -3.75;
117 4 0.35 -3.75;
MEMBER INCIDENCES
1 13 105; 2 105 3; 3 14 106; 4 106 5; 5 15 107; 6 107 4; 7 16 108; 8 108 6;
13 105 106; 14 107 108; 16 105 107; 18 106 108; 19 11 114; 20 114 1; 21 12 117;
22 117 2; 23 114 117; 24 114 105; 25 117 107;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
13 14 24 25 PRIS YD 0.5 ZD 0.3
MEMBER PROPERTY AMERICAN
16 18 23 PRIS YD 0.6 ZD 0.3
MEMBER PROPERTY AMERICAN
1 TO 8 19 TO 22 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
11 TO 16 FIXED
*****************************************
LOAD 1 DEAD LOAD
SELFWEIGHT Y -1
JOINT LOAD
1 FX -0.3 FY -33.1 FZ -0.1
2 FX 0.3 FY -31.5 FZ 0
3 FX -20.4 FY -251.3 FZ -10.6
4 FX -19 FY -94.5 FZ -2.6
5 FX -21.1 FY -254.9 FZ 15.7
6 FX -19.5 FY -70.5 FZ 9.8
```

```
LOAD 2 LIVE LOAD
JOINT LOAD
1 FX 0 FY -59.2 FZ -0.1
2 FX 0 FY -60.9 FZ 0
3 FX -1 FY -182.5 FZ -10.3
4 FX 0 FY -190.9 FZ -10.5
5 FX 0 FY -164.3 FZ 9.3
6 FX 0.9 FY -191.6 FZ 11.5
LOAD 3 MATERIAL LOAD
JOINT LOAD
1 FX 0 FY 0 FZ 0
2 FX 0 FY 0 FZ 0
3 FX -0.1 FY -21.5 FZ -1.3
4 FX -0.1 FY -32.3 FZ -1.8
5 FX 0.1 FY -19.8 FZ 1.2
6 FX 0.1 FY -34.1 FZ 2
LOAD 4 WIND LOAD +Z
JOINT LOAD
1 FX 0 FY 0.4 FZ 0.6
2 FX 0 FY 0.3 FZ 0.9
3 FX 1.5 FY 191.2 FZ 43.4
4 FX 0.7 FY 202.9 FZ 48.6
5 FX -1.5 FY -191.6 FZ 43.4
6 FX -0.7 FY -203.2 FZ 49.4
LOAD 5 WIND LOAD -Z
JOINT LOAD
1 FX 0 FY -0.35 FZ -0.62
2 FX 0 FY -0.28 FZ -0.93
3 FX -1.5 FY -191.22 FZ -43.39
4 FX -0.69 FY -202.89 FZ -48.63
5 FX 1.5 FY 191.57 FZ -43.38
6 FX 0.68 FY 203.18 FZ -49.41
LOAD 6 WIND LOAD (+X)
JOINT LOAD
1 FX 13.89 FY 43.8 FZ 0.08
2 FX 13.88 FY -43.94 FZ -0.02
3 FX 50.64 FY 231.95 FZ 9.3
4 FX 51.05 FY -231.68 FZ -8.61
5 FX 53.42 FY 260.54 FZ -12.25
6 FX 53.59 FY -260.66 FZ 11.51
LOAD 7 WIND LOAD (-X)
JOINT LOAD
1 FX -13.89 FY -43.8 FZ -0.08
2 FX -13.88 FY 43.94 FZ 0.02
3 FX -50.64 FY -231.95 FZ -9.3
4 FX -51.05 FY 231.68 FZ 8.61
5 FX -53.42 FY -260.54 FZ 12.25
6 FX -53.59 FY 260.66 FZ -11.51
LOAD 8 TEMP LOAD 1
JOINT LOAD
1 FX -0.06 FY -0.01 FZ -2.65
2 FX 0.04 FY 0.03 FZ -1.42
3 FX -0.28 FY 10.09 FZ -34.35
4 FX 2.27 FY -7.11 FZ -38.04
```

```
5 FX -2.17 FY -10.08 FZ 39.06
6 FX 0.19 FY 7.08 FZ 37.38
LOAD 9 TEMP LOAD 2
JOINT LOAD
1 FX 0.06 FY 0.01 FZ 2.65
2 FX -0.04 FY -0.03 FZ 1.42
3 FX 0.28 FY -10.09 FZ 34.35
4 FX -2.27 FY 7.11 FZ 38.04
5 FX 2.17 FY 10.08 FZ -39.06
6 FX -0.19 FY -7.08 FZ -37.38
LOAD 10 EARTHQUAKE LOAD +X
JOINT LOAD
1 FX 0.07 FY 0.31 FZ 0
2 FX 0.07 FY -0.32 FZ 0
3 FX 1.79 FY 8.71 FZ 0.34
4 FX 1.82 FY -8.46 FZ -0.29
5 FX 1.65 FY 8.69 FZ -0.43
6 FX 1.67 FY -8.77 FZ 0.41
LOAD 11 EARTHQUAKE LOAD +Z
JOINT LOAD
1 FX 0 FY 0.02 FZ -0.01
2 FX 0 FY 0.02 FZ -0.01
3 FX 0.08 FY 9.11 FZ 1.71
4 FX 0.04 FY 9 FZ 1.99
5 FX -0.08 FY -9.12 FZ 1.72
6 FX -0.04 FY -9.02 FZ 2
*****************************************
*****Load combination for bearing capacity check
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 8 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 9 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 4 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 5 0.75 2 0.75 3 0.75
LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
1 1.0 6 0.75 2 0.75 3 0.75
LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
1 1.0 7 0.75 2 0.75 3 0.75
LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
1 1.0 10 0.525 2 0.75 3 0.75
LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
1 1.0 11 0.525 2 0.75 3 0.75
LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
1 0.6 4 1.0
LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
1 0.6 5 1.0
```

```
LOAD COMB 314 0.6 DL + 1.0 WL(+X)
1 0.6 6 1.0
LOAD COMB 315 0.6 DL + 1.0 WL(-X)
1 0.6 7 1.0
LOAD COMB 316 0.6 DL + 0.7 EL(+X)
1 0.6 10 0.7
LOAD COMB 317 0.6 DL + 0.7 EL(-X)
1 0.6 10 -0.7
LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
1 0.6 11 0.7
LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
1 0.6 11 -0.7
*****************************************
*****Load combination for foundation design
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 8 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 9 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
1 1.2 4 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
1 1.2 5 1.6 2 0.5 3 0.5
LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
1 1.2 6 1.6 2 0.5 3 0.5
LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
1 1.2 7 1.6 2 0.5 3 0.5
LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
1 1.2 10 1.0 2 0.5 3 0.5
LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
1 1.2 11 1.0 2 0.5 3 0.5
LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
1 0.9 4 1.6
LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
1 0.9 5 1.6
LOAD COMB 414 0.9 DL + 1.6 WL(+X)
1 0.9 6 1.6
LOAD COMB 415 0.9 DL + 1.6 WL(-X)
1 0.9 7 1.6
LOAD COMB 416 0.9 DL + 1.0 EL(+X)
1 0.9 10 1.0
LOAD COMB 417 0.9 DL + 1.0 EL(-X)
1 0.9 10 -1.0
LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
1 0.9 11 1.0
LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
1 0.9 11 -1.0
*****************************************
PERFORM ANALYSIS
```

```
LOAD LIST 301 TO 319
PRINT SUPPORT REACTION ALL
**************************************************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 1 TO 8 13 14 16 18 TO 23
FYMAIN 420000 MEMB 1 TO 8 13 14 16 18 TO 23
TRACK 1 MEMB 1 TO 8 13 14 16 18 TO 23
CLB 0.05 MEMB 1 TO 8 13 14 16 18 TO 23
CLT 0.05 MEMB 1 TO 8 13 14 16 18 TO 23
CLS 0.05 MEMB 1 TO 8 13 14 16 18 TO 23
MAXMAIN 25 MEMB 1 TO 8 13 14 16 18 TO 23
MINMAIN 16 MEMB 1 TO 8 13 14 16 18 TO 23
MINSEC 8 MEMB 1 TO 8 13 14 16 18 TO 23
REINF 0 MEMB 1 TO 8 19 TO 22
DESIGN COLUMN 1 TO 8 19 TO 22
DESIGN BEAM 13 14 16 18 23
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro CONNECT Edition            *
*           Version  21.00.01.12                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR 17, 2020                *
*           Time=    17: 8:52                    *
*                                                *
*  Licensed to: MMH KOLKATTA                     *
**************************************************
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```
     1. STAAD SPACE
INPUT FILE: D:\F Drive\Maaden Gold\Conveyor CNV-3210-001\Civil\Documents\Staad\3210-CNV-001-VGTU-Fdn.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 16-APR-20
     4. JOB NAME MMGP-SAUDI ARABIA
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045/900707
     7. JOB REV 0
     8. JOB PART GROUND PORTAL FOUNDATIONS
     9. JOB REF FOUNDATIONS FOR CONVEYOR 3130-CNV-003
    10. ENGINEER NAME RPK
    11. CHECKER NAME SBY
    12. APPROVED NAME PRI
    13. END JOB INFORMATION
    14. INPUT WIDTH 79
    15. UNIT METER KN
    16. JOINT COORDINATES
    17. 1 0 1 -3.75; 2 4 1 -3.75; 3 0 1 0; 4 4 1 0; 5 0 1 4.5; 6 4 1 4.5
    18. 11 0 -3 -3.75; 12 4 -3 -3.75; 13 0 -3 0; 14 0 -3 4.5; 15 4 -3 0; 16 4 -3 4.5
    19. 105 0 0.35 0; 106 0 0.35 4.5; 107 4 0.35 0; 108 4 0.35 4.5; 114 0 0.35 -3.75
    20. 117 4 0.35 -3.75
    21. MEMBER INCIDENCES
    22. 1 13 105; 2 105 3; 3 14 106; 4 106 5; 5 15 107; 6 107 4; 7 16 108; 8 108 6
    23. 13 105 106; 14 107 108; 16 105 107; 18 106 108; 19 11 114; 20 114 1; 21 12 117
    24. 22 117 2; 23 114 117; 24 114 105; 25 117 107
    25. DEFINE MATERIAL START
    26. ISOTROPIC CONCRETE
    27. E 2.17185E+007
    28. POISSON 0.17
    29. DENSITY 23.5616
    30. ALPHA 1E-005
    31. DAMP 0.05
    32. TYPE CONCRETE
    33. STRENGTH FCU 27579
    34. END DEFINE MATERIAL
    35. MEMBER PROPERTY AMERICAN
    36. 13 14 24 25 PRIS YD 0.5 ZD 0.3
    37. MEMBER PROPERTY AMERICAN
    38. 16 18 23 PRIS YD 0.6 ZD 0.3
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39. MEMBER PROPERTY AMERICAN
40. 1 TO 8 19 TO 22 PRIS YD 0.6 ZD 0.6
41. CONSTANTS
42. MATERIAL CONCRETE ALL
43. SUPPORTS
44. 11 TO 16 FIXED
45. ****************************************
46. LOAD 1 DEAD LOAD
47. SELFWEIGHT Y -1
48. JOINT LOAD
49. 1 FX -0.3 FY -33.1 FZ -0.1
50. 2 FX 0.3 FY -31.5 FZ 0
51. 3 FX -20.4 FY -251.3 FZ -10.6
52. 4 FX -19 FY -94.5 FZ -2.6
53. 5 FX -21.1 FY -254.9 FZ 15.7
54. 6 FX -19.5 FY -70.5 FZ 9.8
55. LOAD 2 LIVE LOAD
56. JOINT LOAD
57. 1 FX 0 FY -59.2 FZ -0.1
58. 2 FX 0 FY -60.9 FZ 0
59. 3 FX -1 FY -182.5 FZ -10.3
60. 4 FX 0 FY -190.9 FZ -10.5
61. 5 FX 0 FY -164.3 FZ 9.3
62. 6 FX 0.9 FY -191.6 FZ 11.5
63. LOAD 3 MATERIAL LOAD
64. JOINT LOAD
65. 1 FX 0 FY 0 FZ 0
66. 2 FX 0 FY 0 FZ 0
67. 3 FX -0.1 FY -21.5 FZ -1.3
68. 4 FX -0.1 FY -32.3 FZ -1.8
69. 5 FX 0.1 FY -19.8 FZ 1.2
70. 6 FX 0.1 FY -34.1 FZ 2
71. LOAD 4 WIND LOAD +Z
72. JOINT LOAD
73. 1 FX 0 FY 0.4 FZ 0.6
74. 2 FX 0 FY 0.3 FZ 0.9
75. 3 FX 1.5 FY 191.2 FZ 43.4
76. 4 FX 0.7 FY 202.9 FZ 48.6
77. 5 FX -1.5 FY -191.6 FZ 43.4
78. 6 FX -0.7 FY -203.2 FZ 49.4
79. LOAD 5 WIND LOAD -Z
80. JOINT LOAD
81. 1 FX 0 FY -0.35 FZ -0.62
82. 2 FX 0 FY -0.28 FZ -0.93
83. 3 FX -1.5 FY -191.22 FZ -43.39
84. 4 FX -0.69 FY -202.89 FZ -48.63
85. 5 FX 1.5 FY 191.57 FZ -43.38
86. 6 FX 0.68 FY 203.18 FZ -49.41
87. LOAD 6 WIND LOAD (+X)
88. JOINT LOAD
89. 1 FX 13.89 FY 43.8 FZ 0.08
90. 2 FX 13.88 FY -43.94 FZ -0.02
91. 3 FX 50.64 FY 231.95 FZ 9.3
92. 4 FX 51.05 FY -231.68 FZ -8.61
93. 5 FX 53.42 FY 260.54 FZ -12.25
94. 6 FX 53.59 FY -260.66 FZ 11.51
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 95. LOAD 7 WIND LOAD (-X)
 96. JOINT LOAD
 97. 1 FX -13.89 FY -43.8 FZ -0.08
 98. 2 FX -13.88 FY 43.94 FZ 0.02
 99. 3 FX -50.64 FY -231.95 FZ -9.3
100. 4 FX -51.05 FY 231.68 FZ 8.61
101. 5 FX -53.42 FY -260.54 FZ 12.25
102. 6 FX -53.59 FY 260.66 FZ -11.51
103. LOAD 8 TEMP LOAD 1
104. JOINT LOAD
105. 1 FX -0.06 FY -0.01 FZ -2.65
106. 2 FX 0.04 FY 0.03 FZ -1.42
107. 3 FX -0.28 FY 10.09 FZ -34.35
108. 4 FX 2.27 FY -7.11 FZ -38.04
109. 5 FX -2.17 FY -10.08 FZ 39.06
110. 6 FX 0.19 FY 7.08 FZ 37.38
111. LOAD 9 TEMP LOAD 2
112. JOINT LOAD
113. 1 FX 0.06 FY 0.01 FZ 2.65
114. 2 FX -0.04 FY -0.03 FZ 1.42
115. 3 FX 0.28 FY -10.09 FZ 34.35
116. 4 FX -2.27 FY 7.11 FZ 38.04
117. 5 FX 2.17 FY 10.08 FZ -39.06
118. 6 FX -0.19 FY -7.08 FZ -37.38
119. LOAD 10 EARTHQUAKE LOAD +X
120. JOINT LOAD
121. 1 FX 0.07 FY 0.31 FZ 0
122. 2 FX 0.07 FY -0.32 FZ 0
123. 3 FX 1.79 FY 8.71 FZ 0.34
124. 4 FX 1.82 FY -8.46 FZ -0.29
125. 5 FX 1.65 FY 8.69 FZ -0.43
126. 6 FX 1.67 FY -8.77 FZ 0.41
127. LOAD 11 EARTHQUAKE LOAD +Z
128. JOINT LOAD
129. 1 FX 0 FY 0.02 FZ -0.01
130. 2 FX 0 FY 0.02 FZ -0.01
131. 3 FX 0.08 FY 9.11 FZ 1.71
132. 4 FX 0.04 FY 9 FZ 1.99
133. 5 FX -0.08 FY -9.12 FZ 1.72
134. 6 FX -0.04 FY -9.02 FZ 2
135. ****************************************
136. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
137. LOAD COMB 301 1.0 DL
138. 1 1.0
139. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
140. 1 1.0 8 1.0 2 1.0 3 1.0
141. LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
142. 1 1.0 9 1.0 2 1.0 3 1.0
143. LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
144. 1 1.0 4 0.75 2 0.75 3 0.75
145. LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
146. 1 1.0 5 0.75 2 0.75 3 0.75
147. LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
148. 1 1.0 6 0.75 2 0.75 3 0.75
149. LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
150. 1 1.0 7 0.75 2 0.75 3 0.75
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151. LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
152. 1 1.0 10 0.525 2 0.75 3 0.75
153. LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
154. 1 1.0 10 -0.525 2 0.75 3 0.75
155. LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
156. 1 1.0 11 0.525 2 0.75 3 0.75
157. LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
158. 1 1.0 11 -0.525 2 0.75 3 0.75
159. LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
160. 1 0.6 4 1.0
161. LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
162. 1 0.6 5 1.0
163. LOAD COMB 314 0.6 DL + 1.0 WL(+X)
164. 1 0.6 6 1.0
165. LOAD COMB 315 0.6 DL + 1.0 WL(-X)
166. 1 0.6 7 1.0
167. LOAD COMB 316 0.6 DL + 0.7 EL(+X)
168. 1 0.6 10 0.7
169. LOAD COMB 317 0.6 DL + 0.7 EL(-X)
170. 1 0.6 10 -0.7
171. LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
172. 1 0.6 11 0.7
173. LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
174. 1 0.6 11 -0.7
175. ****************************************
176. *****LOAD COMBINATION FOR FOUNDATION DESIGN
177. LOAD COMB 401 1.4 DL
178. 1 1.4
179. LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
180. 1 1.4 8 1.4 2 1.7 3 1.7
181. LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
182. 1 1.4 9 1.4 2 1.7 3 1.7
183. LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
184. 1 1.2 4 1.6 2 0.5 3 0.5
185. LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
186. 1 1.2 5 1.6 2 0.5 3 0.5
187. LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
188. 1 1.2 6 1.6 2 0.5 3 0.5
189. LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
190. 1 1.2 7 1.6 2 0.5 3 0.5
191. LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
192. 1 1.2 10 1.0 2 0.5 3 0.5
193. LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
194. 1 1.2 10 -1.0 2 0.5 3 0.5
195. LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
196. 1 1.2 11 1.0 2 0.5 3 0.5
197. LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
198. 1 1.2 11 -1.0 2 0.5 3 0.5
199. LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
200. 1 0.9 4 1.6
201. LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
202. 1 0.9 5 1.6
203. LOAD COMB 414 0.9 DL + 1.6 WL(+X)
204. 1 0.9 6 1.6
205. LOAD COMB 415 0.9 DL + 1.6 WL(-X)
206. 1 0.9 7 1.6
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207. LOAD COMB 416 0.9 DL + 1.0 EL(+X)
208. 1 0.9 10 1.0
209. LOAD COMB 417 0.9 DL + 1.0 EL(-X)
210. 1 0.9 10 -1.0
211. LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
212. 1 0.9 11 1.0
213. LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
214. 1 0.9 11 -1.0
215. ****************************************
216. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**18**	**NUMBER OF MEMBERS**	**19**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**6**

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 11, TOTAL DEGREES OF FREEDOM = 72
TOTAL LOAD COMBINATION CASES = 38 SO FAR.

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217. LOAD LIST 301 TO 319
218. PRINT SUPPORT REACTION ALL
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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
11	301	3.88	83.90	0.60	-1.86	2.18	-6.66
	302	3.84	145.66	0.86	-2.38	2.08	-6.56
	303	3.98	143.00	1.92	0.32	2.58	-6.86
	304	3.72	114.96	-23.60	-58.98	2.55	-6.29
	305	4.09	143.44	25.98	56.51	2.03	-7.11
	306	-9.37	84.88	0.76	-1.39	-0.51	25.13
	307	17.18	173.56	1.63	-1.08	5.09	-38.52
	308	3.77	128.93	1.17	-1.25	2.20	-6.39
	309	4.03	129.52	1.21	-1.22	2.38	-7.01
	310	3.90	128.82	0.50	-2.83	2.30	-6.68
	311	3.91	129.62	1.88	0.36	2.28	-6.71
	312	2.08	31.32	-32.69	-78.12	1.65	-3.45
	313	2.58	69.30	33.41	75.88	0.96	-4.54
	314	-15.37	-8.78	-0.22	-1.33	-2.42	38.44
	315	20.03	109.46	0.94	-0.91	5.04	-46.43
	316	2.16	49.94	0.33	-1.14	1.18	-3.58
	317	2.51	50.73	0.38	-1.09	1.43	-4.41
	318	2.32	49.80	-0.56	-3.24	1.32	-3.97
	319	2.34	50.87	1.28	1.01	1.30	-4.01
12	301	0.02	77.47	-1.40	-5.54	2.18	-2.61
	302	0.02	141.42	-0.23	-4.58	2.07	-2.54
	303	0.09	137.21	-2.29	-8.08	2.58	-2.78
	304	-0.15	108.57	-28.69	-69.86	2.55	-2.24
	305	0.23	139.14	26.11	57.63	2.02	-3.06
	306	-13.23	168.27	-0.86	-5.98	-0.51	29.18
	307	13.32	79.44	-1.73	-6.28	5.09	-34.47
	308	-0.09	124.15	-1.28	-6.12	2.19	-2.34
	309	0.18	123.56	-1.31	-6.13	2.38	-2.96
	310	0.04	123.42	-2.07	-7.93	2.30	-2.63
	311	0.05	124.29	-0.52	-4.33	2.28	-2.66
	312	-0.24	26.10	-37.37	-88.29	1.65	-1.01
	313	0.27	66.86	35.70	81.69	0.96	-2.12
	314	-17.69	105.70	-0.26	-3.12	-2.43	40.87
	315	17.71	-12.74	-1.42	-3.52	5.04	-44.00
	316	-0.16	46.88	-0.82	-3.32	1.18	-1.15
	317	0.19	46.09	-0.86	-3.33	1.43	-1.98
	318	0.00	45.90	-1.88	-5.72	1.32	-1.54
	319	0.02	47.06	0.20	-0.93	1.30	-1.58
13	301	19.54	321.24	-3.47	-6.22	2.03	-44.86
	302	18.89	511.16	-13.94	-18.31	2.00	-43.64
	303	20.92	537.20	2.91	1.56	2.14	-47.89
	304	19.26	333.16	-30.68	-67.69	2.37	-44.17
	305	20.37	613.74	20.67	52.02	1.75	-46.91
	306	-15.57	272.75	-3.71	-6.10	-0.68	40.36
	307	55.20	674.15	-6.30	-9.57	4.80	-131.44
	308	18.97	468.24	-4.98	-7.80	1.98	-43.48

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	309	20.66	478.65	-5.03	-7.87	2.14	-47.60
	310	19.79	468.75	-5.71	-9.48	2.07	-45.49
	311	19.84	478.14	-4.30	-6.19	2.05	-45.59
	312	10.98	5.70	-36.32	-83.54	1.63	-25.08
	313	12.46	379.80	32.15	76.07	0.81	-28.74
	314	-35.46	-74.86	-0.36	-1.42	-2.43	87.62
	315	58.91	460.34	-3.81	-6.05	4.87	-141.45
	316	10.60	185.80	-2.05	-3.69	1.11	-24.17
	317	12.86	199.68	-2.12	-3.77	1.33	-29.67
	318	11.70	186.48	-3.02	-5.92	1.24	-26.84
	319	11.75	199.00	-1.15	-1.54	1.21	-26.99
14	301	22.18	322.17	-0.78	-4.21	0.20	-51.15
	302	22.19	517.99	8.72	1.61	0.21	-51.57
	303	21.08	494.61	-5.97	-6.85	0.23	-48.01
	304	22.52	614.81	-15.09	-53.44	0.71	-51.92
	305	21.04	305.75	16.76	47.40	-0.29	-48.36
	306	-18.29	233.12	-0.66	-3.66	0.22	46.95
	307	61.83	687.42	2.33	-2.38	0.21	-147.21
	308	20.89	455.01	0.80	-3.04	0.22	-48.00
	309	22.65	465.52	0.87	-3.00	0.20	-52.26
	310	21.80	465.35	0.40	-4.40	0.23	-50.20
	311	21.74	455.18	1.27	-1.64	0.20	-50.06
	312	14.30	399.36	-21.70	-69.75	0.78	-33.07
	313	12.32	-12.71	20.77	64.70	-0.55	-28.34
	314	-40.11	-109.56	-2.46	-3.38	0.12	98.75
	315	66.72	496.17	1.52	-1.68	0.12	-160.13
	316	12.13	186.29	-0.52	-2.56	0.13	-27.85
	317	14.48	200.31	-0.42	-2.50	0.10	-33.54
	318	13.34	200.09	-1.04	-4.37	0.14	-30.78
	319	13.27	186.52	0.11	-0.69	0.09	-30.60
15	301	15.85	137.66	-4.04	-8.38	2.03	-40.96
	302	15.94	363.94	-14.69	-20.05	1.99	-40.45
	303	16.90	353.82	0.44	-5.18	2.14	-43.69
	304	15.60	155.84	-34.70	-77.60	2.37	-40.31
	305	16.95	451.30	22.01	54.53	1.75	-43.27
	306	-19.05	504.11	-7.54	-13.17	-0.68	44.05
	307	51.60	103.04	-5.17	-9.94	4.80	-127.63
	308	15.43	308.65	-6.38	-11.60	1.98	-39.73
	309	17.12	298.50	-6.32	-11.51	2.14	-43.85
	310	16.25	298.98	-7.15	-13.41	2.07	-41.74
	311	16.30	308.17	-5.56	-9.70	2.05	-41.85
	312	8.60	-114.39	-40.22	-93.09	1.63	-22.59
	313	10.41	279.56	35.40	83.08	0.81	-26.55
	314	-37.59	349.98	-4.00	-7.19	-2.43	89.88
	315	56.61	-184.79	-0.84	-2.87	4.87	-139.03
	316	8.38	89.36	-2.46	-5.09	1.11	-21.83

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	317	10.63	75.83	-2.38	-4.98	1.33	-27.32
	318	9.47	76.47	-3.48	-7.50	1.23	-24.50
	319	9.54	88.72	-1.37	-2.56	1.21	-24.66
16	301	18.53	106.15	-3.11	-7.78	0.20	-47.30
	302	19.24	325.50	7.09	-0.72	0.21	-48.38
	303	17.12	339.84	-9.22	-14.11	0.23	-43.86
	304	19.13	439.07	-19.16	-63.18	0.71	-48.32
	305	17.40	113.02	16.03	48.20	-0.28	-44.53
	306	-21.76	503.27	-0.19	-6.95	0.22	50.64
	307	58.30	48.81	-2.95	-8.06	0.21	-143.47
	308	17.39	281.34	-1.54	-7.50	0.22	-44.29
	309	19.14	270.74	-1.60	-7.51	0.20	-48.55
	310	18.30	281.07	-2.06	-9.07	0.23	-46.49
	311	18.23	271.01	-1.08	-5.94	0.20	-46.34
	312	12.27	281.06	-25.32	-78.90	0.78	-30.92
	313	9.96	-153.67	21.61	69.61	-0.55	-25.86
	314	-42.26	366.66	-0.03	-3.93	0.12	101.02
	315	64.49	-239.28	-3.70	-5.40	0.11	-157.78
	316	9.95	70.76	-1.82	-4.66	0.13	-25.54
	317	12.28	56.62	-1.91	-4.68	0.10	-31.22
	318	11.16	70.39	-2.52	-6.75	0.14	-28.48
	319	11.07	56.99	-1.21	-2.58	0.09	-28.28

********** END OF LATEST ANALYSIS RESULT **********

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219. ******************************************************************************
220. LOAD LIST 401 TO 419
221. START CONCRETE DESIGN
222. CODE ACI 2008
223. FC 35000 MEMB 1 TO 8 13 14 16 18 TO 23
224. FYMAIN 420000 MEMB 1 TO 8 13 14 16 18 TO 23
225. TRACK 1 MEMB 1 TO 8 13 14 16 18 TO 23
226. CLB 0.05 MEMB 1 TO 8 13 14 16 18 TO 23
227. CLT 0.05 MEMB 1 TO 8 13 14 16 18 TO 23
228. CLS 0.05 MEMB 1 TO 8 13 14 16 18 TO 23
229. MAXMAIN 25 MEMB 1 TO 8 13 14 16 18 TO 23
230. MINMAIN 16 MEMB 1 TO 8 13 14 16 18 TO 23
231. MINSEC 8 MEMB 1 TO 8 13 14 16 18 TO 23
232. REINF 0 MEMB 1 TO 8 19 TO 22
233. DESIGN COLUMN 1 TO 8 19 TO 22
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COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	630.69	44.39	0.02101

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	630.69	11.66	0.00552

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    541.26      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    541.26      0.00    0.00000
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     3  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
---------------------------------------------------------

 12 - 20 MM               1.047      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    632.71     49.83    0.02351
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    632.71      3.45    0.00163
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     4  DESIGN PER ACI 318-08 - AXIAL + BENDING
```

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	549.02	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	549.02	0.00	0.00000

==========

COLUMN NO. 5 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	235.29	26.12	0.03313

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    235.29     11.07    0.01404
 ---------------------------------------------------------
```

==========

COLUMN NO. 6 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
---------------------------------------------------------

 12 - 20 MM               1.047      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    203.54      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    203.54      0.00    0.00000
 ---------------------------------------------------------
```

==========

COLUMN NO. 7 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM
```

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	167.43	31.79	0.05667

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	167.43	5.68	0.01013

==========

COLUMN NO. 8 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	151.85	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	151.85	0.00	0.00000

==========

COLUMN NO. 19 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	119.50	13.69	0.03420

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	119.50	8.33	0.02082

==========

COLUMN NO. 20 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37     71.29      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37     71.29      0.00    0.00000
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.    21  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-----------------------------------------------------------

 12 - 20 MM               1.047      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    105.65      5.44    0.01538
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    105.65      1.83    0.00516
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.    22  DESIGN PER ACI 318-08 - AXIAL + BENDING
```

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	67.85	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	67.85	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

234. DESIGN BEAM 13 14 16 18 23

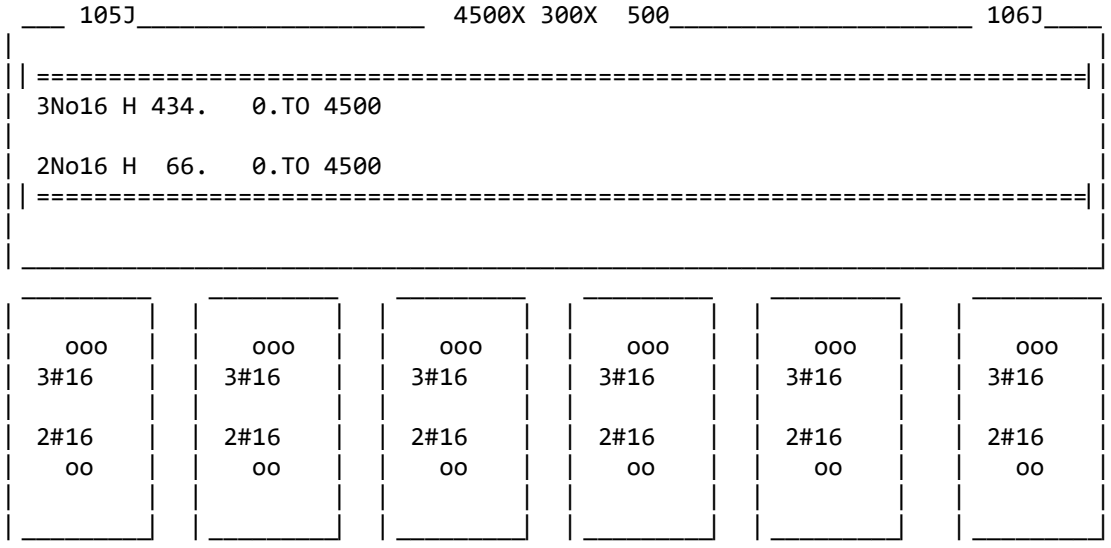
BEAM NO. 13 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	4500.	YES	YES
	CRITICAL POS MOMENT= 44.33 KN-MET AT 4500.MM, LOAD 413					
	REQD STEEL= 365.MM2, RHO=0.0028, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					
2	434.	3 - 16MM	0.	4500.	YES	YES
	CRITICAL NEG MOMENT= 62.99 KN-MET AT 4500.MM, LOAD 404					
	REQD STEEL= 457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					

B E A M N O. 13 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 7.70 KNS Vc= 122.73 KNS Vs= 0.00 KNS
Tu= 0.22 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 10.27 KNS Vc= 122.73 KNS Vs= 0.00 KNS
Tu= 0.22 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 14 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	4500.	YES	YES

CRITICAL POS MOMENT= 49.56 KN-MET AT 4500.MM, LOAD 413
REQD STEEL= 409.MM2, RHO=0.0031, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 58050.0 cm^4

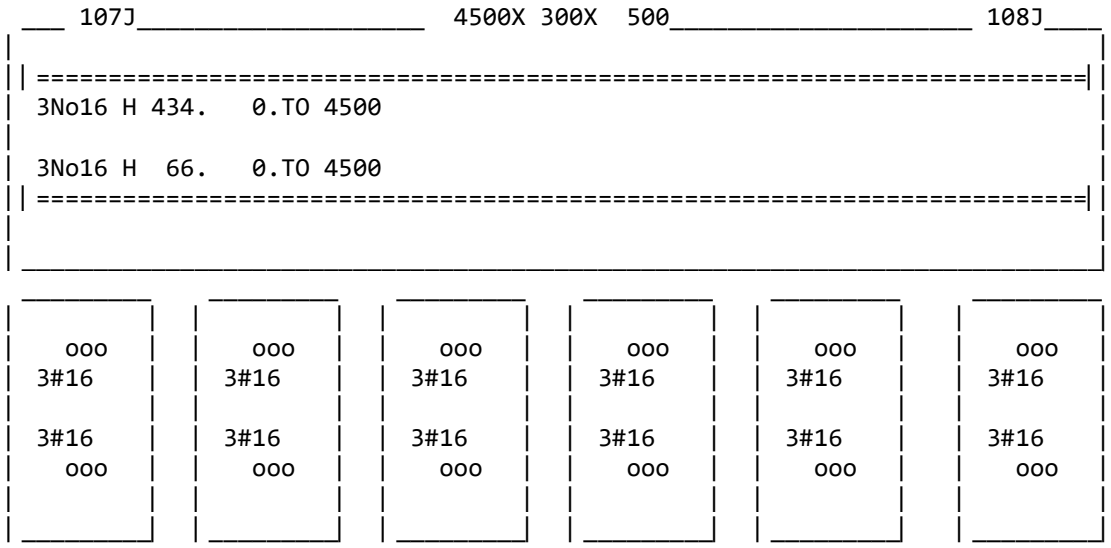
2	434.	3 - 16MM	0.	4500.	YES	YES

CRITICAL NEG MOMENT= 69.61 KN-MET AT 4500.MM, LOAD 404
REQD STEEL= 457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 58050.0 cm^4

B E A M N O. 14 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 7.19 KNS Vc= 125.62 KNS Vs= 0.00 KNS
Tu= 0.22 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 10.78 KNS Vc= 125.62 KNS Vs= 0.00 KNS
Tu= 0.22 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 16 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	4 - 16MM	0.	4000.	YES	YES
	CRITICAL POS MOMENT= 147.91 KN-MET AT 4000.MM, LOAD 407					
	REQD STEEL= 758.MM2, RHO=0.0047, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 56. MMS					
	REQD. DEVELOPMENT LENGTH = 586. MMS					
	Cracked Moment of Inertia Iz at above location = 115622.7 cm^4					
2	532.	3 - 20MM	0.	2334.	YES	NO

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=    159.15 KN-MET  AT     0.MM, LOAD  407|
|   REQD STEEL=   821.MM2, RHO=0.0051, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   45./   82. MMS         |
|   REQD. DEVELOPMENT LENGTH =   626. MMS                       |
|-------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 130657.1 cm^4

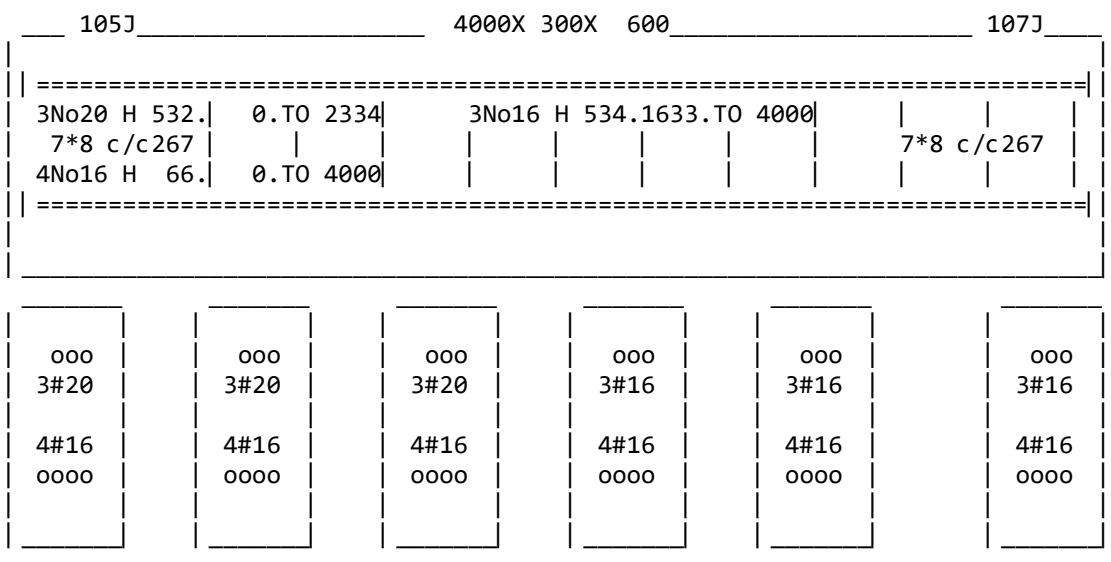
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  3        534.     3 - 16MM      1633.         4000.        NO   YES
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     98.27 KN-MET  AT  4000.MM, LOAD  414|
|   REQD STEEL=   562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./   84. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 90761.5 cm^4

B E A M N O. 16 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 84.22 KNS Vc= 154.72 KNS Vs= 0.00 KNS
Tu= 0.41 KN-MET Tc= 6.6 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 267. MM C/C FOR 1466. MM

AT END SUPPORT - Vu= 69.30 KNS Vc= 153.87 KNS Vs= 0.00 KNS
Tu= 0.41 KN-MET Tc= 6.6 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 267. MM C/C FOR 1466. MM



BEAM NO. 18 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

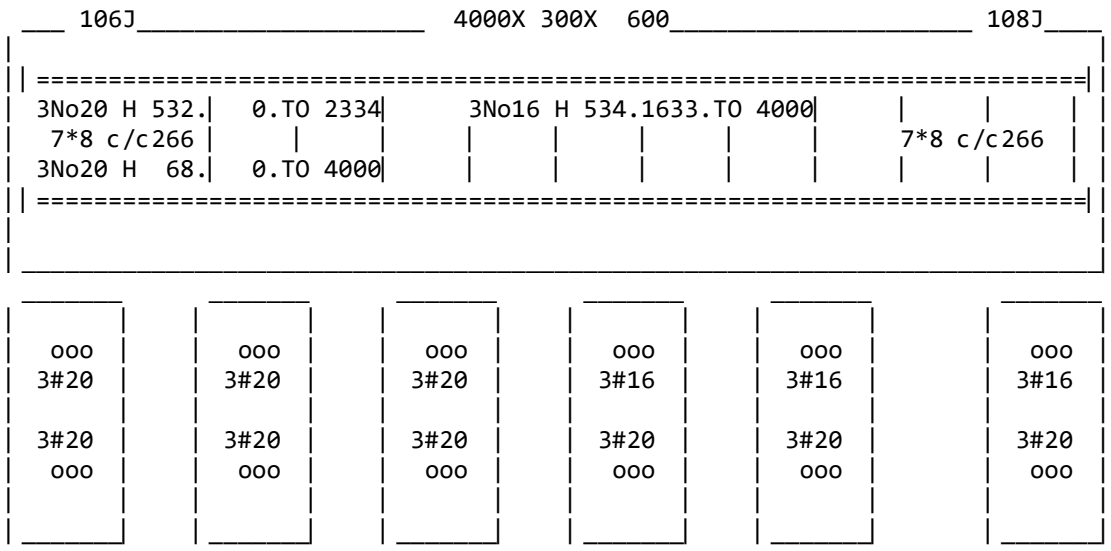
LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	68.	3 - 20MM	0.	4000.	YES	YES
	CRITICAL POS MOMENT= 166.64 KN-MET AT 4000.MM, LOAD 407					
	REQD STEEL= 861.MM2, RHO=0.0054, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 45./ 82. MMS					
	REQD. DEVELOPMENT LENGTH = 626. MMS					
	Cracked Moment of Inertia Iz at above location = 130657.1 cm^4					
2	532.	3 - 20MM	0.	2334.	YES	NO
	CRITICAL NEG MOMENT= 177.96 KN-MET AT 0.MM, LOAD 407					
	REQD STEEL= 923.MM2, RHO=0.0058, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 45./ 82. MMS					
	REQD. DEVELOPMENT LENGTH = 626. MMS					
	Cracked Moment of Inertia Iz at above location = 130657.1 cm^4					
3	534.	3 - 16MM	1633.	4000.	NO	YES
	CRITICAL NEG MOMENT= 109.64 KN-MET AT 4000.MM, LOAD 414					
	REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90761.5 cm^4					

B E A M N O. 18 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 93.61 KNS Vc= 154.78 KNS Vs= 0.00 KNS
Tu= 0.93 KN-MET Tc= 6.6 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 266. MM C/C FOR 1466. MM

AT END SUPPORT - Vu= 78.69 KNS Vc= 153.94 KNS Vs= 0.00 KNS
Tu= 0.93 KN-MET Tc= 6.6 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 266. MM C/C FOR 1466. MM



BEAM NO. 23 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	4000.	YES	YES

CRITICAL POS MOMENT= 43.66 KN-MET AT 4000.MM, LOAD 415
REQD STEEL= 290.MM2, RHO=0.0018, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 63898.6 cm^4

2	534.	2 - 16MM	0.	2200.	YES	NO

CRITICAL NEG MOMENT= 54.36 KN-MET AT 0.MM, LOAD 407
REQD STEEL= 363.MM2, RHO=0.0023, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 63898.6 cm^4

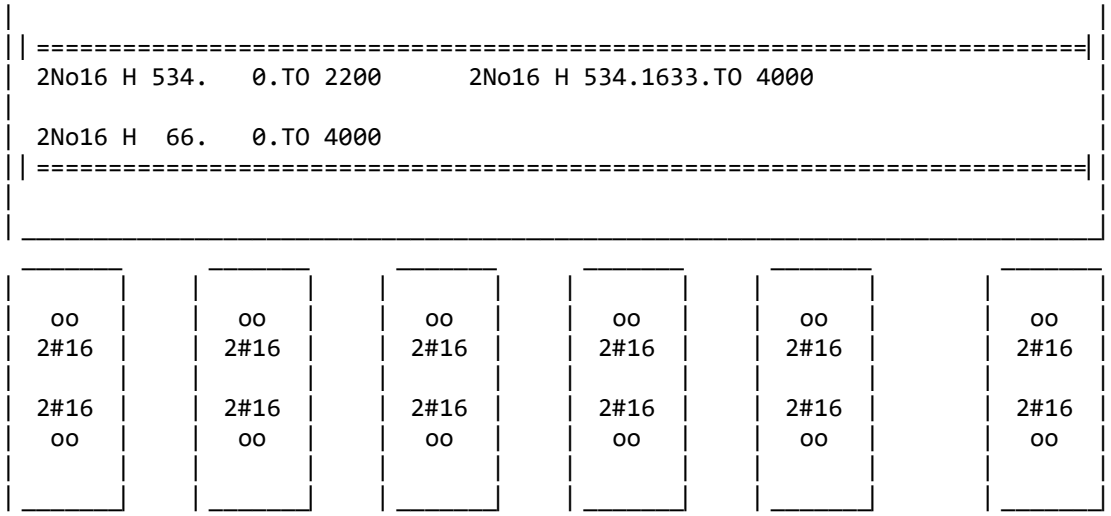
3	534.	2 - 16MM	1633.	4000.	NO	YES

CRITICAL NEG MOMENT= 46.87 KN-MET AT 4000.MM, LOAD 406
REQD STEEL= 312.MM2, RHO=0.0019, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 63898.6 cm^4

B E A M N O. 23 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 10.87 KNS Vc= 157.31 KNS Vs= 0.00 KNS
Tu= 0.06 KN-MET Tc= 6.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 6.53 KNS Vc= 160.69 KNS Vs= 0.00 KNS
Tu= 0.06 KN-MET Tc= 6.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

235. END CONCRETE DESIGN
236. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 17,2020 TIME= 17: 8:53 ****

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* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
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